

# MOLECULAR TAXONOMY AND NATURAL HISTORY COLLECTIONS

DICK GROENENBERG

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# **Molecular Taxonomy and Natural History Collections**

Dirk Schilman Jakob Groenenberg

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GROENENBERG, DIRK SCHILMAN JAKOB

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# **Molecular Taxonomy and Natural History Collections**

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*Voor mijn opa  
en zij die mij dierbaar zijn.*



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## INLEIDING EN SAMENVATTING

Dit proefschrift is de reflectie van een enigszins onconventieel promotie-traject. In dit boekje zullen diverse diergroepen de revue passeren, in de meerderheid van de gevallen om een beter beeld te krijgen van de kleinste voor natuurhistorische collecties relevante eenheden. Hierbij zal het enerzijds gaan om duiding van operationele taxonomische eenheden (OTU's - binnen natuurhistorische collecties gaat het dan vrijwel altijd om identificatie van soorten) en anderzijds om de systematische positie van deze eenheden binnen een cladistische of fylogenetische context (wat zijn zuster-soorten, tot welk genus (geslacht) behoort een soort, uit hoeveel soorten bestaat een genus, of welke genera worden aan een bepaalde familie toegeschreven?).

Van oudsher worden soorten onderscheiden en beschreven aan de hand van uiterlijke kenmerken (*habitus* en *anatomie*). Echter, met de komst van nieuwe moleculair biologische technieken zoals PCR (polymerase ketting reactie) en Sanger sequenzen (een methode om de basenvolgorde van het DNA te bepalen) werd het ook mogelijk om direct het genotype te bestuderen. Dit proefschrift geeft een reeks van voorbeelden waarbij hypothesen op basis van morfologische kenmerken, werden getoetst met additionele datasets verkregen door middel van deze nieuwe technieken.

De moleculaire data in dit proefschrift werden primair verkregen om het morfologisch bewijs te verifiëren, maar leverden vaak ook additionele ondersteuning voor identificaties. Zoals verwacht blijkt er vaak een grote mate van overeenkomst te bestaan tussen de morfologisch en genetisch onderscheiden groepen. Zo nu en dan laten de resultaten van het moleculaire onderzoek echter meer diversiteit zien dan in eerste instantie door de specialist werd herkend (hoofdstuk 2). In die gevallen gaat het meestal om cryptische soorten en vormt het verkregen inzicht aanleiding om te zoeken naar nieuwe morfologische kenmerken, of om in gebruik zijnde kenmerken te herwaardenen. Anderzijds komt het ook voor dat specialisten, op basis van morfologische verschillen, taxa onderverdelen, terwijl er geen verschil in afstamming blijkt te bestaan. Genetisch zullen deze groepen of soorten dan niet van elkaar te verschillen (hoofdstuk 4 en 5).

Ogenschijnlijke discrepanties tussen morfologische determinaties en groepen die genetisch worden onderscheiden, kunnen helaas het gevolg zijn van menselijke fouten in het traject van het verzamelen van het materiaal tot aan het publiceren van sequenties in publieke databases zoals GenBank (hoofdstuk 2 en 3). Om deze reden is het belangrijk dat er altijd een terugkoppeling plaatsvindt tussen de moleculaire uitkomsten en het verwachtingspatroon op basis van de morfologie, alvorens de gegevens publiekelijk beschikbaar te stellen. Hierbij is terughoudendheid niet alleen geboden voor de onderzoeker die de data beschikbaar stelt, maar ook voor diegene die de data uit deze bronnen voor eigen onderzoek wil gebruiken (bijvoorbeeld voor de plaatsing van taxa in een ruimere, fylogenetische context).



In de hoofdstukken 2 en 3 worden voorbeelden van fouten in GenBank besproken. Hoofdstuk 2 beschrijft de moleculaire systematiek van een onderfamilie van onder andere in de Noordzee voorkomende dwerginktvissen, de Sepiolinae. Op GenBank zijn binnen deze groep verschillende voorbeelden te vinden van COI sequenties (*Cytochrome Oxidase subunit I* - een veelgebruikte merker voor barcoding studies: [www.boldsystems.org](http://www.boldsystems.org)), waarvan de vouchers (het collectiemateriaal dat zowel gebruikt werd om de soort te determineren als voor de DNA-extractie) als dezelfde soort werden aangemerkt, terwijl de sequenties aanzienlijk verschilden (Tabel 3, blz. 45), en ook fylogentisch niet dicht bij elkaar zaten (bijvoorbeeld de vertegenwoordigers van *Sepiola affinis* of *Sepiola ligulata*, Fig. 5, blz. 44). Indien voldoende sequenties van een betreffende soort beschikbaar zijn (bij voorkeur een combinatie van zowel sequenties uit eigen onderzoek als uit publieke databases), kan het op basis van de consensus duidelijk zijn in welke gevallen de vouchers waarschijnlijk onjuist gedetermineerd werden (vergelijk ter illustratie sequentie DQ646730 van '*Euprymna tasmanica*' Fig. 4, blz. 40/41 of sequentie AY293722 van *Sepietta 'neglecta'* met de overige vertegenwoordigers van die betreffende soorten). Het in hoofdstuk 2 beschreven onderzoek bracht tevens een nog onbeschreven, cryptische *Sepiola* soort aan het licht, die bij publicatie in het *Journal of Molluscan studies* gemakshalve werd aangeduid als *Sepiola* sp. nov. Deze soort werd door de Heij & Goud (2010) beschreven als *Sepiola tridens*.

Hoofdstuk 3 beschrijft de rectificatie van een studie gepubliceerd in het tijdschrift MPE (*Molecular Phylogenetics and Evolution*), waar onderzoek werd gedaan naar de moleculaire fylogenie van de West-Palearctische Helicidae (een familie van op het land levende huisjesslakken, waartoe ook de 'gewone' tuinslak, *Cepaea nemoralis* behoort). Ondanks het feit dat het zojuist genoemde tijdschrift peer-review (ook wel collegiale toetsing) toepast, bleken er op tal van plaatsen ernstige fouten in de studie te zijn gemaakt. Een zgn. Blast-search op basis van sequenties uit de betreffende publicatie, bracht een aantal determinatiefouten of andere vergissingen aan het licht. Omdat niet kan worden aangenomen dat soorten als *Trochoidea pyramidata* en *Otala punctata*, alsmede *Oestophora turriplana* en *Candidula codia* nauw aan elkaar verwant zijn. Inspectie van het voucher-materiaal toonde tevens andere fouten; de voucher voor *Caracollina lenticula* bleek een juveniel exemplaar van *Helicigona lapicda andorrica* en de voucher voor *Chilostoma cingulatum* bleek tot *Marmorana signata* te behoren. Op basis van door ons gedetermineerd en gesequenced materiaal en sequenties afkomstig uit betrouwbare studies, konden diverse fouten in GenBank worden aangetoond. Hierbij ging het niet alleen om onjuist gedetermineerde vouchers, maar ook om verwisselingen (of contaminatie) die in het lab, of bij het analyseren van de data, dan wel bij het deponeren van de sequenties, moeten zijn opgetreden. Naast de rectificatie van een malacologisch (weekdierkundig) artikel (zodat anderen de betreffende publicatie op haar merites kunnen beoordelen), stelt dit hoofdstuk ook de kwaliteitswaarborg van publieke databases als GenBank ter discussie. De kracht van een publieke database (een schijnbaar onuitputtelijke bron van informatie) is ook gelijk haar zwakte (wie controleert die bron?). Maatregelen



om de betrouwbaarheid van data in GenBank beter te waarborgen zoals opname van vouchers (of foto's daarvan indien de exemplaren voor de analyses volledig vernietigd dienden te worden) en DNA-extracten in natuurhistorische collecties, zijn op zijn minst wenselijk. Het grootste probleem met aantoonbare fouten (zie hoofdstuk 2 en 3) in GenBank-bestanden is dat deze alleen gecorrigeerd kunnen worden door de oorspronkelijke auteurs. Rectificatie van de betreffende gegevens is dus niet mogelijk zonder hun medewerking. Om deze reden pleiten sommigen voor 'wikificatie' van GenBank. Hierdoor zouden andere onderzoekers in staat moeten zijn om gegevens te corrigeren of te annoteren, uiteraard niet anoniem. Daarbij moet de oorspronkelijke, onjuiste gegevens, wel beschikbaar blijven om daarop mogelijk gebaseerde vergissingen te kunnen achterhalen.

In tegenstelling tot het in hoofdstuk 2 beschreven voorbeeld van cryptische soorten zien we in de praktijk ook voorbeelden van situaties waarbij men op basis van veelal subtiele morfologische verschillen nauwverwante 'soorten' meent te kunnen onderscheiden, die genetisch niet als zodanig worden herkend. Hoofdstuk 4 beschrijft een voorbeeld van een mosselsoort, *Mytilus edulis*, die in Nederland commercieel gekweekt wordt. In de Oosterschelde zijn van deze soort van dezelfde vindplaats zowel exemplaren met een puntige schelp als exemplaren met een brede schelp bekend. Hierbij tonen exemplaren met een brede schelp een grote morfologisch overeenkomst met *Mytilus galloprovincialis*, een soort die met name in het Mediterrane gebied sterk vertegenwoordigd is. Naar aanleiding van deze overeenkomst en op basis van het voorkomen van vergelijkbare schelpfragmenten uit het Eemien (ca. 113.00 tot 114.000 jaar geleden) in de Noordzee, rees de vraag of deze brede vorm niet toegeschreven kon worden aan *M. galloprovincialis*. Om dit te testen werden zowel schelp-morfologische kenmerken (Lengte/Breedte ratio's en Verduin-paramters) als moleculaire merkers (lengte polymorfismen van Me 15/16, COIII sequenties en AFLP's) met elkaar vergeleken. Omdat de vormverschillen ook een uiting van fenotypische plasticiteit (het vermogen van een organisme, dan wel het genotype, om uiterlijke kenmerken, dan wel het fenotype, aan te passen als reactie op zijn milieu of omgeving) kan zijn, is er tevens gekeken naar de verdeling van puntige en brede schelpen over verschillende diepten (rekening houdend met de mogelijkheid dat op verschillende diepten het leefmilieu ook zou kunnen verschillen). Een verband tussen de schelpvorm en de diepte kon echter niet worden aangetoond, evenmin als een verband tussen de (geringe) genetische verschillen (die 'an sich' al duiden op een zeer monomorfe populatie) en de schelpvorm, dan wel diepte. Op basis van deze resultaten hebben we geen bewijs kunnen vinden voor de aanwezigheid van meer dan één *Mytilus* soort in de Oosterschelde.

Een vergelijkbaar vraagstuk (één of twee soorten?) wordt beschreven in hoofdstuk 5. Hier wordt getest of twee vermeende zandbij-soorten (*Andrena rosae* en *Andrena stragulata*), met verschillende vluchtperiodes (respectievelijk zomer en lente) en met kleine morfologische verschillen, waaronder een sterker ontwikkelde tand (Fig. 5, blz. 114) bij mannetjes van de voorjaars-generatie, ook moleculair te onderscheiden zijn.



*Andrena rosae*, de roodrandzandbij, is in Nederland altijd al een vrij zeldzame soort geweest, maar de laatste decennia is het aantal waarnemingen sterk achteruit gegaan. Inmiddels is het een ernstig bedreigde soort, die opgenomen is in de Nederlandse Rode Lijst voor bijen. *Andrena rosae* komt in Nederland nagenoeg uitsluitend in en rondom de Biesbosch (een nationaal park op de grens van Zuid-Holland en Noord-Brabant) voor. Naar schatting heeft slechts een vijfde van de Nederlandse *Andrena* soorten een bivoltiene levenscyclus, d.w.z. twee generaties per jaar (atlas van de Nederlandse bijen, in prep.). Om inzicht te krijgen in de genetische variatie binnen *Andrena rosae* s.l. zijn een aantal mitochondriale merkers, *COI*, *COII* & *Cytochrome B* (*Cyt B*) en een kern-merker, *Elongation Factor 1 alpha* (*EF-1a*) gesequenced. Op basis daarvan konden de zomer- en de voorjaarsgeneratie niet van elkaar onderscheiden worden. Er werd dus geen bewijs gevonden voor de hypothese dat *A. rosae* en *A. stragulata* twee univoltiene (één generatie per jaar) soorten zouden zijn. Dus is de jongste naam, *Andrena stragulata* Illiger 1806, synoniem voor *Andrena rosae* Panzer 1801.

Met name wanneer taxa weinig van elkaar verschillen en het aantal bruikbare kenmerken beperkt is, kan moleculair onderzoek een oplossing bieden. Dit is bijvoorbeeld het geval bij de Ariantinae (hoofdstuk 6), een subfamilie van de landslakkenfamilie Helicidae (hoofdstuk 3). De Ariantinae komen voor in het West-Palearctisch gebied, waarbij de grootste radiatie is te vinden in het Mediterrane deel van Zuid-Europa. Op een aantal, nader te noemen interessante uitzonderingen na, is het een vrij monomorfe groep slakken (de huisjes verschillen niet veel van elkaar; Fig. 1, blz.122/123), hetgeen heeft geresulteerd in enkele grote geslachten (zoals *Campylaea*, *Chilostoma* en *Helicigona*) met een schijnbaar arbitraire hoeveelheid subgenera. Van de soorten die louter op schelp-morfologie te onderscheiden zijn, is *Cylindrus obtusus*, een Oostenrijks endem (een soort die alleen in het betreffende gebied voorkomt) het meest excentriek. *Cylindrus obtusus* (Fig. 1-3, p. 122) is de enige soort binnen de Ariantinae met een langgerekt cilindrisch huisje (vandaar de naam) en doet sterk denken aan soorten van de Enidae, een totaal andere familie waarbinnen vrijwel alle soorten (aanvankelijk vaak tot het geslacht *Pupa* gerekend) huisjes bezitten die (veel) hoger dan breed zijn. Het is dan ook niet verbazingwekkend dat dit karakteristieke slakje in een postuum werk van Draparnaud (1805) beschreven werd als *Pupa obtusus*. Inmiddels heeft onderzoek aangetoond dat het monotypische geslacht (een genus waarvoor slechts één soort bekend is) *Cylindrus* anatomisch in ieder geval tot de Ariantinae gerekend moet worden. Andere soorten met binnen deze subfamilie afwijkende huisjes zijn bijvoorbeeld *Isognomostoma isognomostomos* (Fig. 1-4, p. 122; samen met *Causa holosericea*, Fig. 1-5, p. 122 de enige vertegenwoordigers van de Ariantinae met tanden in de mondopening) en *Helicigona lapicida* (Fig. 1-25, herkenbaar aan een scherpe kiel en een duidelijk vlekkenpatroon). Schelp-morfologie en genitaal-anatomische kenmerken zijn praktisch voor het herkennen en beschrijven van soorten en geslachten, maar geven geen (of hoogstens beperkt) inzicht in de fylogenetische relaties binnen de Ariantinae. Om daar meer duidelijkheid over te krijgen werden de mitochondriale merkers *COI*, *Cyt B* &



ribosomaal 16S, alsmede de kern-merker *Histone H3* gesequenced. De stambomen op basis van de sequenties van zojuist genoemde merkers, afzonderlijk, maar ook na samenvoeging, werden met de bestaande taxonomische indeling vergeleken. Aan de hand hiervan kunnen we concluderen dat aan veruit het grootste deel van de subgenera (zoals *Causa*, *Dinarica*, *Josephinella*, *Faustina*, *Liburnica*, *Kosicia* en *Thiessea*) generieke status toegekend moet worden. Tevens kon de fylogenetische positie van een aantal morfologisch lastig te plaatsen soorten waaronder *Cylindrus obtusus* en *Ariantopsis pelia* worden bepaald.

Het materiaal dat voor de eerste zes hoofdstukken is gebruikt, werd bij voorkeur 'recent' (ca. binnen de afgelopen 10 jaar) verzameld en in geconcentreerde alcohol (ethanol 96%) bewaard. Helaas bleek dit niet altijd mogelijk en moest in sommige gevallen minder recent, of vanuit DNA-perspectief minder goed geconserveerd (ethanol 70% of lager) collectiemateriaal worden gebruikt. Van oudsher worden de mollusken (weekdieren) op 70% ethanol bewaard om anatomisch onderzoek mogelijk te houden (in geconcentreerde ethanol [96%] wordt te veel water aan het weefsel onttrokken, waardoor het stug en voor anatomisch onderzoek onhandelbaar wordt). Van de soorten die we in hoofdstuk 6 wilden opnemen, had het beschikbare materiaal een gemiddelde leeftijd van 20 jaar (het oudste materiaal is afkomstig uit 1957; Fig. 2, blz. 125). Het DNA in objecten van deze ouderdom is over het algemeen (sterk) gedegradeerd (uitzonderingen daargelaten; collectiemateriaal bewaard op spiritus bleek boven verwachting goed geconserveerd DNA te bevatten), waardoor bij de PCR vaak alleen kortere fragmenten geamplificeerd kunnen worden.

Objecten die niet geconserveerd zijn met als doel om daar moleculair onderzoek aan te doen vormen een rode draad in de vier voorlaatste hoofdstukken van dit proefschrift. Zo wordt in hoofdstuk 7 DNA geïsoleerd uit een recent gemummificeerde muis, terwijl de objecten in de hoofdstukken 8-10 bijna anderhalve eeuw oud zijn (allen afkomstig uit de periode 1864-1865). Doordat er uit zulke oude objecten in de regel weinig DNA geïsoleerd kan worden, van bovendien een slechte kwaliteit, dient bij dergelijk onderzoek met een aantal voorwaarden en risico's rekening gehouden te worden. Zo is de kans op contaminatie (vervuiling) met hedendaags DNA alleen al op basis van kwantiteit (de geringe hoeveelheid 'oud DNA'-moleculen ten opzichte van verse monsters) een probleem. In laboratoria waar regelmatig PCRs uitgevoerd worden, zullen altijd amplicons (DNA fragmentjes die ontstaan zijn als gevolg van natuurlijke of kunstmatige vermenigvuldiging) rondzwerven. Maatregelen om vervuiling door amplicons te voorkomen, zoals een fysieke scheiding tussen pre- en post-PCR labs, vormen een goede basis, maar zijn op zich niet voldoende om contaminatie uit te sluiten. Beter is het om gebruik te maken van speciaal voor 'oud DNA'-onderzoek toegeruste laboratoria, waar niet aan modern (recent verzameld) materiaal wordt gewerkt. Voorzorgsmaatregelen om vervuiling te voorkomen zijn bijvoorbeeld het dragen van beschermende (of eigenlijk afschermdende) kleding, decontaminatie van gebruiksvoorwerpen en oppervlakten door afname met een bleekoplossing (natrium hypochloriet) of bestraling met ultraviolet licht



en het gebruik van filtertips. Een andere maatregel die het risico op contaminatie vermindert is het ontwikkelen van soortspecifieke primers, die alleen DNA amplificeren van de betreffende soort of groep (één geslacht of een bepaalde familie). Juist omdat 'oud DNA' over het algemeen gedegradeerd is, kunnen alleen kortere fragmenten middels PCR vermenigvuldigd worden. Daarom worden in praktijk bij voorkeur primers ontworpen die korte fragmenten amplificeren én taxon-specifiek zijn. Een laatste risico dat in het kader van 'oud DNA'-onderzoek niet ongenoemd mag blijven, is de aanwezigheid van inhibitoren. Hierbij gaat het om stoffen die bij de DNA-isolatie mede geëxtraheerd worden en die vervolgreacties (zoals de PCR) kunnen verhinderen. Om van inhibitoren af te komen (of hun effect te onderdrukken) kunnen de DNA-extracten worden verdund (met bij dit type onderzoek een groter risico dat er onvoldoende DNA overblijft voor amplificatie). Ook kan er gekozen worden voor een andere extractie-methode of kan de extractie wellicht opnieuw uitgevoerd worden op andere, mogelijk beter geconserveerde delen van het bemonsterde object. Helaas valt op voorhand vaak niet of nauwelijks te voorspellen of oude monsters goed geconserveerd DNA of eventuele inhibitoren bevatten.

Hoofdstuk 7 beschrijft een casus van een verzekeringsmaatschappij waarbij, in een container met steriele goederen die verscheept werd van China naar Nederland, op de plaats van bestemming een dode muis tussen het verpakkingsmateriaal werd aangetroffen. Door deze vondst kon niet langer gegarandeerd worden dat de goederen daadwerkelijk steriel waren, waardoor de inhoud van de container onbruikbaar was geworden. In verband met de aansprakelijkheid wilde de verzekeraar weten of op basis van de identiteit van de muis kon worden vastgesteld waar deze in de container moest zijn gekomen (sommige soorten bos- of veldmuis van het genus *Apodemus* komen wel in Europa voor, maar niet in China en visa versa). Doordat de muis nog niet volledig volgroeid en deels gemummificeerd was, ontbraken de diagnostische morfologische kenmerken (of waren die niet langer te onderscheiden) om de soort op naam te brengen. Op basis van *Cyt B* sequenties kon de muis als *Apodemus sylvaticus* worden geïdentificeerd. Ondanks het feit dat de muis recentelijk was gestorven (dus in dat opzicht niets van doen had met 'oud DNA' zoals beschreven in de hoofdstukken 8-10) was het exemplaar niet geconserveerd 'met als doel daar moleculair onderzoek mee te doen'. Het verwachtingspatroon was derhalve dat het DNA gedegradeerd zou zijn, waardoor er op voorhand soortspecifieke primers voor de amplificatie van korte fragmentjes (zoals gebruikelijk bij 'oud DNA' onderzoek) werden ontworpen. Als onverwacht resultaat bleek de 'Apodemus draco specifieke' primerset een goed gedocumenteerd pseudogen (een defecte kopie van het functionele gen) van *A. sylvaticus* te amplificeren. Hierdoor werd de conclusie dat het een exemplaar van *A. sylvaticus* betrof zowel door *Cyt B* als het *Cyt B* pseudogen onderbouwd.

De drie voorlaatste hoofdstukken kunnen zowel op basis van de conditie van het DNA als op basis van de ouderdom van de objecten als 'oud DNA' studies worden aangemerkt (al zijn sommige onderzoekers van mening dat alleen (sub)fossiel



materiaal tot deze categorie gerekend zou moeten worden). Hoofdstuk 8 beschrijft het onderzoek naar de identiteit van de eerste neushoorn in het bezit van de Dublin Zoo (1864). Helaas overleed het jonge mannetje afkomstig uit Calcutta, al in 1865 op drie-jarige leeftijd. Zoals destijds gebruikelijk bij exotische soorten, werd het kadaver opgekocht door een wetenschappelijk instituut (in dit geval Trinity College Dublin) om er anatomisch onderzoek op te verrichten. Na de dissectie zijn de opgezette huid en het skelet in het Zoologisch Museum van hetzelfde instituut opgenomen, waar ze nog steeds gehuisvest zijn. Stilzwijgend werd altijd aangenomen dat het hier om een exemplaar van de Indische neushoorn ging. Doordat de huidplooien (vermoedelijk door de wijze van prepareren) zeer slecht bewaard zijn gebleven is het Trinity-exemplaar op basis van morfologische kenmerken echter niet met zekerheid te determineren. Daar komt bij dat de afmetingen van dit dier uiterst gering zijn voor een Indische neushoorn van deze leeftijd. Bovendien kwamen in 1865 in Calcutta zowel de Indische, *Rhinoceros unicornis*, als de kleinere, inmiddels zo goed als uitgestorven Javaanse neushoorn, *Rhinoceros sondaicus* voor. Theoretisch valt dus niet uit te sluiten dat de eerste neushoorn van de Dublin Zoo een exemplaar van de uiterst zeldzame Javaanse neushoorn is geweest. Om deze hypothese te testen werden op basis van beschikbare *Cyt B* sequenties van zowel de Indische als de Javaanse neushoorn primersets ontworpen, voor korte elkaar overlappende DNA fragmentjes (lengte 73 tot 396 basenparen). De sequenties, verkregen op basis van de in het 'oud DNA' laboratorium (LAF, Leiden Ancient DNA facility) uitgevoerde DNA extracties, tonen aan dat het Trinity-exemplaar, ondanks zijn geringe afmetingen, toch behoort tot de grote éénhornige neushoorn, *R. unicornis*.

In de twee voorlaatste hoofdstukken van dit proefschrift wordt DNA uit type-exemplaren (bij een type gaat het om één of meerdere exemplaren die gebruikt zijn voor de formele beschrijving van een soort) gesequenced. Hoofdstuk 9 beschrijft een onderzoek waarin de controverse rondom de waterhoentjes van Tristan da Cunha (TdC) wordt beslecht. In 1861 werd door Sclater *Gallinula nesiotis* beschreven, een waterhoentje dat niet kon vliegen en voorkwam op TdC. Enkele decennia later (1892) beschreef Allen een vergelijkbaar niet-vliegend waterhoentje, *Gallinula comeri*, van het 'nabij' gelegen (ca. 400 km) eiland Gough. Ten tijde van Allen's beschrijving waren de waterhoentjes op TdC al sterk bedreigd en sinds het begin van de twintigste eeuw werden ze er zelfs niet meer waargenomen. Totdat in 1972 de waterhoentjes 'herontdekt' werden op een afgelegen en ontoegankelijk deel van het eiland. Voor zover bekend zijn de waterhoentjes op Gough nooit bedreigd en is die populatie altijd stabiel geweest. Elf jaar voor de 'herontdekking' van de waterhoentjes op TdC had Eber (1961) op basis van metingen aan de typen van *G. nesiotis* en extante hoentjes van Gough, de discussie op scherp gezet door te stellen dat de waterhoentjes van TdC en Gough onvoldoende van elkaar verschilden om ze als aparte soorten aan te duiden. De ontstane controverse (zijn de waterhoentjes van TdC en Gough aparte soorten, of niet?) alsmede de vraag of *G. nesiotis* nu wél of niet is uitgestorven, wordt in dit hoofdstuk beantwoord met behulp van een drietal mitochondriale merkers: sequenties van *Cyt B*, tRNA *Lysine* / *Adenosine Triphosphate*



*subunit 8* (tRNA-Lys/ATP8) en de D-loop van de control region (*D-loop*). De sequenties van het type-materiaal van *G. nesiotis* verschillen duidelijk van *G. comeri*. Bovendien blijken de huidige waterhoentjes van TdC voor de geteste merkers identiek aan *G. comeri* van Gough. Op basis van deze resultaten dienen *G. nesiotis* en *G. comeri* inderdaad als aparte (onder)soorten te worden beschouwd en kunnen we concluderen dat *G. nesiotis* vermoedelijk is uitgestorven, terwijl op TdC tegenwoordig de uit Gough stammende soort voorkomt.

Hoofdstuk 10 gaat over de goudstreepsalamander, een soort die met name voorkomt in het noorden van Portugal en de noordwestelijke regio's van Spanje (Galicië en Asturië). Deze soort, *Chioglossa lusitanica*, werd in 1864 beschreven door Bocage. Recent onderzoek (Arntzen, 2007) toont binnen het genoemde verspreidingsgebied een morfologisch verschillende noordelijke en zuidelijke groep aan, met daartussen in centraal Portugal een zone waarin de vormen in elkaar overgaan. Doordat de twee groepen niet volledig reproductief van elkaar gescheiden zijn, worden ze als ondersoorten van *C. lusitanica* beschouwd. Het probleem bij de naamgeving van deze ondersoorten (waarvan één nieuw te beschrijven) is dat het type-materiaal dat gebruikt werd door Bocage afkomstig is uit Buçaco, midden in het overgangsgebied tussen de noordelijke en zuidelijke vorm. Op basis van alleen de morfologie kan het type-materiaal dus niet toegeschreven worden aan de noordelijke of de zuidelijke ondersoort. Omdat beide groepen genetisch goed te onderscheiden zijn, werd besloten om een fragmentje van *Cyt B* te sequencen voor één van de twee syntypen die toch al licht beschadigd was (gebroken staart). Hiervoor werden twee *Chioglossa* specifieke primersets ontworpen, die fragmentjes van 159 en 220 baseparen amplificeren met 3 en 2 diagnostische posities voor respectievelijk de noordelijke en zuidelijke ondersoort. Vermoedelijk als gevolg van DNA degradatie was het alleen mogelijk om het kortste fragment te amplificeren. Op basis hiervan moet het type-materiaal aan de zuidelijke ondersoort worden toegeschreven. De noordelijke ondersoort werd door Arntzen *et al.* (2007) vervolgens beschreven als *Chioglossa lusitanica longipes*.

Het laatste hoofdstuk van dit proefschrift blijkt vooruit op het gebruik van nieuwe technieken binnen de moleculaire taxonomie en het potentieel hiervan bij het ontsluiten van tot nog toe verborgen informatie in natuurhistorische collecties. In dit hoofdstuk wordt een pilot-studie beschreven waarbij getest wordt of met behulp van 'next generation sequencing' (NGS) in één keer een mitochondriaal genoom gesequenced kan worden op basis van collectiemateriaal. De NGS-techniek die in dit hoofdstuk wordt getest is DNA sequencing met een Genome Analyser IIx (GAIIx) van Illumina. Het voordeel van deze nieuwe manier van sequencen ten opzichte van single read sequencing (ook wel Sanger sequencen; de techniek die in alle voorgaande hoofdstukken werd gebruikt) is dat theoretisch elk willekeurig DNA-fragment gesequenced kan worden. Onderzoekers zijn dus niet langer beperkt tot die merkers waar een PCR-produkt voor verkregen kan worden. Het collectiemateriaal dat hier getest wordt betreft een monster van *Cylindrus obtusus* (zie hoofdstuk 6) dat 8,5 jaar



in ethanol [70%] was bewaard. De vraag bij deze pilot was niet of het met NGS mogelijk zou zijn om in één keer een compleet mitochondrium (ca. 15.000 basenparen) te sequencen. Een volledige run op een GAIIx genereert 50 Gbp (50.000.000.000 basenparen bij een 2x100 bp paired-end run) en extranucleaire genomen (zoals van mitochondria of chloroplasten) worden regelmatig als 'bijproducten' van dergelijke runs verkregen. De uitdaging zat in de vraag of het mogelijk zou zijn om zonder voorselectie (isolatie van mitochondria door middel van long-range PCR of ultracentrifuge) de mitochondriale sequenties (met voldoende 'coverage') uit de overvloed van gegenereerde sequenties te verkrijgen. Het niet kunnen maken van de genoemde voorselectie is inherent aan het werken met collectiemateriaal en de aard van de vraagstelling. Collectiemateriaal bevat over het algemeen gedegradieerd DNA, waardoor long-range PCRs (reacties waarbij in één keer amplicons van 10 kb of groter worden gegenereerd) niet meer mogelijk zijn. Degradatie van DNA is ook een probleem wanneer men mitochondria op basis van fysieke eigenschappen (ultracentrifuge) wil scheiden. Bovendien zijn voor ultracentrifuge ook behoorlijke DNA concentraties nodig (vaak ook problematisch bij collectiemateriaal) waardoor de extracten van meerdere individuen bij elkaar gevoegd zouden moeten worden, hetgeen niet wenselijk is bij moleculair taxonomische (of fylogenetische) vraagstukken. Analyse van de verkregen sequenties resulteerde in een 'contig' (consensus) sequentie van 14.610 basen met een gemiddelde 'coverage' van 27 keer. Doordat voor 3 mitochondriale merkers (*COI*, *Cyt B* & *16S*) in hoofdstuk 6 de sequenties al waren bepaald kon eenvoudig aangetoond worden dat het hier om de sequentie van het complete mitochondrium ging. Hoofdstuk 11 laat zien dat het met NGS mogelijk is om op basis van collectiemateriaal dat meerdere jaren op alcohol heeft gestaan (DNA-extract van één exemplaar) zonder PCR een compleet mitochondriaal genoom te sequencen. Hiermee biedt NGS toegang tot een niet eerder beschikbare schat aan informatie binnnen natuurhistorische collecties en opent het de deuren voor moleculair, fylogenetisch onderzoek in de 21e eeuw.







## **CHAPTER 1**

# **Introduction and summary**

## INTRODUCTION AND SUMMARY

This thesis is a reflection of a somewhat unconventional PhD-trajectory. Within this small book various animal groups will pass in revue, in the majority of cases to broaden our understanding of the smallest units that are relevant for natural history collections. On the one hand this is a study of 'operational taxonomic units' (OTU's - within natural history collections this, nearly always, is about the identification of species) and, on the other hand, it is to understand the systematic position of these units in a cladistic or phylogenetic context (what are sister-species, to which genus does a species belong, how many species does a genus contain, or which genera belong to a certain family?).

Traditionally species were described and distinguished based on phenotypic characters ('habitus' or anatomy). However, with the advent of new molecular biological techniques such as PCR (polymerase chain reaction) and Sanger sequencing (a method for determining the order of nucleotide bases in a DNA molecule) it became possible to directly study the genotype as well. This thesis provides a number of examples in which hypotheses based on morphological characters, were reevaluated with additional datasets obtained with these new techniques.

The molecular data gathered in this thesis are primarily used for verification of the morphological evidence, but can also provide additional support for identifications. As expected, there is often a lot of agreement between the morphological and genetically discerned units. Occasionally the molecular results show more diversity than initially recognized by the expert (chapter 2). In those cases, it usually concerns cryptic species and the new insights can trigger the search for new morphological characters, or a reassessment of the characters already in use. It also happens that experts subdivide taxa based on differences in morphology that do not, in fact, reflect differences in relationship; or create groups that turn out to be genetically indistinguishable (chapter 4 and 5).

Apparent discrepancies between identifications based on morphology and groups that can be distinguished genetically, are sometimes caused by human errors in the process from collecting the specimens to the publication of sequences in public databases, such as GenBank (chapter 2 and 3). For this reason it is important to always have a feedback between the molecular results and the morphological expectations, before the data are made publicly available. Prudence is not only required for the researcher who puts his data in the public domain, but also for those who want to use data from these sources for their own research (e.g. for positioning taxa in a broader, phylogenetic context).

Chapters 2 and 3 exemplify errors in GenBank. Chapter 2 describes the molecular systematics of a subfamily of pygmy squids, the Sepiolinae, which among others



occurs in the North Sea. Within this group several examples were shown in GenBank in which the vouchers (specimens that were used for the identification and DNA extraction) were identified as the same species, whereas the COI sequences (*Cytochrome Oxidase subunit I* - a common marker used in barcoding studies: [www.boldsystems.org](http://www.boldsystems.org)) differed considerably (Table 3, page 45). These sequences did not group close to each other phylogenetically either; e.g. compare the positions of the representatives of *Sepiola affinis* and *Sepiola ligulata* (Fig. 5, page 44). If, for a certain species, an adequate number of sequences is available (preferably from both a personal investigation as well as from public databases), the majority consensus might show which vouchers most likely were identified incorrectly (e.g. compare sequence DQ646730 for '*Euprymna tasmanica*' and sequence AY293722 of '*Sepietta neglecta*' with the other representatives of those species; Fig. 4, page 40/41). Chapter 2 also revealed an undescribed, cryptic species of *Sepiola*, which for the sake of convenience, was referred to as *Sepiola* sp. nov. when published in the *Journal of Molluscan studies*. This species was described by de Heij & Goud (2010) as *Sepiola tridens*.

Chapter 3 reports research errors in MPE (*Molecular Phylogenetics and Evolution*; a science citation indexed journal), wherein the molecular phylogeny of the West-Palearctic Helicidae (a family of terrestrial snails, which also includes the 'common' garden snail, *Cepaea nemoralis*) was investigated. Despite the fact that the just mentioned journal is subject to peer-review, many serious errors were made in that article. A Blast-search with sequences from that study demonstrated incorrect identifications or other errors. Because it could not be true that species such as *Trochoidea pyramidata* and *Otala punctata*, as well as *Oestophora turriplana* and *Candidula codia* are closely related. Inspection of the voucher specimens showed more errors; the voucher for *Caracollina lenticula* turned out to be a juvenile specimen of *Helicigona lapicida andorrica*, and the voucher for *Chilostoma cingulatum* turned out to be *Marmorana signata*. Based on material that was identified and sequenced by ourselves and on sequences from reliable studies, errors in multiple GenBank records were shown. These were not confined to merely misidentified vouchers, but also included mistakes (or contamination) that must have occurred in the lab, or with submission of the sequences. In addition to rectifying a malacological publication (which allows other researchers to judge that article on its merits), chapter 3 also brings to discussion the quality safeguard of public databases such as Genbank. The power of a public database (a seemingly inexhaustible source of information) at the same time is also its weakness (who supervises this source?). Measures to safeguard the data that are being deposited in GenBank, such as the uptake of voucher-specimens (or photographs thereof, in case the specimens had to be completely destroyed for the molecular study) and DNA-extracts in natural history collections, are at least desirable. The biggest issue is that records with demonstrable errors can only be corrected by the original author. Correction of these records is thus not possible without their cooperation. For this reason some have plead to 'wikify' GenBank. Other researchers should be able to correct or annotate the data,



of course not anonymously. In addition, the original, erroneous records should still be accessible, to allow the correction of potential mistakes thereupon.

In contrast to the cryptic species described in chapter 2, it can also be the case that subtle morphological differences are used to discriminate between closely related ‘species’, which can’t be distinguished genetically (chapter 4 and 5). Chapter 4 is about a mussel species, *Mytilus edulis*, which is commercially bred in the Netherlands. From a single locality in the Oosterschelde, specimens with a pointed umbo, as well as specimens with a broad shell, are known. The broad-shelled specimens morphologically show a strong resemblance with *Mytilus galloprovincialis*, a species that is especially well represented in the Mediterranean area. Based on this resemblance and the existence of similar broad shell fragments from the Eemian (ca. 113.000 to 114.000 years ago) from the North Sea, the question was raised whether this broad form might represent *M. galloprovincialis*. To test this hypothesis, both shell-morphological characters (Length/Width ratios and ‘Verduin’-parameters) as well as molecular markers (length polymorphism of Me 15/16, *COIII* sequences and AFLP’s) were compared. Because the differences in shell morphology could also be caused by phenotypic plasticity (the ability of an organism, or genotype, to adapt its morphology, or phenotype, as a reaction to its environment), we also investigated the distribution of shell forms over different depths (anticipating that at different depths the environmental conditions could differ). No association could be shown between shell form and depth. Neither could an association be shown between the (scant) genetic differences (which in itself are indicative for a monomorphic population) and shell form, or depth. Our results provide no evidence to support the presence of more than a single species of *Mytilus* in the Oosterschelde.

A similar issue is described in chapter 5. Here we test whether two supposed species of sandbee (*Andrena rosae* and *Andrena stragulata*) can be distinguished genetically. Both species have different flight periods (summer and spring, respectively) and show subtle morphological differences. For instance, males of *A. stragulata* possess a more strongly developed occipital spine (Fig. 5, p. 112). *Andrena rosae* has always been a rare species in the Netherlands, but in the last decades the number of observations strongly declined. By now it is a severely threatened species which is on the red list of Dutch bees. In the Netherlands it only occurs in the Biesbosch (a national park on the border of Zuid-Holland and Noord-Brabant). It is estimated that only a fifth of the Dutch *Andrena* species has a bivoltine life cycle, i.e. two generations per year (Atlas van de Nederlandse bijen, in prep.). To get a grasp of the genetic variation in *Andrena rosae* s.l. a number of mitochondrial markers, *COI*, *COII* & *Cytochrome B* (*CytB*) and a nuclear marker *Elongation Factor 1 alpha* (*EF-1a*) were sequenced. The results however did not show any differentiation between the spring and summer generation. Therefore no evidence was found to support the hypothesis that *A. rosae* and *A. stragulata* are two univoltine (one generation per year) species. Consequently the youngest name, *Andrena stragulata* Illiger 1806, has to be synonymized with *Andrena rosae* Panzer 1801.



Indistinct taxa with few obvious characters stand to benefit the most from molecular analysis. Such is the case in the Ariantinae (chapter 6), a subfamily of the Helicidae (see chapter 3). The Ariantinae inhabit most of the West-Palearctic, but have a main radiation the Mediterranean area in southern Europe. Besides a number of interesting exceptions (discussed hereafter), it is a rather monomorphic group of snails (overall the shell are very similar; Fig. 1, p. 122/123), in a few large genera (such as *Campylaea*, *Chilostoma* and *Helicigona*) with a seemingly arbitrary number of subgenera. Of those species that can be identified on just shell morphology, *Cylindrus obtusus*, a species endemic (i.e. native or restricted to an area or country) to Austria is most eccentric. *Cylindrus obtusus* (Fig. 1-3, p. 122) is the only species of Ariantinae with a vertically extended cylindrical shell (hence its name). It strongly resembles species of Enidae, a completely different family in which nearly all of the species (initially assigned to the genus *Pupa*) have shells that are much higher than broad. Hence it is not surprising that this characteristic snail was described as *Pupa obtusus* in a posthumous work of the French malacologist Draparnaud (1805). Based on genital-anatomy, it has been shown that the monotypic genus (a genus for which only one species is known) *Cylindrus* belongs to the subfamily Ariantinae. Other species of Ariantinae with clearly distinct shells are for example *Isognomostoma isognomostomos* (Fig. 1-4, p. 122; together with *Causa holosericum*, Fig. 1-5, p. 122, the only Ariantinae that have palatal teeth in their aperture) and *Helicigona lapicida* (Fig. 1-25, p. 125, which is characterized by a sharp keeled shell with a blotchy pattern). Shell-morphology and genital-anatomy are a practical means to recognize and describe species or genera, but give none (or very limited) insight into the phylogenetic relationships within the Ariantinae. To better understand those relationships the mitochondrial markers *COI*, *Cyt B* & ribosomal *16S*, as well as the nuclear marker *Histone H3* (*H3*) were sequenced. The phylogenies based on the combined as well as separate datasets were compared with the existing classification. This comparison led to the conclusion that most of the so-called subgenera (such as *Causa*, *Dinarica*, *Josephinella*, *Faustina*, *Liburnica*, *Kosicia* and *Thiessea*) should be promoted to the generic level. Also the phylogenetic position of a number of taxa that were difficult to elucidate morphologically, such as *Cylindrus obtusus* and *Ariantopsis pelia*, was determined.

The material used in the first six chapters was, preferably, collected ‘recently’ (roughly within the last decade) and stored in concentrated alcohol (ethanol 96%). Unfortunately this was not always possible and in a number of instances much older, or (with regard to the DNA) less well preserved material (ethanol 70%) had to be used. Traditionally molluscs are kept in ethanol 70% to allow anatomical studies; when stored in concentrated ethanol [96%] too much water gets withdrawn from the tissue, causing it to become stiff and unsuitable for anatomical research. In particular some of the species that we wanted to include in chapter 6 were old (up to 1957; Fig. 2, p. 125). The average age of the material at our disposal for chapter 6 was 20 years. DNA within such aged objects generally is (severely) degraded; with the exception of material kept on ‘brandspiritus’ (methylated spirits), which yielded unexpectedly well conserved DNA. The biggest issue with degraded DNA is that



only short fragments can be PCR amplified.

Objects that have not been preserved for molecular studies are the leitmotif of the four but last chapters of this thesis. In chapter 7 DNA is extracted from a mouse that was recently mummified, whereas the objects from chapters 8-10 are from the period 1864-1865). Generally speaking, only poor quality DNA in small quantities can be obtained from objects of this age. Therefore there are a number of risks inherent to this type of research, for which precautions have to be taken. For instance, the limited number of ancient DNA molecules (the quantity) compared to that of DNA extracts from fresh specimens makes contamination a severe issue. There will always be amplicons (small DNA fragments that are the product of natural or artificial amplification events) in laboratories in which PCRs are performed on a regular basis. Measures to prevent contamination caused by amplicons, such as a physical separation between pre- and post-PCR labs are a good start, but by itself is insufficient to rule out contamination in ancient DNA studies. Therefore dedicated laboratories are required, in which work on fresh specimens is prohibited. Precautions taken within these ancient DNA facilities are the use of forensic coveralls (suits to protect the experiment), decontamination of tools and surfaces with diluted bleach or by ultraviolet irradiation, and the use of filtertips. Another frequently taken measure to decrease the risk of contamination is the use of specific primers, which will only amplify DNA fragments of a single species or group (a specific genus or family). Because 'ancient DNA' is generally degraded, the designed primers preferably amplify short fragments, and are taxon-specific. A last issue that affects ancient DNA studies is the presence of inhibitors. These are chemical substances that are co-extracted with the DNA isolation and that can hamper subsequent reactions (such as PCR). A simple method to get rid of inhibitors (or suppress their effect) is to dilute the DNA-extracts (though this increases the risk of having too little DNA leftover for successful PCR amplification). Other solutions might be the use of a different extraction method (some methods are clearly better in removing certain types of inhibitors) or the use of another, possibly better conserved, part of the sampled object. Unfortunately, it is rarely possible to predict whether samples from old specimens will contain a sufficient amount of properly preserved DNA, or whether they are likely to contain inhibitors.

In chapter 7 a case-study of an insurance company is described. A sea container with sterile goods was shipped from China to the Netherlands. Upon inspection at its final destination, a dead mouse was found between packing material within this container. Consequently the contents of the container, which were no longer guaranteed to be sterile, were declared lost. Some species of *Apodemus* (the genus of wood- or fieldmouse to which the specimen belonged) are known to occur in Europe, but not in China and visa versa. Because of liability issues, the insurer wanted to know if the specimen could be identified to the species level. Since the mouse was not fully-grown and partly mummified, diagnostic characters for a morphological identification were either missing or not distinguishable. Based on *Cyt B* sequences,



the mouse was identified as *Apodemus sylvaticus*. Despite the fact that this mouse died recently (and in this sense had nothing to do with ‘ancient DNA’ as described in the three following chapters), the specimen also had not been preserved for the purpose of molecular investigation. Therefore we expected the DNA within this specimen to be degraded, and species specific primers (aiming on short fragments) were designed in advance. The ‘*Apodemus draco* specific’ primerset unexpectedly amplified a well documented pseudogene (a dysfunctional copy of an existing gene, likely to have a different evolutionary history because it is not restrained by functionality) of *A. sylvaticus*. Consequently the identification of the specimen as *A. sylvaticus* was supported by both the *Cyt B* and the *Cyt B* pseudogene sequences.

The three last but one chapters can be classified as ‘ancient DNA’ studies, both on the overall state of conservation as well as by the age of the objects (although some researchers are of opinion that the term ‘ancient DNA’ should only be used for (sub)fossil material). Chapter 8 describes a study on the identity of the first rhinoceros acquired by the Dublin Zoo in 1864. Unfortunately, the young male which originated from Calcutta, died in 1865 on the age of three. As common in those days with exotic animals, the carcass was purchased by a scientific institute (in this case Trinity College Dublin) for anatomical studies. After dissection the mounted skin and skeleton were admitted to the Zoological Museum of the same institute, where they still are today. It has always been assumed that this specimen belonged to the Indian rhinoceros. Due to the poor state of preservation, most taxidermic characters (such as neckfolds and skin details) got lost, hampering the morphological identification of this rhinoceros. Also, the size of the Trinity specimen is very small for a three year old Indian rhinoceros (*Rhinoceros unicornis*). Moreover, in those days the Javan rhinoceros (*Rhinoceros sondaicus*) also occurred near Calcutta where this specimen was caught. Therefore it can’t be ruled out that the first rhinoceros owned by the Dublin Zoo was a representative of the nowadays extremely rare Javan rhinoceros. To test this hypothesis, rhino-specific primersets were designed to amplify short overlapping DNA fragments (length 73-396 basepairs) based on available *Cyt B* sequences from both the Indian and Javan rhinoceros. DNA extractions for this study (as well as for the chapters 9-10) were carried out in a dedicated ‘ancient DNA lab’ (LAF, Leiden Ancient DNA facility). The obtained sequences showed that, despite its small size, the Trinity-specimen indeed belongs to the great one-horned rhinoceros, *R. unicornis*.

Chapter 9 and 10 both are examples of studies in which the DNA of type-specimens (a type is the specimen, or group of specimens, used for the formal description of a species) is sequenced. Chapter 9 describes a study in which the controversy about the moorhens of Tristan da Cunha (TdC) is settled. In 1861 Sclater described a flightless moorhen, *Gallinula nesiotis*, from the island of TdC. A few decades later (1892), Allen described a similar flightless moorhen, *Gallinula comeri*, from the ‘nearby’ (ca. 400 km) island of Gough. At the time when Allen described this Gough-moorhen, *G. nesiotis* was severely threatened and by the beginning of the twentieth century the latter



species was no longer observed. Until 1972 when the moorhens were rediscovered on a remote and inaccessible part of TdC. As far as known the moorhens on Gough were never threatened and that population has always been stable. Eleven years before the ‘rediscovery’ of the moorhens on TdC, Eber (1961) concluded, based on measurements of the type-specimens of *G. nesiotis* and a series of extant moorhens from Gough, that the moorhens from both islands were too similar to regard them as separate species. This is where the controversy arose: are the moorhens from TdC and Gough separate species, or not? and did *G. nesiotis* become extinct, or not? These hypotheses were tested with partial sequences of three mitochondrial markers: *Cyt B*, tRNA *Lysine* / Adenosine Triphosphate subunit 8 (tRNA-*Lys*/ATP8) and the D-loop of the control region (*D-loop*). The sequences for the type-specimen of *G. nesiotis* turned out to be clearly different from the sequences for *G. comeri*. Furthermore, it was shown that the contemporary moorhens of TdC are identical to *G. comeri* from Gough. These results indicate that *G. nesiotis* and *G. comeri* indeed should be regarded as different (sub)species, that *G. nesiotis* most likely became extinct and that the species now occurring on TdC in fact originated from Gough.

Chapter 10 is about the golden-striped salamander, a species that is mainly dispersed in northern Portugal and the northeastern districts of Spain (Galicia and Asturia). This species, *Chioglossa lusitanica*, was described in 1864 by Bocage. A morphologically distinguishable northern and southern group has been shown within the mentioned area of distribution; in between (central Portugal) there is a zone of integradation (Arntzen *et al.*, 2007). Because the reproductive isolation between these two groups is incomplete, both are regarded as subspecies of *C. lusitanica*. The main difficulty with the description of these subspecies (of which one, new to describe) is that the type-specimens used by Bocage originate from Buçaco, which is located within the zone of integradation between the northern and southern form. Based on morphology, the type-specimen can’t be assigned to the northern nor the southern group. Since both groups are well characterized genetically, we decided to sequence a small fragment of *Cyt B* from one of the syntypes that was slightly damaged already (it had a broken tail). To this end two *Chioglossa* specific primersets were designed which amplified fragments with a length of 159 and 220 bp, that should yield 3 and 2 diagnostic positions for the northern and southern subspecies, respectively. Only the shortest fragment could be amplified (most likely due to DNA degradation). The resulting sequence showed that this type-material belonged to the southern subspecies. The northern subspecies was subsequently described as *Chioglossa lusitanica longipes* by Arntzen *et al.* (2007).

The last chapter of this thesis looks ahead at new sequencing technologies and their potential to unlock data from natural history collections. This chapter describes a pilot-study in which ‘next generation sequencing’ (NGS) was used to sequence a complete mitogenome from a single collection specimen. The tested NGS platform was the Genome Analyser IIx (GAIIx) from Illumina. The main advantage of NGS compared to traditional Sanger sequencing (the technique used in all the preceding



chapters) is that theoretically any DNA fragment (any part of the genome) can be sequenced. Researchers are therefore no longer restricted to those markers for which a PCR product can be obtained. For this study one specimen of *Cylindrus obtusus* (see chapter 6) which had been kept in ethanol [70%] for 8,5 years was tested. The aim of this pilot was not to show that a complete mitochondrion (ca. 15.000 basepairs) could be sequenced in a single NGS run. A complete run on a GAIIX will generate 50 Gbp (50.000.000.000 basepairs with a 2 x 100 bp paired-end run) and extranuclear genomes (e.g. mitochondria or chloroplasts) are regularly obtained as a residual product of such runs. The challenge here was to see if the mitogenome could be obtained without any form of preselection ('enrichment'), such as (long-range) PCR or ultracentrifugion, from the surplus of generated sequences with sufficient coverage. The inability to make the just mentioned preselection is inherent to working with collection material and the type of research (sequences have to trace back to individual specimens). DNA extracted from collection specimens is generally degraded, which causes that long-range PCRs (reactions in which fragments of 10 kb or larger are generated) are no longer possible. DNA degradation is also a problem when mitogenomes are to be isolated based on physical properties (i.e. ultracentrifugion). Moreover the latter method requires substantial amounts of DNA (probably difficult to obtain for collection specimens) and might only be possible if DNA extracts from multiple specimens are pooled (undesirable for taxonomic or phylogenetic studies). Analysis of the GAIIX data resulted in a 'contig' sequence of 14,610 basepairs with an average coverage of 27 times. Because sequences were obtained for three mitochondrial markers (*COI*, *Cyt B* and *16S*) in chapter 6, it was easily shown that the 'contig' sequences represented the mitochondrial genome. Thus, chapter 11 shows that NGS can be used to sequence a complete mitogenome without PCRs from a single collection specimen that had been in ethanol [70%] for multiple years. Therefore NGS will be elementary in unlocking thus far inaccessible data from natural history collections and will open the doors for molecular, phylogenetic studies in the 21st century.





CHAPTER 2

**Molecular phylogeny of North Sea  
Sepiolinae (Cephalopoda: Sepiolidae)  
reveals an overlooked  
*Sepiola* species**

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## ABSTRACT

Recent bottom trawl surveys in the North Sea have broadened our understanding of an obscure group of bobtail squids (Sepiolidae, Sepiolinae). There are at least two *Sepietta* species in this region, viz. *Sepietta oweniana* and *Sepietta neglecta*, and three *Sepiola* species, one of which is undescribed, viz. *Sepiola atlantica*, *Sepiola pfefferi* and *Sepiola* sp. nov. *Sepiola pfefferi* is distinct from *Sepiola aurantiaca* because of differences in the hectocotylus; the type localities of both species are far apart and data in GenBank indicate a substantial genetic difference between them. It is unclear how to distinguish *Sepiola* sp. nov. morphologically from its sister species *Sepiola atlantica*, but molecular phylogenetic analyses and distributional data readily set them apart. The occurrence of *Sepiola rondeleti* in the northeastern Atlantic could not be confirmed. Several obviously incorrect or dubious identifications regarding GenBank data are listed.

## INTRODUCTION

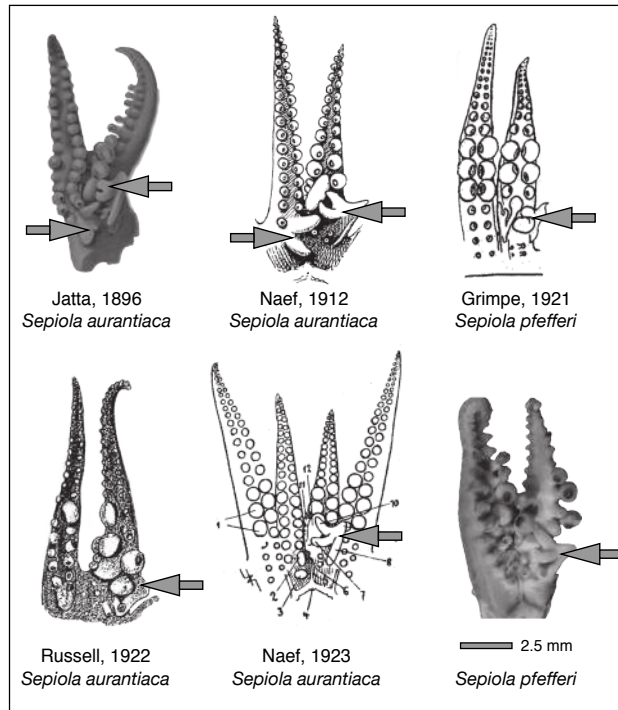
The Sepiolinae are small, poorly known, so-called bobtail squids, with a mantle length of up to five cm at most for the largest species used in this study. The subfamily encompasses five of the fifteen known genera of Sepiolidae, viz. *Euprymna*, *Inioteuthis*, *Rondeletiella*, *Sepietta* and *Sepiola*. Of these, only *Sepiola* and *Sepietta* are known from the North Sea, where they are represented by seven species (Reid & Jereb, 2005). These are *Sepiola atlantica* d'Orbigny, 1839-1842 (it is unknown in what year the plate with legends which validate this name were published; see Tillier & Boucher-Rodoni, 1993), *Sepiola intermedia* Naef, 1912, *Sepiola rondeleti* Leach, 1817, *Sepiola aurantiaca* Jatta, 1896, *Sepiola pfefferi* Grimpe, 1921, *Sepietta neglecta* Naef, 1916, and *Sepietta oweniana* (d'Orbigny, 1839-1841) (the plate with legends, which could validate this name, was published in 1839-1842, whereas the text was published in 1841).

Although Reid & Jereb (2005) include northeastern Atlantic in the geographical distribution of *S. intermedia*, there is a query about its distribution west of the Strait of Gibraltar (Reid & Jereb, 2005: 163, fig. 236). The occurrence of *S. rondeleti* in the North Sea is doubtful as well, since the recent records (Reid & Jereb, 2005: 168, fig. 244) all refer directly or indirectly to relatively old data in the literature, i.e. Joubin (1895, 1902) and Naef (1912). Apart from that, Reid & Jereb (2005) did not cite a critical paper by Grimpe (1925), in which he attributed the records by Joubin and Naef to *S. oweniana*. Later, Grieg (1933) came to the same conclusion for records of so-called *S. rondeleti* from the Scandinavian part of the North Sea. *Sepiola rondeleti* is also missing from comprehensive overviews of the German, Dutch and British areas of the North Sea (Jaekel, 1958; Janssen, 1975; Hayward & Ryland, 1990).

*Sepiola aurantiaca* was described from the Tyrrhenian Sea by Jatta (1896). At the beginning of the twentieth century, Russell collected twenty specimens of sepiolids from the North Sea, off the northeast coast of Scotland (ca. 56°N 3°W). Despite some



details of the hectocotylus, i.e. the lack of a peculiar lobe at the base of the first arm right (or right dorsal arm - Fig. 1; illustration in Russell, 1922), Russell (1909) attributed these bobtail squids to *S. aurantiaca*. Grimpe (1921) introduced *S. pfefferi* as a new species for sepiolids from the east coast of England (ca. 53°N 1°E). *Sepiola pfefferi*



**Figure 1.** Ventral view of dorsal arms of *Sepiola aurantiaca* and *Sepiola pfefferi*. Arrows indicate basal lobes on the hectocotylus. The scale refers only to the photo in the lower right corner of a hectocotylus of *Sepiola pfefferi*, collected in 1931 from Cove Bay, off Aberdeen (57°06' N 2°03' W).

*feri* could allegedly be discerned from *S. aurantiaca* by a deep V-shaped (instead of U-shaped) mantle outline (Fig. 2) and a basal lobe only on the left dorsal arm of the hectocotylus. According to Naef (1923) however, the differences between *S. pfefferi* and *S. aurantiaca* are so minimal, that he considered *S. pfefferi* a form of *S. aurantiaca*.

Clearly, the Sepiolinae represent a controversial group in which species are often difficult to identify morphologically (especially juvenile and female specimens). Sibling species are perhaps most easily distinguished genetically. In such cases a database like GenBank will be most valuable. However, the utility of such databases depends

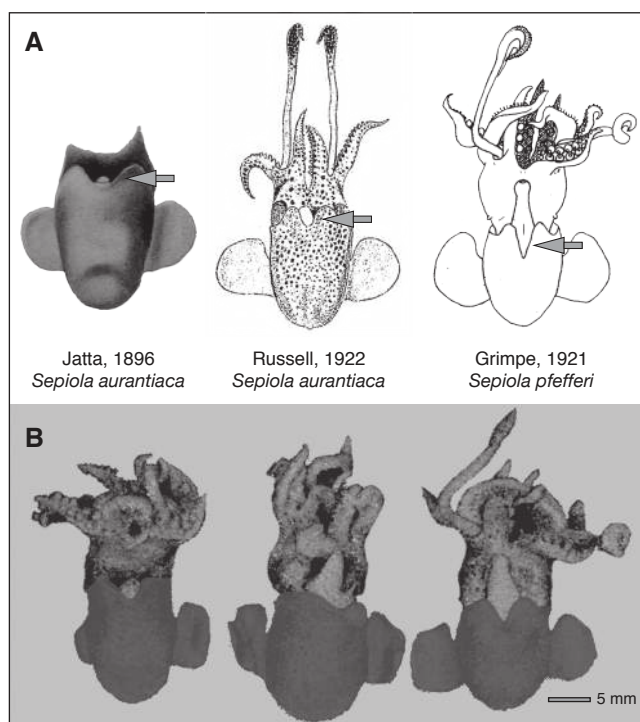


on how well the taxonomic group under study is represented and whether reliably identified voucher specimens are available for more detailed morphological studies. While trying to characterize the North Sea Sepiolinae by DNA analyses ('barcoding') and to unravel their phylogenetic relationships, our initial views, largely based on data in the literature, had to be modified.

## MATERIAL AND METHODS

### Sampling

Sepiolinae in general are small, obscure, bottom-dwelling organisms, that are most commonly known from the by-catch of fisheries. Except for SCUBA diving, which



**Figure 2.** A. U- and V-shaped mantle outline of *Sepioloidea aurantiaca* and *Sepioloidea pfefferi*. B. Illustration after Grimpe (1921) showing the type-specimens of *Sepioloidea pfefferi* with their mantles highlighted. The scale only refers to the photographed part of this figure.

is not a feasible means to survey larger areas, there does not seem to be an easy way specifically to catch these small cuttlefish. We therefore made use of trawl surveys in this study. For a number of years, IBTS (International Bottom Trawl Survey)



and BTS (Beam Trawl Survey) surveys have been conducted in the North Sea by IMARES (the Netherlands Institute for Fisheries Research) under auspices of ICES (the International Council for the Exploration of the Sea). The sepiolids from these surveys were stored in 96% ethanol and added to the collection of the National Museum of Natural History Naturalis (Leiden, The Netherlands). Our identifications were primarily based on morphological data published by Grimpe (1925) and Reid & Roper (2005), i.e. size, characters of the hectocotylus, the number and arrangement of suckers, and the presence and position of light organs. All species that were indisputably known to occur in the North Sea were represented in our material: *Sepiola atlantica* (21 specimens) *Sepiola aurantiaca* / *pfefferi* (5), *Sepietta oweniana* (4), and *Sepietta neglecta* (8). Hence we assume that the IMARES surveys yielded a representative sample of this subfamily for this area. Specimens and associated data (depth, gender) are listed in Table 1 and sampling sites are illustrated in Fig. 3.

#### *Type specimens and vouchers.*

The syntypes of neither *S. aurantiaca* nor *S. pfefferi* could be located in the institutes where they reportedly had been deposited, i.e. the Stazione Zoologica di Napoli, Naples, (A. Travaglini, personal communication) and the Museum für Naturkunde der Humboldt-Universität, Berlin (M. Glaubrecht, personal communication) respectively. We are not aware of any recently collected material of undisputable *Sepiola aurantiaca* and a voucher specimen for the sequence in GenBank (AF035708) was not available (M.E. Nishiguchi, personal communication).

#### *DNA isolation, PCR and sequencing*

Total genomic DNA was extracted from up to 15 mg of muscle tissue, using a DNeasy® Tissue Kit (Qiagen), following the manufacturer's protocol. Specimen vouchers were deposited at the National Museum of Natural History Naturalis, Leiden, The Netherlands. A 658-bp fragment of cytochrome oxidase subunit I (COI), that was shown to be informative in other cephalopod studies (Nishiguchi *et al.*, 1998; Lindgren *et al.*, 2004; Nishiguchi *et al.*, 2004) was amplified with primers L1490 5'-GGTCAACAAATCATAAAGATATTGG-3' and H2198 5'-TAAACTTCAGGGT-GACCAAAAAATCA-3' (Folmer *et al.*, 1994). PCR conditions were 1.5 mM MgCl<sub>2</sub>, 0.2 mM dNTPs, 0.4 µM of each primer and 5 U of Taq DNA polymerase (Qiagen) per reaction. Reactions were always carried out in a volume of 25 µl using a thermocycle profile of 3 minutes at 94°C, followed by 40 cycles of 15 seconds at 94°C, 30 seconds at 50°C, and 40 seconds at 72°C, and a final extension of 5 minutes at 72°C. PCR products were purified with Nucleospin® Extract II columns (Macherey-Nagel) and both strands were directly sequenced using the same primers. Sequencing was done on an ABI 3730 automated sequencer (Applied Biosystems) at MacroGen Corp. (Korea). Forward and reverse sequences were assembled and checked using Sequencher version 4.2 (Gene Codes Corp.). Sequences were aligned manually using MacClade 4.08 (Maddison & Maddison, 2003).



**Table 1.** Specimens, sampling information and GenBank accessionnumbers (GB ACCN).

| Species                                         | Lat (Deg) | Long (Deg) | Depth (m) | Sex | Date         | GB ACCN  | RMNH_MOL |
|-------------------------------------------------|-----------|------------|-----------|-----|--------------|----------|----------|
| <i>Sepietta neglecta</i> Naef, 1916             | 56.33653  | 4.5567     | 62        | F   | 2.ii.2005    | FJ231301 | 105647   |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | F   | 8.ii.2008.   | FJ231312 | 110308   |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | F   | 8.ii.2008.   | FJ231324 | 110308.1 |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | F   | 8.ii.2008.   | FJ231325 | 110308.2 |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | F   | 8.ii.2008.   | FJ231326 | 110308.3 |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | M   | 8.ii.2008.   | FJ231327 | 110309.1 |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | M   | 8.ii.2008.   | FJ231328 | 110309.2 |
| <i>Sepietta neglecta</i> Naef, 1916             | 56.66683  | -1.61233   | 58        | M   | 8.ii.2008.   | FJ231329 | 110309.3 |
| <i>Sepietta oveniana</i> (d'Orbigny, 1839-1841) | 55.73613  | -0.27547   | 73        | M   | 9.ix.2005    | FJ231297 | 107426   |
| <i>Sepietta oveniana</i> (d'Orbigny, 1839-1841) | 57.82507  | -2.42627   | 89        | M   | 26.viii.2005 | FJ231298 | 108685   |
| <i>Sepietta oveniana</i> (d'Orbigny, 1839-1841) | 55.68268  | 6.40243    | 43        | -   | 23.viii.2007 | FJ231299 | 110057   |
| <i>Sepietta oveniana</i> (d'Orbigny, 1839-1841) | 54.64303  | 0.52227    | 70        | F   | 12.ix.2005   | FJ231300 | 108941   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 53.4611   | 0.9181     | 20        | F   | 15.ii.2006   | FJ231302 | 102844   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 53.106798 | 1.7576     | 38        | M   | 26.i.2005    | FJ231303 | 105641   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 53.106798 | 1.7576     | 20        | M   | 26.i.2005    | FJ231304 | 105658   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 51.57346  | 2.77096    | 32        | M   | 25.i.2005    | FJ231308 | 105638   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 53.62875  | 2.46205    | 25        | M   | 11.ix.2007   | FJ231309 | 110059   |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 52.2099   | 2.99283    | 38        | F   | 23.8.2004    | FJ231310 | 99500    |
| <i>Sepioloa atlantica</i> d'Orbigny, 1839-1842  | 54.67276  | 2.57207    | 20        | F   | 9.ii.2005    | FJ231311 | 105655   |

|                                                   |          |          |    |   |              |          |          |
|---------------------------------------------------|----------|----------|----|---|--------------|----------|----------|
| <i>Sepiola atlantica</i> d'Orbigny, 1839-1842     | 54.70469 | 0.37152  | 68 | M | z2006        | FJ231314 | 105637   |
| <i>Sepiola atlantica</i> d'Orbigny, 1839-1842     | 52.68033 | 4.04217  | 24 | F | 28.i.2008    | FJ231316 | 110291   |
| <i>Sepiola atlantica</i> d'Orbigny, 1839-1842     | 54.28333 | 8.03333  | 19 | M | 29.i.2008    | FJ231317 | 110293   |
| <i>Sepiola atlantica</i> d'Orbigny, 1839-1842     | 54.12033 | 4.90917  | 44 | M | 30.i.2008    | FJ231318 | 110297   |
| <i>Sepiola atlantica</i> d'Orbigny, 1839-1842     | 53.8275  | 4.21467  | 29 | F | 13.ii.2008   | FJ231322 | 110316   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 56.78483 | -0.23631 | 79 | F | 7.ii.2005    | FJ231305 | 105650   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 57.82509 | -2.42626 | 63 | M | 26.viii.2005 | FJ231306 | 108684   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 55.68268 | 6.40243  | 43 | M | 21.viii.2007 | FJ231307 | 110058   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 54.77194 | -0.25279 | 66 | M | z2006        | FJ231313 | 105636   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 55.21893 | -0.50832 | 75 | F | z2005        | FJ231315 | 108938   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 55.83367 | 3.35133  | 64 | M | 5.ii.2008    | FJ231319 | 110298   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 56.27517 | 4.45883  | 65 | F | 6.ii.2008    | FJ231320 | 110299   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 56.09333 | 1.19967  | 83 | M | 11.ii.2008   | FJ231321 | 110311   |
| <i>Sepiola</i> sp. nov. (in prep. Goud & de Heij) | 54.30417 | -1.27367 | 94 | M | 7.ii.2008    | FJ231323 | 110301   |
| <i>Sepiola pfefferi</i> Grimpe, 1921              | 57.74243 | -1.3547  | 90 | M | 26.viii.2005 | FJ231292 | 108846.1 |
| <i>Sepiola pfefferi</i> Grimpe, 1921              | 57.74243 | -1.3547  | 90 | - | 26.viii.2005 | FJ231293 | 108846.2 |
| <i>Sepiola pfefferi</i> Grimpe, 1921              | 53.44067 | 0.96167  | 20 | F | 26.i.2005    | FJ231294 | 105642   |
| <i>Sepiola pfefferi</i> Grimpe, 1921              | 53.4611  | 0.9181   | 38 | F | 15.ii.2006   | FJ231295 | 110056   |
| <i>Sepiola pfefferi</i> Grimpe, 1921              | 53.11671 | 1.7827   | 38 | M | 26.i.2005    | FJ231296 | 108845   |

RMNH\_MOL refers to the voucher reference number (Molluscan collection) of the National Museum of Natural History, Naturalis (Leiden, The Netherlands). Sex determination: M = male, F = female and "-" = juvenile (or undetermined).



### Phenetic analysis

A neighbor-joining (NJ) phylogram (optimality criterion = distance, total character difference, distance measure = uncorrected  $p$ ) was constructed with Paup 4.0b10 (Swofford, 2003), including sequences from this study, all Sepiolineae sequences that had been previously deposited in GenBank and four outgroup sequences (representing Rossinae, Heteroteuthinae and Sepiadariidae), also from GenBank. Genetic distances (uncorrected  $p$ -distances; absolute number of differences, Table 2) were calculated with Paup 4.0b10 as well. Although corrected distances (such as the General Time Reversible (GTR) model, see the paragraph below) will be more realistic (e.g. they compensate for multiple substitutions at the same position), we choose to use uncorrected distances, because these at least show the minimum amount of change.

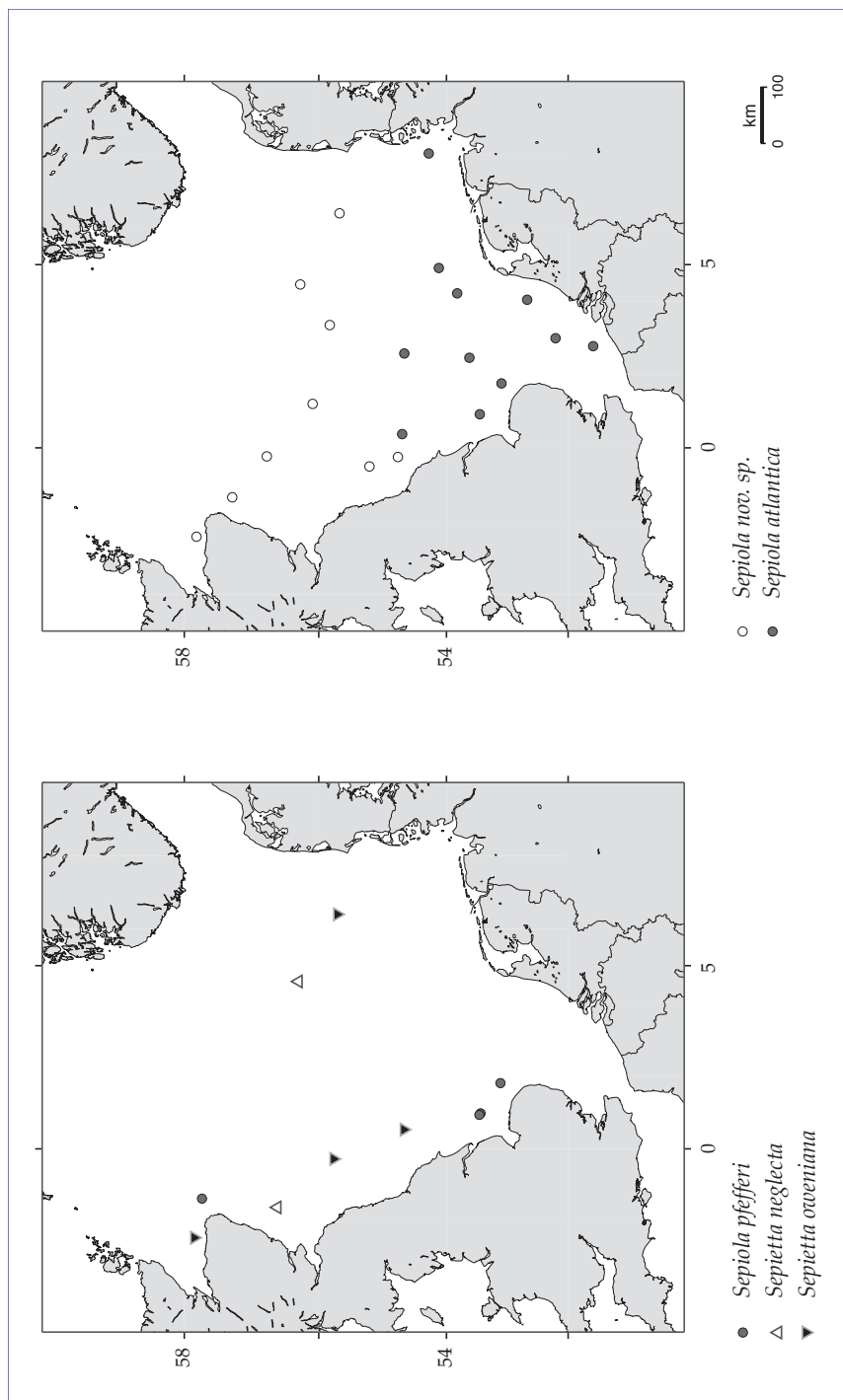
### Phylogenetic analysis

To reduce computational time and focus on the genera present in the area of study, we retained one representative of the sequences from GenBank for *E. tasmanica* (AY293713), *E. scolopes* (AF035701) and *E. hyllebergi* (AY293714) for Bayesian analysis. The sequence in GenBank for *Rondeletiola minor* (AF035714) was also excluded, as we suspect that it represents a specimen of *E. scolopes* (see discussion). For the Bayesian analysis, the best fit model of sequence evolution was selected with MrModeltest version 2.2 (Nylander, 2004). Both hLRT and AIC selected a GTR model (Nst = 6) with gamma distribution rate variation across sites and a proportion of invariable sites (rates = invgamma; GTR+I+G). A flat Dirichlet distribution was selected as prior on state frequencies (statefreqpr = dirichlet (1,1,1,1)). Bayesian inference was done with MrBayes 3.1.2. (Ronquist & Huelsenbeck, 2001), using two independent runs of four Markov chains with 10.000.000 generations. Sampling was done every 100 generations and the first 5.000.000 generations were discarded (burnin = 50.000; standard deviation of split frequencies was 0.0176 and dropped below 0.01 after 8.775.000 generations). *Sepioloidea lineolata* (AF000064) was specified as outgroup.

## RESULTS

Both phenetic and cladistic analyses (Figs 4 and 5, respectively) distinguished *Sepietta neglecta* (group F) and *Sepietta oweniana* (group G). No intraspecific variation was observed in either species according to our own data, whereas the interspecific divergence between these species was 7.6% (Table 2). Sequences of specimens provisionally named *Sepiola pfefferi* (group E) differed substantially (14.5% sequence divergence) from the only available alleged *Sepiola aurantiaca* sequence (AF035708, GenBank), while intraspecific variation within *S. pfefferi* was low (0.15% or less, Table 2). Specimens initially identified as *Sepiola atlantica*, turned out to represent two monophyletic groups: *S. atlantica* and *Sepiola* sp. nov. (groups A and B respectively; Fig. 4 and 5) with a minimum sequence divergence of 3.50% (Table 2). There was no intraspecific sequence divergence within *Sepiola* sp. nov. and our data show low intraspecific sequence divergence (0.15%) within *S. atlantica* (Table 2). Numerous





**Figure 3.** North Sea sampling localities of *Sepioida* and *Sepietta*.



**Table 2.** Sequence divergence percentages (calculated as the uncorrected *p*-distance times hundred) between sequences used in this study.

| Group | Taxon                              | Group | Taxon                                | Sdv % |
|-------|------------------------------------|-------|--------------------------------------|-------|
| A     | <i>Sepiolo sp. nov.</i> FJ231306   | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 0     |
| B     | <i>Sepiolo atlantica</i> FJ231317  | B     | <i>Sepiolo atlantica</i> FJ231302    | 0.15  |
| B     | <i>Sepiolo atlantica</i> AY293721  | B     | <i>Sepiolo atlantica</i> FJ231317    | 0.77  |
| A     | <i>Sepiolo sp. nov.</i> FJ231305   | B     | <i>Sepiolo atlantica</i> FJ231302    | 3.5   |
| B     | <i>Sepiolo atlantica</i> FJ231317  | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 3.65  |
| C     | <i>Sepiolo rondeleti</i> AY293720  | C     | <i>Sepiolo affinis</i> AF035706      | 0.16  |
| C     | <i>Sepiolo rondeleti</i> AY293720  | C     | <i>Sepiolo intermedia</i> AF035709   | 0.99  |
| C     | <i>Sepiolo intermedia</i> AF035709 | C     | <i>Sepiolo affinis</i> AF035706      | 0.83  |
| C     | <i>Sepiolo rondeleti</i> AY293720  | B     | <i>Sepiolo atlantica</i> FJ231317    | 1.67  |
| C     | <i>Sepiolo rondeleti</i> AY293720  | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 3.96  |
| C     | <i>Sepiolo affinis</i> AF035706    | B     | <i>Sepiolo atlantica</i> FJ231317    | 1.81  |
| C     | <i>Sepiolo affinis</i> AF035706    | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 3.96  |
| C     | <i>Sepiolo intermedia</i> AF035709 | B     | <i>Sepiolo atlantica</i> FJ231317    | 2.65  |
| C     | <i>Sepiolo intermedia</i> AF035709 | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 4.78  |
| -     | <i>Sepiolo rondeleti</i> AF035712  | C     | <i>Sepiolo rondeleti</i> AY293720    | 4     |
| -     | <i>Sepiolo rondeleti</i> AF035712  | C     | <i>Sepiolo affinis</i> AF035706      | 3.84  |
| -     | <i>Sepiolo rondeleti</i> AF035712  | C     | <i>Sepiolo intermedia</i> AF035709   | 3.67  |
| -     | <i>Sepiolo rondeleti</i> AF035712  | B     | <i>Sepiolo atlantica</i> FJ231318    | 5.5   |
| -     | <i>Sepiolo rondeleti</i> AF035712  | A     | <i>Sepiolo sp. nov.</i> FJ231305     | 7.5   |
| D     | <i>Sepiolo robusta</i> AY293719    | B     | <i>Sepiolo atlantica</i> AF035707    | 0     |
| D     | <i>Sepiolo robusta</i> AY293719    | D     | <i>Sepiolo intermedia</i> AY293718   | 0.46  |
| D     | <i>Sepiolo robusta</i> AY293719    | D     | <i>Heteroteuthis dispar</i> AF035713 | 0     |
| D     | <i>Sepiolo robusta</i> AF035711    | D     | <i>Sepiolo ligulata</i> AF035710     | 2.66  |
| D     | <i>Sepiolo ligulata</i> AF035710   | D     | <i>Sepiolo intermedia</i> AY293718   | 1.06  |
| D     | <i>Sepiolo ligulata</i> AF035710   | D     | <i>Sepiolo affinis</i> AY293716      | 1.77  |
| D     | <i>Sepiolo robusta</i> AY293719    | D     | <i>Sepiolo aurantiaca</i> AF035708   | 7.85  |
| E     | <i>Sepiolo pfefferi</i> FJ231296   | E     | <i>Sepiolo pfefferi</i> FJ231295     | 0     |
| E     | <i>Sepiolo pfefferi</i> FJ231296   | E     | <i>Sepiolo pfefferi</i> FJ231294     | 0.15  |
| F     | <i>Sepietta neglecta</i> FJ231301  | E     | <i>Sepiolo pfefferi</i> FJ231294     | 10.03 |
| F     | <i>Sepietta neglecta</i> FJ231329  | F     | <i>Sepietta neglecta</i> FJ231301    | 0     |
| -     | <i>Rondeletiola minor</i> AY293725 | F     | <i>Sepietta neglecta</i> FJ231301    | 11.57 |
| -     | <i>Sepiolo ligulata</i> AY293717   | E     | <i>Sepiolo pfefferi</i> FJ231294     | 9.29  |
| D     | <i>Sepiolo aurantiaca</i> AF035708 | E     | <i>Sepiolo pfefferi</i> FJ231294     | 14.48 |



Continued

|   |                                       |   |                                       |       |
|---|---------------------------------------|---|---------------------------------------|-------|
| G | <i>Sepietta oweniana</i> FJ231300     | G | <i>Sepietta oweniana</i> FJ231297     | 0     |
| F | <i>Sepietta neglecta</i> AY293722     | G | <i>Sepietta oweniana</i> FJ231297     | 0.31  |
| F | <i>Sepietta neglecta</i> AY293722     | - | <i>Euprymna stenodactyla</i> AF035704 | 3.22  |
| - | <i>Euprymna stenodactyla</i> AF035704 | G | <i>Sepietta oweniana</i> FJ231297     | 2.92  |
| G | <i>Sepietta obscura</i> AF036912      | F | <i>Sepietta neglecta</i> AY293722     | 0.46  |
| F | <i>Sepietta neglecta</i> FJ231329     | G | <i>Sepietta oweniana</i> FJ231297     | 7.6   |
| D | <i>Sepiola robusta</i> AF035711       | D | <i>Sepiola aurantiaca</i> AF035708    | 7.13  |
| D | <i>Sepiola intermedia</i> AY293718    | D | <i>Sepiola aurantiaca</i> AF035708    | 7.85  |
| - | <i>Sepietta obscura</i> AY293723      | F | <i>Sepietta neglecta</i> FJ231301     | 9.3   |
| - | <i>Sepietta obscura</i> AY293723      | G | <i>Sepietta oweniana</i> FJ231300     | 9.45  |
| B | <i>Euprymna tasmanica</i> DQ646730    | B | <i>Sepiola atlantica</i> FJ1231302    | 1.52  |
| B | <i>Euprymna tasmanica</i> DQ646730    | - | <i>Euprymna tasmanica</i> AY293713    | 12.94 |

Sepiolinae species were represented by more than one COI sequence in GenBank, but rarely did allegedly conspecific individuals form monophyletic groups (Fig. 4, Table 3).

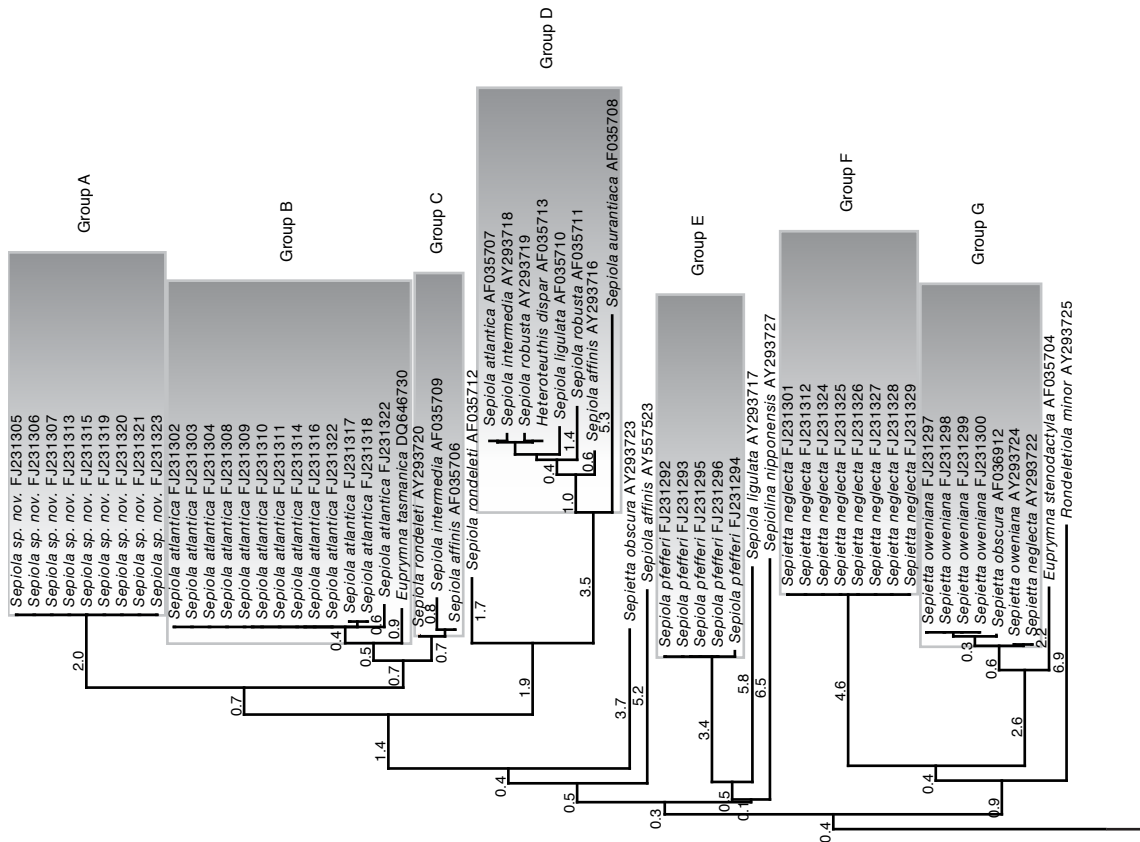
## DISCUSSION

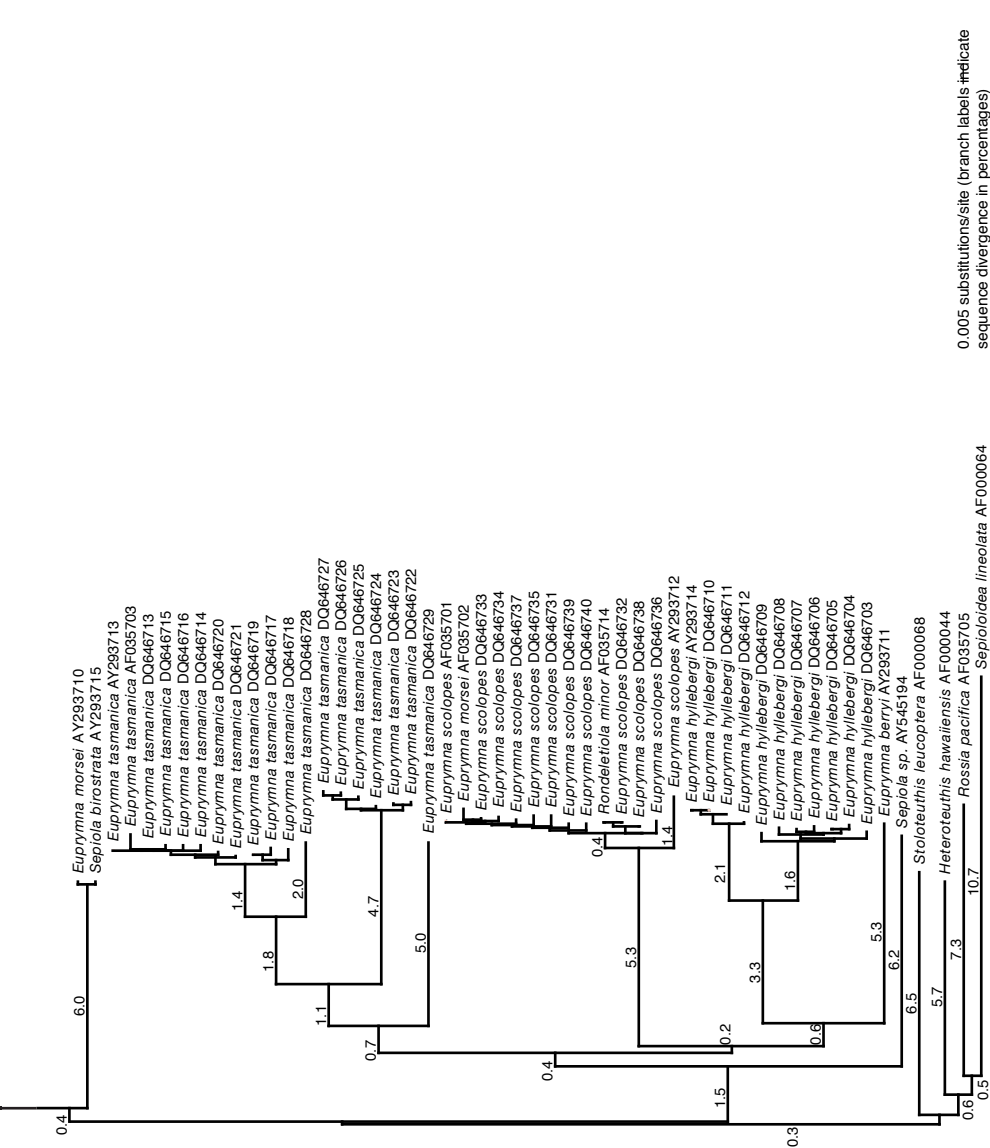
Five species of Sepiolidae occur in the North Sea, representing two genera, viz. *Sepietta oweniana* (group G) and *Sepietta neglecta* (group F), and *Sepiola atlantica* (group B), *Sepiola pfefferi* (group E) and *Sepiola* sp. nov. (group A) with different distributional patterns (Fig. 3). For the moment, *Sepiola atlantica* and *Sepiola* sp. nov. are considered sibling species. Based on DNA analyses of COI, a provisional phylogeny reconstruction (Fig. 5) is given, including sepiolid taxa from elsewhere using data in GenBank. Our data lead us to conclude that *Sepiola* is paraphyletic, whereas *Sepietta* is monophyletic. This study indicates that a high percentage of the species, genera and in one case subfamilies, of Sepiolidae in GenBank are most probably misidentified. This hampered our conclusions, in particular since voucher specimens were not available for inspection.

There is a striking contrast between sequences obtained with this study, that clearly show distinct species that are well-separated from each other (small intra-specific sequence divergences and large interspecific divergences) and sequences deposited in GenBank, which showed two other extremes: either sequences of the same so-called species that showed large interspecific divergences (Table 3) or sequences with limited intraspecific divergences of which the vouchers were identified as different species (group D, Figs. 4 and 5).

Both this incongruence between our data and the data from GenBank, as well as the high number of likely misidentifications (Fig. 4, Table 3), limited our ability to evaluate whether the sequence of *Sepiola aurantiaca* is actually distinct from group D







**Figure 4.** Neighbor-joining phylogram (branch lengths indicate sequence divergence percentages). Grey shading indicates groups of sequences of which the voucher specimens are considered conspecific in this study.



(Fig. 4 and 5). According to Nishiguchi *et al.* (1998, 2004) group D represents seven species. Among these all except, *Sepiolo atlantica* (AF035707), were collected in the Mediterranean Sea; most came from the same locality (Banyuls-sur-Mer: Nishiguchi *et al.*, 1998, 2004). Comparing the topology of Figures 4 and 5 and divergence among sequences (Table 2) referred to as *Sepiolo atlantica* d'Orbigny, 1839-1842, *Sepiolo intermedia* Naef, 1912, *Sepiolo affinis* Naef, 1912, and *Sepiolo ligulata* Naef, 1912, led us to conclude that GenBank entries AF035707, AY293718, AY293716 and AF035710, respectively (group D), were probably misidentified and likely all pertain to the species *Sepiolo robusta*.

Considering that *Heteroteuthis dispar* (Rüppell, 1844) (AF035713, group D) belongs to the subfamily Heteroteuthinae instead of Sepiolinae, and does not group with either its congener *Heteroteuthis hawaiiensis* (Berry, 1909) (AY293728) or with other representatives of the Heteroteuthinae, like *Stoloteuthis leucoptera* (Verrill, 1878) (AF000044), this specimen may also have been misidentified. One *S. robusta* Naef, 1912 sequence (AF035711, group D) is problematic in several ways: it has an insertion that causes a frame shift, coding region information is not in GenBank and it has four amino acid substitutions that are otherwise unique among sepiolid COI sequences in GenBank.

The sequence assigned to *S. aurantiaca* in GenBank (AF035708) differs considerably from other members of group D (7.85% sequence divergence between AF035708 and AY293718; Table 2). However, AF035708 is of unusual length and bears four insertions and two deletions, which are unique among sepiolid COI sequences (GenBank). AF035708 also contains stopcodons regardless of the chosen reading frame and even using an optimized alignment to minimize the number of amino acid substitutions, it has fourteen unique amino acids at positions that do not differ through sequences in the family. A BLAST search showed that the this sequence is cephalopod in origin. In our opinion the large number of point mutations, indels and stopcodons could indicate a nuclear mitochondrial pseudogene (NUMT), though no bias in nucleotide composition was observed. NUMTs can also be detected by unexpected phylogenetic placement of the taxon under study (Bensasson *et al.*, 2001), but the unclear phylogenetic relationships within the Sepiolinae, the availability of only a single sequence (AY035708) and the absence of a voucher (to check the morphological identification of the specimen in question) preclude use of these methods. Since syntypes of *S. aurantiaca* are missing, whereas according to Jatta *et al.* (1997) the records of the species are limited to those published by Jatta (1896) and Naef (1912), we consider *Sepiolo aurantiaca* a nomen dubium, referring to a species described from the Tyrrhenian Sea, most probably not occurring in the North Sea.

After exclusion of the dubious sequences AY035708 and AF035711, the maximum sequence divergence within group D is 1.77% (between AF035710 and AY293716, Table 2). Accepting a threshold of 3% sequence divergence for species status, as suggested by Hebert *et al.* (2003), this supports our assumption that group D represents only a single species, instead of seven according to Nishiguchi *et al.* (1998, 2004). Assuming that incorrect names were assigned to GenBank entries AF035707,



AY293718, AY293716, AF035710 and AF035713, group D only encompasses *S. robusta* (AY293719), which is one of the most frequently captured Mediterranean *Sepiola* species (Jereb *et al.*, 1997; Reid & Jereb, 2005) and is suggested to be the sister species (Fig. 5) of *S. rondeleti* according to Bello (1998).

In our view, Grimpe's (1921) description of *S. pfefferi* was inaccurate. The photograph of the type specimen (Fig. 2B; Grimpe, 1925) clearly showed a V-shaped mantle outline (specimen on the right), but the two syntypes on the left of it, seemed to have that structure more U- than V-shaped (as indicated for *S. aurantiaca*). Our specimens did not show the typical V-shaped mantle outline cited by Grimpe (1921). The drawing that was added (illustration on the right of Fig. 2A) is further misleading in that the size of the V-shaped incision extends up to one third of the mantle length, whereas on the photograph the length is about one fifth (Grimpe, 1925: specimen on the right of Fig. 2B). Even though the shape of the mantle outline is a dubious character, a clear-cut difference exists in the hectocotylus between *S. pfefferi* and *S. aurantiaca*. There are no lobes on the base of the right dorsal arm of the hectocotylus in *S. pfefferi* (Grimpe, 1921), while both Jatta (1896) and Naef (1912) depict lobes on the right dorsal arm of *S. aurantiaca* (Fig. 1). The geographical distance between the type localities of *S. aurantiaca* (Tyrrhenean sea) and *S. pfefferi* (North Sea, East coast of England) is noteworthy as well. Despite our doubts on the correctness of the alleged *S. aurantiaca* sequence (AF035708), we have included it in our table of sequence divergences (Table 2). The *S. pfefferi* sequences (group E) differ from the sequences of all other *Sepiola* species included in this study, in particular from the alleged *S. aurantiaca* sequence. With 9.29% minimum sequence divergence (Table 2) with its sister species (Fig. 4) *S. ligulata* (AY293717), *S. pfefferi* warrants species status if the same threshold of 3% sequence divergence (Hebert *et al.*, 2003) is imposed.

Only two *Sepietta* species, i.e. *S. neglecta* and *S. oweniana*, are known from the North Sea. Reid and Jereb (2005) indicated that these species can be difficult to distinguish morphologically; both are easily separated by their COI sequences (group F and G, Figs 4 and 5, Table 2), however. Therefore, we accept *S. neglecta* and *S. oweniana* as separate taxa (Bayesian posterior probability of 1.0, Fig. 5). Our data indicated no intraspecific variation in the small samples of *S. neglecta* ( $n = 8$ ) and *S. oweniana* ( $n = 4$ , Table 1, Fig. 3). Addition of the GenBank *Sepietta* sequences (AY293722, AY293724 and AF036912) once more brought misidentifications to light. These sequences form a monophyletic group with our sequences of *S. oweniana* (group G), only slightly increasing the sequence divergence within that clade to 0.31% (Table 2). If we again impose a divergence threshold of 3%, group G represents a single species. This implies that the specimens belonging to GenBank sequences AF036912 (Nishiguchi *et al.*, 1998) and AY293722 (Nishiguchi *et al.*, 2004) identified as *S. obscura* and *S. neglecta*, most likely are *S. oweniana*.

A second sequence (AY293723) is referred to as *S. obscura* in GenBank. This is the only *Sepietta* sequence that indicates that this genus is not monophyletic. Except for sequences of *Rondeletiola minor* (AY293725), *Euprymna stenodactyla* (AF035704) and *S. obscura* (AY293723), the other 15 *Sepietta* sequences used in this study form a monophyletic group (Figs 4 and 5). Consequently, we are sceptical about the iden-



tification of AY293723 as *Sepietta obscura* by Nishiguchi *et al.* (2004); we suggest it might belong to a *Sepiolo* species. More sequences of reliably identified *S. obscura* are required to clarify this point.



**Figure 5.** Bayesian cladogram with posterior probabilities. Grey shading indicates groups of sequences of which the voucher specimens are considered conspecific in this study.



The phylogenetic analysis (Fig. 5) and genetic distances (Table 2) show that individuals which we initially identified as *S. atlantica* represent two groups, viz. *S. atlantica* (group B) and a cryptic sister species, *Sepiola* sp. nov. (group A). Although we cannot (yet) distinguish the latter species from *S. atlantica* on morphological characters such as size, the number and arrangement of suckers and comparison of the hectocotyli, we consider it justified to report it as a separate taxon. Even within the relatively small area of the North Sea, there seems to be a sharp geographic delineation (Fig. 3) between *S. atlantica* (group B, the southern species) and *Sepiola* sp. nov. (group A, the northern species). Further investigation has to show to what extent the sharp delineation between these groups should be attributed to either different geographical distributions of both taxa indeed (southern and northern North Sea, respectively; Fig. 3) or to a preference for different depths (on average 31 versus 70 meter, respectively – Table 1). The separation between groups A and B is supported by high Bayesian posterior probabilities (Fig. 5). Both analyses (Figs. 4 and 5) show that not group A, but group C is the sister taxon of group B. Due to the inclusion of group C and D (Fig. 5), group B is paraphyletic. According to our data ( $n = 21$ ) no intermediates were detected between groups A and B. The minimum sequence divergence between these clades (3.5%, Table 2) exceeds the species threshold of 3% used here, while intraspecific divergences were low (0.15% and 0% within *S. atlantica* and *Sepiola* sp. nov., respectively). Due to a transversion (T to A), all individuals of *S. sp. nov.* have a methionine residue at amino acid position 160, which apparently is typical for this species (all other sepiolid COI sequences in GenBank show a leucine residue at that position).

**Table 3.** Accession numbers of conspecific specimens (according to GenBank registration) that do not form monophyletic groups.

|                             |                                              |
|-----------------------------|----------------------------------------------|
| <i>Sepiola atlantica</i> :  | AY293721 and AF035707                        |
| <i>Sepila rondeleti</i> :   | AY293720 and AF035712                        |
| <i>Sepiola affinis</i> :    | AF035706, AY557523 and AY293716              |
| <i>Sepiola intermedia</i> : | AF035709 and AY293718                        |
| <i>Sepiola ligulata</i> :   | AF035710 and AY293717                        |
| <i>Sepietta obscura</i> :   | AY293723 and AF036912                        |
| <i>Euprymna tasmanica</i> : | DQ646730 and all other conspecifics (Fig. 4) |
| <i>Euprymna morsei</i> :    | AY293710 and AF035702                        |
| <i>Rondeletiola minor</i> : | AY293725 and AF035714                        |

*tica* and *Sepiola* sp. nov., respectively). Due to a transversion (T to A), all individuals of *S. sp. nov.* have a methionine residue at amino acid position 160, which apparently is typical for this species (all other sepiolid COI sequences in GenBank show a leucine residue at that position).

In both analyses two GenBank sequences (AY293721 and DQ646730) are placed in group B. The specimen belonging to sequence AY293721 was collected near Vigo, Spain and (unlike the voucher for AF035707, Table 3) seems to be correctly identified as *S. atlantica*. Inclusion of AY293721 slightly increases the maximum divergence



within group B to 0.77% (Table 2). According to Jones *et al.* (2006), the voucher for sequence DQ646730 was collected near Adelaide, Australia and represents *Euprymna tasmanica* (Pfeffer, 1884). This is surprising, since the other nineteen *E. tasmanica* sequences in GenBank form a monophyletic, if heterogeneous, group (Fig. 4). Possibly an invading specimen of *S. atlantica* (e.g. traveling in the ballast water of a ship from the North Sea) was caught and not recognized as such, this far outside its normal geographic distribution. Including DQ646730 in group B increases its maximum divergence to 1.52%. For the moment, we conclude that the sequence DQ646730 is actually from *S. atlantica* (Table 3).

The position of group C (Figs 4 and 5) is unclear. The maximum sequence divergence between group B and C (2.65%, Table 2) is inflated as both AF035709 and AF035706 have an insertion that causes a frame shift and coding region information is not provided to GenBank. As group B and C cluster on the NJ-tree (Fig. 4), it is tempting to regard them as a single species, *S. atlantica*. The Bayesian analysis (Fig. 5), on the other hand, shows that group C and D are monophyletic. According to the species identifications now registered in GenBank, there is an allegedly conspecific specimen elsewhere in the tree (Table 3) for each so-called species in group C. Among the three so-called *S. affinis* sequences in GenBank, AY557523 is basal to the groups A, B, C and D and AF035706 and AY293716 belong to group C and D, respectively. The minimum sequence divergence between group C and D (between AY293720 and AY293716) is 9.74% (Table 2), which (with a 3% threshold for species status) suggests that these groups are not conspecific. As shown, (see the paragraph on group D) sequence AY293716 most likely belongs to *S. robusta*. Sequence AY557523 appears to be the sole representative of a separate group (sequence divergence above 3%), maybe *S. affinis*, indicating that sequence AF035706 probably does not belong to this species.

*Sepiola rondeleti* and *S. intermedia* are each represented by two sequences, viz. AY293720 & AF035712 and AY293718 & AF035709. Of the former, AY293720 belongs to group C, whereas sequence AF035712 probably represents a separate group (minimum distance to group C, *S. rondeleti* AY293720 is 3.84 % and to group D, *S. affinis* AY293716 is 7.97%, Table 2), maybe *S. rondeleti* indeed. Of the latter species, sequence AY293718 likely belongs to *S. robusta* (see the paragraph on group D). Consequently, the only sequence of group C that does not refer to a conspecific, more reliably identified specimen elsewhere in the tree (AF035709) was identified as *S. intermedia* by Nishiguchi *et al.* (1998). Hence we tentatively label group C as *S. intermedia*.

There is a discrepancy, which cannot yet be resolved, between the relationships shown in the NJ (Fig. 4) and the Bayesian (Fig. 5) figures, in particular regarding the position of group D relative to *S. atlantica*. We propose this can be attributed to a difference between phenetic and cladistic methods. The topology of a minimum evolution (ME) analysis (results not shown) is identical to Fig. 4, whereas the topology of a maximum parsimony (MP) analysis (results not shown) is similar to that obtained by Bayesian inference (Fig. 5). As to the sequences identified as *R. minor* in GenBank, we can only conclude that they are not conspecific (Table 3) and that AF035714 is from *E. tasmanica* (Fig. 4). Given the consistency between the molecu-



lar phylogeny and our morphological determinations, and the observation that the intraspecific sequence divergences of these specimens are generally small (Table 2), we are convinced that other explanations for the observed discrepancies between the molecular phylogeny and the species identifications registered in GenBank, like incomplete lineage sorting or mitochondrial introgression, are all far less probable than our suggestion of misidentification.

In addition to contributing to our knowledge of the systematics and phylogeny of the Sepiolidae, in particular of the genera *Sepiola* and *Sepietta*, this study should serve as a warning against indiscriminant use of GenBank data, as well as being a plea for the deposition of voucher specimens in easily accessible institutional collections. DNA barcoding can be advantageous when dealing with cryptic species or for an initial assessment of biodiversity. However, to avoid confusion and error, it should be preceded or combined with a critical taxonomic analysis.

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## REFERENCES

- BELLO, G. 1998. Cladistic analysis of the Atlanto-Mediterranean sepiolines (Cephalopoda: Sepiolidae) based on morphological characters. *Memorie del Museo Civico di Storia Naturale di Verona (2nd series), Sezione Scienze della Vita* 13: 81-85.
- BENSASSON D., ZHANG D., HARTL D.L. & HEWITT G.M. 2001. Mitochondrial pseudogenes: evolution's misplaced witnesses. *Trends in Ecology and Evolution* 16 (6): 314-321.
- FOLMER, O., BLACK M., HOEH W., LUTZ R. & VRIJENHOEK R. 1994. DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. *Molecular Marine Biology and Biotechnology* 3, no. 5: 294-9. doi:7881515.
- GRIEG, J.A. 1933. Cephalopods from the west coast of Norway. In: *Communication No. 79 from the Bergen Museum, Biological Station*.
- GRIMPE, G. 1921. 1. Teuthologische Mitteilungen. Die Sepiolinen der Nordsee. *Zoologischer Anzeiger* 53: 1-12.
- GRIMPE, G. 1925. Zur Kenntnis der Cephalopodenfauna der Nordsee. - *Wissenschaftliche Meeresuntersuchungen*, 16: 1-122.
- HAYWARD, P.J. & RYLAND, J.S. 1990. In: *The marine fauna of the British Isles and North-West Europe*, Volume 2: Molluscs to Chordates: 787-793.
- HEBERT, P.D.N., CYWINSKA, A., BALL, L.S. & DEWAARD J.R. 2003. Biological identifica-



- tions through DNA barcodes. *Proceedings of the Royal Society of London Series B, Biological Sciences*. 270: 313-321.
- JAECKEL, S.G.A. 1958. Cephalopoda. In: *Die Tierwelt der Nord- und Ostsee* 37 (G. Grimpe, G. & E. Wagner, eds): 479-723.
- JANSSEN, A.W. 1975. Systematische lijst van Nederlandse recente en fossiele mollusken. *Mededelingen van de Werkgroep Tertiaire en Kwartaire Geologie* 12: 115-170.
- JATTA, G. 1896. Cefalopodi viventi nel Golfo di Napoli (Sistematica). *Fauna und Flora des Golfes von Neapel*, 23: 1-268.
- JEREB, P., MAZZOLA, A. & DI STEFANO, M. 1997. Sepiolineae (Mollusca: Cephalopoda) from the Strait of Sicily. *Scientia marina* 61: 459-470.
- JONES, B.W., LOPEZ, E.J., HUTTENBURG, J. & NISHIGUCHI, M.K. 2006 Population structure between environmentally transmitted vibrios and bobtail squids using nested clade analysis. *Molecular Ecology* 15, 4317-4329.
- JOUBIN, L. 1895. Cephalopods from the scientific expedition of Prince Albert I of Monaco: Contribution to the study of cephalopods of the North Atlantic. English translation Washington D.C., Smithsonian Institution Libraries and the National Science Foundation.
- JOUBIN, L. 1902. Révision des Sepiolidae. Mémoires. *Société Zoologique de France*, 15 : 80-145.
- LINDGREN, A.R., GIRIBET, G. & NISHIGUCHI, M.K. 2004. A combined approach to the phylogeny of Cephalopoda (Mollusca). *Cladistics* 20, no. 5: 454-486.
- MADDISON, D.R. & MADDISON, W.P. 2003. MacClade 4: Analysis of phylogeny and character evolution. Version 4.08.- Sunderland, Massachusetts.
- NAEF, A. 1912. Teuthologische Notizen, 3. Die Arten der Gattungen *Sepiola* und *Sepietta*. *Zoologischer Anzeiger* 39: 262-271.
- NAEF, A., 1923. Die Cephalopoden. Fauna e Flora del Golfo di Napoli, monograph 35, 1(1) part 2: 149-863.
- NESIS, K.N. 1987. Cephalopods of the World: Squids, Cuttlefishes, Octopuses, and Allies. Neptune City, New Jersey, T.F.H. Publications Inc. Ltd., 351 pp.
- NISHIGUCHI, M.K., RUBY, E.G. & MCFALL-NGAI, M.J. 1998. Competitive dominance among strains of luminous bacteria provides an unusual form of evidence for parallel evolution in sepiolid squid-vibrio symbioses. *Applied and Environmental Microbiology* 64 (9): 3209-3213.
- NISHIGUCHI, M.K., LOPEZ, J.E. & BOLETZKY, S. 2004. Enlightenment of old ideas from new investigations: more questions regarding the evolution of bacteriogenic light organs in squids. *Evolution & Development* 6, no. 141-49.
- NYLANDER, J.A.A. 2004. MrModeltest v2. Evolutionary Biology Centre, Uppsala University. Program distributed by the author.
- PFEFFER, G. 1884. Die Cephalopoden des Hamburger Naturhistorischen Museums. *Abhandlungen aus dem Gebiete der Naturwissenschaften, Hamburg*. 8(1): 1-30.
- RONQUIST, F. & HUELSENBECK, J.P. 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics* 19, no. 12: 1572-1574.
- REID, A. & JEREB, P. 2005. Family Sepiolidae. In P. JEREB, P. & ROPER, C.F.E, eds. Cephalopods of the world. An annotated and illustrated catalogue of species known to date. Volume 1. Chambered nautilus and sepioids (Nautilidae, Sepiidae, Sepiolidae, Sepiadari-



- idae, Idiosepiidae and Spirulidae). FAO Species Catalogue for Fishery Purposes. No. 4, Vol. 1. Rome, FAO. pp. 153–203.
- RUSSELL, E.S. 1909. Preliminary notice of the Cephalopoda collected by the fishery cruiser “Goldseeker”, 1903-1908. *Annals and Magazine of Natural History*. Series 8, 3:446-455.
- RUSSELL, E.S., 1922. Report on the Cephalopoda collected by the research steamer “Goldseeker”, during the years, 1903-1908. *Fishery Board for Scotland, Scientific Investigations*. No. 3:1-45.
- SWOFFORD, D. L. 2003. PAUP\*. Phylogenetic Analysis Using Parsimony (\*and Other Methods). Version 4.0 beta 10.- Massachusetts.
- TILLIER, A. & BOUCHER-RODONI, R. 1993. Férussac and d’Orbigny’s “Histoire naturelle générale et particulière des Céphalopodes acétabulifères”: Dates of publication of plates and text. *The Nautilus* 107: 97-103.





CHAPTER 3

**Reappraisal of the “Molecular phylogeny  
of Western Palearctic Helicidae s.l.  
(Gastropoda: Stylommatophora)”: when  
poor science meets GenBank**

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## ABSTRACT

A paper on the 'Molecular phylogeny of Western Palaearctic Helicidae *s.l.*', published by Steinke *et al.* (2004) in this journal, is critically analysed. Several obvious errors are corrected and methodological weaknesses are revealed. BLAST searches on the sequences published in that paper and now in GenBank, showed high percentages of similarity of the alleged species with taxa that are considered only distantly related in the literature. Inspection of the so-called voucher specimens showed that some shells were misidentified, whereas others contained dirt or were bleached, indicating that these had been collected empty. Obviously the sequences published for those species could not have originated from those specimens, which cannot be considered vouchers therefore, even if they are from the same locality. In other instances, spurious sequences were published for correctly identified voucher specimens. For several species for which we collected specimens ourselves, the *COI* or the *16S* sequence, or both, clearly differed from the results published by Steinke *et al.* The consequences of our results for the molecular data on helicid gastropods and their classification are listed.

## 1. INTRODUCTION

Some years ago a paper which was published in this journal, provided an interesting scope on what was referred to as the 'Molecular phylogeny of the Western Palaearctic Helicidae *s.l.*' (Steinke *et al.*, 2004). The authors claimed to present the first comprehensive molecular phylogeny of this family. With a skewed taxon sampling (arguably overrepresenting the genera *Candidula* and *Trochoidea*) and only about a fifth of the genera of Helicidae (according to Zilch, 1959) included, this claim seemed somewhat far-fetched. Nevertheless, the paper did include representatives of the subfamilies Helicodontinae, Ariantinae, Helicinae, Hygromiinae and Helicellinae and as such could have provided a scaffold for the molecular phylogeny of this family.

Public databases (like GenBank) provide a great resource nowadays for the comparison of sequence-data. NCBI-BLAST (Basic Local Alignment Search Tool) searches (Zhang *et al.*, 2000), for instance, can confirm species identifications or indicate sources of contamination and incomplete taxon-sampling might be reduced or overcome by adding sequences obtained through GenBank. However, the added value of these open source databases strongly depends on the trustworthiness of the data that have been put into them. Usage of GenBank sequences which have not been safeguarded during each step in the process from the collection of the specimens all the way to the submission of the resulting sequences is a potential source of error. GenBank contains sequences of which the vouchers were misidentified, or worse, sequences that do not correspond to their supposed voucher specimens. Occasionally, researchers may not be able to unequivocally identify a species, or it



can be impossible to preserve a voucher (destructive sampling for DNA extractions). But even when errors in GenBank records are evident, GenBank only allows the primary authors to reannotate those data. Moreover, removal of corrupt sequences might be undesirable, once they have been used in the literature. Therefore it has been suggested to 'wikify' GenBank, in order to allow for the correction of inaccuracies in GenBank records (Pennisi, 2008).

Initial BLAST searches with the *COI* and *16S* sequences of Steinke *et al.* (2004) in several cases revealed high percentages (> 90%) of sequence similarity between taxa that are expected to be only distantly related (e.g. *Oestophora turriplana* AY546289 and *Candidula codia* AY238622, or *Trochoidea pyramidata* AY546297 and *Otala punctata* AY546290). In other instances the sequences of Steinke *et al.* (2004) differed substantially from the sequences on GenBank for which the vouchers were identified as conspecific (e.g. *Helicigona lapicida* AY546280 and AF297000; *Trochoidea elegans* AY546295 and FJ627176). Based on the remarkable results for some initial BLAST-searches, our own interest in the Ariantinae (Gittenberger *et al.*, 2004) and the peculiar position of *Caracollina lenticula* (Fig. 1 in Steinke *et al.*, 2004), we decided to obtain sequences for a number of species ourselves.

We compared the sequence data published by Steinke *et al.* (2004) with our data and those published by other research groups (in particular Manganelli *et al.*, 2005). For the Hygromiidae we restrict ourselves here to indicating anomalies or clear errors. Going into more detail would take things too far in this speciose and partly still problematic family. For the Helicidae *sensu stricto* however, we checked all the *COI* and *16S* sequence data published by Steinke *et al.* (2004).

## 2. MATERIAL AND METHODS

### 2.1. Taxon sampling

In order to verify some of the results of Steinke *et al.* (2004) the following taxa were used: *Arianta arbustorum* (Linnaeus, 1758), *Caracollina lenticula* (Michaud, 1831), *Chilostoma cingulatum* (Studer, 1820), *Helicigona lapicida andorraca* (Bourguignat, 1876), *Helicigona lapicida lapicida* (Linnaeus, 1758), *Helicodonta obvoluta* (Müller, 1774), *Isognomostoma isognomostomos* (Schröter, 1874), *Marmorana signata* (A. Férussac, 1821), *Otala punctata* (Müller, 1774) and *Zonites algirus* (Linnaeus, 1758). For sampling localities, see Table 1. For the various species, the nomenclature proposed by CLECOM (Bank *et al.*, 2001) is adopted, but while referring to sequence data in GenBank, the generic classification of the species is not altered.

### 2.2. Voucher material

Next to sequencing specimens of the above mentioned (sub)species, the following voucher specimens of Steinke *et al.* (2004), which are kept in the molluscan collection of the Forschungsinstitut Senckenberg, Frankfurt am Main, Germany (= SMF), were studied by all authors of this paper: *Caracollina lenticula* (SMF325448), *Xerosecta*



**Table 1.** Taxa and sampling localities.

| Species                               | RMNH        | Localitiy                                  | Country     |
|---------------------------------------|-------------|--------------------------------------------|-------------|
| <i>Arianta arbustorum</i>             | 139193      | Zuid-Holland, on the banks of the wijde Aa | Netherlands |
| <i>Arianta arbustorum</i>             | 139194      | Aude, Gorges du Rébenty                    | France      |
| <i>Caracollina lenticula</i>          | 100400      | I. di Lampedusa, Punta Sottile residence   | Italy       |
| <i>Chilostoma cingulatum</i>          | 114170      | Halltal, Tirol, 9 km NE Innsbruck          | Austria     |
| <i>Chilostoma cingulatum</i>          | 62177       | Belluno, Valle d'Ampezzo, near Rio Felizon | Italy       |
| <i>Helicigiona lapicida andorrica</i> | 116481/1    | Catalunya, Barcelona, Fonis de Liobregat   | Spain       |
| <i>Helicigiona lapicida andorrica</i> | 116481/2    | Catalunya, Barcelona, Fonis de Liobregat   | Spain       |
| <i>Helicigiona lapicida andorrica</i> | 116481/3    | Catalunya, Barcelona, Fonis de Liobregat   | Spain       |
| <i>Helicigona lapicida lapicida</i>   | 96401       | Luxembourg, La Roche-en-Ardenne            | Belgium     |
| <i>Helicigona lapicida lapicida</i>   | 99439       | Nordrhein-Westfalen, Blankenheim           | Germany     |
| <i>Helicodonta obvoluta</i>           | 139195      | Limburg, Magraten, Savelsbos               | Netherlands |
| <i>Isognomostoma isognomostomos</i>   | 139196      | Trento, between Predazzo and Bellamonte    | Italy       |
| <i>Isognomostoma isognomostomos</i>   | 139197      | Trento, between Predazzo and Bellamonte    | Italy       |
| <i>Marmorana signata</i>              | 139198      | Umbria, Perugia, Bevagna town wall         | Italy       |
| <i>Otala punctata</i>                 | 139199      | Valencia, Carlet, in an orange grove       | Spain       |
| <i>Otala punctata</i>                 | 139200      | Valencia, Carlet, in an orange grove       | Spain       |
| <i>Otala punctata</i>                 | 139201      | Valencia, Carlet, in an orange grove       | Spain       |
| <i>Zonites algirus</i>                | pers. coll. | Peloponnisos, Lakonia, Gerolimenas         | Greece      |

*cespitum* (Draparnaud, 1801) (SMF325417), *Chilostoma cingulatum* (SMF325420), *Helicigona lapicida* (SMF325426), *Helicopsis striata* (Müller, 1774) (SMF325428), *Iberus gualtierianus* (Linnaeus, 1758) (SMF325431), *Gittenbergeria turriplana* (Morelet, 1845) (SMF325435), *Otala punctata* (SMF325436), *Trochoidea elegans* (Gmelin, 1791) (SMF325441), *Trochoidea pyramidata* (Draparnaud, 1805) (SMF325447) and *Zonites algirus* (SMF325447). The remainder 29 vouchers were inspected by Neubert only.

### 2.3. DNA-isolation, PCR and sequencing

Tissues were always stored in ethanol (70% or 96%)), except for *Chilostoma cingulatum* from Italy (Table 1) which was stored in methylated spirits. Genomic DNA was extracted with a DNeasy blood and tissue kit of Qiagen, following the manufacturer's protocol. PCRs were carried out in 25 µl volumes using 1.25 units of Taq DNA polymerase from Qiagen, 0.4 mM of each primer and 0.2 mM dNTPs. For the amplification reactions (*COI* and *16S*) the same primers were used as described



| Lat/Long                 | Date        | Collectors                           | COI      | 16S      |
|--------------------------|-------------|--------------------------------------|----------|----------|
| 52°11'00" N, 4°38'2" E   | 3.viii.1992 | E. Gittenberger                      | AF296940 | JF717809 |
| -                        | 15.vi.1985  | A.J. de Winter                       | AF296945 | JF717810 |
| -                        | 2005        | A.W. Jansen                          | JF717793 | JF717811 |
| 47°19'29" N, 5°45'24" E  | 1.v.2009    | D.S.J. Groenenberg & M. Kolet        | JF717794 | JF717812 |
| -                        | 18.vii.1968 | W.H. Neuteboom                       | JF717795 | JF717813 |
| 42°16'59" N, 2°0'47" E   | 11.vi.2009  | B. Kokshoorn & N. Snijders           | JF717796 | JF717814 |
| 42°16'59" N, 2°0'47" E   | 11.vi.2009  | B. Kokshoorn & N. Snijders           | JF717797 | JF717815 |
| 42°16'59" N, 2°0'47" E   | 11.vi.2009  | B. Kokshoorn & N. Snijders           | JF717798 | JF717816 |
| 50°10'56" N, 5°34'35" E  | ix.2003     | D.S.J. Groenenberg                   | JF717799 | JF717817 |
| 50°26'16"N, 6°39'4"E     | 24.vii.1983 | P. Subai                             | JF717800 | JF717818 |
| 50°47'21" N, 5°45'24" E  | 30.vii.2009 | D.S.J. Groenenberg & P.J.L. Cuijpers | JF717801 | JF717819 |
| 46°18'37" N, 11°37'30" E | 9.ix.2009   | D.S.J. Groenenberg & I.M. Voermans   | JF717802 | JF717820 |
| 46°18'37" N, 11°37'30" E | 9.ix.2009   | D.S.J. Groenenberg & I.M. Voermans   | JF717803 | JF717821 |
| 42°55'51" N, 12°36'16"E  | 25.v.2010   | A.S.H. Breure                        | JF717804 | JF717822 |
| -                        | 31.v.2010   | A. Martínez-Ortí                     | JF717805 | JF717823 |
| -                        | 31.v.2010   | A. Martínez-Ortí                     | JF717806 | JF717824 |
| -                        | 31.v.2010   | A. Martínez-Ortí                     | JF717807 | JF717825 |
| 36°28'55" N, 22°24'02" E | 29.5.2009   | Pall-Gergely, G. Kornilios           | JF717808 | JF717826 |

in Steinke *et al.* (2004). We concentrated on mtDNA, in particular *COI* and *16S*, since these can be considered as a single locus and most data in GenBank refer to those markers.

For both markers we used the following PCR thermoprofile: initial denaturation 3 min. @ 94°C, followed by 40 cycles of – denaturation 15 sec. @ 94°C, annealing 30 sec. @ 50°C, extension 40 sec. @ 72°C – and a final extension of 5 min. @ 72°C. The obtained PCR products were send to Macrogen Inc. Europe (Amsterdam), where they were purified with a Montage purification kit (Millipore) and subsequently sequenced in both directions on an ABI3730XL using the same primers as used for PCR. The resulting forward and reverse sequences were assembled with Sequencher 4.10.1 (Gene Codes Corporation), checked for irregularities (including amino acid translation for *COI*) and submitted to GenBank.

#### 2.4. Alignment

Our sequences were aligned with the sequences of Steinke *et al.* (2004) and other



sequences from GenBank. For the analyses that focussed on the Helicidae s.s. (Ariantinae, Helicinae, Helicodontinae), *COI* sequences were aligned directly in MacClade 4.08 (Maddison and Maddison, 2005) and *16S* sequences were aligned with MAFFT v.6.847b (Katoh *et al.*, 2002). For an analysis that focussed on the Hygromiidae we used all of the sequences of Manganelli *et al.* (2005) and some noteworthy *16S* sequences of Steinke *et al.* (2004). Because the Hygromiidae sequences were more diverged than the Helicidae sequences, this alignment clearly improved when structural information was included. Therefore MAFFT-Q-INS-i (Katoh and Toh, 2008) was used for a structural alignment of the *16S* Hygromiidae dataset.

### 2.5. Phylogenetic analyses

For the Helicidae s.s. *COI* and *16S* were analysed separately to test if both datasets would yield similar overall topologies and to check for inconsistencies within taxa. In order to make a comparison with the phylogeny presented by Steinke *et al.* (2004), the same 526 bases of *COI* were included; our *16S* alignment had a length of 326 bases (compared to 323 = 380-57 of Steinke *et al.*, 2004). For both datasets a NJ (Neighbor-Joining) analysis was performed in PAUP v. 4.0b10 for Unix (Swofford, 2002). We imposed a 3:3:1 weighting scheme on the *COI*-dataset to compensate for the relaxed mutation rate at third base positions, rather than completely excluding these positions (as done by Steinke *et al.*, 2004). For the analysis that focussed on the Hygromiidae 342 presumably homologous positions (on par with Manganelli *et al.*, 2005) were included. Two Markov chain Monte Carlo (MCMC) runs were executed simultaneously in MrBayes 3.1.2 (Ronquist and Huelsenbeck, 2003) using the general time reversible model with a proportion of invariant sites and Gamma-distributed among site rate variation (GTR+I+G) as selected by MrModeltest 2.2 (Nylander, 2004). These analyses were set to run for 10,000,000 generations or until the MCMC runs converged (diagnostic stop value 0.009).

### 2.6 BLAST searches

MegaBLAST searches (that look for highly similar sequences) were performed on all of the *COI* (AY546262–AY546301) and *16S* (AY546342–546381) sequences from Steinke *et al.* (2004) as currently included in GenBank. By default the output of a BLAST search is sorted by the bit score of high-scoring segment pairs (HSPs). Using any sequence from GenBank as a query sequence, obviously will result in that query sequence showing the largest bit score. When referring to ‘most similar’ sequences, we do not show the query sequences, but the second-best matches based on bit score. Since bit scores by themselves are not very informative between different BLAST searches, we report the identity scores (which are more intuitive for comparisons between sequences) for these second-best matches instead. The results for a selection of sequences (including all the Helicidae s.s. from Steinke *et al.*, 2004) are summarized. Throughout this paper, whenever we refer to a BLAST search, we automatically imply MegaBLAST.



### 2.7. Genetic distances

Uncorrected *p*-distances (all positions included and weighted equally) were calculated with PAUP v. 4.0b10 for both the *COI* and *16S* dataset (length 526 and 326 nucleotides, respectively, see above). These distances were converted into “similarity scores” (calculated as 100 minus the uncorrected *p*-distances times 100), so the shown values can be interpreted as percentages.

## 3. RESULTS

Figure 1 shows a compilation of some of the inspected vouchers of Steinke *et al.* (2004) and a selection of shells from the conchological collection of NCB Naturalis. Figure 2 focusses mainly on the Helicidae s.s. and shows the *COI* and *16S* NJ phylogenies based on sequences of Steinke *et al.* (2004; in blue), sequences from various studies (GenBank; in green) and from this study (in black). Figure 3 depicts a majority rule consensus tree (showing all compatible partitions) of a Bayesian analysis that mainly focussed on the Hygromiidae. After 3,340,000 generations both MCMC runs converged (the average standard deviation of split frequencies reached 0.008965), resulting in two tree files, each consisting of 33,401 trees of which the first 10,020 trees (30%) were discarded as burnin. Only posterior probabilities higher than 0.5 have been indicated. In this figure, sequences of Steinke *et al.* (2004) are in blue, sequences from various studies (Genbank) are in green and sequences from Manganelli *et al.* (2005) are in black. Table 2 gives an overview of some of the noteworthy MegaBLAST results based on the *COI* and *16S* sequences (including all the Helicidae s.s.) of Steinke *et al.* (2004). Table 3 gives an overview of the genetic distances (in percentages, calculated as described above).

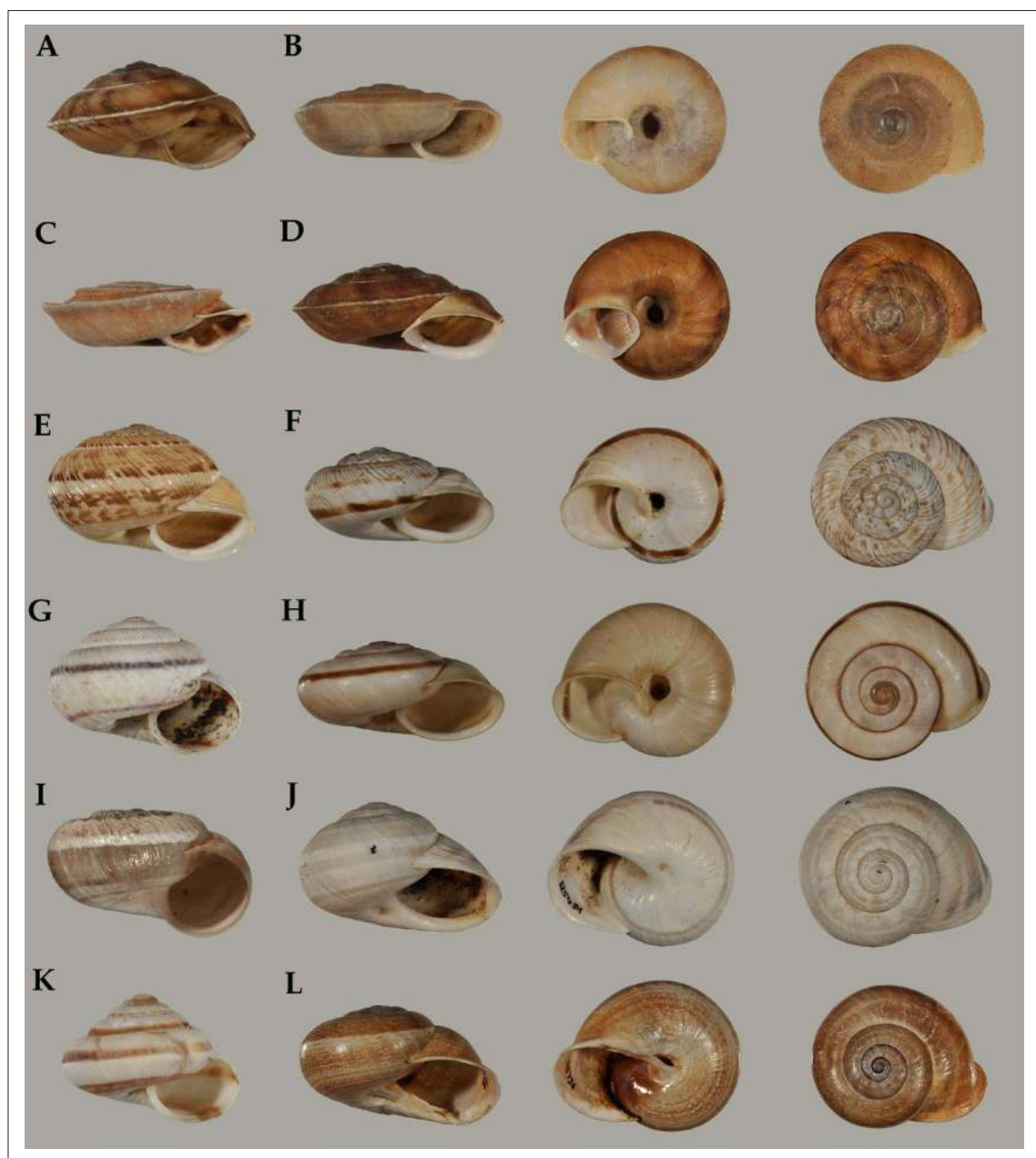
## 4. DISCUSSION

### *Helicidae, Ariantinae*

#### 4.1 *Caracollina lenticula* & *Helicigona lapicida andorrica*

A BLAST search with the so-called *C. lenticula* *COI* sequence (AY546265) showed a relative high similarity (BLAST identity-score of 90%, Table 1) with sequence AF297000 for *Helicigona lapicida* (Gittenberger *et al.*, 2004). The voucher for *C. lenticula* (SMF325448, not SMF325411 as stated in Appendix A by Steinke *et al.* [2004]: Fig. 1A) is a juvenile shell of *Helicigona lapicida andorrica*, which differs from real *C. lenticula* (Fig. 1B) by its much larger size, sharper and more centrally situated peripheral keel and a granular surface. Despite the fact that Steinke *et al.* (2004) did include a correctly identified fully grown specimen of *H. lapicida lapicida* (Fig. 1D) in their study (see below), they did not recognize a juvenile specimen of the subspecies from Andorra and parts of the surrounding Pyrenees. The *COI* and *16S* sequences (JF717796-JF717798 and JF717814-JF717816) that we independently obtained for





**Figure 1.** Compilation of shells including some of the inspected vouchers of Steinke *et al.* (2004) and a selection of shells from the conchological collection of NCB Naturalis.



- 
- A. "*Caracollina lenticula* sensu Steinke *et al.* 2004" (published as SMF 325411)  
= *Helicigona lapicida andorrnica* (Bourguignat, 1876), juvenile (without apertural lip)  
B 15.0 mm [SMF 325448/1] Spain, "Seteases"; M. Pfenninger leg.
  - B. *Caracollina lenticula* (Michaud, 1831) B 7.2 mm [RMNH 100372]  
Italy, Lampedusa island, Punta Sotille; A.W. Janssen leg., 18.v.2005.
  - C. "*Oestophora turriplana*" = *Gittenbergeria turriplana* (Morelet, 1845) B 14.7 mm  
[SMF 325435/1] Portugal, "Boliouenne"; M. Pfenninger leg.
  - D. *Helicigona lapicida* (L., 1758) B 17.6 mm [SMF325426/1] Germany, Hessen, Schlüchtern;  
M. Pfenninger leg.
  - E. *Candidula codia* (Bourguignat, 1859) B 12.0 mm [RMNH G1980] Portugal, Algarve,  
5 km NE of Boliouime; E. Gittenberger leg., iv.1980.
  - F. "*Chilostoma cingulatum* sensu Steinke *et al.* 2004" = *Marmorana (Ambigua) signata* Férussac,  
1821. B 20.2 mm [SMF 325420/1]; Italy, Passo di Furlo; M. Pfenninger leg.
  - G. "*Helicopsis striata*" (Müller, 1774)? B 7.0 mm [SMF 325428/1] *T. geyeri* (Soós, 1926)?  
Germany, "Thüringen, Kyffhäuser"; D. Steinke leg.
  - H. *Chilostoma cingulatum* (Studer, 1820) B 20.5 mm [RMNH H1938] Switzerland, Tessin,  
Melide along Lago di Lugano; J.T. Henrard leg., 28.viii.1938.
  - I. *Xerotricha apicina* (Lamarck, 1822) B 7.8 mm (after Soes & de Winter, 2005) Netherlands,  
Ijmuiden; A.J. de Winter leg.
  - J. "*Iberus gualtierianus*" = *Iberus gualtierianus sensu lato* (L., 1758) B 29.6 mm [SMF 325431/1]  
Spain, "Ronda"; M. Pfenninger leg.
  - K. *Trochoidea pyramidata* (Draparnaud, 1805) B 7.4 mm [SMF325443/1] France,  
"St. Maximin"; A. Eppenstein leg.
  - L. *Otala punctata* (Müller, 1774) B 31.3 mm [SMF 325436/1] Spain, "Zaragoza";  
M. Pfenninger leg.
- 

three specimens of *H. lapicida andorrnica* were on average 97.2% and 98.0% identical (Table 2) to the sequences AY546265 and AY546345 of *C. lenticula* sensu Steinke *et al.* (2004). The COI and 16S sequences that we ourselves obtained for the real *C. lenticula* (JF717793 and JF717811) differed 19.4% and 29.5%, respectively, from the alleged *C. lenticula* sequences of Steinke *et al.* (2004). Moreover, Fig. 2 shows that according to our COI data *C. lenticula* belongs to the Helicodontinae or Trissexodontinae, in accordance with the usual classification (e.g. Bank *et al.*, 2001). The 16S dataset is somewhat indeterminate regarding the position of *C. lenticula*, but that taxon clearly does not belong to the monophyletic subfamily Ariantinae.

We consider it a methodological weakness that Steinke *et al.* (2004), instead of reconsidering their identification on the basis of their voucher specimen (or checking their surprising results by sequencing additional specimens), asserted that *C. lenticula* belongs to the Ariantinae instead of the Helicodontinae.



#### 4.2. *Chilostoma cingulatum* & *Marmorana signata*

BLAST searches with the COI (AY546274) and 16S (AY546354) sequences of Steinke *et al.*'s (2004) so-called *Chilostoma cingulatum* show a striking similarity (Table 2) with sequences GU391375 (100%) and GU391404 (89%), respectively, of *Marmorana signata strigata* (see Fiorentino, 2010). The voucher of *C. cingulatum* sensu Steinke *et al.*, 2004 (SMF325420: Fig. 1F), is a species of *Marmorana* indeed and not *C. cingulatum* (Fig. 1H). The COI and 16S sequences (JF717804 and JF717822) that we obtained for a specimen which we identified as *M. signata* (Table 1) ourselves, were 100% and 94.9% identical (Table 3) to sequences AY546274 and AY546354, respectively. The average sequence divergences (Table 3) between our own *C. cingulatum* sequences (JF717794-JF717795 and JF717811-JF717812) and *C. cingulatum* sensu Steinke (AY546274 and AY546354) are 18.8% (COI) and 23.1% (16S, Table 3). The phylogeny reconstructions for both COI and 16S datasets show that our specimens of *C. cingulatum* belong to the Ariantinae clade (Fig. 2), in conformity with the usual classification (Bank *et al.*, 2001), whereas *M. signata*, or *C. cingulatum* sensu Steinke *et al.* (2004), has to be classified with another subfamily, possibly the Helicinae.

#### 4.3. *Helicigona lapicida* & *Xerotricha* sp.

Even though the shell voucher for *Helicigona lapicida* (SMF325426: Fig. 1D) was correctly identified, it remains unclear what species or even genus Steinke *et al.* (2004) have actually sequenced here. A BLAST search with sequences AY546280 (COI) and AY546360 (16S), published and deposited in GenBank as connected with voucher SMF325426, showed a similarity of 84% with a sequence of *Lozekia deubeli* (M. von Kimakowicz, 1890) (EU182491, Fehér *et al.*, 2009) and 94% with *Xerotricha conspurcata* (Draparnaud, 1801) (AY741439, Manganelli *et al.*, 2005), respectively (Table 2). A phylogeny reconstruction based on a larger COI dataset (results not shown) indicates that it might belong to a *Leptaxis* species (Van Riel *et al.*, 2005), but the sequences from that study are considerably shorter. No COI sequences have yet been deposited in Genbank for *Xerotricha*. However, the 16S data show (Fig. 3) that sequence AY546360 is situated in between *X. conspurcata* and *X. apicina* (Lamarck, 1822) (Manganelli *et al.*, 2005). On that basis we conclude that contrary to its supposed shell voucher, AY546360 belongs to a *Xerotricha* species (Fig. 1I), most certainly not to *H. lapicida* (Fig. 1D). The COI and 16S sequences (JF717799-JF7177800 and JF717817-JF717818, Table 1) that we independently obtained for *H. l. lapicida* on average differed 23.3% and 26.0% from sequence AY546280 and AY546360, respectively (Table 3). Moreover, our COI sequences are on average 98.7% identical to the *H. lapicida* sequences (EF398129-EF398131) of Haase & Misof (2009). The NJ phylogeny reconstructions on the basis of both COI and 16S (Fig. 2) show that *H. lapicida lapicida* and *H. lapicida andorrlica* form a monophyletic group, which is not closely related to *H. lapicida* sensu Steinke *et al.* (2004).

#### 4.4. *Arianta arbustorum* and *Isognomostoma isognomostomos*

The only COI and 16S sequences in GenBank from representatives of the subfamily Ariantinae that were deposited by research groups other than Steinke *et al.* (2004)



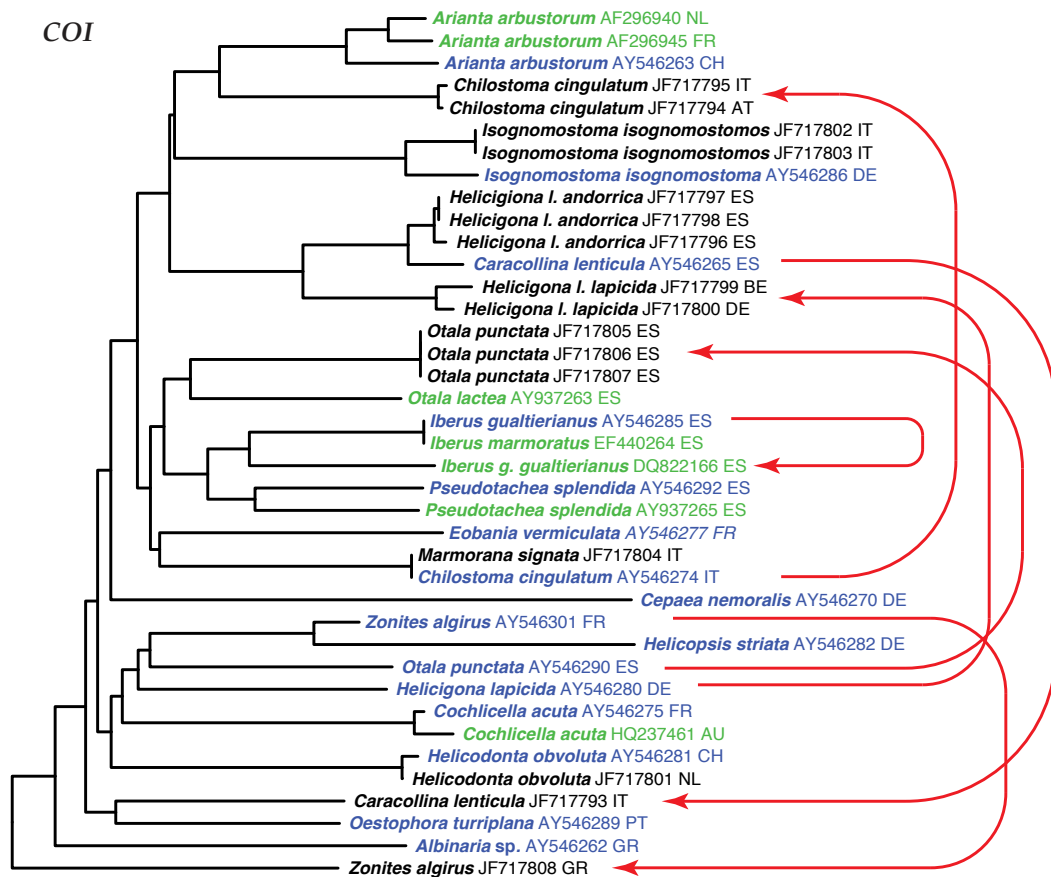
are COI sequences for *A. arbustorum* (Gittenberger *et al.*, 2004; Haase & Misof, 2009). Hence only the BLAST search for sequence AY546263 (*A. arbustorum*, COI) yielded a high similarity with other sequences in GenBank (94% with sequence AF296945 of *A. arbustorum*, Table 2). The other three sequences (*A. arbustorum* 16S and *I. isognomostomos* COI and 16S) were less similar (BLAST identity scores of less than 83%, Table 2) to best-matching sequences in GenBank, but were very similar to our sequences for those species. Our 16S sequences for *A. arbustorum* (JF717809-JF717810) were on average 95.1% identical to sequence AY546343, whereas our COI and 16S sequences for *I. isognomostomos* (JF717802-JF717803 and JF717820-JF717821) were on average 94.1% and 88.2% identical to sequences AY546286 and AY546366 (Steinke *et al.*, 2004), respectively (Table 3). Given the fact that the vouchers for *A. arbustorum* (SMF325410) and *I. isognomostomos* (SMF325432) were identified correctly and that the sequences deposited by Steinke *et al.* (2004) are consistent with our data, we assert that sequences AY546263, AY546286, AY546343 and AY546366 indeed belong to the indicated species.

#### *Helicidae, Helicinae*

##### 4.5 *Iberus gualtierianus* & *Iberus marmoratus* (A. Férussac, 1821)

A BLAST search with sequences AY546285 (COI) and AY546365 (16S) for *Iberus gualtierianus* sensu Steinke *et al.* (2004) showed a very high similarity (Table 2) with sequences EF440264 (100% identical) and EF440214 (BLAST identity score of 91%) for *I. marmoratus* (Elejalde *et al.*, 2008a). Elejalde *et al.* (2008b) showed that the *I. gualtierianus* species complex is monophyletic and consists of at least six major clades, whereas *I. marmoratus* belongs to a different *Iberus* lineage and constitutes a separate clade with at least three other species (clade 5 in Elejalde *et al.*, 2008a). An analysis with all of the sequences currently available for *I. gualtierianus* (78 for each COI and 16S) and *I. marmoratus* (11 for each marker), resulted in two major clades, i.e. one for *I. gualtierianus* and one for *I. marmoratus*. The only exception are the alleged *I. gualtierianus* sequences of Steinke *et al.* (2004) (AY546285 and AY546365) which group within the *I. marmoratus*-clade (for both COI and 16S) sensu Elejalde *et al.* (2008a). Inspection of the voucher for *I. gualtierianus* (SMF325431: Fig. 1J), according to the label collected near Ronda, Spain, brings another 'problem' to light. The shell in question is bleached and full of mud. Clearly, the snail that once sheltered inside cannot have been used for a molecular analysis. Shells that were severely damaged during the research process were replaced by better ones later on (pers. com. by Prof. Dr. M. Pfenninger to Dr. R. Janssen, Dr. E. Neubert and H. Nordsieck at a joint meeting on April 20th, 2005, which took place to discuss the observed problems with the voucher specimens). This procedure is fundamentally incorrect and potentially misleading however. A voucher should represent a part of the specimen that was actually sequenced. Occasionally labwork indeed is destructive and we empathise efforts that are being made to at least have a reference specimen, but if not the real voucher these should be marked as such (e.g. para-voucher) and originate from the same sample or population. In this case, the 'pseudo-voucher' is most probably from





**Figure 2.** A comparison of COI (this page) and 16S (page on the right) NJ phylogenies with a focus on the Helicidae s.s. In blue are sequences from Steinke *et al.* (2004), in green sequences from various studies (GenBank) and in black sequences from this study. Red arrows indicate erroneous or misidentified sequences and the correct sequences for the corresponding closest taxa in this figure.

even another locality. From near Ronda, *I. gualtierianus* is not known (T.E.J. Ripken, pers. com.), but *I. marmorana* occurs in that area indeed and, obviously, the sequences AY546285 and AY546365 belong to that species. Apparently, Steinke *et al.* (2004) sequenced *I. marmorana* from near Ronda, recorded that locality for their species, and deposited the shell of a conchologically similar, somewhat larger species, viz. *I. gualtierianus*, from another locality in southern Spain as its voucher.





would have been swapped, each sequence should still be unique and at least one of them should be most closely related to *O. lactea* (AY937263). Since no other sequences of *O. punctata* are available to support this explanation, we sequenced some specimens of *O. punctata* (kindly put at our disposal by A. Martínez-Ortí) ourselves. It turned out that our COI sequences for *O. punctata* (JF717805-JF717807) differed on average 19.2 % from the *O. punctata* sequence of Steinke *et al.*, whereas our 16S sequences (JF717823-JF717825) were nearly identical (on average 99.6%, Table 3) to AY546370. In this case Steinke *et al.*'s (2004) 16S sequence for *O. punctata* is correct, but the COI sequence is from another species. Moreover, inclusion of our *O. punctata* sequences showed (Fig. 2) that *O. punctata* and *O. lactea* form a monophyletic group and that the COI and 16S datasets are congruent.

#### 4.7. Other Helicinae in Steinke *et al.* (2004)

Sequences of the four remaining Helicinae used by Steinke *et al.* (2004), showed most significant alignments in a BLAST search with sequences for the same species as indicated in that paper (Table 2). In fact, except for *Pseudotachea splendida* (Draparnaud, 1801) these sequences all have an identity score of 96% or higher (Table 2). Neither these BLAST results, nor the inspected vouchers indicate other errors within the Helicinae, than those here reported for *Iberus* and *Otala*.

#### *Helicidae, Helicodontinae*

#### 4.8 *Gittenbergeria turriplana* & *Candidula codia* (Bourguignat, 1859)

The only sequences in GenBank for *Oestophora*, or *Gittenbergeria* according to Schileyko (1991), are those deposited by Steinke *et al.* (2004). Consequently, we expected to yield only moderate similarities with our BLAST searches. Nevertheless, sequence AY546289 (COI) for *G. turriplana* was (100%, Table 2) identical to sequence AY238622 of *Candidula codia* (published by Pfenniger *et al.*, 2003, according to the GenBank record, but not mentioned in that paper). Inspection of the voucher for *G. turriplana* (SMF325435: Fig. 1C) showed that it was identified correctly. To illustrate that the two species are strikingly different, a specimen of *C. codia* is also depicted (Fig. 1E). Clearly, the COI sequence cannot belong to both species. Given the position of sequence AY546289 in the COI based phylogeny reconstruction (the left tree in Fig. 2), i.e. next to our sequence for *Caracollina lenticula* (both belonging to the Trissexodontidae; Nordsieck, 1987), together with the fact that voucher SMF325435 for *G. turriplana* was identified correctly (we did not inspect the voucher for *C. codia*), we presume that sequence AY546289 belongs to *G. turriplana* indeed and hence that sequence AY238622 does not represent *C. codia*. Maybe the fact that both *G. turriplana* and *C. codia* occur syntopically near Boliqueime, Portugal ('Boliuquenne' after Steinke *et al.*, 2004: Appendix A), partly explains this confusion.

Given the extreme similarity of the *H. obvoluta* sequences (99.6% for both datasets, Table 3) with our sequences for that species and the fact that voucher SMF325427 was identified correctly, the data deposited for *H. obvoluta* are shown to be correct.



#### *Anomalies within the remainder data*

##### 4.9 *Zonites algirus* & *Helicella itala* (Linnaeus, 1758)

Although voucher SMF325447 was correctly identified as *Z. algirus*, a BLAST search showed that sequence AY546301 (COI) is strikingly similar (95%, Table 2) to sequence AY546282 for *Helicopsis striata* from the same publication. For neither species are additional COI sequences available at this moment, but for *Zonites* several 16S sequences have been deposited in GenBank (among which *Z. algirus*; Kornilios *et al.*, 2009). A BLAST search with sequence AY546381 (16S) supposedly also belonging to voucher SMF325447 did not show any of the just mentioned *Zonites* sequences as most similar. Instead sequence AY485907 for *Zebrina detrita* (Müller, 1774) (Parmakelis *et al.*, 2005) was retrieved as being most similar (78%, Table 2). Particularly given the surplus of 16S data it was a pity not to have any additional COI sequences of *Zonites*. Upon request P. Kornilios and S. Giokas kindly put at our disposal a COI and 16S sequence for a specimen which they identified as *Z. algirus*. These sequences were used in the NJ phylogenies depicted in Fig. 2. The 16S phylogeny (the tree on the right side in Fig. 2) shows that sequence AY546381 does not belong to the monophyletic (after Kornilios *et al.*, 2009) taxon *Zonites*. Moreover, these *Z. algirus* sequences (JF717808 and JF717826, Table 1) differed 21.6% and 34.3% (COI and 16S, respectively, Table 3) from the sequences for the same species (AY546301 and AY546381) according to Steinke *et al.* (2004). These results indicate that sequence AY546301 (COI) must be a contaminant, possibly originating from *H. striata*. This definitely needs confirmation however, since the ‘voucher’ for *H. striata* (SMF325428) is another example of a ‘pseudo-voucher’ (Fig. 1G). Given the fact that sequence AY546381 is unique within the 16S dataset of Steinke *et al.* and very different (40.0%, Table 3) from sequence AY546362 for *H. striata*, we cannot suggest to which species it actually belongs. It is certainly not a *Zonites* species.

##### 4.10. *Trochoidea elegans* & *Xerosecta cespitum*

A BLAST search with sequence AY546295 (COI) for *Trochoidea elegans* sensu Steinke *et al.* (2004) yielded an identity score of 99% (Table 2) with sequence AY546271 for “*Cernuella*” *cespitum*, published in the same paper. The only other *T. elegans* sequence on GenBank (FJ627176; Sauer and Hausdorf, 2009) differs 20.9% from sequence AY546295.

Moreover, a NJ analysis with all of the COI sequences of Steinke *et al.* (2004), thus including the five *Trochoidea* sequences and additional sequences for *T. elegans* (FJ627176), *T. sprati* (L. Pfeiffer, 1846) (FJ627177) and *T. geyeri* (Soós, 1926) (AY240948) never showed *Trochoidea* as a monophyletic group (phylogeny not depicted). The group of *Trochoidea* sequences that did form a clade always excluded *T. elegans* (AY546295), *T. trochoides* (Poiret, 1789) (AY546299) and *T. seetzeni* (L. Pfeiffer, 1847) (AY546298). This contrasts conspicuously with the monophyly of the genus as depicted in Fig. 1 in Steinke *et al.* (2004). As for *T. seetzeni* it might be true that this species indeed does not belong to *Trochoidea*; nowadays it is classified under the



**Table 2.** MegaBLAST search results for a selection of COI and 16S sequences of Steinke *et al.* (2004).

| MegaBlast searches based on COI sequences of Steinke <i>et al.</i> (2004) |          |                                   |          | Date on which the search was performed: 28-03-2011. |      |          |                                   |
|---------------------------------------------------------------------------|----------|-----------------------------------|----------|-----------------------------------------------------|------|----------|-----------------------------------|
| Query taxon                                                               | ACCN     | Most significant alignment        | ACCN     | Score                                               | Covg | Identity | Reference                         |
| <i>Arianta arbustorum</i>                                                 | AY546263 | <i>Arianta arbustorum</i>         | AF296945 | 800                                                 | 100% | 94%      | Gittenberger <i>et al.</i> , 2004 |
| <i>Caracollina lenticula</i>                                              | AY546265 | <i>Helicigona lapicida</i>        | AF297000 | 660                                                 | 95%  | 90%      | Gittenberger <i>et al.</i> , 2004 |
| <i>Cepaea nemoralis</i>                                                   | AY546270 | <i>Cepaea nemoralis</i>           | U23045   | 902                                                 | 100% | 97%      | Yamazaki <i>et al.</i> , 1997     |
| <i>Chilostoma cingulatum</i>                                              | AY546274 | <i>Marmorana signata strigata</i> | GU391375 | 654                                                 | 67%  | 100%     | Fiorentino <i>et al.</i> , 2010   |
| <i>Cochlicella acuta</i>                                                  | AY546275 | <i>Cochlicella acuta</i>          | HQ237463 | 950                                                 | 100% | 99%      | unpublished                       |
| <i>Eobania vermiculata</i>                                                | AY546277 | <i>Eobania vermiculata</i>        | JF277391 | 881                                                 | 100% | 98%      | unpublished                       |
| <i>Helicella itala</i>                                                    | AY546278 | <i>Helicella obovia</i>           | AY546279 | 822                                                 | 95%  | 96%      | Steinke <i>et al.</i> , 2004      |
| <i>Helicigona lapicida</i>                                                | AY546280 | <i>Lozekia deubeli</i>            | EU182491 | 523                                                 | 100% | 84%      | Fehér <i>et al.</i> , 2009        |
| <i>Helicodontia obvoluta</i>                                              | AY546281 | <i>Lozekia deubeli</i>            | EU182493 | 440                                                 | 99%  | 81%      | Fehér <i>et al.</i> , 2009        |
| <i>Helix aspersa</i>                                                      | AY546283 | <i>Helix aspersa</i>              | AY345052 | 913                                                 | 100% | 98%      | Grande <i>et al.</i> , 2004       |
| <i>Iberus gualtierianus</i>                                               | AY546285 | <i>Iberus marmoratus</i>          | EF440264 | 972                                                 | 100% | 100%     | Elejalde <i>et al.</i> , 2008a    |
| <i>I. isognomostoma</i>                                                   | AY546286 | <i>Eobania vermiculata</i>        | JF277394 | 444                                                 | 98%  | 82%      | unpublished                       |
| <i>Monacha cantiana</i>                                                   | AY546287 | <i>Helix aspersa</i>              | HM179144 | 375                                                 | 90%  | 81%      | unpublished                       |
| <i>Monacha martensiana</i>                                                | AY546288 | <i>Helix aspersa</i>              | HM179144 | 497                                                 | 99%  | 84%      | unpublished                       |
| <i>Oestophora turriplana</i>                                              | AY546289 | <i>Candidula codia</i>            | AY238622 | 891                                                 | 93%  | 100%     | Pfenninger <i>et al.</i> , 2003   |
| <i>Otala punctata</i>                                                     | AY546290 | <i>Trochoidea pyramidata</i>      | AY546297 | 894                                                 | 99%  | 97%      | Steinke <i>et al.</i> , 2004      |
| <i>Pseudotachea splendida</i>                                             | AY546292 | <i>Pseudotachea splendida</i>     | AY937265 | 617                                                 | 100% | 87%      | Elejalde <i>et al.</i> , 2008b    |
| <i>Trochoidea elegans</i>                                                 | AY546295 | <i>Cernuella cespitum</i>         | AY546271 | 824                                                 | 98%  | 99%      | Steinke <i>et al.</i> , 2004      |
| <i>Trochoidea pyramidata</i>                                              | AY546297 | <i>Otala punctata</i>             | AY546290 | 894                                                 | 98%  | 97%      | Steinke <i>et al.</i> , 2004      |
| <i>Zonites algerius</i>                                                   | AY546301 | <i>Helicopsis striata</i>         | AY546282 | 675                                                 | 81%  | 95%      | Steinke <i>et al.</i> , 2004      |

| MegaBlast searches based on 16S sequences of Steinke <i>et al.</i> (2004) |          |                               |          | Date on which the search was performed: 28-03-2011. |      |          |                                 |
|---------------------------------------------------------------------------|----------|-------------------------------|----------|-----------------------------------------------------|------|----------|---------------------------------|
| Query taxon                                                               | ACCN     | Most significant alignment    | ACCN     | Score                                               | Covg | Identity | Reference                       |
| <i>Arianta arbustorum</i>                                                 | AY546343 | <i>Helix aspersa</i>          | AF126131 | 241                                                 | 87%  | 83%      | Guiller <i>et al.</i> , 2001    |
| <i>Caracollina lenticula</i>                                              | AY546345 | <i>Cochlicella acuta</i>      | AY546355 | 385                                                 | 100% | 97%      | Steinke <i>et al.</i> , 2004    |
| <i>Cepaea nemoralis</i>                                                   | AY546350 | <i>Cepaea nemoralis</i>       | AF249259 | 612                                                 | 100% | 99%      | Wollschei <i>et al.</i> , 2001  |
| <i>Cernuella neglecta</i>                                                 | AY546352 | <i>Hygromia limbata</i>       | AY546364 | 569                                                 | 100% | 96%      | Steinke <i>et al.</i> , 2004    |
| <i>Chilostoma cingulatum</i>                                              | AY546354 | <i>Marmorana signata</i>      | GU391404 | 370                                                 | 91%  | 89%      | Fiorentino <i>et al.</i> , 2010 |
| <i>Cochlicella acuta</i>                                                  | AY546355 | <i>Caracollina lenticula</i>  | AY546345 | 385                                                 | 100% | 87%      | Steinke <i>et al.</i> , 2004    |
| <i>Eobania vermiculata</i>                                                | AY546357 | <i>Eobania vermiculata</i>    | JF277389 | 459                                                 | 100% | 96%      | unpublished                     |
| <i>Helicella itala</i>                                                    | AY546358 | <i>Hygromia limbata</i>       | AY546364 | 599                                                 | 100% | 95%      | Steinke <i>et al.</i> , 2004    |
|                                                                           |          | <i>Cernuella neglecta</i>     | AY546352 | 542                                                 | 92%  | 95%      | Steinke <i>et al.</i> , 2004    |
|                                                                           |          | <i>Helicella itala</i>        | AY741441 | 442                                                 | 94%  | 89%      | Manganelli <i>et al.</i> , 2005 |
| <i>Helicella obvia</i>                                                    | AY546359 | <i>Hygromia limbata</i>       | AY546364 | 387                                                 | 100% | 87%      | Steinke <i>et al.</i> , 2004    |
| <i>Helicigona lapicida</i>                                                | AY546360 | <i>Xerotracha conspurcata</i> | AY741439 | 499                                                 | 100% | 94%      | Manganelli <i>et al.</i> , 2005 |
| <i>Helicodonta obvoluta</i>                                               | AY546361 | <i>Helicodonta obvoluta</i>   | AY741446 | 457                                                 | 100% | 94%      | Manganelli <i>et al.</i> , 2005 |
| <i>Helix aspersa</i>                                                      | AY546363 | <i>Helix aspersa</i>          | AF126128 | 580                                                 | 100% | 99%      | Guiller <i>et al.</i> , 2001    |
| <i>Iberus gualtierianus</i>                                               | AY546365 | <i>Iberus marmoratus</i>      | EF440214 | 416                                                 | 100% | 91%      | Elejalde <i>et al.</i> , 2008   |
| <i>I. isognonostoma</i>                                                   | AY546366 | <i>Chilostoma cingulatum</i>  | AY546354 | 255                                                 | 96%  | 82%      | Steinke <i>et al.</i> , 2004    |
| <i>Monacha cantiana</i>                                                   | AY546367 | <i>Monacha cartusiana</i>     | AY741416 | 276                                                 | 88%  | 96%      | Manganelli <i>et al.</i> , 2005 |
| <i>Monacha martensiana</i>                                                | AY546368 | <i>Candidula spadæ</i>        | AY741435 | 377                                                 | 89%  | 88%      | Manganelli <i>et al.</i> , 2005 |
| <i>Oestophora turriplana</i>                                              | AY546369 | <i>Monacha martensiana</i>    | AY741420 | 145                                                 | 92%  | 78%      | Manganelli <i>et al.</i> , 2005 |
| <i>Otala punctata</i>                                                     | AY546370 | <i>Otala lactea</i>           | AY937264 | 425                                                 | 100% | 89%      | Elejalde <i>et al.</i> , 2008b  |
| <i>Pseudotachea splendida</i>                                             | AY546372 | <i>Pseudotachea splendida</i> | AY937266 | 499                                                 | 100% | 93%      | Elejalde <i>et al.</i> , 2008b  |
| <i>Trochoidaea elegans</i>                                                | AY546375 | <i>Xerosecta aragonis</i>     | AY741428 | 254                                                 | 100% | 81%      | Manganelli <i>et al.</i> , 2005 |
| <i>Trochoidaea pyramidata</i>                                             | AY546377 | <i>Trochoidaea pyramidata</i> | AY741444 | 459                                                 | 94%  | 90%      | Manganelli <i>et al.</i> , 2005 |
| <i>Zonites algirus</i>                                                    | AY546381 | <i>Zebrina detrita</i>        | AY485907 | 191                                                 | 99%  | 78%      | Parmakelis <i>et al.</i> , 2005 |

ACCN = GenBank Accession Numbers, Covg = Coverage.

genus *Xerocrassa*.

The 16S data show a slightly different pattern (also indicating however, that *Trochoidea* is not monophyletic); here *T. elegans* does group with *T. pyramidata* (the identity of the latter sequence is confirmed by sequence AY741444 of Manganelli *et al.*, 2005). Inspection of voucher SMF325441 for *T. elegans* showed that this specimen was identified correctly. For now we can only conclude that the COI sequence AY546295 does not belong to *T. elegans* and possibly represents *Xerosecta cespitum* (SMF325417; Steinke *et al.*, 2004).

#### 4.11. *Cochlicella acuta* (Müller, 1774) & *Helicigona lapicida*

A BLAST search with sequence AY546275 (COI) for *Cochlicella acuta* shows that it is nearly identical (average identity score of 99%, Table 2) to sequences HQ237461-HQ237463 for the same species (a study by Bon *et al.*; data unpublished). However, the 16S sequence (AY546355) is very different (on average 30.9%, Table 3) from other *C. actuta* 16S sequences (AY741442 and AY741443; Manganelli *et al.*, 2005). The similarity of sequence AY546355 to our sequences (JF717817-JF717818) for *Helicigona lapicida lapicida* (average similarity score of 95.2%, Table 3) and the topological position of that sequence within the 16S NJ phylogeny (Fig. 2) are striking. Based on these results we conclude that the 16S sequence for *C. acuta* sensu Steinke *et al.* (2004) exemplifies contamination and actually originated from *H. lapicida* (voucher SMF325426; Fig. 1D). Herewith showing yet another inconsistency between the COI and the 16S datasets of Steinke *et al.* (2004).

#### 4.12. *Monacha* species

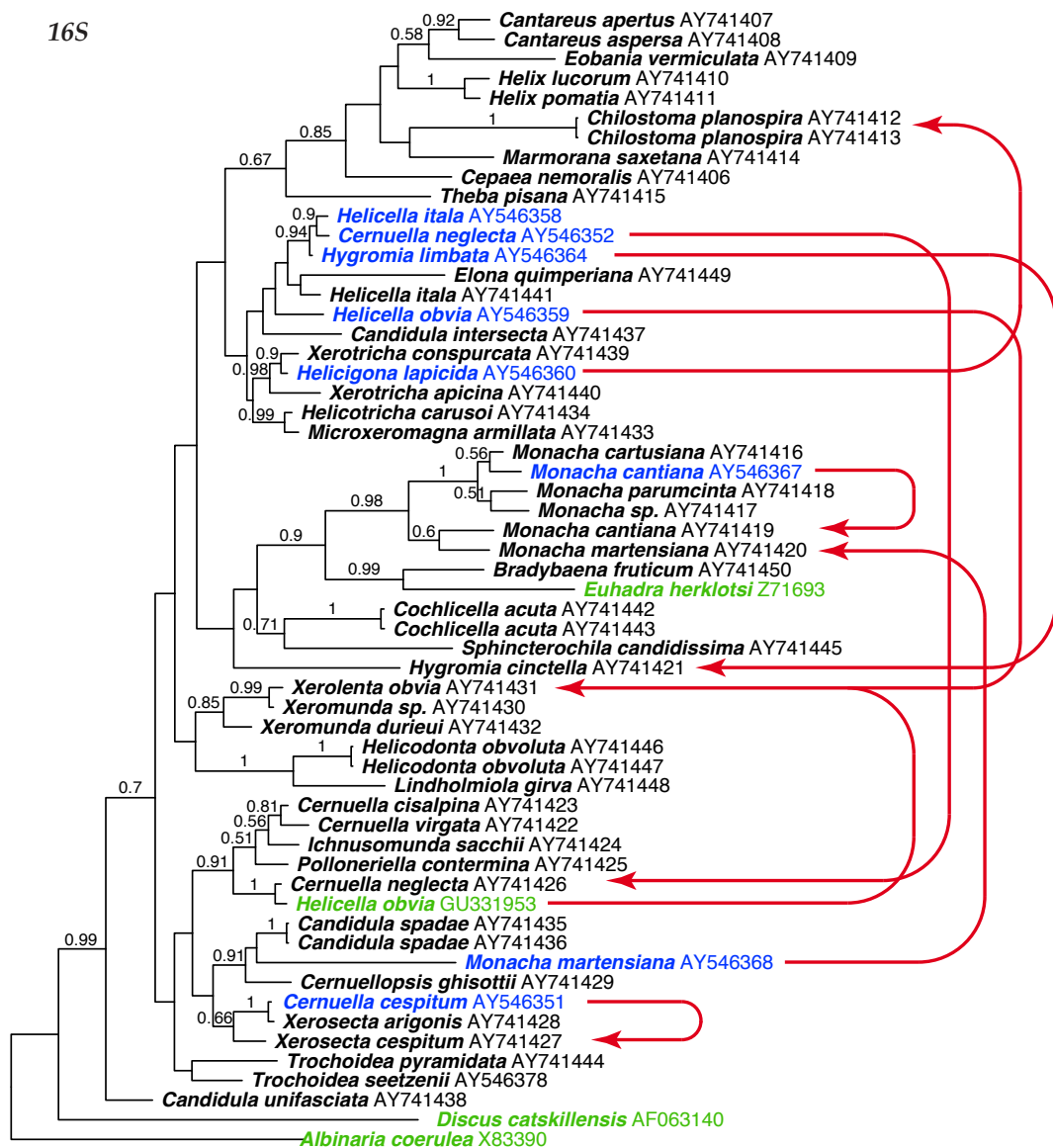
A BLAST search with Steinke *et al.*'s (2004) COI (AY546287) and 16S (AY546367) sequences for *Monacha cantiana* (Montagu, 1803) (voucher SMF325433) did not show sequences HQ204502 and HQ204543 (Duda *et al.*, 2010) for the same marker and species, respectively, as most similar. By sequencing a number of *Monacha* species, including *Monacha cartusiana* (Müller, 1774) and *M. cantiana*, Manganelli *et al.* (2005) have shown that the genus is monophyletic (also Fig. 3; posterior probability 0.98). The BLAST search with sequence AY546367 (16S), however, did not show *M. cantiana* (AY741419), but *M. cartusiana* (AY741416) as most similar (identity score of 96%, Table 2). Since two research groups (Manganelli *et al.*, 2005 and Duda *et al.*, 2010) independently obtained similar 16S sequences for *M. cantiana* (AY741419 and HQ204543) that clearly differ (Fig. 3) from the *M. cantiana* sequence (AY546367) sensu Steinke *et al.* (2004), we conclude that the latter authors did not sequence *M. cantiana*, but most likely *M. cartusiana*. The BLAST search with sequence AY546287 (COI) was

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**Figure 3.** Bayesian consensus tree based on 16S with a focus on the Hygrommidae. This cladogram shows all compatible partitions and posterior probabilities of 0.5 and higher. In blue are sequences from Steinke *et al.* (2004), in green sequences from various studies (Genbank)

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and in black sequences from Manganelli *et al.* (2005). Red arrows indicate erroneous or misidentified sequences and the correct sequences for the corresponding most closely related taxa in this figure.

most similar (81%, Table 2) to sequence HM179144 for *Cornu aspersum* (Müller, 1774). At this moment it is unclear if COI sequence AY546287 indeed belongs to a species of *Monacha*. In agreement with the BLAST result for 16S, the shell voucher (SMF325433) belongs to *M. cartusiana*.

A BLAST search with sequences AY546288 (COI) and AY546368 (16S), allegedly belonging to *M. martensiana* (Tiberi, 1869) (Steinke *et al.*, 2004), showed the highest similarity with sequences HM179144 (84%) and AY741435 (88%) of *Cornu aspersum* (unpublished) and *Candidula spadae* (Calcara, 1845) (Manganelli *et al.*, 2005), respectively. Currently the only other *M. martensiana* sequence available on GenBank is sequence AY741420 (16S) of Manganelli *et al.* (2005), which differs 22.5% from sequence AY546368. Hence the *M. martensiana* sequences of Steinke *et al.* (2004) are unlikely to belong to *Monacha* (see Fig. 3). However, voucher SMF 325434 indeed belongs to *Monacha*, possibly *M. cartusiana* as well.

#### 4.13. *Helicella itala* and *Xerolenta obvia* (Menke, 1828)

At this moment, three research groups deposited clearly different sequences in Genbank for *Xerolenta obvia* (Steinke *et al.*, 2004; Manganelli *et al.*, 2005 and Dinapoli *et al.*, 2010), suggesting that the identification of this species requires specialist knowledge. Given the close relation of *Xerolenta* with *Xeromunda*, we assume that only the identification of Manganelli will be correct (see Fig. 3). In agreement with Steinke *et al.* (2004, Fig. 1), separate NJ analyses (including all taxa from that paper) of both the COI and 16S datasets (results not depicted) showed that *H. itala* (AY546358) and *X. obvia* (AY546359) form a clade (which is also shown in Fig. 3, albeit with a low posterior probability). In comparison with the Helicellinae sensu Steinke *et al.* (2004), the helicelline hygromiids and monachine hygromiids sensu Manganelli *et al.* (2005) form a broader sampling of the taxonomic diversity within this group. Manganelli *et al.* (2005) showed that different reconstruction methods yield different topologies (based on 16S), but each of their phylogeny reconstructions shows that *H. itala* and *X. obvia* do not constitute a monophyletic group. We used Fig. 3 to assess how the sequences for *H. itala* and *X. obvia* from the different studies relate to each other and conclude that *X. obvia* (GU331953) sensu Dinapoli *et al.* is most probably the same species that Manganelli *et al.* identified as *Cernuella neglecta* (Draparnaud, 1805) (AY741426); the BLAST identity score between both sequences is 97%. As far as only the above mentioned sequences are concerned, Steinke *et al.* and Manganelli *et al.* seem to recognize the same taxon as *H. itala* (AY546359 and AY741441). Nevertheless BLAST identity scores (Table 2) for the 16S sequence of *H. itala* (AY546358) sensu Steinke *et al.* (2004) AY546359 are higher with *Hygromia limbata* (Draparnaud, 1805) (AY546364; 95%) and *C. neglecta* (AY546352; 95%) from the same study than with *H. itala* (AY741441; 87%) from Manganelli *et al.* (2005). This is also shown in Fig. 3 where *H. itala*, *H. limbata* and *C. neglecta* sensu Steinke *et al.* (2004) constitute a clade with a high posterior probability (0.94). Given the monophyly of *Cernuella* (as supported by *C. cisalpina* (Rossmässler, 1837) and *C. virgata* (Da Costa, 1778); Fig.3), we may conclude that *C. neglecta* (AY546352) sensu Steinke *et al.* (2004) does not represent a



**Table 3.** Sequence similarities based on uncorrected *p*-distances.

| From taxon (COI sequence / 16S sequence):              | To taxon (COI sequence / 16S sequence):                                        | Similarity score (%) |      |
|--------------------------------------------------------|--------------------------------------------------------------------------------|----------------------|------|
|                                                        |                                                                                | COI                  | 16S  |
| <i>Arianta arbustorum</i> AY546263 / AY546343          | <i>Arianta arbustorum</i> (AF296940+AF296945) / (JF717809-JF717810)            | 93.2                 | 95.1 |
| <i>Caracollina lenticula</i> AY546365 / AY546345       | <i>Helicigona lapicida andorrica</i> (JF717796-JF717798) / (JF717814-JF717816) | 97.2                 | 98.0 |
| <i>Caracollina lenticula</i> AY546365 / AY546345       | <i>Caracollina lenticula</i> JF717793 / JF717811                               | 80.6                 | 70.5 |
| <i>Chilostoma cingulatum</i> AY546354 / AY54635        | <i>Chilostoma cingulatum</i> (JF717794-JF717795) / (JF717812-JF717813)         | 81.2                 | 76.9 |
| <i>Chilostoma cingulatum</i> AY546354 / AY54635        | <i>Marmorana signata</i> JF717804 / JF717822                                   | 100.0                | 94.9 |
| <i>Cochlicella acuta</i> AY546275 / AY546355           | <i>Cochlicella acuta</i> HQ237461 / (AY741442-AY741443)                        | 98.9                 | 69.1 |
| <i>Cochlicella acuta</i> AY546275 / AY546355           | <i>Helicigona lapicida lapicida</i> (JF717799-JF717800) / (JF717817-JF717818)  | 75.9                 | 95.2 |
| <i>Helicigona lapicida</i> AY546280 / AY546360         | <i>Helicigona lapicida lapicida</i> (JF717799-JF717800) / (JF717817-JF717818)  | 76.7                 | 74.0 |
| <i>Helicodonta obvoluta</i> AY546281 / AY546361        | <i>Helicodonta obvoluta</i> JF717801 / JF717819                                | 99.6                 | 99.6 |
| <i>Isognomostoma isognomostoma</i> AY546286 / AY546366 | <i>Isognomostoma isognomostomos</i> (JF717802-JF717803) / (JF717820-JF717821)  | 94.1                 | 88.2 |
| <i>Otala punctata</i> AY546290 / AY546370              | <i>Otala punctata</i> (JF717805-JF717807) / (JF717823-JF717825)                | 80.8                 | 99.6 |
| <i>Zonites algirus</i> AY546301 / AY546381             | <i>Zonites algirus</i> JF717808 / JF717826                                     | 78.4                 | 65.7 |
| <i>Zonites algirus</i> AY546301 / AY546381             | <i>Helicopsis striata</i> AY546282 / AY546362                                  | 93.2                 | 60.0 |

\* Similarity score = 100 - uncorrected *p*-distance \* 100

*Cernuella* species. Given the limited data on *Hygromia* (Fig. 3 presumably includes only two species) we can not draw firm conclusions about sequence AY546364 (*H. limbata* sensu Steinke *et al.*, 2004). However we can conclude that the 16S sequences of *H. limbata*, *C. neglecta* and *H. obvia* sensu Steinke *et al.* (2004), are all more similar to *H. itala* (based on sequences of both Steinke *et al.*, 2004 and Manganelli *et al.*, 2005) than each of them is to the most closely related taxon in Fig. 3 (red arrows). In general sequences and phylogenies published by Manganelli *et al.* (2005) strongly contrast with the data of Steinke *et al.* (2004) and much better reflect current anatomy-based classifications.

#### 4.14 Accuracy of deposited sequences

In order to align the COI sequences, gaps had to be postulated in sequences of *Trochoidea pyramidata* (AY546297), *Trochoidea geyeri* (AY546296) and *Otala punctata* (AY546290) and one base had to be deleted from the sequence of *Hygromia limbata* (AY546284). Since all of these indels will cause frame-shifts, we conclude that the sequences were not properly checked before submission to GenBank.

## CONCLUSION

This study shows a variety of errors and methodological weaknesses in the 2004 paper of Steinke *et al.* published in Molecular Phylogenetics and Evolution. In that paper *Helicigona lapicida andorrica* and *Marmorana signata* were incorrectly identified as *Caracollina lenticula* and *Chilostoma cingulatum*, respectively. Here we add data (COI and 16S) for the real *Caracollina lenticula* and *Chilostoma cingulatum* to GenBank. Furthermore we correct the wrong conclusions drawn by Steinke *et al.* (2004) on the bases of their misidentifications. *Caracollina* does not belong to the Ariantinae but most likely to the Trissexodontinae, in agreement with the usual classification. The so-called *C. lenticula* sequences of Steinke *et al.* (2004) show that *H. lapicida andorrica* belongs to the Ariantinae, again in agreement with the usual classification.

In other cases we have shown that the sequences deposited in GenBank by Steinke *et al.* (2004) may not belong to the specified taxon, even in cases where the vouchers (shells) were identified correctly. The COI and 16S sequences that are allegedly of *H. lapicida* do certainly not refer to that species, despite the correctly identified shell voucher. In other instances (e.g. *Otala lactea*, *Cochlicella acuta*) only one of the two sequences (of the COI and 16S datasets examined here) turned out to be correct, so errors had to be made either with the molecular analyses or with submission of the sequences. For a number of taxa (e.g. *Trochoidea elegans*, *Monacha cartusiana*, *Xerolenta obvia* and *Cernuella neglecta*) the sequences deposited in GenBank by Steinke *et al.* (2004) differ strongly from those published by other authors, which are in conformity then with the usual classification, whereas the results published by Steinke *et al.* (2004) are not. A fundamentally incorrect and potentially misleading procedure employed by Steinke *et al.* (2004) is the use of what is here introduced as a 'pseudo-voucher', i.e. a part (like a shell) of an individual that differs from the specimen that



was actually used for DNA analysis. Steinke *et al.* (2004) apparently sequenced *Iberus marmorana* from Ronda, but while referring to that locality, deposited a shell of a conchologically somewhat similar species from elsewhere, i.e. *I. gualtierianus*, as its 'voucher'. As an alternative for the lack of an actual voucher (which indeed might be destructed during DNA extractions) we here introduce the term 'para-voucher', which is (a part of) a different specimen than the one used for DNA analysis, but from the same sample or population. Although for the Ariantinae sensu Steinke *et al.* (2004) the sequences for *A. arbustorum* and *I. isognomostomos* are after all correct, still three out of the five taxa of that subfamily were either identified incorrectly or were published with incorrect sequences.

Steinke *et al.* (2004) published a major paper on the phylogeny of the Gastropoda, Pulmonata.

We intend to prevent the further spreading of an inexcusably high number of errors of various kinds in that article, because these incorrect data may result in wrong conclusions, such as already happened in Steinke *et al.*, 2005. In the latter paper COI and 16S sequences of Steinke *et al.* (2004) were used as a reference dataset (required by the program TaxI) for the identification of juvenile snails. Four of these snails being *C. acuta*, according to the authors. Remarkably the 16S "divergence-value" (Steinke *et al.*, 2005) between sequences from these snails and the reference dataset was only 0.4%. Given the fact that the 2004 study of Steinke *et al.* did not include any 16S sequence of *C. acuta* (this paper), for this to be true would either mean that the authors identified a juvenile *H. lapicida andorrica* as *C. acuta* four times, or their reference dataset (Steinke *et al.*, 2005) did not consist of sequences AY546342–546381 and AY546262–AY546301 as currently included in GenBank. Either way, we strongly advise to be cautious when using sequence data from Steinke *et al.* (2004), if any.

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## REFERENCES

- BANK, R.A., BOUCHET, P., FALKNER, G., GITTENBERGER, E., HAUSDORF, B., VON PROSWITZ, T. & RIPKEN, T.E.J. 2001. Supraspecific classification of European non-marine Mollusca (CLECOM Sections I+II). *Heldia* 4, 77–128.



- DINAPOLI, A., ZINSSMEISTER, C. & KLUSSMANN-KOLB, A. 2011. New insights into the phylogeny of the Pyramidellidae (Gastropoda). *Journal of Molluscan Studies* 77, 1-7. doi:10.1093/mollus/eyq027
- DUDA, M., SATTMANN, H., HARING, E., BARTEL, D., WINKLER, H., HARL, J. & KRUCKENHAUSER, L. 2011. Genetic differentiation and shell morphology of *Trochulus oreinos* (Wagner, 1915) and *T. hispidus* (Linnaeus, 1758) (Pulmonata: Hygromiidae) in the northeastern Alps. *Journal of Molluscan Studies* 77, 30-40.
- ELEJALDE, M.A., MADEIRA, M.J., ARREBOLA, J.R., MUNOZ, B. & GOMEZ-MOLINER, B.J. 2008a. Molecular phylogeny, taxonomy and evolution of the land snail genus *Iberus* (Pulmonata: Helicidae). *Journal of Zoological Systematics and Evolutionary Research* 46, 193–202.
- ELEJALDE, M.A., MADEIRA, M.J., MUÑOZ, B., ARRÉBOLA, J.R. & GÓMEZ-MOLINER, B.J. 2008b. Mitochondrial DNA diversity and taxa delineation in the land snails of the *Iberus gualtieranus* (Pulmonata, Helicidae) complex. *Zoological Journal of the Linnean Society* 154, 722–737.
- FEHÉR, Z., SZABÓ, K., BOZSÓ, M. & PÉNZES, Z. 2009. Phylogeny and phylogeography of the *Lozekia*–*Kovacsia* species group (Gastropoda: Hygromiidae). *Journal of Zoological Systematics and Evolutionary Research* 47, 306–314.
- FIORENTINO, V., SALOMONE, N., MANGANELLI & G., GIUSTI, F. 2010. Historical biogeography of Tyrrhenian land snails: The Marmorana–Tyrrheniberus radiation (Pulmonata, Helicidae). *Molecular Phylogenetics and Evolution* 55, 26–37.
- GITTENBERGER, E., PIEL W.H. & GROENENBERG D.S.J. 2004. The Pleistocene glaciations and the evolutionary history of the polytypic snail species *Arianta arbustorum* (Gastropoda, Pulmonata, Helicidae). *Molecular Phylogenetics and Evolution* 30, 64–73.
- HAASE M. & MISOF B. 2009. Dynamic gastropods: stable shell polymorphism despite gene flow in the land snail *Arianta arbustorum*. *Journal of Zoological Systematics and Evolutionary Research* 47, 105–114. doi: 10.1111 /j.1439-0469.2008.00488.x
- KATOH, K., MISAWA, K., KUMA, K.I. & MIYATA, T. 2002. MAFFT: a novel method for rapid multiple sequence alignment based on fast Fourier transform. *Nucleic Acids Research* 30, 3059-3066.
- KATOH, K. & TOH, H. 2008. Improved accuracy of multiple ncRNA alignment by incorporating structural information into a MAFFT-based framework. *BMC Bioinformatics*, 9, 13.
- KORNILIOS, P., POULAKAKIS, N., MYLONAS, M. & VARDINOYANNIS, K. 2009. The phylogeny and biogeography of the genus *Zonites* Montfort, 1810 (Gastropoda: Pulmonata): preliminary evidence from mitochondrial data. *Journal of Molluscan Studies* 75, 109-117. doi:10.1093/mollus/eyp003
- MADDISON, W.P. & MADDISON, D.R. 2005. *MacClade 4.08: Analysis of Phylogeny and Character Evolution*. Sinauer Associates, Sunderland, Massachusetts.
- MANGANELLI G., SALOMONE N. & GIUSTI, F. 2005. A molecular approach to the phylogenetic relationships of the western palearctic Helicoidea (Gastropoda: Stylommatophora). *Biological Journal of the Linnean Society* 85, 501–512.
- NORDSIECK, H. 1987. Revision des Systems der Helicoidea (Gastropoda: Stylommatophora). *Archiv für Molluskenkunde* 118, 9-50.



- NYLANDER, J.A.A. 2004. MrModeltest v. 2.2. Evolutionary Biology Centre, Uppsala University.
- PARMAKELIS, A., PFENNINGER, M., SPANOS, L., PAPAGIANNAKIS, G., LOUIS & C., MYLONAS, M. 2005. Inference of a radiation in *Mastus* (Gastropoda, Pulmonata, Enidae) on the island of Crete. *Evolution* 59, 991–1005.
- PFENNINGER, M., EPPENSTEIN, A. & MAGNIN, F. 2003. Evidence for ecological speciation in the sister species *Candidula unifasciata* (Poiret, 1801) and *C. rugosiuscula* (Michaud, 1831) (Helicellinae, Gastropoda). *Biological Journal of the Linnean Society* 79, 611–628.
- PENNISI, E. 2008. DNA data. Proposal to 'Wikify' GenBank meets stiff resistance. *Science* 319, 1598–1599.
- RONQUIST, F. & HUELSENBECK, J.P. 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics*, 19: 1572–1574.
- SAUER, J. & HAUSDORF, B. 2009. Sexual selection is involved in speciation in a land snail radiation on crete. *Evolution* 63, 2535–2546.
- SCHILEYKO, A. A. 1991. Taxonomic status, phylogenetic relations and system of the Helicoidea sensu lato (Pulmonata). *Archiv für Molluskenkunde* 120, 187–236.
- SOES, D.M. & WINTER, DE, A.J., 2005. *Xerotricha apicina* (Lamarck, 1822), een nieuwe immigrant in Nederland (Gastropoda, Pulmonata, Hygromiidae). *Basteria* 69, 81–86.
- STEINKE, D., ALBRECHT, C. & PFENNINGER, M. 2004. Molecular phylogeny and character evolution in the western Palaearctic Helicidae s.l. (Gastropoda: Stylommatophora). *Molecular Phylogenetics and Evolution* 32, 724–734.
- STEINKE, D., VENCES, M., SALZBURGER & W., MEYER A. 2005. TaxI: a software tool for DNA barcoding using distance methods. *Philosophical Transactions of the Royal Society B* 360, 1975–1980. doi:10.1098/rstb.2005.1729
- SWOFFORD, D.L. 2002. PAUP\*. Phylogenetic Analysis Using Parsimony (\* and other methods). Version 4.0b10. Sinauer Associates, Sunderland, Massachusetts.
- VAN RIEL, P., JORDAENS K., HOUTTE, N. VAN, FRIAS MARTINS, A.M., VERHAGEN R. & BACKELJAU T. 2005. Molecular systematics of the endemic Leptaxini (Gastropoda: Pulmonata) on the Azores islands. *Molecular Phylogenetics and Evolution* 37, 132–143.
- ZHANG, Z., SCHWARTZ, S., WAGNER, L. & MILLER, W. 2000. A greedy algorithm for aligning DNA sequences. *Journal of Computational Biology* 7, 203–214.





CHAPTER 4

**On the identity of broad-shelled mussels  
(Mollusca, Bivalvia, *Mytilus*) from the  
Dutch delta region**

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## ABSTRACT

Late Quaternary (Eemian) deposits of the Netherlands contain shells that resemble those of living *Mytilus galloprovincialis*. Similar broad-shelled mytilids also occur in estuaries of the southwestern Netherlands together with slender individuals typical of *M. edulis*. We sampled living mussels along a depth gradient in the Oosterschelde to a) investigate whether a relation exists between shell shape and depth, b) test if the broad-shelled specimens might represent *M. galloprovincialis* (or a hybrid with *M. edulis*) and c) assess by inference if the Quaternary specimens might be attributed to *M. galloprovincialis* as well. In order to do so, we compared genetic (length polymorphism of *Me 15/16*, *COIII* sequences and AFLPs) and shell-morphological characteristics (juvenile L/W ratios and so-called Verduin parameters) of the same specimens. The obtained dataset indicates that all studied mussels from the Oosterschelde should be attributed to *M. edulis*, including those with broad shell outlines. No correlation of shell-morphology and depth-distribution was found. The worn and generally damaged state of the Eemian specimens precluded measurement of the Verduin parameters, while juvenile L/W ratios turned out not to be diagnostic. Therefore the shell characters examined in this study are insufficient to demonstrate the possible presence of *M. galloprovincialis* shells in Quaternary deposits of the Netherlands.

## INTRODUCTION

The taxonomy of common European *Mytilus* species is complex. The wide variety of shell shapes as well as the common occurrence of genetic interchange between populations shows that the discrimination between species is not straightforward (Filipowicz *et al.*, 2008, Śmietanka *et al.*, 2009). Conventionally, three *Mytilus* species are distinguished within the European seas, viz. *Mytilus edulis* Linné, 1758, *M. galloprovincialis* Lamarck, 1819 and *M. trossulus* Gould, 1850.

Mussels living along the coast of the southwestern Netherlands show a wide variety of shell shapes. On exposed North Sea localities, populations are entirely dominated by slender shells typically identified as *Mytilus edulis*. Broad specimens with a relatively sharp dorsal ridge and a pointed umbo are very rare on these exposed shores, although flat and broad specimens with a characteristic lightbrown periostracum do occur on floating material (Entrop, 1965) and are also known from wrecks of the North Sea floor (pers. obs.). The identity of these lightbrown mussels with a flat and broad shell shape remains to be established. Bluish broad-shelled *Mytilus* specimens are not uncommon in harbour basins and estuaries. Intermediate forms between broad and slender-shelled bluish mussels also occur (Fig. 1). Some of the broad forms resemble shells of *Mytilus galloprovincialis* Lamarck, 1819, a species well known from the Mediterranean but also occurring in southwestern Europe (SanJuan *et al.*, 1994, 1997). Similar broad-shelled mussels with a pointed umbo are also known from Eemian interglacial faunas of the Netherlands (Fig. 1). The closest populations of *M. galloprovincialis*, as confirmed by genetic analyses, are found in



the western Channel region (Hilbish *et al.*, 2002). In Dutch *Mytilus* populations, *M. galloprovincialis*-type alleles have been found at very low frequencies (Luttikhuisen *et al.*, 2002; Kijewski *et al.*, 2009). The center of the modern distribution of *M. galloprovincialis* is to the south of The Netherlands. Since the Eemian interglacial fauna contains various taxa with a more southerly center of distribution than the present Dutch fauna (e.g. *Nucula nucleus*, *Lucinella divaricata*, *Solen marginatus*, *Pholas dactylus*, *Gibbula magus*) it would be realistic to assume that the Eemian broad-shelled mussels might belong to *M. galloprovincialis*. However, the exact identity of the Eemian as well as the modern broad-shelled mussels in The Netherlands remains to be investigated. The generally damaged or worn preservation state of Eemian mussels precludes assessment of shell morphological characters including relative adductor scar lengths, as shells are incomplete (suppleted material) or adductor scars are worn (fossils washed ashore). Their identification can only be based on indirect comparison with living broad-shelled mussels.

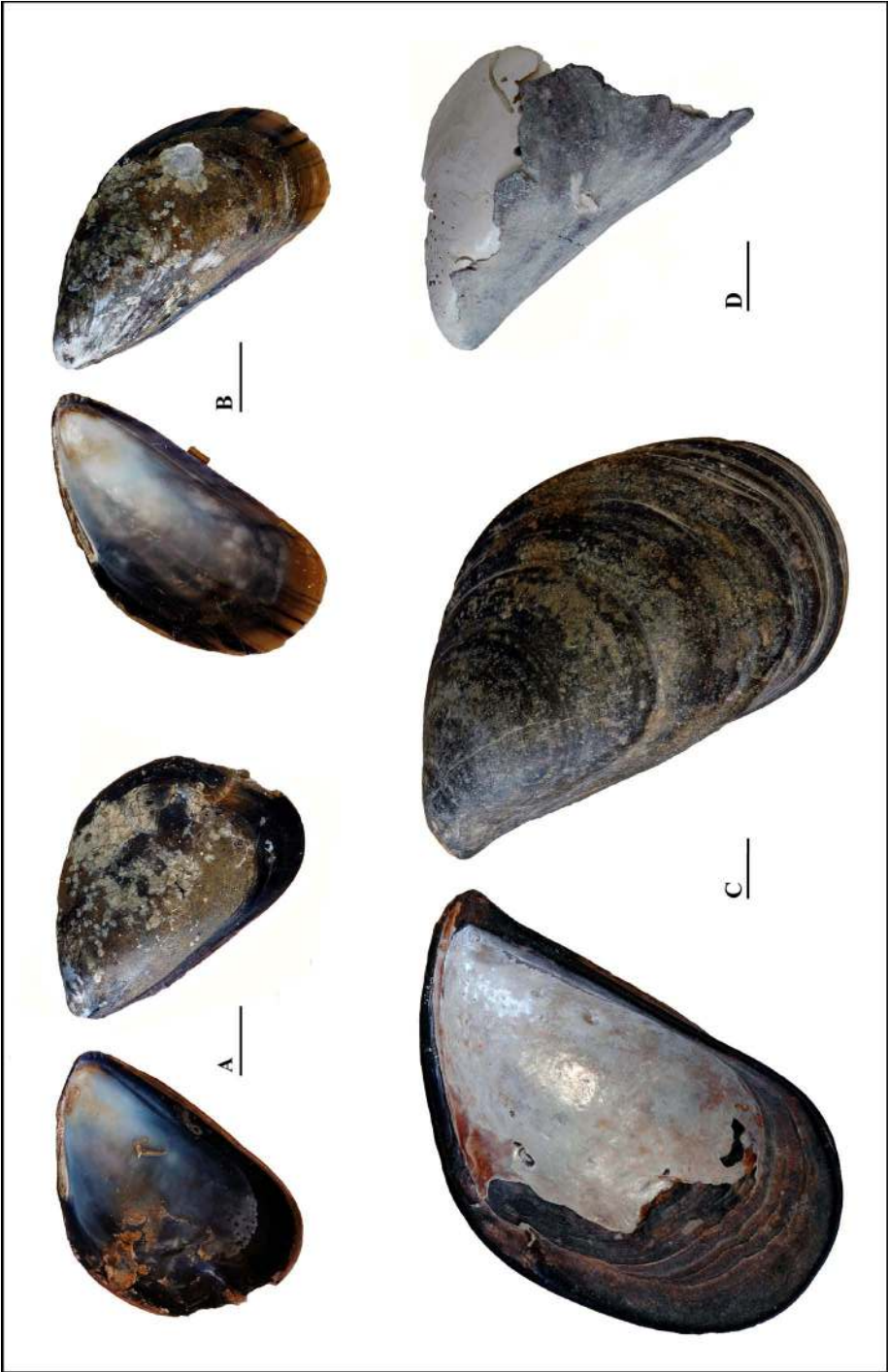
In this paper we investigate the identity of extant broad-shelled specimens of *Mytilus* from the southwestern Netherlands by comparing their shell morphological and genetic data and we indirectly evaluate whether broad-shelled Eemian mussels could be attributed to *M. galloprovincialis*.

#### *Distribution of modern European mussel species*

The three European *Mytilus* species have different, partially overlapping distribution ranges. *Mytilus trossulus* Gould, 1850 has a subarctic-high boreal distribution (also outside Europe). It is well adapted to cool temperatures and wide salinity variations. *Mytilus edulis* Linné, 1758 is recorded to be common along the entire European Atlantic coasts, southwards to Morocco (Jaziri *et al.*, 2003), and also occurs in the western part of the Mediterranean (Quesada *et al.*, 1995). *Mytilus galloprovincialis* Lamarck, 1819 is regarded as the dominating mussel species in the Mediterranean, but also lives along the Atlantic coast of Europe, as far north as northwestern Ireland (e.g. Gardner, 1992). The species appears the least adapted to salinity variations, and in general thrives in higher temperature settings than the other two species.

Hybridization occurs in virtually every known case where *Mytilus* taxa occur sympatrically, or where they are in geographic contact (Koehn, 1991). This resulted in *M. edulis-galloprovincialis* and *M. edulis-trossulus* hybrids in Europe (Hilbish *et al.*, 2002; Rawson and Hilbish, 1998; Riginos *et al.*, 2002; Kijewski *et al.*, 2009) and *M. trossulus-galloprovincialis* hybrids outside Europe (e.g. Inoue *et al.*, 1997; Rawson *et al.*, 1999). The common occurrence of hybrids questions the validity of the taxonomic status of *Mytilus* species. In convention with common terminology, we use the before mentioned names, as if they were well-established species, despite these taxonomic uncertainties (also see the Discussion at the end of this paper).





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**Figure 1.** *Mytilus* specimens. **A** RMNH 122086. *Mytilus edulis*. Oosterschelde (Zeeland, The Netherlands). Broad subadult specimen nr.140. **B** RMNH 122087. *Mytilus edulis*. Same locality, slender subadult specimen nr.47. **C** RGM 466.689. *Mytilus* sp., adult broad-shelled form washed ashore at the beach of Domburg (Zeeland, The Netherlands). The preservation state (lack of translucent margins, the crudely worn shell's exterior and the worn and stained nacre) is indicative of a likely Eemian (Late Pleistocene) age. **D**. RGM 607.080. *Mytilus* sp., adult broad-shelled form showing the typical defect preservation state of Eemian mussels precluding assessment of Verduin characters. Slotervaartse Plas, Amsterdam, The Netherlands, Eemian (Late Pleistocene). Scalebar 1 cm.

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### Identification

The discrimination of *Mytilus* species based on shell characters is problematic. Historically, identification was based on the combination of geographic provenance and the overall shape of the shell. No single shell-character (Koehn, 1991; Gosling, 1992) nor a combination of shell characters (McDonald *et al.*, 1991) discriminates completely between *M. edulis* and *M. galloprovincialis*. Transplantation experiments as well as field surveys have shown that shell outlines in *Mytilus* species are strongly influenced by environmental settings, such as the degree of exposure to waves (Seed, 1968; Akester and Martel, 1999), or population density and type of predators (Innes and Bates, 1999 and references therein). Multivariate approaches based on large character data sets, independently verified by molecular data, have yielded incomplete species discrimination (McDonald *et al.*, 1991; Innes and Bates, 1999). However, a set of complex shell characters (including shapes and sizes of various adductor and retractor muscle scars) proposed by Verduin (1979) might discriminate between *M. edulis* and *M. galloprovincialis*, although this is in need of confirmation by molecular data. Furthermore, the Verduin study used few populations located far apart, and lacked areas in between from where we now know that hybrid populations exist.

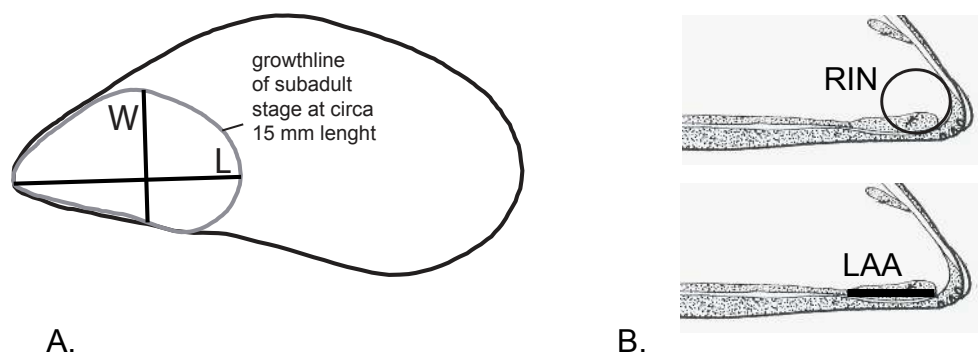
The identity of broad-shelled mussels found in Late Quaternary deposits of the Netherlands and extant ones living in the southwestern Dutch coastal region still needs to be established. In this paper, we examine shell-characters (L/W ratios and parameters after Verduin, 1979) and genetic markers (*Me15/16*, *COIII* and AFLP) of slender- and broad-shelled specimens of *Mytilus*, from the Oosterschelde estuary in the southwestern Netherlands, in order to establish whether the broad-shelled mussels may include *M. galloprovincialis* or hybrids.



## MATERIAL AND METHODS

### Sample collection

In total 229 specimens of *Mytilus* sp. were collected by SCUBA diving at nine different depths at the second northernmost pillar at the north side of the Zeelandbrug (Oosterschelde, the Netherlands, N 51,37.705 / E 3,54.817; WGS84, Table 1). We sampled at different depths in order to establish the distribution of morphological variation or potential distribution of species. Using regular scissors, individual animals were removed from the hard substrate by cutting their byssus threads. Under water, the mussels were collected in separate, labeled bags. Subsequently, the mussels were transported to the Netherlands Institute of Ecology (NIOO, Yerseke) for registration and dissection. Using a scalpel and pincers, mantle and anterior muscle tissue were separated from the rest of the animal. Half of each animal was stored in absolute ethanol (from here on denoted by suffix A) and the other half (denoted by suffix B) was stored in a saturated CTAB – 20% DMSO solution. The corresponding shells were labeled, cleaned and stored for shape analysis. Tissue samples were numbered as to enable recognition of the various parts of each individual. Shells and tissue samples in ethanol (A) were transported to the Netherlands Centre for Biodiversity, Naturalis, The Netherlands, for the study of shell morphometrics (shells were kept as voucher specimens) as well as for genetic analysis of markers *Me 15/16* and *COIII*. Tissues of the same specimens stored in CTAB (B) were sent to the Radboud University Nijmegen for AFLP analysis.



**Figure 2.** A. Establishment of L/W ratios on subadult shell stages. Such stages are recognizable through the growth lines on the shell. B. The two Verduin parameters used are the length of the anterior adductor scar (LAA) and the radius of the circle (RIN) that fits the shell's interior below the apex.



### *Morphometrics*

Two sets of shell parameters were measured, a length/width (L/W) set and a 'Verduin' parameter set. Juvenile shell outlines (shell L circa 15 mm) were drawn along growth lines preserved in adult mussel shells using a stereomicroscope equipped with a drawing attachment. From these outlines the maximum length and width (the latter perpendicular to length axis) were measured (Fig. 2A). These subadult outlines were used in order to minimise eco-phenotypic effects occurring in later growth stages (Seed, 1968). Based on the total distribution of L/W ratios, twenty-one specimens that represented slender and broad morphologies were selected for molecular analyses (described below). For these specimens, the length of the anterior adductor muscle scar and of the internal radius of the hinge plate (parameters utilized by Verduin, 1979, illustrated in Figure 2B) were measured and standardized for the height (H) of the shell (in this case of the subadult outline).

### *Sample treatment and DNA extraction*

Total genomic DNA from tissue stored in ethanol (A) was extracted with a general CTAB protocol, using prolonged incubation and precipitation times. Approximately 3 mm<sup>3</sup> of tissue was digested with 20 µl proteinase K (20 mg/ ml) overnight at 60°C in 500 µl CTAB-buffer. Extractions were done with 500 µl of chloroform : isoamylalcohol (24:1) and DNA was precipitated overnight at -20°C in 350 µl of isopropanol. The precipitate was washed with 500 µl ethanol/ammonium acetate, air-dried and resuspended in 50 µl MQ. DNA extractions on tissues stored in CTAB (B) were done with a DNeasy Tissue Kit (Qiagen), following the manufacturer's protocol. Different extraction protocols were used, according to the expertise within the institutes at that time (2003). If the extraction method used would in any way affect the laboratory outcome, this would be equal for all the specimens analysed. Therefore, we deduce that the data from the separate analyses trajectories are compatible and that a comparison is legitimate.

### *Adhesive protein gene (also referred to as Me15/16, nuclear marker)*

Inoue *et al.* (1995) have shown that species in the *Mytilus* complex can be identified by a length polymorphism in the nucleotide sequence of the adhesive protein gene. Fragments amplified with sense primer Me15 CCAGTATACAAACCTGTGAAGA and antisense primer Me16 TGTTGTCTTAATAGGTTTGTAAGA have a length of 180, 168 and 126 bp for *M. edulis*, *M. trossulus* and *M. galloprovincialis* respectively (Inoue *et al.*, 1995). Polymerase chain reactions were carried out in 25 µl volumes using 1 µl (approx. 150 ng) of DNA extract (A). Reaction conditions were 2.5 mM MgCl<sub>2</sub>, 0.2 mM dNTPs, 0.4 µM of each primer and 5 units (0.25 µl) of Taq Polymerase (Qiagen). PCR thermoprofile was: 3 min. initial denaturation at 94°C, followed by 40 amplification cycles (denature at 94°C (30 sec.), annealing at 56°C (30 sec.) and extension at 72°C (90 sec.)) with a final extension of 5 min. at 72 °C. Amplification products were separated on a 4.5% agarose gel.



**Table 1.** Sample details. All samples were taken from the second pillar on the northern side of the Zeelandbrug (Prov. Zeeland, The Netherlands: N 51,37.705 / E 3,54.817; WGS84)

| Specimen numbers | Sampling depth            |
|------------------|---------------------------|
| 0-25             | at low tide mark          |
| 31-55            | 1.5 m below low tide mark |
| 61-85            | 3 m below low tide mark   |
| 91-115           | 4.5 m below low tide mark |
| 121-145          | 6 m below low tide mark   |
| 151-175          | 9 m below low tide mark   |
| 181-210          | 12 m below low tide mark  |
| 211-235          | 15 m below low tide mark  |
| 241-265          | 20 m below low tide mark  |

*Cytochrome c oxidase subunit III (COIII: mitochondrial marker)*

In order to assess the recent history within species or populations of *Mytilus*, we sequenced a 416 bp fragment of *Cytochrome c oxidase subunit III (COIII)*, a mitochondrial gene that shows high substitution rates. *Mytilus*, like other members of the Mytilidae and Unionidae (Curole and Kocher, 2002) transmits its mitochondria by means of doubly uniparental inheritance (DUI: Zouros *et al.*, 1994ab, also known as gender associated inheritance: Skibinski *et al.*, 1994ab). This results in two independently evolving mitochondrial lineages (termed F and M). Males possess both lineages, whereas females only possess the F-lineage. Our shell morphometric and molecular analyses were carried out without prior anatomical gender assessment. Although all the DNA extracts had to contain F-lineage mitochondria, M-lineage mitochondria were potentially present as well. At the time the lab work for this study was carried out (2003) this was a technical difficulty, because only generic or M-specific *COIII* primers (*M. edulis*) were available (nowadays both F- and M-specific primers are available; Śmietanka *et al.*, 2009).

To remove potentially present M-lineage *COIII* products, we slightly modified the nested-PCR procedure described by Stewart *et al.* (1995). Initial amplification was done with primer For1 5'-TATGTACCAGGTCCAAGTCCGTG-3' and Rev1 5'-ATGCTCTTCTTGAATATAAGCGTACC-3' using the same reaction conditions as described for *Me15/Me16*, but with an annealing temp. of 54°C and a final extension at 70°C. Amplification products were split and half of each product was digested with Mbo I, while the other half was digested with Ssp I (New England Biolabs, following manufacturer's protocol). Recognition sites for these restriction enzymes are located on the M-lineage *COIII* sequences of *M. edulis* and *M. trossulus*, respectively. In 2003,



it was not known if M-mitotype *COIII* sequences of *M. galloprovincialis* possessed a recognition site for these enzymes. Digestion products were separated on a 1% agarose gel and the largest fragment of the MboI digest was extracted from the gel and purified with a Qiaquick Gel Extraction Kit (Qiagen), following manufacturer's protocol. A 1:100 dilution (MQ) of the purified product was used as a template (1  $\mu$ l) for a half nested PCR with primers For2 5'-GTAAGTCAAGCCCCATAAGAG-3' and Rev1. Products of this second PCR were cleaned with a Qiaquick PCR purification kit (Qiagen) and sequenced (BigDye Terminator, Applied Biosystems) in both directions, using the same primers. Sequence products were purified using AutoSeq G-50 columns (Amersham) and run on an ABI 377 sequencer (Applied Biosystems). Forward and reverse sequences were assembled and edited with Sequencher version 4.2 (Gene Codes corp.) and the resulting contig sequences were aligned manually in MacClade version 4.08 (Maddison and Maddison, 2005). A minimum spanning network was created using SplitsTree4 (Huson and Bryant, 2006).

#### *Amplified Fragment Length Polymorphism (AFLP)*

An AFLP analysis was performed after Vos *et al.* (1995) with modifications as described by De Roos (2003). Restriction-ligation reactions were performed in a single reaction at 37°C for 2 hours, in a total volume of 10  $\mu$ l containing 100 ng DNA, 1 $\times$  T4 Ligase Buffer (Life Technology, Invitrogen), 0.05 M NaCl, 5 U EcoRI - Enzyme (New England BioLabs), 5 U MseI - Enzyme (New England BioLabs), 0.045 M bovine serum albumine (BSA, New England BioLabs), 0.2  $\mu$ M EcoRI - adapter (5'-CTCGTAGACTGCGTACC, CATCTGACGCATGGTTAA-5'), 2.0  $\mu$ M MseI - adapter (5'-GACGATGAGTCCTGAG, TACTCAGGACTCAT-5') and 1 U T4 - Ligase (Life Technology, Invitrogen). PCR amplifications were performed on a T-gradient thermocycler (Biometra) in two separate amplification steps. Pre-amplification PCR reactions were conducted in a 20  $\mu$ l volume containing 4  $\mu$ l 20 $\times$  diluted Restriction-Ligation mix, 0.5  $\mu$ M EcoRI / +A primer, 0.5  $\mu$ M MseI / +C primer and 15  $\mu$ l AFLP Amplification Core Mix (all from Applied Biosystems), with the following temperature profile: an initial denaturation step of 2 min. at 94°C; 20 cycles with 20 sec. at 94°C, 30 sec. at 56°C, 2 min. at 72°C; followed by 2 min. at 72°C and 30 min. at 60°C. Selective amplifications were performed in a 10  $\mu$ l volume (containing 1.5  $\mu$ l 10 $\times$  diluted Pre-amplification product, 0.05  $\mu$ M Dye labeled EcoRI / +3 primer, 0.25  $\mu$ M MseI / +3 primer and 7.5  $\mu$ l AFLP Amplification Core Mix (all from Applied Biosystems), with the following temperature profile: an initial denaturation step of 2 min. at 94°C; 10 cycles with 20 sec. at 94°C, 30 sec. at 66°C decreasing with 1°C per cycle, 2 min. at 72°C; 25 cycles with 20 sec. at 94°C, 30 sec. at 56°C, 2 min. at 72°C; followed by 30 min. at 60°C. Selective amplifications were performed using two primer pairs EcoRI / +ACC-MseI / +CTA and EcoRI / +AAC - MseI / +CAG. Amplified fragments were analysed on a CEQ™ 8000 Genetic Analysis System (Beckman Coulter Inc. 2002). Loci with a band frequency below 5% (Beckman Coulter, 2002) were considered as potential artefacts and omitted from further analysis. Based on unpublished data on *Mytilus* s.l. from a larger area (Rajagopal and van der Velde, pers. com.), 248 fragments between 50 and 400 bp were scored as



present (1) or absent (0) using the Fragment Analysis Software Module of the CEQ™ 8000 (AFLP settings: bin width = 1.0 nt; Ythreshold = 400 RFU). Gene loci that were putatively informative (non-monomorphic fragments) were assembled in a binary data matrix.

#### *Statistical analysis*

In order to see if the L/W ratios were distributed normally a Shapiro-Wilk test was done. To test the association between L/W ratios and depth, a univariate ANOVA (sum of squares type III) was performed in GLM (General Linear Model), in which the L/W ratios (continuous) were used as the dependent variable and depth as a covariate. A Fisher exact test was used to test for an association between L/W ratios (categorized as either slender, < 1.65 or broad, > 1.65) and COIII groups (this covariate was categorized as group A or B; see the results section). These statistical tests were performed in SPSS release 11.0.4 (SPSS Inc.).

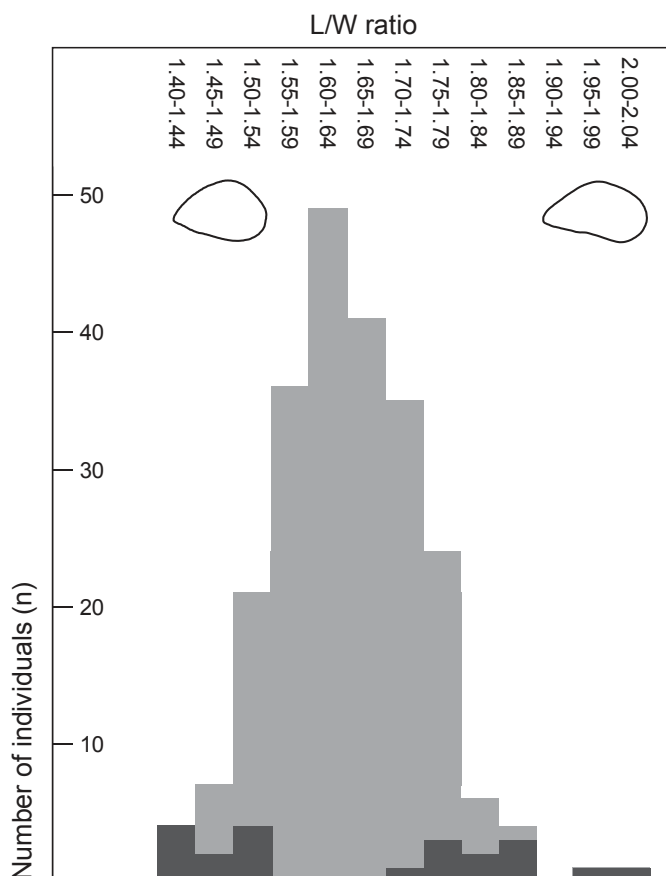
To test for a relation between AFLP-data and L/W-ratios a Principal Component Analysis (PCA) and Correspondence Analysis (CA, Nenadic and Greenacre, 2007) were performed in R (R foundation for statistical computing, Ihaka and Gentleman, 1996). Using scatter plots, we inspected whether the two L/W-ratio categories had very different combinations of scores of the new independent variables obtained in either PCA or CA. Using logistic regression, we tested whether the scores of new variables significantly explained the probability of being either broad or not.

## RESULTS

#### *Shell morphology*

The L/W ratios of the 229 measured specimens (Fig. 3) range between 1.42 and 2.00 (mean 1.65, standard deviation 0.10). The L/W ratios are continuous with the exception of two specimens that have a slender subadult shell outline. Slender- and broad-shelled forms are connected through intermediates and the entire L/W ratio range describes an almost a perfect normal distribution (Fig. 3). After exclusion of the two just mentioned outliers, the Shapiro-Wilk test statistic becomes 0.994 and  $p = 0.430$ . No correlation between L/W ratios and sampling depth was observed (GLM univariate ANOVA,  $F = 1.026$ ,  $p = 0.324$ ); distribution of L/W ratios over different depths is illustrated by Fig. 4. The combined Verduin parameters of the studied Oosterschelde samples fall within the range of *Mytilus edulis* (Fig. 5; Table 2). The range of morphological variation of the 21 Oosterschelde shells covers a large part of the total morphological variation found by Verduin (1979) for various Western European *Mytilus edulis* populations. *Mytilus* shells from the Zeeland bridge show considerable morphological variation in L/W ratios as well as Verduin parameters. The latter fall within the range of *M. edulis* (Fig. 5). Hence the Verduin parameters imply that the broad morphs should be attributed to *Mytilus edulis*.





**Figure 3.** L/W ratios of subadult stages in *Mytilus* shells collected from different depths along a pillar of the Zeelandbrug. In dark are the specimens used for further analyses. The L/W ratios show a normal distribution with two outliers representing slender shells.

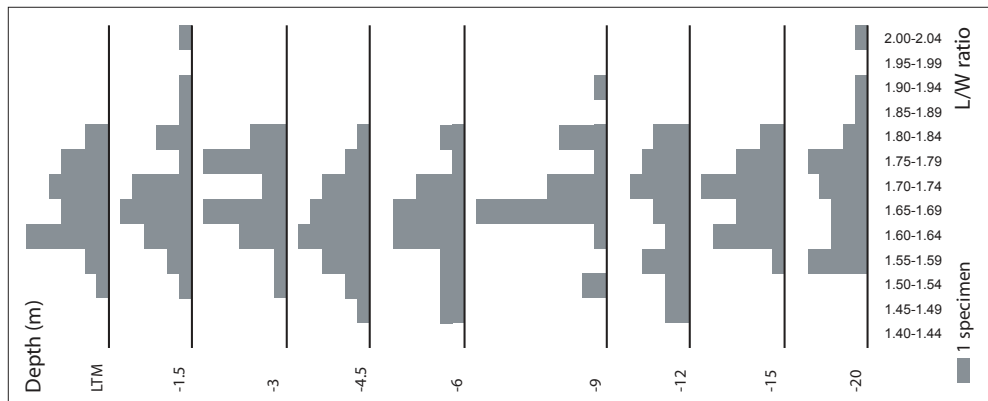
#### *Adhesive protein gene*

The *Me15/16*-PCR products for the 21 selected specimens all had a length of 180 bp and belong to *M. edulis*, following Inoue *et al.* (1997).

#### *COIII*

Digestion of the initial PCR product (primers For1 and Rev1) with *MboI* resulted in one fragment if only product from the F-lineage was present (or if the *MboI* recognition site would have been lost in the M-lineage). If next to F-lineage product, M-lineage product was present as well, three fragments were observed. Digestion of initial PCR product with *SspI* resulted in two fragments if *COIII* was amplified





**Table 2.** Verduin parameters. Rin denotes internal radius of hinge area. Laa denotes length of anterior adductor muscle scar (illustrated in Figure 2B; after Verduin, 1979).

| Specimen | H (mm) | Rin (mm) | Laa (mm) | Laa/H | Rin/H | Laa/H + 1,5 Rin/H | L/H  |
|----------|--------|----------|----------|-------|-------|-------------------|------|
| 8A       | 17     | 2        | 4        | 0.24  | 0.12  | 0.42              | 1.5  |
| 22A      | 17     | 1.6      | 4        | 0.24  | 0.09  | 0.38              | 1.78 |
| 38A      | 21     | 2        | 3        | 0.14  | 0.1   | 0.28              | 1.88 |
| 47A      | 19     | 1.5      | 5        | 0.26  | 0.08  | 0.38              | 2    |
| 52A      | 13     | 1        | 2        | 0.15  | 0.08  | 0.27              | 1.5  |
| 61A      | 21     | 1.75     | 3        | 0.14  | 0.08  | 0.26              | 1.48 |
| 81A      | 18     | 1.5      | 5        | 0.28  | 0.08  | 0.4               | 1.77 |
| 95A      | 15     | 1.6      | 4        | 0.27  | 0.11  | 0.43              | 1.73 |
| 106A     | 17     | 1.25     | 4        | 0.24  | 0.07  | 0.35              | 1.42 |
| 124A     | 20     | 1.75     | 4        | 0.2   | 0.09  | 0.33              | 1.42 |
| 128A     | 18     | 2        | 3        | 0.17  | 0.11  | 0.34              | 1.8  |
| 140A     | 21     | 2.25     | 4        | 0.19  | 0.11  | 0.35              | 1.44 |
| 162A     | 20     | 1.75     | 5        | 0.25  | 0.09  | 0.38              | 1.47 |
| 172A     | 21     | 2        | 5        | 0.24  | 0.1   | 0.38              | 1.87 |
| 196A     | 17     | 1.5      | 4        | 0.24  | 0.09  | 0.37              | 1.44 |
| 203A     | 20     | 2.25     | 5        | 0.25  | 0.11  | 0.42              | 1.78 |
| 222A     | 20     | 2.25     | 5        | 0.25  | 0.11  | 0.42              | 1.8  |
| 225A     | 20     | 2.4      | 4        | 0.2   | 0.12  | 0.38              | 1.53 |
| 247A     | 18     | 2.25     | 4        | 0.22  | 0.13  | 0.41              | 1.51 |
| 248A     | 18     | 1.6      | 5        | 0.28  | 0.09  | 0.41              | 1.98 |
| 254A     | 20     | 2        | 4        | 0.2   | 0.1   | 0.35              | 1.88 |

from the F-lineage, and in four fragments if it was amplified from the M-lineage. Hence products from both lineages possess an *SspI* recognition site, whereas only the M-lineage possesses a restriction site for *MboI*. Subsequent nested PCR on the largest fragment of the *MboI*-digest with primers For2 and Rev1 resulted in a 462 bp product for all specimens. A minimum spanning haplotype network (Fig. 6) was created based on the nucleotide sequences from this product (primersites were excluded, length: 416 bp). The average uncorrected P-distance between the obtained sequences was 0.012 (standard deviation 0.011). This relatively high average can mostly be ascribed to the large distances added by the specimens 203, 247 and 248 (indicated as group B). These three specimens were separated from the most common haplotype (largest circle, Fig. 6) by 14 mutational steps or more, whereas all other specimens (group A) were separated from this haplotype by 6 mutational steps or less. If the sequences from group B are excluded, the average uncorrected P-distance decreases to 0.007 (standard deviation 0.006). All substitutions were synonymous and the largest uncorrected P-distance was 0.036 (between 203 and 81, 162 & 222). No association between L/W ratio and COIII groups A and B could be shown (Fisher's exact test  $p$  (2-sided) = 1,000). Sequences were deposited in GenBank (accession numbers DQ353870-DQ353890).

#### AFLP

Of the 248 putative gene loci that were scored, only 13 were non-monomorphic, indicative of a homogeneous population. Only the first four principal components each explained more than 10% of the total variation in the AFLP sample. Cumulatively these four principal components explained 72% of the total variance. In the correspondence analysis, only the first two principal inertias explain more than 10% of the summed inertias, cumulatively they contribute 61 %. Both the PCA and CA scatter plots (results not shown) suggested some separation in L/W groups based on the AFLP-data, but always with an intricate dependence on the scores. Backward model selection using likelihood ratio tests, showed that none of the principal components or none of the coordinates of the principal inertias significantly explained the variation in L/W ratios. Therefore we decided to depict the AFLP results by means of a minimal spanning network (Fig. 7, SplitsTree4, Huson and Bryant, 2006), which more clearly illustrates the lack of separation of L/W groups. Also Fig. 7 shows a single population in which slender-shelled specimens (L/W ratio > 1.65; represented by grey circles) and broad-shelled specimens (L/W ratio < 1.65; represented by white circles) are connected repeatedly by only a single step.

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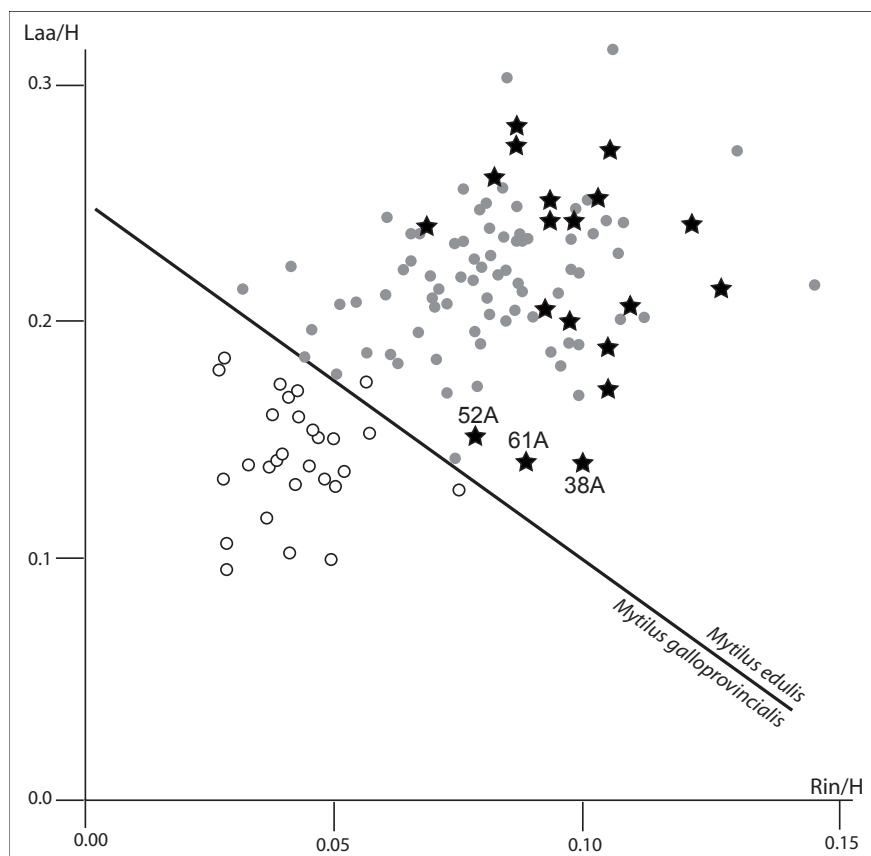
**Figure 4.** L/W ratios of *Mytilus* shells from the various depths. LTM refers to lower tide mark. No trends in L/W ratios are discernable.

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## DISCUSSION

The combined Verduin parameters and molecular data (Table 3) show that the analysed *Mytilus* specimens from the Zeelandbrug should be attributed to *Mytilus edulis*. Three specimens (38, 52 and 61) have Verduin parameter values that lie within the *M. edulis* range, but are close to the *M. edulis* - *M. galloprovincialis* border (Fig. 5). None of the other datasets (*Me15/16* and the minimum spanning networks based on *COIII* sequences and AFLP-data) indicate that these specimens are any different from the remainder eighteen.



**Figure 5.** Verduin characters of *Mytilus* individuals from the Zeelandbrug (this study, stars) versus European shells (dots) analysed by Verduin (1982). Line depicts the boundary between *Mytilus galloprovincialis* (open dots) and *M. edulis* (filled grey dots):  $((Laa/H) + 1.5 (Rin/H) = 0.25)$ . Stars denote Zeelandbrug samples. Three specimens that are close to the *M. edulis* - *M. galloprovincialis* -boundary are tagged.



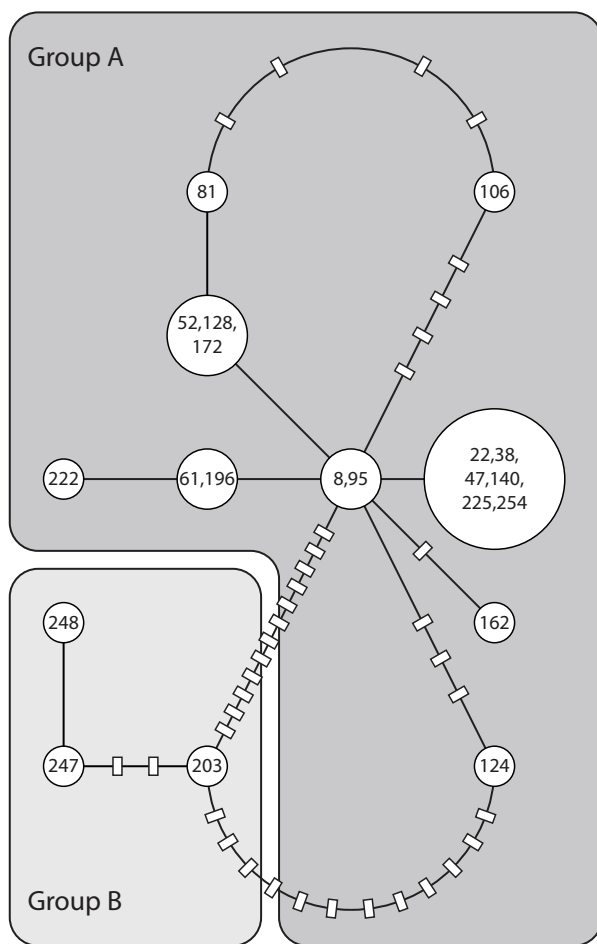
**Table 3.** Morphometric and molecular data of fully analysed mussels. Depth below low tide mark. VI denotes identity as inferred from Verduin parameters. E denotes *Mytilus edulis*. “A” and “B” refer to two groups of haplotypes (see text) based on *COIII*. The columns for MboI and SspI denote the number of fragments after digestion of the PCR product obtained with primers For1 and Rev1.

| Specimen | Depth | L/W  | VI | Me 15-16 | MboI | SspI | AFLP | <i>COIII</i> |
|----------|-------|------|----|----------|------|------|------|--------------|
| 106A     | 4.5 m | 1.42 | E  | 180 bp   | 3    | 4    | E    | A            |
| 124A     | 6 m   | 1.42 | E  | 180 bp   | 1    | 2    | E    | A            |
| 140A     | 6 m   | 1.44 | E  | 180 bp   | 3    | 4    | E    | A            |
| 196A     | 12 m  | 1.44 | E  | 180 bp   | 1    | 2    | E    | A            |
| 162A     | 9 m   | 1.47 | E  | 180 bp   | 1    | 2    | E    | A            |
| 61A      | 3 m   | 1.48 | E  | 180 bp   | 3    | 4    | E    | A            |
| 8A       | 0 m   | 1.5  | E  | 180 bp   | 1    | 2    | E    | A            |
| 52A      | 1.5 m | 1.5  | E  | 180 bp   | 3    | 4    | E    | A            |
| 247A     | 20 m  | 1.51 | E  | 180 bp   | 1    | 2    | E    | B            |
| 225A     | 15 m  | 1.53 | E  | 180 bp   | 1    | 2    | E    | A            |
| 95A      | 3 m   | 1.73 | E  | 180 bp   | 1    | 2    | E    | A            |
| 81A      | 3 m   | 1.77 | E  | 180 bp   | 3    | 4    | E    | A            |
| 22A      | 0 m   | 1.78 | E  | 180 bp   | 1    | 2    | E    | A            |
| 203A     | 12 m  | 1.78 | E  | 180 bp   | 1    | 2    | E    | B            |
| 128A     | 6 m   | 1.8  | E  | 180 bp   | 3    | 4    | E    | A            |
| 222A     | 15 m  | 1.8  | E  | 180 bp   | 3    | 4    | E    | A            |
| 172A     | 9 m   | 1.87 | E  | 180 bp   | 3    | 4    | E    | A            |
| 38A      | 1.5 m | 1.88 | E  | 180 bp   | 3    | 4    | E    | A            |
| 254A     | 20 m  | 1.88 | E  | 180 bp   | 3    | 2    | E    | A            |
| 248A     | 20 m  | 1.98 | E  | 180 bp   | 1    | 2    | E    | B            |
| 47A      | 1.5 m | 2    | E  | 180 bp   | 3    | 4    | E    | A            |

As for the molecular datasets, *Me15/16* unanimously shows the presence of *M. edulis* and does not indicate potential *M. edulis* – *M. galloprovincialis* hybrids, let alone ‘pure’ *M. galloprovincialis*. The AFLP data show a random distribution of morphotypes over the network (Fig. 7). Only the network based on *COIII* sequences (Fig. 6) points to the presence of more than a single group; group B contains the haplotypes obtained from specimens 203, 247 and 248 and group A contains nine other haplotypes. All haplotypes in group A are within six mutational steps from the main haplotype



(Fig. 6). At least fourteen steps separate the haplotypes of group B from the main haplotype. A BLAST-search (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) showed that sequence DQ353885 (specimen 203, group B) was most similar to F-lineage *COIII* sequences of specimens identified as either *M. galloprovincialis* (from Greece and the Black Sea: accession numbers DQ403170, AY130171, AY130172, AY130174, DQ445474 and DQ445468) or *M. edulis* (from Ireland and France, accession numbers AY130154 and AY130161). Śmietanka *et al.* (2009) also obtained partial *COIII* sequences for *Mytilus* sp. with a reasonable coverage of the entire European coast, including



**Figure 6.** Minimum Spanning Network based on *COIII* sequences. Haplotype sizes are proportional to the number of sequences. Lines between haplotypes represent one mutational step. Boxes on these lines represent additional mutational steps.



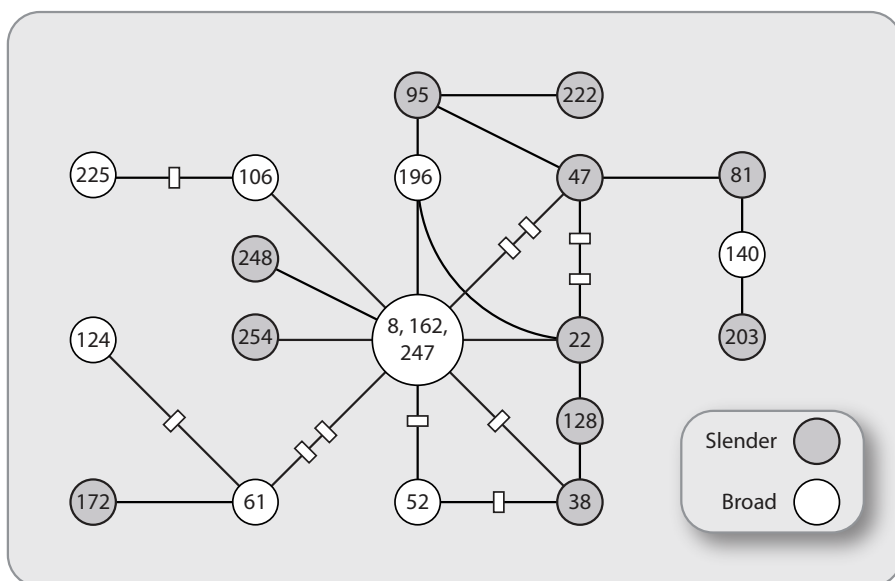
Westerschelde. Hence we expected that sequences from this study would be most similar to those of Śmietanka *et al.* This was not the case, due to the fact that the forward primer used by Śmietanka *et al.* (5'-TCTTGGTACAACCTGCGGGAA-3', Skibinski *et al.*, 1994b) is located ca. 115 bp downstream from For2. A BLAST-search with only the overlapping part (301 bp) of sequences from this study and that of Śmietanka *et al.* (2009), indeed shows that sequence DQ353885 (specimen 203) is most similar to sequence FJ549913 from the Westerschelde. A minimum spanning network based on the 21 *COIII* sequences obtained with this study and the additional 26 *COIII* sequences of Śmietanka *et al.* (network not depicted) reveals seven additional haplotypes. Nevertheless, all of these haplotypes are within six mutational steps from either the main haplotype of group A or B. Based on these results we have no indication for the presence of more than two groups of haplotypes in the Dutch delta region. Since in our Oosterschelde material *Me15/16* shows no indication for hybridisation and the AFLP-results do not indicate more than a single population, we assume incomplete lineage sorting (ancestral polymorphism) is a more plausible explanation for these *COIII* results, than introgression.

Despite the presence of two distinct groups (A and B, see text) of *COIII* haplotypes in our material, all molecular results indicate that the Oosterschelde mussels belong to a single population of *Mytilus edulis*. This also corresponds to our identifications based on the Verduin parameters. The existence of broad-shelled *M. edulis* and the lack of *M. galloprovincialis* in the Oosterschelde, makes it for now impossible to establish the identity of similar broad-shelled *Mytilus* shells from Eemian (Late Quaternary) deposits of the North Sea Basin. Moreover it shows that merely the relative width (L/W ratio) of broad-shelled mytilids, is not sufficient to discriminate between *M. edulis* and *M. galloprovincialis*. We call for further studies using molecular analyses to investigate the ability of Verduin parameters to discriminate between *M. edulis* and *M. galloprovincialis* shells.

Whether the taxonomic units, jointly known as the *Mytilus* species complex, are distinct enough to merit full specific status has been a matter of debate since the 1860s (Gosling, 1992). The choice of species concept is of importance in such discussions. Nowadays it becomes common practice to refer to 'pure' *M. edulis* and 'pure' *M. galloprovincialis* solely based on genetic characters (Kijewski *et al.*, 2009) and species are 'taxonomically identified' by the length of PCR products (Śmietanka *et al.*, 2009), regardless of the morphology (shell shape, colouration of periostracum and mantle, etc.) that was classically used to distinguish *Mytilus* species.

In the fossil shells, the Verduin parameters could not be established due to wear of critical characters (adductor scars). Furthermore, the attribution of extant mussels to *M. galloprovincialis* as proposed by Verduin (1979) still needs to be validated in populations by molecular data. All of the specimens from the Oosterschelde (this study), including broad-shelled ones (low L/W ratio), were attributed to *M. edulis* based on the Verduin parameters.





**Figure 7.** Minimum Spanning Network based on AFLP-data. Haplotype sizes are proportional to the number of sequences. Lines between haplotypes represent the number of steps under the parsimony criterion. Boxes on these lines represent additional steps. The color of the circles corresponds to the relative shell shapes (see inset).

## CONCLUSION

Based on an assessment and comparison between shell-morphological and molecular characteristics from single mussel specimens (*Mytilus* sp.) collected along a depth profile in the Oosterschelde we conclude that:

- (1) broad-shelled specimens from this location (identified by low L/W ratios) do not belong to *Mytilus galloprovincialis*, but instead all of the analysed Oosterschelde specimens (both slender and broad-shelled) must be attributed to *M. edulis*.
- (2) based on *COIII* sequence data, two groups of haplotypes can be discerned, whereas *Me15/16* and AFLP data do not show any structure within this mussel population.
- (3) there is no association between L/W ratio (broad- or slender-shelled specimens) and the depth at which specimens were collected.
- (4) there is no association between the two *COIII* haplotype groups and shell morphological variation (*COIII* haplotype groups and L/W ratios are independent variables, whereas the Verduin parameters indicate only a single group: *M. edulis*).
- 5) due to the existence of broad-shelled *M. edulis* (low L/W ratio), we cannot use



the L/W ratio to classify similar broad-shelled specimens from Late Quarternary (Eemian) deposits of the North Sea Basin as either *M. edulis* or *M. galloprovincialis*.

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## REFERENCES

- AKESTER, R.J. & MARTEL, A.L. 2000. Shell shape, dysodont tooth morphology, and hinge-ligament thickness in the bay mussel *Mytilus trossulus* correlate with wave exposure. *Canadian Journal of Zoology* 78: 240-253.
- BECKMAN COULTER. 2002. CEQ 8000 Genetic Analysis System User's Guide. Beckman Coulter Inc., Fullerton, California, U.S.A.
- CUROLE, J.P. & KOCHER, T.D. 2002. Ancient sex-specific extension of the *Cytochrome c Oxidase II* gene in bivalves and the fidelity of doubly uniparental inheritance. *Molecular Biology and Evolution* 19: 1323-1328.
- DE ROOS, K. 2003. CEQ™ 8000 AFLP Protocol. Beckman Coulter Netherlands BV, Mijdrecht, The Netherlands.
- ENTROP, B. 1965. Schelpen vinden en herkennen. Zutphen: Thieme.
- FILIPOWICZ, M., BURZYJSKI, A., ŚMIETANKA, B. & WENNE, R. 2008. Recombination in Mitochondrial DNA of European Mussels *Mytilus*. *Journal of Molecular Evolution* 67:377–388. doi:10.1007/s00239-008-9157-6
- GARDNER, J.P.A. 1992. *Mytilus galloprovincialis* (Lmk) (bivalva, mollusca): the taxonomic status of the mediterranean mussel. *Ophelia* 35:219-243.
- GOSLING, E.M. 1992. Systematics and geographic distribution of *Mytilus*. Pp. 1-20. In: The mussel *Mytilus*: ecology, physiology, genetics and culture. Amsterdam: Elsevier.
- HILBISH, T.J., CARSON, E.W., PLANTE, J.R., WEAVER, L.A. & GILG, M.R. 2002. Distribution of *Mytilus edulis*, *M. galloprovincialis*, and their hybrids in open-coast populations of mussels in southwestern England. *Marine Biology* 140:137-142.
- HUSON, D.H. & BRYANT, D. 2006. Application of Phylogenetic Networks in Evolutionary Studies. *Molecular Biology and Evolution* 23(2): 254-267.
- IHAKA, R. & GENTLEMAN, R. 1996. R: A language for data analysis and graphics. *Journal of Computational and Graphical Statistics* 5(3): 299-314.
- INNES, D.J. & BATES, J.A. 1999. Morphological variation of *Mytilus edulis* and *Mytilus trossulus* in eastern Newfoundland. *Marine Biology* 133: 691-699.



- INOUE, K., WAITE, J.H., MATSUOKA, M., ODO, S. & HARAYAMA, S. 1995. Interspecific variations in adhesive protein sequences of *Mytilus edulis*, *M. galloprovincialis*, and *M. trossulus*. *The Biological Bulletin* 189: 370-375.
- INOUE, K., ODO, S., NODA, T., NAKAO, S., TAKEYAMA, S., YAMAHA, E., YAMAZAKI, S. & HARAYAMA, S. 1997. A possible hybrid zone in the *Mytilus edulis* complex in Japan revealed by PCR markers. *Marine Biology* 128: 91-95.
- JAZIRI, H. & SBABOU, L. 2003. Variabilité morphologique et systématique de la moule méditerranéenne *Mytilus galloprovincialis* Lmk. (Bivalvia, Mytilidae) au Maroc. *Haliotis* 32: 61-72.
- KIJEWski, T., WIJSMAN, J.W.M., HUMMEL, H. & WENNE, R. 2009. Genetic composition of cultured and wild mussels *Mytilus* from The Netherlands and transfers from Ireland and Great Britain. *Aquaculture* 287: 292-296.
- KOEHN, R.K. 1991. The genetics and taxonomy of species in the genus *Mytilus*. *Aquaculture* 94: 125-145.
- LUTTIKHUIZEN, P.C., KOOLHAAS, A., BOL, A. & PIERSMA, T. 2002. *Mytilus galloprovincialis*-type foot-protein-1 alleles occur at low frequency among mussels in the Dutch Wadden Sea. *Journal of Sea Research* 48: 241-245.
- MADDISON, D.R. & MADDISON, W.P. 2005. *MacClade* 4.08. Sunderland, Massachusetts: Sinauer Associates.
- MCDONALD, J.H., SEED, R. & KOEHN, R.K. 1991. Allozymes and morphometric characters of three species of *Mytilus* in the Northern and Southern Hemispheres. *Marine Biology* 111: 323-333.
- NENADIC, O. & GREENACRE, M. 2007. Correspondence analysis in R, with two- and three-dimensional graphics: The ca package. *Journal of Statistical Software* 20 (3).
- QUESADA, H., BEYNON, C.M. & SKIBINSKI, D.O.F. 1995. A mitochondrial DNA discontinuity in the mussel *Mytilus galloprovincialis* Lmk: Pleistocene vicariance biogeography and secondary intergradation. *Molecular Biology and Evolution* 12: 521-524.
- RAWSON, P.D. & HILBISH, T.J. 1998. Asymmetric introgression of mitochondrial DNA among European populations of blue mussels (*Mytilus* spp.). *Evolution* 52: 100-108.
- RAWSON P.D., AGRAWAL, V., HILBISH, T.J. 1999. Hybridization between the blue mussels *Mytilus galloprovincialis* and *M. trossulus* along the Pacific coast of North America evidence for limited introgression. *Marine Biology* 134: 201-211.
- RIGINOS, C., SUKHDEO, K. & CUNNINGHAM, C.W. 2002. Evidence for selection at multiple allozyme loci across a mussel hybrid zone. *Molecular Biology and Evolution* 19: 347-351.
- SANJUAN, A., ZAPATA, C. & ALVAREZ, G. 1994. *Mytilus galloprovincialis* and *M. edulis* on the coasts of the Iberian peninsula. *Marine Ecology Progress Series* 113: 131-146.
- SANJUAN, A., ZAPATA, C. & ALVAREZ, G. 1997. Genetic differentiation in *Mytilus galloprovincialis* Lmk. throughout the world. *Ophelia* 47: 13-31.
- SEED, R. 1968. Factors influencing shell shape in the mussel *Mytilus edulis*. *Journal of the Marine Biology Association of the United Kingdom* 48: 561-584.
- SKIBINSKI, D.O.F., GALLAGHER, C. & BEYNON, M. 1994a. Mitochondrial DNA inheritance. *Nature* 368: 817-818.
- SKIBINSKI, D.O.F., GALLAGHER, C. & BEYNON, M. 1994b. Sex-limited mitochondrial DNA transmission in the marine mussel *Mytilus edulis*. *Genetics* 138: 801-809.



- ŚMIETANKA, B., BURZYNSKI, A. & WENNE, R. 2009. Molecular population genetics of male and female mitochondrial genomes in European mussels *Mytilus*. *Marine Biology* 156: 913-925. DOI 10.1007/s00227-009-1137-x.
- SPSS. 2005. SPSS for Mac OS X. Release 11.0.4. Chicago: SPSS Inc.
- STEWART, D.T., SAAVEDRA, C., STANWOOD, R.R., BALL, A.O. & ZOUROS, E. 1995. Male and female mitochondrial DNA lineages in the Blue Mussel (*Mytilus edulis*) species group. *Molecular Biology and Evolution* 12(5): 735-747.
- VERDUIN, A. 1979. Conchological evidence for the separate specific identity of *Mytilus edulis* L. and *M. galloprovincialis* Lmk. *Basteria* 43: 61-80.
- VOS, P., HOGERS, R., BLEEKER, M., REIJANS, M., VAN DE LEE, T., HORNES, M., FRIJSTERS, A., POT, J., PELEMAN, J., KUIPER, M. & ZABEAU, M. 1995. AFLP: a new technique for DNA fingerprinting. *Nucleic Acids Research* 23: 4407-4414.
- ZOUROS, E., BALL, A.O., SAAVEDRA, C. & FREEMAN, K.R. 1994a. Mitochondrial DNA inheritance. *Nature* 368: 818.
- ZOUROS, E., BALL, A.O., SAAVEDRA, C. & FREEMAN, K.R. 1994b. An unusual type of mitochondrial DNA inheritance in the blue mussel *Mytilus*. *Proceedings of the National Academy of Sciences of the United States of America* 91: 7463-7467.





CHAPTER 5

**Taxonomic assessment of *Andrena rosae*  
and *Andrena stragulata* by  
DNA-sequencing (Hymenoptera:  
Apoidea: Andrenidae)**

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## ABSTRACT

Two taxa of the genus *Andrena* Fabricius 1775 (subgenus *Hoplandrena* Pérez 1890): *A. rosae* Panzer 1801 and *A. stragulata* Illiger 1806, are considered species by some authors; others consider *A. stragulata* as the early spring generation of *A. rosae*. In this paper the taxonomical status of both taxa is studied by comparing nuclear and mitochondrial DNA-sequences of spring and summer specimens from the Biesbosch National Park in The Netherlands. Despite the differences in morphology, behaviour and nesting sites, no consistent differences could be found in the DNA-sequences of both supposed species. Therefore, *Andrena rosae* is considered to be a bivoltine species: with a spring and a summer generation. *Andrena stragulata* is considered a junior synonym of *A. rosae*. This view is also supported by the distributional data of spring and summer specimens in The Netherlands.

## INTRODUCTION

The solitary mining bees *Andrena stragulata* Illiger 1806 (= *A. eximia* Smith 1847) and *A. rosae* Panzer 1801 are West-Palaeartic taxa, occurring in Central and South-Europe. *A. stragulata* can be found in early spring, while the flight period of *A. rosae* is in summer. Some authors regard both taxa as two distinct, univoltine (= one generation per year) species (Westrich 1989, Schmid-Egger & Scheuchl, 1997 and Gusenleitner & Schwarz, 2002). Despite small differences in the male morphology (Schmid-Egger & Scheuchl, 1997), other authors disagree and consider *A. stragulata* and *A. rosae* to be a single species with two generations per year (bivoltine: Van der Vecht, 1928 and Peeters *et al.*, 1999).

In Dutch literature only the name *Andrena rosae* has been used, both for spring and summer specimens. In the Netherlands it is a rare species (Fig 1). It used to be more widely distributed before 1980, but after a strong decline it is now regarded as severely threatened in the national Red List of threatened species (Peeters *et al.*, 1999 and Peeters & Reemer, 2003). In recent years, the Dutch population is concentrated in a limited area in the southwestern part of the Netherlands (Fig 1). In this area, both spring (*A. stragulata*) and summer generations (*A. rosae*) have been found in large numbers. Therefore this population provided a good opportunity to investigate the taxonomic status of both taxa. Additional information on the ecology of the population, including descriptions of nesting sites and notes on pollen collection and behaviour, can be found in Van der Meer *et al.* (2006).

This study was conducted in two parts. In the pilot study, a limited number of specimens was studied to determine which of four genetic markers was most informative. In the main study, only mitochondrial *COI*-sequences (*Cytochrome Oxidase Subunit I* gene) were analyzed for a larger number of specimens. It is hypothesized that if the total number of 39 specimens of spring and summer



generations indeed represents different species, then cladistic or haplotype analyses should likewise group these species in separate clades.



**Figure 1:** Distribution of *Andrena rosae* Panzer 1801 in the Netherlands. Squares: records before 1980. Dots: records between 1980 and 2006. Source: database of European Invertebrate Survey - the Netherlands.

## MATERIAL AND METHODS

### *Research area and collection of specimens*

Male and female specimens of both spring and summer generations of *Andrena rosae* were collected from four localities in the National Park “De Biesbosch”, an area in the southwest of The Netherlands. Table 1 lists the names and mutual distances of



these localities. Specimens of *Andrena carantonica* Pérez 1902 were used as outgroup. Table 2 lists the collection data of both ingroup and outgroup specimens. After capturing by net the specimens were stored in 96% ethanol and were identified using the keys of Schmid-Egger & Scheuchl (1997). Voucher specimens were deposited in the collection of the National Centre for Biodiversity (NCB) Naturalis, Leiden, the Netherlands. Specimen registration codes for the DNA voucher specimens are included in Table 2.

**Table 1:** Distances in kilometers between research localities in the Biesbosch (Netherlands). Values in parentheses indicate the coordinates in the “Amersfoort”-grid, a national geographical grid-system in the Netherlands.

|    | DE  | LD  | TP  | ZP |
|----|-----|-----|-----|----|
| DE | -   | -   | -   | -  |
| LD | 0,1 | -   | -   | -  |
| TP | 1,7 | 1,8 | -   | -  |
| ZP | 7,7 | 7,7 | 7,3 | -  |

DE= De Elzen (106,3-419,2);  
LD = Louisapolderse Dijk (106,3-419,3);  
TP = Zuiddijk bij Tongplaat (107,8-418,3);  
ZP = Zuilespolder (112,2-424,2).

#### *DNA extraction, PCR and sequencing*

For DNA extraction of specimens collected in 2005 and 2006 (Table 2) the abdomens were partially used; of the specimens collected in 2007, 2-3 legs were used. Total genomic DNA was extracted with a DNeasy® Tissue Kit (Qiagen) following the manufacturers protocol. To concentrate the DNA, it was eluted in a final volume of 100 µl (buffer AE).

In the pilot study six specimens of “*A. rosae*” (three of each generation, all females) and four outgroup specimens of *A. carantonica* Pérez 1902 were studied (Table 2). For these six specimens, fragments of *COI*, *COII*, *CytB* and *EF-1a* were sequenced. For the 33 additional specimens (2006 and 2007) only *COI* was sequenced. Primer information and references are listed in Table 3. For all markers the PCR conditions were 1.5 mM MgCl<sub>2</sub>, 0.2 mM DNTPs, 0.4 µM of each primer and 5 U of Taq DNA polymerase (Qiagen) per reaction. The only exception being the use of 2.0 mM MgCl<sub>2</sub> for *EF-1a*. PCR amplifications were always carried out in a volume of 25 µl. For all amplifications the following thermocycle profile was used: 3 minutes at 94°C followed by 40 cycles of 30 seconds at 94°C, 1 minute at 50°C and 1 minute at 72°C and a final extension of 5 minutes at 72°C. PCR products were directly purified with Nucleospin® Extract II columns from Macherey-Nagel and both strands were sequenced using the same primers (Table 3). Sequencing was either done on an ABI 3730 automated sequencer (Applied Biosystems) by Macrogen Corp. in Korea or at Leiden University on a Megabace™ 1000 DNA Analysis System (Amersham). Forward and reverse sequences were assembled and checked using Sequencher



**Table 2A.** Collection and registration data of specimens from the pilot study.

| Specimen    | Species               | RMNH       | Season | Sex    | Location     | Date        | Leg |
|-------------|-----------------------|------------|--------|--------|--------------|-------------|-----|
| DE-01-Spr-F | <i>A. stragulata</i>  | INS.100001 | spring | Female | De Elzen     | 2.iv.2005   | FM  |
| DE-02-Spr-F | <i>A. stragulata</i>  | INS.100002 | spring | Female | De Elzen     | 2.iv.2005   | FM  |
| DE-03-Spr-F | <i>A. stragulata</i>  | INS.100003 | spring | Female | De Elzen     | 2.iv.2005   | MR  |
| ZP-04-Sum-F | <i>A. rosae</i>       | INS.100027 | summer | Female | Zuilespolder | 14.vii.2005 | MR  |
| ZP-05-Sum-F | <i>A. rosae</i>       | INS.100028 | summer | Female | Zuilespolder | 14.vii.2005 | MR  |
| ZP-06-Sum-F | <i>A. rosae</i>       | INS.100029 | summer | Female | Zuilespolder | 14.vii.2005 | MR  |
| ANDRCARA1   | <i>A. carantonica</i> | INS.100040 | -      | Male   | Steenbergen  | 30.iv.2005  | TP  |
| ANDRCARA2   | <i>A. carantonica</i> | INS.100041 | -      | Female | Steenbergen  | 30.iv.2005  | TP  |
| ANDRCARA3   | <i>A. carantonica</i> | INS.100042 | -      | Male   | Rijswijk     | 14.v.2005   | FM  |
| ANDRCARA4   | <i>A. carantonica</i> | INS.100043 | -      | Female | Rijswijk     | 14.v.2005   | FM  |

| Specimen    | GenBank Accession Number |          |            |
|-------------|--------------------------|----------|------------|
|             | COI                      | COI-COII | CytB EF-1a |
| DE-01-Spr-F | EU374640                 | EU374680 | EU374700   |
| DE-02-Spr-F | EU374641                 | EU374681 | EU374701   |
| DE-03-Spr-F | EU374642                 | EU374682 | EU374702   |
| ZP-04-Sum-F | EU374661                 | EU374683 | EU374703   |
| ZP-05-Sum-F | EU374662                 | EU374684 | -          |
| ZP-06-Sum-F | EU374663                 | EU374685 | EU374704   |
| ANDRCARA1   | -                        | EU374686 | EU374705   |
| ANDRCARA2   | -                        | EU374687 | EU374706   |
| ANDRCARA3   | -                        | EU374688 | EU374707   |
| ANDRCARA4   | EU374679                 | EU374689 | -          |



**Table 2B.** Collection and registration data of specimens from the main study.

| Specimen Nr. | Species              | RMNH       | Season | Sex    | Location               | Date        | Leg     | GB Accession |
|--------------|----------------------|------------|--------|--------|------------------------|-------------|---------|--------------|
| DE-16-Spr-F  | <i>A. stragulata</i> | INS.100004 | Spring | Female | De Elzen               | 2.iv.2007   | FM & JS | EU374659     |
| DE-17-Spr-F  | <i>A. stragulata</i> | INS.100005 | Spring | Female | De Elzen               | 2.iv.2007   | FM & JS | EU374660     |
| LD-06-Spr-M  | <i>A. stragulata</i> | INS.100006 | Spring | Male   | Louisapolderse Dijk    | 2.iv.2007   | FM & JS | EU374649     |
| LD-07-Spr-M  | <i>A. stragulata</i> | INS.100007 | Spring | Male   | Louisapolderse Dijk    | 2.iv.2007   | FM & JS | EU374650     |
| LD-08-Spr-M  | <i>A. stragulata</i> | INS.100008 | Spring | Male   | Louisapolderse Dijk    | 2.iv.2007   | FM & JS | EU374651     |
| LD-09-Spr-M  | <i>A. stragulata</i> | INS.100009 | Spring | Male   | Louisapolderse Dijk    | 2.iv.2007   | FM & JS | EU374652     |
| LD-10-Spr-M  | <i>A. stragulata</i> | INS.100010 | Spring | Male   | Louisapolderse Dijk    | 2.iv.2007   | FM & JS | EU374653     |
| LD-65-Sum-F  | <i>A. rosae</i>      | INS.100011 | Summer | Female | Louisapolderse Dijk    | 23.vii.2007 | MR      | EU374665     |
| LD-66-Sum-F  | <i>A. rosae</i>      | INS.100012 | Summer | Female | Louisapolderse Dijk    | 23.vii.2007 | MR      | EU374666     |
| LD-67-Sum-F  | <i>A. rosae</i>      | INS.100013 | Summer | Female | Louisapolderse Dijk    | 23.vii.2007 | MR      | EU374667     |
| LD-71-Sum-F  | <i>A. rosae</i>      | INS.100014 | Summer | Female | Louisapolderse Dijk    | 23.vii.2007 | MR      | EU374671     |
| LD-72-Sum-F  | <i>A. rosae</i>      | INS.100015 | Summer | Female | Louisapolderse Dijk    | 23.vii.2007 | MR      | EU374672     |
| TP-01-Spr-F  | <i>A. stragulata</i> | INS.100016 | Spring | Female | Zuiddijk bij Tongplaat | 2.iv.2007   | FM & JS | EU374644     |
| TP-02-Spr-F  | <i>A. stragulata</i> | INS.100017 | Spring | Female | Zuiddijk bij Tongplaat | 2.iv.2007   | FM & JS | EU374645     |
| TP-03-Spr-M  | <i>A. stragulata</i> | INS.100018 | Spring | Male   | Zuiddijk bij Tongplaat | 2.iv.2007   | FM & JS | EU374646     |
| TP-04-Spr-M  | <i>A. stragulata</i> | INS.100019 | Spring | Male   | Zuiddijk bij Tongplaat | 2.iv.2007   | FM & JS | EU374647     |
| TP-05-Spr-F  | <i>A. stragulata</i> | INS.100020 | Spring | Female | Zuiddijk bij Tongplaat | 2.iv.2007   | FM & JS | EU374648     |
| TP-64-Sum-M  | <i>A. rosae</i>      | INS.100021 | Summer | Male   | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374664     |
| TP-70-Sum-F  | <i>A. rosae</i>      | INS.100022 | Summer | Female | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374670     |

| Specimen Nr. | Species              | RMNH          | Season | Sex    | Location               | Date        | Leg     | GB Accession |
|--------------|----------------------|---------------|--------|--------|------------------------|-------------|---------|--------------|
| TP-73-Sum-F  | <i>A. rosae</i>      | INS.100023    | Summer | Female | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374673     |
| TP-76-Sum-F  | <i>A. rosae</i>      | INS.100024    | Summer | Female | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374676     |
| TP-78-Sum-F  | <i>A. rosae</i>      | INS.100025    | Summer | Female | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374677     |
| TP-79-Sum-F  | <i>A. rosae</i>      | INS.100026    | Summer | Female | Zuiddijk bij Tongplaat | 23.vii.2007 | MR      | EU374678     |
| ZP-07-Spr-M  | <i>A. stragulata</i> | Specimen lost | Spring | Male   | Zuilespolder           | 21.iv.2006  | FM      | EU374643     |
| ZP-11-Spr-F  | <i>A. stragulata</i> | INS.100030    | Spring | Female | Zuilespolder           | 2.iv.2007   | FM & JS | EU374654     |
| ZP-12-Spr-F  | <i>A. stragulata</i> | INS.100031    | Spring | Female | Zuilespolder           | 2.iv.2007   | FM & JS | EU374655     |
| ZP-13-Spr-F  | <i>A. stragulata</i> | INS.100032    | Spring | Female | Zuilespolder           | 2.iv.2007   | FM & JS | EU374656     |
| ZP-14-Spr-F  | <i>A. stragulata</i> | INS.100033    | Spring | Female | Zuilespolder           | 2.iv.2007   | FM & JS | EU374657     |
| ZP-15-Spr-M  | <i>A. stragulata</i> | INS.100034    | Spring | Male   | Zuilespolder           | 2.iv.2007   | FM & JS | EU374658     |
| ZP-68-Sum-M  | <i>A. rosae</i>      | INS.100035    | Summer | Male   | Zuilespolder           | 23.vii.2007 | MR      | EU374668     |
| ZP-69-Sum-F  | <i>A. rosae</i>      | INS.100036    | Summer | Female | Zuilespolder           | 23.vii.2007 | MR      | EU374669     |
| ZP-74-Sum-F  | <i>A. rosae</i>      | INS.100037    | Summer | Female | Zuilespolder           | 23.vii.2007 | MR      | EU374674     |
| ZP-75-Sum-F  | <i>A. rosae</i>      | INS.100038    | Summer | Female | Zuilespolder           | 23.vii.2007 | MR      | EU374675     |

The GenBank Accession Numbers of the specimens from the main study (Table 2B) belong to marker *COI*. Leg (collector): FM = Frank van der Meer; JS = John T. Smit; MR = Menno Reemer; TP = Theo M.J. Peeters. Voucher specimens are deposited in the collection of NCB Naturalis (RMNH) in Leiden.



4.2 (Gene Codes Corp.). Sequences were aligned manually using MacClade 4.08 (Maddison & Maddison, 2003).

Sequence analysis

The datasets were analysed with PAUP 4.0 beta 10 (Swofford, 2003). To make a comparison between sequence variation within supposed species of *Andrena* (“*A. rosae*”) and between accepted species of *Andrena*, each of the four datamatrices of the pilot study was checked for the number of variable and parsimony-informative sites within “*A. rosae*” and between “*A. rosae*” and *A. carantonica* Pérez 1902. To visually cluster the specimens based on sequence similarity, an exhaustive search (optimality criterion: parsimony) and Neighbor-Joining (NJ) analysis (a distance method) were performed on each of the pilot-datasets. Because a univocal split between spring and summer specimens of “*A. rosae*” was only observed for marker *COI*, it was necessary to check whether the observed grouping was not caused by sampling error (due to the restricted number of specimens). Hence the *COI* dataset was extended to 40 taxa (including two outgroup specimens). On this extended dataset a NJ bootstrap analysis (1000 replicates) was carried out and a strict consensus tree was calculated. A Minimum Spanning Network (MSN) analysis was performed in order to visualize how generation and locality relate to the detected haplotypes.

Table 3. Primer information.

| Primer name | Sequence                         | Marker | Source                         |
|-------------|----------------------------------|--------|--------------------------------|
| LCO1490 F   | 5'-GGTCAACAAATCATAAAGATATTGG-3'  | COI    | Folmer <i>et al</i> , 1994.    |
| HCO2198 R   | 5'-TAAACTTCAGGGTGACCAAAAAATCA-3' |        | Folmer <i>et al</i> , 1994.    |
| 'Jack'      | 5'-AGATCACTTGAATGATCACAAAAT-3'   | COI-II | Larkin <i>et al</i> , 2006.    |
| C2-N-3661   | 5'-CCACAAATTCTGAACATTGACCA-3'    |        | Simon <i>et al</i> , 1994.     |
| CytB F      | 5'-CGWTTAATTCATATAAATGG-3'       | Cyt B  | Koulianos <i>et al</i> , 1999. |
| CytB R      | 5'-TATCATTCWGGTTTAATATG-3'       |        | Koulianos <i>et al</i> , 1999. |
| EF-1a F     | 5'-GGACACAGAGATTTTCATCAARAA-3'   | EF 1a  | Kawakita <i>et al</i> , 2003.  |
| EF-1a R     | 5'-TTGCAAAGCTTCRTGRTGCATTT-3'    |        | Kawakita <i>et al</i> , 2003.  |

RESULTS

Pilot study: selection of informative markers

Of the six *Andrena* specimens from the Bieschbosch (pilot set), sequences were obtained for each of the tested markers, except for an *EF-1a* sequence for specimen ZP-05-Sum-F. Of the ougroup specimens marker *COI* and *EF-1a* occasionally failed,



but each of the pilot datasets contained sequences of at least two *A. carantonica* specimens. For each dataset the results of an exhaustive search are shown in Fig 2. Only the exhaustive search on the *CytB* dataset resulted in more than a single shortest tree; Fig 2c shows the 50% majority rule consensus tree calculated of these five shortest trees. For the parsimony analysis the distance-measure was set to absolute.

A summary of the number of variable and parsimony-informative sites for each dataset is given in Table 4. This table also shows the effect of inclusion of *A. carantonica* sequences on the total number of informative sites. The variation within *COI*-sequences of *A. carantonica* was high compared to the variation of the other datasets analysed. Here a pseudogene might have been sequenced accidentally. Besides an unusually high sequence divergence for *COI* within *A. carantonica* there were no other signs (i.e. frame-shift mutations or stopcodons) that indicated the presence of possible pseudogenes.

Table 4 shows that the variation within "*A. rosae*" is very limited. In fact, all sequences of "*A. rosae*" in the *COI-COII* dataset are identical, as are those in the *EF-1α* dataset. The *COI* and *CytB* datasets each have two parsimony informative sites within "*A. rosae*", but the *CytB* dataset results in a grouping of spring and summer specimens that is incompatible with our null-hypothesis (Fig 2c).

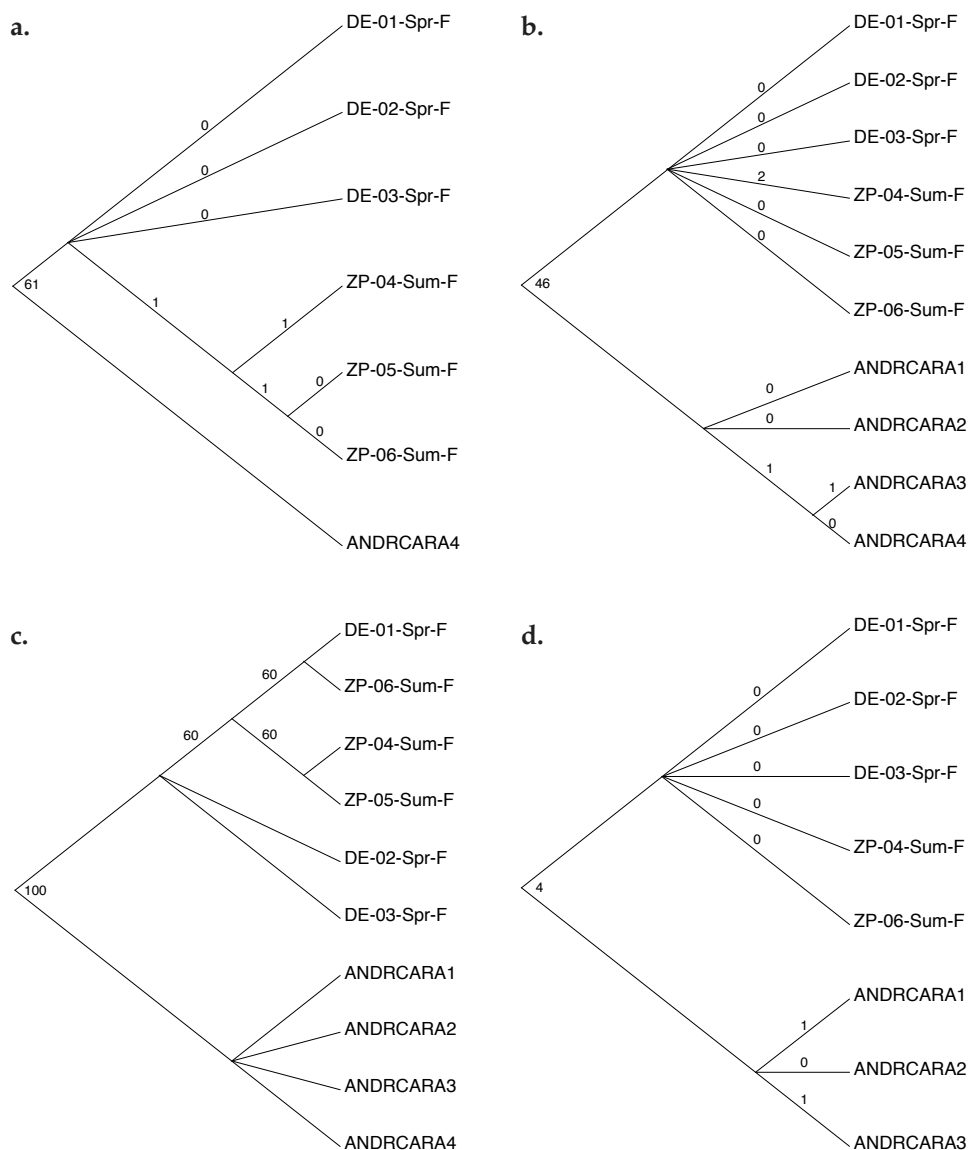
**Table 4.** Character status.

| Dataset                 | Number of characters | Constant characters | Uninformative characters | Parsimony informative |
|-------------------------|----------------------|---------------------|--------------------------|-----------------------|
| <i>COI</i> (pilot)      | 658                  | 655 / 595           | 1 / 61                   | 2 / 2                 |
| <i>COI-COII</i> (pilot) | 689                  | 687 / 639           | 2 / 3                    | 0 / 47                |
| <i>CytB</i> (pilot)     | 541                  | 539 / 498           | 0 / 0                    | 2 / 43                |
| <i>EF-1α</i> (pilot)    | 782                  | 782 / 776           | 0 / 2                    | 0 / 4                 |
| <i>COI</i> (main)       | 658                  | 655 / 595           | 0 / 60                   | 3 / 3                 |

Numbers before the forward slash were calculated using only specimens of the ingroup (*A. rosae* and *A. stragulata*); values after the slash show these same numbers after inclusion of outgroup (*A. carantonica*) specimens.

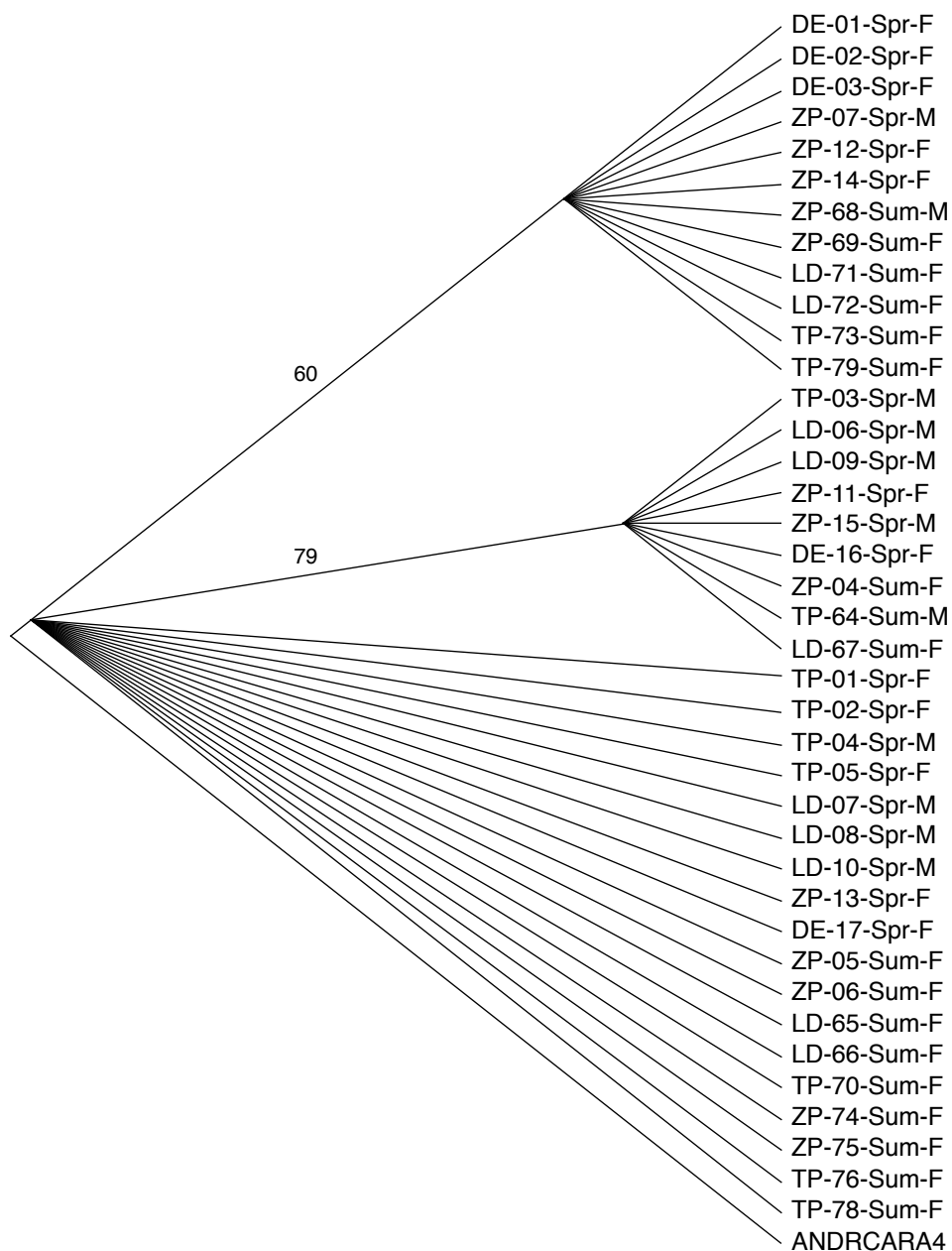
Although *COI-COII* sequences can discriminate between a large number of *Andrena*-species (Larkin *et al.*, 2006), these sequences did not allow us to distinguish between spring and summer generations of the pilot specimens. In fact this pilot showed that except for *COI* none of the tested markers could univocally discriminate between spring and summer generations. Since *COI* was the only marker that was potentially informative, we selected this marker for the analyses of the main study.





**Fig 2.** Cladistic analyses of sequences of four different genetic markers, as determined during the pilot-study for six specimens of *Andrena rosae* Panzer 1801 and one to four outgroup specimens of *A. carantonica* Pérez 1902. **(a)** *Cytochrome c oxidase subunit I* (exhaustive search, shortest tree), **(b)** *Cytochrome c oxidase subunit I-II* (exhaustive search), **(c)** *Cytochrome B* (50% majority rule consensus, based on five shortest trees), **(d)** *Elongation Factor 1α* (exhaustive search shortest tree).



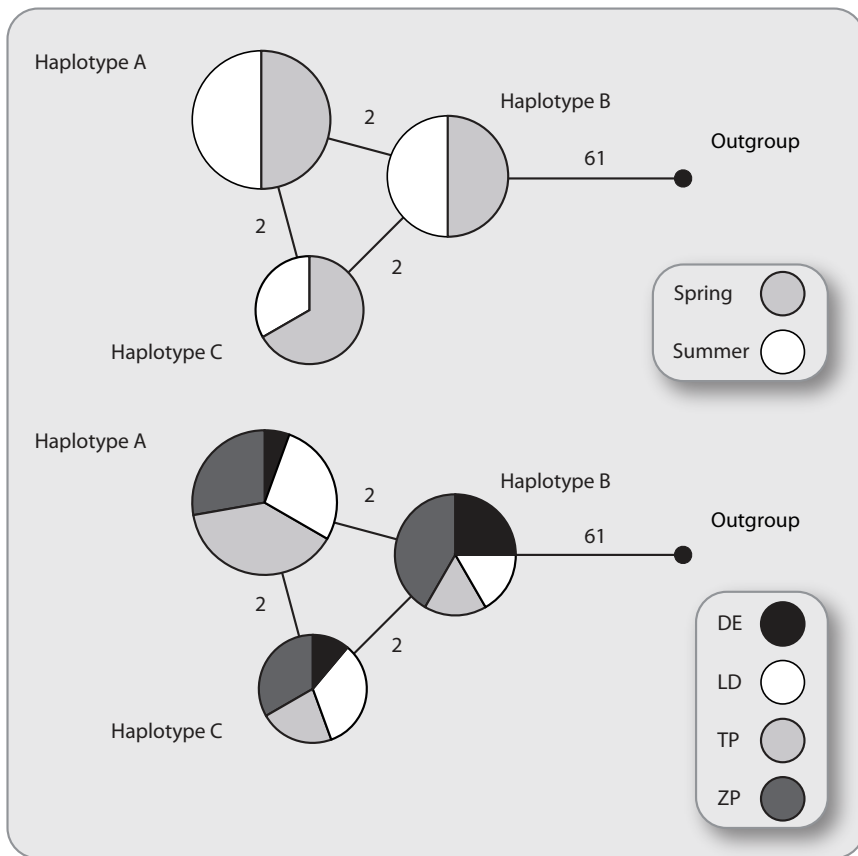


**Fig 4.** 50% majority rule consensus tree of a bootstrap analysis (1000 replicates, criterion parsimony), based on COI-sequences of 39 specimens of *Andrena rosae* Panzer 1801 from both spring and summer generations, and from four different localities. .



*Main study: analysis of the extended COI dataset*

Although the topology obtained with marker *COI* univocally supported our null-hypothesis (Fig 2a), the limited number of taxa made this dataset prone to sampling error. To check if there was any support for these clades, 33 specimens were added to the dataset. Fig 3 shows the 50% majority rule consensus tree of a neighbor joining bootstrap analysis (1000 replicates) with this extended *COI*-dataset. This cladogram shows that spring and summer specimens do not form separate clades. Moreover, even specimens from a single locality (presumably a single population) do not group. Each clade contains a mixture of specimens from different generations and localities, which is also shown by the Minimum Spanning Network (Fig. 4)



**Fig. 4:** Minimum Spanning Network based on the extended *COI* dataset (40 taxa). Haplotype size are proportional to the number of sequences. Lines between haplotypes represent one mutational step; the numbers next to these lines represent additional steps. The upper half shows the distribution of spring and summer generations within each haplotype; the lower half shows the distribution of localities (abbreviation as in Table 1) within each haplotype.



## DISCUSSION

Our null-hypothesis predicts that specimens of spring and summer generations would form different clades if they represented different species of *Andrena*. However, our results neither show grouping of specimens according to generation, nor according to location.

An explanation for not observing grouping according to generation could be incomplete lineage sorting; causing the gene-tree to differ from the species-tree. Although this is an argument that needs to be considered, marker *COI-COII* has shown to be discriminative for other species within this genus (Larkin *et al.*, 2006), while it clearly showed to be uninformative in this study. In fact none of the markers tested here enabled us to discriminate between *A. rosae* and *A. stragulata*. Theoretically, it is conceivable that we are dealing with two very recently diverged species, in which genetic differences are too small to detect with the used methods. Considering the differences found by Larkin *et al.* (2006) between *Andrena* species, however, this does not seem likely.

Since there is no reason to assume that *A. rosae* from the Netherlands is different from *A. rosae* elsewhere in Europe, we conclude that *A. stragulata* and *A. rosae* should be regarded respectively as the spring and summer generation of the same species. The name that should be used for this species is *A. rosae* PANZER, because this is the oldest available name.

Remarkably, there are considerable morphological differences between spring and summer specimens, especially in the males (Schmid-Egger & Scheuchl, 1997). The spring males (March-May) have a ventral occipital spine, mandibles are longer and cross each other for more than 60 %, the hypopygium is comparatively robust, the eyes are smaller, and more than half of the second metasomal tergite is reddish. The summer males (July-August) have no ventral occipital spine (but see remarks below), the mandibles are shorter and cross each other for about 10%, the hypopygium is comparatively slender, and less than half of the second metasomal tergite is reddish. A similar case seems to be the species couple *Andrena spinigera* (Kirby 1802) (spring generation with occipital tooth) and *A. trimmerana* (Kirby, 1802) (summer generation without occipital tooth). According to Tadauchi & Hirashima (1984) the occipital spine and the modified mandibles of the males of the spring generation are typical for six Japanese species of the subgenus *Hoplاندrena*, to which *A. rosae* also belongs. They name the spring and summer generations “seasonal forms” and consider seasonal dimorphism normal for the subgenus in Japan. The males of the first generation in several species of this subgenus have a distinct occipital spine. It is the main character used by Schmid-Egger & Scheuchl (1997) to separate *A. stragulata* and *A. rosae*.

Although in-depth morphological analyses are beyond the scope of the present paper, some illustrations are included to demonstrate that the character states of the ventral occipital spine are not always clear-cut (Fig 5). In spring specimens this spine is always well-developed (Fig 5a), while in most summer specimens it is not (Fig 5b). However, in some summer specimens the state of this character is intermediate



a.



b.



c.



between spring specimens and usual summer specimens, as can be seen in Fig 5c. The variability of this character can be considered as additional support for our conclusion that there is only one species involved.

It is not clear how these morphological differences can develop among males of different generations within a single population. Another issue is whether these differences are functionally relevant, e.g. in mating behaviour or ecology. Ecological research in the Netherlands has revealed ecological differences between spring and summer generations (Van der Meer *et al.*, 2006). For instance, females collect pollen on *Salix* catkins in spring and on flowers of Apiaceae in summer. According to Westrich (1989) spring females also collect pollen on other flowers, but this could not be confirmed in the Dutch study. Possibly other flowers are only visited for their nectar. Another ecological difference might be found in the nesting habits: while numerous nests were easy to locate in spring, no nests at all could be found at the same sites during summer, despite thorough searching and despite the presence of hundreds of specimens on nearby flowers. For the time being, however, one can only speculate about the causality of the seasonal morphological differences.

Recently, another paper on the taxonomy of a sibling pair of *Andrena* species (*A. decipiens* Schenck, 1861 and *A. flavilabris* Schenck, 1874) has been published (Mandery *et al.*, 2008). Unlike the present paper, the authors of that paper conclude, based on molecular evidence, that those taxa should be considered as distinct species. This illustrates that each case of sibling bee taxa should be carefully examined, and that (a priori) assumptions always require validation

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**Fig 5:** Head of male *Andrena rosae* Panzer 1801 in frontal view (all specimens from the Netherlands and deposited in collection of RMNH, Leiden). **(a)** Spring specimen with strongly developed ventral occipital spine (indicated by arrow): Maastricht, 22.iv.1954, leg. V. Lefeber. **(b)** Summer specimen without ventral occipital spine: Biesbosch, Hengstpolder, 29.vii.2005 (leg. M. Reemer). **(c)** Summer specimen with moderately developed ventral occipital spine, from same location and date as (b).

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## REFERENCES

- FOLMER, O., BLACK, M., HOEH, W., LUTZ, R. & VRIJENHOEK, R. 1994. DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. *Molecular Marine Biology and Biotechnology* 3: 294-299.
- GUSENLEITNER, F. & SCHWARZ, M. 2002. Weltweite Checkliste der Bienengattung *Andrena* mit Bemerkungen und Ergänzungen zu paläarktischen Arten (Hymenoptera, Apidae, Andreninae, *Andrena*). *Entomofauna*, Supplement 12: 1-1280.
- KOULIANOS, S., SCHMID-HEMPEL, R., ROUBIK, D.W. & SCHMID-HEMPEL, P. 1999. Phylogenetic relationships within the corbiculate Apinae (Hymenoptera) and the evolution of eusociality. *Journal of Evolutionary Biology* 12: 380-384.
- KAWAKITA, A., SOTA, T., ASCHER, J.S., ITO M., TANAKA, H. & KAT, M. 2003. Evolution and phylogenetic utility of alignment gaps within iron sequences of three nuclear genes in bumble bees (*Bombus*). *Molecular Biology and Evolution* 20: 87-92.
- LARKIN, L.L., NEFF, J.L. & SIMPSON, B.B. 2006. Phylogeny of the *Callandrena* subgenus of *Andrena* (Hymenoptera: Andrenidae) based on mitochondrial and nuclear DNA data: polyphyly and convergent evolution. *Molecular Phylogenetics and Evolution* 38: 330-343.
- MADDISON, D.R. & MADDISON, W.P. 2003. MacClade 4: Analysis of phylogeny and character evolution. Version 4.08. – Sunderland, Massachusetts.
- MANDERY, K., KOSUCH, J. & SCHUBERTH, J. 2008. Untersuchungsergebnisse zum Artstatus von *Andrena decipiens* Schenk, 1861, *Andrena flavilabris* Schenck, 1874, und ihrem gemeinsamen Brutparasiten *Nomada stigma* Fabricius, 1804 (Hymenoptera: Apidae). *Nachrichtenblatt bayerischer Entomologen* (München) 57 (1/2): 30-1.
- MEER, VAN DER, F., REEMER, M., PEETERS, T. & NEVE, A. 2006. De roodrandzandbij *Andrena rosae* in de Zuid-Hollandse Biesbosch (Hymenoptera: Apoidea: Andrenidae). *Nederlandse Faunistische Mededelingen* 25: 1-9; Leiden/Nederland.
- PEETERS, T.M.J., RAEMAKERS, I.P. & SMIT, J. 1999. Voorlopige atlas van de Nederlandse bijen (Apidae). European Invertebrate Survey - Nederland, Leiden.
- PEETERS, T.M. J & REEMER, M. 2003. Bedreigde en verdwenen bijen in Nederland (Apidae s.l.). Basisrapport met voorstel voor de Rode Lijst. European Invertebrate Survey – Nederland, Leiden.
- SCHMID-EGGER, C. & SCHEUCHL, E. 1997. Illustrierte Bestimmungstabellen der Wildbienen Deutschlands und Österreichs unter Berücksichtigung der Arten der Schweiz. Band III. *Andrenidae*.
- SIMON, C., FRATI, F., BECKENBACH, A., CRESPI, B., LIU, H. & FLOOK, P. 1994. Evolution, weighting, and phylogenetic utility of mitochondrial gene sequences and a compilation of conserved polymerase chain reaction primers. *Annals of the Entomological Society of America* 87: 651-701.
- SWOFFORD, D.L. 2003. PAUP\*. Phylogenetic Analysis Using Parsimony (\*and Other Methods). Version 4.0 beta 10. – Massachusetts, USA.
- TADAUCHI, O. & HIRASHIMA, Y. 1984. New or little known bees of Japan (Hymenoptera, Apoidea) V. Supplements to *Andrena* (*Hoplandrena*). *Kontyû* 52: 278-285.



- VECHT, VAN DER, J. 1928. Hymenoptera Anthophila (Q XIII m) A. *Andrena*. Fauna van Nederland 4: 1-144.
- WESTRICH, P. 1989. Die Wildbienen Baden-Württembergs. Spezieller Teil: die Gattungen und Arten.





CHAPTER 6

**Molecular phylogeny of the  
Ariantinae  
(Gastropoda, Pulmonata, Helicidae),  
a new approach to an old problem**

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This work is not issued for purposes of zoological nomenclature  
and is not published within the meaning of the  
International Code of Zoological Nomenclature (1999; see article 8.2).

A number of species have provisional names referring to  
a taxonomic revision of the Ariantinae by P. Subai (in prep.).

## ABSTRACT

A new starting-point in Ariantinae systematics is presented by combining traditional shell morphology, genital anatomy and phylogenies based on DNA sequence data. For each genus one or more shells are figured and drawings of the genital organs are shown to illustrate the morphological diversification within the subfamily. For as much as the material allowed it, partial sequences were obtained for *Histone H3* (H3), *Cytochrome c oxidase subunit I* (COI), *Cytochrome B* (CytB) and *16S ribosomal RNA* (16S). Our results show that some of the traditional, speciose genera like *Campylaea*, *Chilostoma*, and *Helicigona*, do not represent monophyletic groups of species, whereas most of the so-called subgenera (e.g. *Causa*, *Dinarica*, *Josephinella*, *Faustina*, *Liburnica*, *Kosicia* and *Thiessea*) warrant generic status from an evolutionary perspective. Besides elucidating sister-group relationships for most of the taxa, this study enables the classification of some enigmatic species such as *Cylindrus obtusus* and *Ariantopsis pelia*. The phylogeny reconstructions indicate that the subfamily originated on the Balkan peninsula, not later than ca. 31 MYA ago, during or prior to the early Oligocene (Rupelian), and subsequently dispersed and radiated over Europe.

## INTRODUCTION

The classification of the Ariantinae Mörch, 1864 (Gastropoda, Helicidae), a subfamily of terrestrial air-breathing snails, with a primary radiation in southern Europe, has been under debate for more than a century. Left aside a few exceptions such as *Cylindrus obtusus*, *Helicigona lapicida* and *Isognomostoma isognomostomos*, most species within this subfamily are conchologically rather close to a basic bauplan (Fig. 1), with shells that are more or less depressed globular, with an open umbilicus and without apertural teeth. All species are characterized by a pair of mucous glands in the genital system, inserting at the female side, between the dart sac and the bursa copulatrix. The mucous glands can either be simple or divided (forked) and are always longer than the dart sac.

The conchological uniformity did not hamper the description of new species, what made the delimitation of genera increasingly subjective. In the literature, more than once, taxon status was changed from a generic to a subgeneric level, or the other way round, without proper argumentation. This led to the confusing situation of today, where there is neither consensus on the number of (sub)genera that the Ariantinae include, nor about their mutual phylogenetic relationships.

Thus, the biggest issue with this subfamily is not the delimitation of species, but that of genera and subgenera. This is illustrated by the number of genera that have been described after Mörch erected the subfamily in 1864, but even more by the number of genera used by different authors to classify the same species. *Ariantopsis pelia*, for example, has also been classified in *Arianta*, *Chilostoma*, *Faustina*, and *Helicigona*. In the taxonomic literature a variety of subdivisions of the Ariantinae has been



proposed (e.g. Hesse, 1931; Knipper, 1939; Subai, 1984, 1996; Bank *et al.*, 2001; Subai, 2002, 2006). Maybe most authoritative is the enumeration by Zilch (1960). In this paper we will compare our results only with that classification (Table 1). This study describes a molecular analysis of the Ariantinae, which are subdivided into 20 genera, 11 subgenera and over 150 species. Our revised classification will be proposed for the Fauna Europaea database (Bank, 2012).

#### *The subgenus*

As pointed out, the main difficulty with the classification of Ariantinae has been the delimitation of genera and subgenera in particular, because of their arbitrary use. In zoological taxonomy, the subgenus is the rank below the genus-level and (unlike in botany) it is the only rank between genus and species. It is a taxonomic rank, not a cladistic one. As such there are no strict rules to consider a taxon either a genus or a subgenus. Already in 1942, Mayr said “.. the subgenus is nearly always used in one of two situations: either when an author would like to make a new genus, but does not quite have the courage to do so... or when an author wants to sink a genus as unnecessary, but does not quite dare to do so because it is particularly old or well-known. The subgenus is a temporary stage in either case.”

Throughout this study we adhere to the following subgenus definition: “A subgenus is a distinct clade, thus characterized by autapomorphies, that is nested within a larger monophyletic group, which is ranked as a genus.” Clearly this definition (a clade within another clade) is imperfect from a cladistic perspective since virtually all phylogenies contain more clades than taxonomic ranks. At least it does provide a mean to tell which groups cannot be ranked as subgenera of a given genus.

## MATERIAL AND METHODS

#### *Taxon sampling*

An overview of the specimens used for molecular analyses, locality and collection data is given in Table 2. The specimens were collected between 1957 and 2010; a summary of the age distribution is shown in Fig. 2. Except for some of the older material collected by W.H. Neuteboom, who stored his specimens in “spiritus” (methylated spirits), most material was fixed and conserved in 70% ethanol; only some of the more recently collected specimens were preserved in 97% ethanol. An overview of the shell shapes of the majority of the recognized (sub)genera is given in Fig. 1.

#### *Genital anatomy*

For nearly all of the presumed (sub)genera drawings of the genital tract were made by P. Subai. These figures (Fig. 3) are arranged according to the type of mucous glands (undivided on the left page, forked on the right) and in alphabetical order. *Josephinella vikosensis* was put on the right page (despite having undivided mucous



**Table 1.** Classification of the Ariantinae (= Helicigoninae, Campylaeinae) after Zilch (1960)

| Genera               | Subgenera of <i>Campylaea</i> | Subgenera of <i>Chilostoma</i> |
|----------------------|-------------------------------|--------------------------------|
| <i>Arianta</i>       | <i>Ariantopsis</i>            | <i>Campylaeopsis</i>           |
| <i>Campylaea</i>     | <i>Campylaea</i>              | <i>Chilostoma</i>              |
| <i>Chilostoma</i>    | <i>Delphinatia</i>            | <i>Cingulifera</i>             |
| <i>Cylindrus</i>     | <i>Dinarica</i>               | <i>Drobacia</i>                |
| <i>Helicigona</i>    | <i>Faustina</i>               | <i>Josephinella</i>            |
| <i>Isognomostoma</i> | <i>Liburnica</i>              | <i>Kosicia</i>                 |
| <i>Vidovicia</i>     | <i>Wladislawia</i>            | <i>Thiessea</i>                |

glands) to keep the three representatives of that genus together. Species in which only one of the mucous glands was forked were put on the right page as well.

#### *DNA extraction, PCR and sequencing*

Total genomic DNA was extracted from small foot tissue samples using a DNeasy Tissue Kit (Qiagen) following the manufacturer's instructions. As a follow-up of an earlier investigation (Gittenberger *et al.*, 2004), this study started with the amplification of *COI*, but due to the poor quality of some of the DNA extracts, mini-barcode primers (Hajibabaei *et al.*, 2006; Meusnier *et al.*, 2008) had to be used to amplify a smaller fragment of *COI* (124 bp fragment excluding primersites). These mini-barcode sequences grouped as expected and nearly all of the recognized (sub) genera resulted in well supported clades (based on Bayesian phylogeny inference; see Phylogenetic analyses), but the relationships between the (sub)genera were still poorly supported. Subsequently, two other mitochondrial markers, *CytB* & *16S*, and nuclear marker *H3*, were added. PCRs were carried out in 25  $\mu$ l volumes using 1.25 units of Taq DNA polymerase (Qiagen), 0.4 mM of each primer and 0.2 mM dNTPs. For *COI* the final  $MgCl_2$  concentration occasionally had to be increased to 2.5 mM (1x PCR buffer contains 1.5 mM; Qiagen). For *16S* Q-solution (final concentration 1x; Qiagen) was added to most of the reactions. PCR primers and references are given in Table 3. The following PCR thermoprofile was used for all markers: initial denaturation 3 min. @ 94°C, followed by 40 cycles of – denaturation 15 sec. @ 94°C, annealing 30 sec. @ AT (Table 3), extension 40 sec. @ 72°C – and a final extension of 5 min. @ 72°C. PCR products were cleaned with a Montage purification kit (Millipore) at Macrgoen Inc. Europe (Amsterdam), where they were sequenced in both directions on an ABI37730XL using the same primers as used for the PCR. Forward and reverse sequences were assembled with Sequencher 4.10.1 (Gene Codes Corporation) and protein coding genes (*H3*, *COI* and *CytB*) were manually aligned in MacClade 4.08 (Maddison & Maddison, 2003). The alignment for *16S* was made with MAFFT v.6.847b (Kato *et al.*, 2002) as implemented in the Bioinformatics



Toolkit website of the Max-Planck institute (<http://toolkit.tuebingen.mpg.de/mafft>). Due to differences in PCR success and restrictions in available resources, there was a substantial difference in the final number of sequences obtained per marker.

### *Phylogenetic analyses*

A nucleotide substitution model was selected for each dataset (containing all sequences for that marker) using MrModeltest 2.2 (Nylander, 2004) in combination with Paup 4.0b10 for Unix (Swofford, 2002). For the mitochondrial datasets the selected model (using the AIC framework) was the same; general time reversible with a proportion of invariant sites and gamma distributed among site rate variation (GTR+I+G). For *H3* the HKY model (2 substitution types; different rates for transitions and transversions) was selected, again with I+G. Each dataset was analysed separately with the selected model in MrBayes 3.1.2 (Ronquist and Huelsenbeck, 2003). Per analysis two Markov Chain Monte Carlo (MCMC) runs (consisting of four chains each) were ran simultaneously. Each analysis was set to run for 10,000,000 generations or until the MCMC runs converged (diagnostic stop value 0.009). After the datasets were analysed separately, taxa for which less than three markers were sequenced were pruned from the individual datasets (resulting in eighty taxa per dataset; for *CytB* and *16S* some taxa had missing sequences). For each of these pruned datasets a substitution model was selected, following the same procedure. The pruned datasets were subsequently concatenated. Despite that the ILD-test (Incongruence Length Difference or Partition Homogeneity test) has been criticized (e.g. Darlu & Lecointre, 2002; Hipp *et al.*, 2004), we do wanted to test for incongruence between the four datapartitions (*H3*, *COI*, *CytB* and *16S*) of the concatenated dataset. Unfortunately the ILD-test as implemented in Paup 4.0b10 (Swofford, 2002), has severe computational limitations to the number of taxa that can be analysed. Moreover, testing for incongruence over all partitions does not show which partition(s) (if any) are incongruent with the others; in this case there could be as little as one and as many as four. To overcome these issues (and to get some indication of the amount of congruence between the different partitions), we selected 18 taxa that represented the majority of the genera of Ariantinae (*Arianta*, *Chilostoma*, *Cingulifera*, *Cylindrus*, *Drobacia*, *Faustina*, *Helicigona*, *Isognomostoma*, *Josephinella*, *Kosicia*, *Liburnica* and *Vidovicica*) and performed an ILD-test on the concatenated dataset (4 partitions) and on the 6 bipartite partition combinations (pairwise ILD-tests; Planet, 2006). The model selected by MrModeltest for each of the pruned datasets was (GTR+I+G). Therefore a MrBayes analysis was set up for the concatenated dataset, specifying the four partitions, each to be run with the the GTR+I+G model, while the other parameters were similar to that of the previous runs. According to Zilch (1960) the oldest known Ariantinae fossils (representing *Arianta* and *Dinarica*) are from the late Miocene ("Sarmat"; ca. 12.7-11.6 MYA). These and the ages of the oldest known fossils of *Cylindrus*, *Faustina* and *Helicigona* (late Pleistocene "Würm" ca. 12,000-110,000 YA, late Pliocene ca. 2.6-1.8 MYA and early Pliocene "Pont" ca. 5.3-3.6 MYA; Frank, 2006; Zilch, 1960) were used as callibration





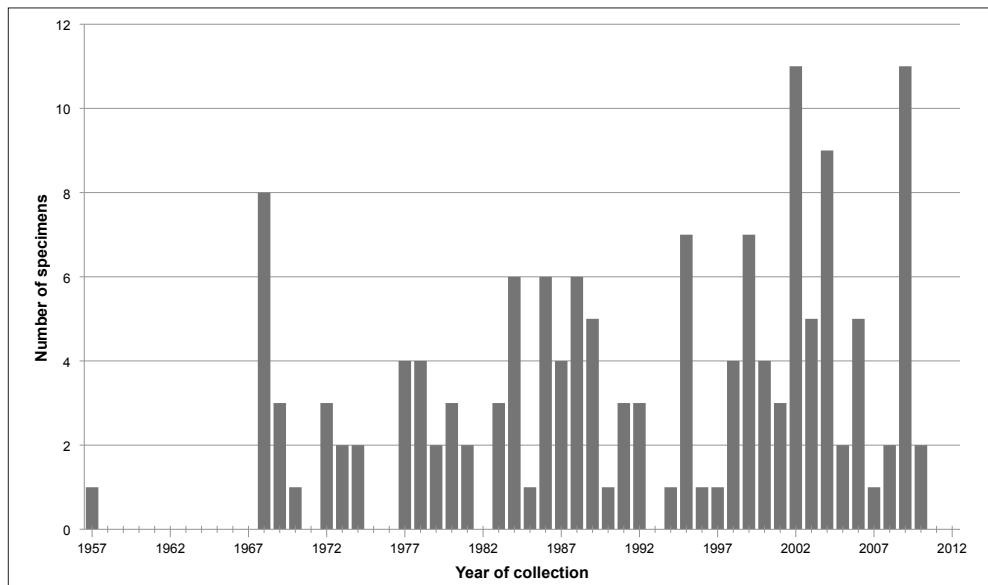


**Figure 1.** Compilation of shells representing most of the currently recognized genera of Ariantinae (measurements in description below; shells do not depict scale).

1. *Arianta arbustorum* (Linnaeus, 1758) B 24.1 mm [RMNH G2131] Austria, Steiermark, near Gstatterboden; E. Gittenberger leg., 10-IX-1964.
2. *Arianta chamaeleon wiedemayeri* (Kobelt, 1903) B 18.3 mm [RMNH G2608] Austria, East Tirol, S of Kartitsch; E. Gittenberger leg., VIII-1974.
3. *Cylindrus obtusus* (Draparnaud, 1805) B 13.7 mm [RMNH 73877] Austria, Oberösterreich, Bledigupf; W.H. Neuteboom leg., 22-VII-1966.
4. *Isognomostoma isognomostomos* (Schröter, 1784) B 9.7 mm [RMNH 74226] Austria, Kärnten, Plöckenpass; W.H. Neuteboom leg., 14-IX-1952.
5. *Causa holosericea* (Studer, 1820) B 10.5 mm [RMNH 74311] Austria, Salzburg, Amerthal; W.H. Neuteboom leg., 16-VII-1968.
6. *Chilostoma (Chilostoma) achates* (Rossmässler, 1835) B 21.9 mm [RMNH G2410] Austria, Steiermark, E of Brandtriedl; A. & E. Gittenberger leg., 19-V-1972.
7. *Chilostoma (Chilostoma) zonata rhaetica* (Strobel, 1857) B 25.2 [RMNH G54412] Zwisserland, E of Martinsbruck; E. Gittenberger leg., IX-1963.
8. *Chilostoma (Cingulifera) cingulatum tigrinum* (Cristofori & Jan, 1832) B 24.7 mm [RMNH 73434] Italy, Como, Pasturo; W.H. Neuteboom leg., 3-VIII-1954.
9. *Chilostoma (Cingulifera) cingulatum cingulatum* (Studer, 1820) B 20.5 mm [RMNH H1938] Switzerland, Tessin, Melide along Lago di Lugano; J.T. Henrard leg., 28.viii.1938.
10. *Chilostoma (Cingulifera) cingulatum gobanzi* (Frauenfeld, 1867) B 22.8 mm [RMNH 73408] Italy, Brescia, Val Toscolano; W.H. Neuteboom leg., 05-VIII-1954.
11. *Delphinatia fontinellii alpina* (Michaud, 1829) B 19.6 mm [RMNH G3646] France, Isère, SSE of Laurent-du-Pont; E. Gittenberger leg., 12-IX-1975.
12. *Faustina faustina* (Rossmässler, 1835) B 19.6 mm [RMNH 53576] Hungary, Bükk, Szalajkavölgy; Agosy leg., 21-V-1921.
13. *Campylaea (Campylaea) planospira planospira* (Lamarck, 1801) B 25.9 mm [RMNH 73625] Italy, Torino, Sant'arvis di Montebruno; W.H. Neuteboom leg., 8-VII-1977.
14. *Campylaea (Campylaea) illyrica* (Stabile, 1864) B 25.4 mm [RMNH 11124] Italy, Friuli, SSE of Tarvisio; E. Gittenberger leg., VI-1992.
15. *Kosicia ambrosi* (Strobel, 1851) B 12.5 mm [RMNH 24940] Italy, Vicenza, Valstagne; W.H. Neuteboom leg., 22-VII-1968.
16. *Kosicia intermedia* (Pfeiffer, 1828) B 15.3 mm [RMNH 54440] Austria, Kärnten, Deuschpeter; E. Gittenberger leg., IX-1964.
17. *Kosicia zieglerei* (Rossmässler, 1836) B 17.7 mm [RMNH G2350] Slovenia, Kamniške Alps, Igla Studenec; A. & E. Gittenberger leg., 26-VIII-1971.
18. *Campylaeopsis moellendorffii* (Kobelt, 1871) B 18.4 mm [RMNH AJW898] Bosnia-Herzegovina, Vrelo Bosne, near Ilidza; A.J. de Winter leg., 17-IX-1980.
19. *Kollarix kollari* (Pfeiffer, 1856) B 24.3 mm [RMNH YU.429] Serbia, along Ovcar Banja; W.J.M. Maassen leg., V-1984.
20. *Liburnica (Liburnica) setosa setosa* (Férussac, 1832) B 23.6 mm [RMNH 94272] Croatia, Lovrec, S of Imotski; W.J.M. Maassen leg., IV-1989.
21. *Dinarica (Sabljaria) stenomphala* (Menke, 1830) B 30.7 mm [col. PS.21228] Croatia, Velebit mountains, near Krasno Polje; P. Subai leg., 29-VII-2002.
22. *Dinarica (Dinarica) pouzolzii* (Deshayes, 1830) B 40.1 mm [RMNH 53508 / 413] Croatia, Dalmatia, E of Biokovo mountains; J.J. ter Pelwijk leg., 12-VIII-1939.
23. *Liburnica (Superba) skipetarica skipetarica* (Subai, 1995) B 19.6 mm [col. PS.20215] Albania, Periferi Berat, Tommorit; P. Subai leg., VIII-2004.
24. *Corneola desmoulinsi* (Farines, 1834) B 17.2 mm [RMNH 93925] Andorra, Canillo, northern wall; W.J.M. Maassen leg., VII-1990.
25. *Helicigona lapicida lapicida* (Linnaeus, 1758) B 17.6 mm [SMF325426/1] Germany, Hessen, Schlüchtern; M. Pfenninger leg.
26. *Drobacia banatica* (Rossmässler, 1838) B 29.5 mm [RMNH 54500 / 485] Rumania, Siebenbürgen; H. de Wever leg.
27. *Cattania (Cattania) trizona*



(Rossmässler, 1835) B 24.4 mm [RMNH 99615] Rumania, Banat mountains, mount Domogled; Kroupa leg., 21-VI-1985. **28.** *Cattania* (*Cattania*) *subai* (Fauer, 1991) B 22.9 mm [RMNH GU.9921 / EK5558] Greece, Makedonia, W of Kozani; E. Gittenberger & D. Uit de Weerd leg., 23-V-1999. **29.** *Vidovicia caeruleans* (Pfeiffer, 1828) B 15.4 mm [RMNH 93836] Croatia, Velebit near Starigrad; W.J.M. Maassen leg., IX-1982. **30.** *Cattania* (*Ariantopsis*) *pelia* (Hesse, 1912) B 17.9 mm [col. PS.23572] Bulgaria, Vitosha, Bistrisko branishte; I. Dedoy leg., 8-VII-2004. **31.** *Cattania* (*Wladislawia*) *polinskii* (Wagner, 1927) B 16.4 mm [RMNH G3749] Bulgaria, Pirin mountains, mount Vihren; A. Riedel leg., 24-VI-1977. **32.** *Cattania* (*Wladislawia*) *sztolcmani* (Wagner, 1927) B 10.6 mm [RMNH G3749] Bulgaria, Pirin mountains, mount Vihren; A. Riedel leg., 24-VI-1977. **33.** *Josephinella vikosensis* (Subai, 1990) B 18.8 mm [RMNH EG.9703 / DK8112] Greece, Ipiros, Vikos valley; E. Gittenberger leg., 23-VII-1997. **34.** *Josephinella hemonica* (Thiesse, 1884) B 19.1 mm [?] Greece, Makedhonia, SE of Grevena; E. Gittenberger leg., 18-VII-1986. **35.** *Thiessea sphaeristoma* (Bourguignat, 1857) B 21.0 mm [RMNH 75078] Greece, Sterea Ellas, SE of Mariolates; E. Gittenberger & D. Uit de Weerd leg., 19-V-2000.



**Figure 2.** Age distribution of studied specimens.

points for an analysis of the concatenated dataset with BEAST v.1.7.0 (Drummond *et al.*, 2012). For each partition the same nucleotide substitution model as used with the MrBayes analysis, was selected. The analysis was set to run for 10,000,000 generations using a relaxed clock model (lognormal uncorrelated) and with the Yule process (Yule, 1926; Gernhard, 2008) specified as the tree prior.



**Table 2.** Taxa, sampling information and GenBank accession numbers.

| Nr. | Species                                      | Locality data                                                           |
|-----|----------------------------------------------|-------------------------------------------------------------------------|
| 1   | <i>Arianta arbustorum alpicola</i>           | Germany, Bayern, Achberg, S-side, 5 km NW of Reit im Winkel             |
| 2   | <i>Arianta arbustorum alpicola</i>           | Germany, Bayern, Achberg, S-side, 5 km NW of Reit im Winkel             |
| 3   | <i>Arianta arbustorum arbustorum</i>         | Austria, Oberösterreich, gardens near Hallstatt                         |
| 4   | <i>Arianta arbustorum arbustorum</i>         | Netherlands, Zuid-Holland, Oestgeest                                    |
| 5   | <i>Arianta arbustorum arbustorum</i>         | Netherlands, Zuid-Holland, Hoogmade, banks of the Wijde Aa river        |
| 6   | <i>Arianta arbustorum arbustorum</i>         | Netherlands, Zuid-Holland, Hoogmade, banks of the Wijde Aa river        |
| 7   | <i>Arianta arbustorum arbustorum</i>         | France, Aude, Gorges du Rébenty                                         |
| 8   | <i>Arianta arbustorum arbustorum</i>         | Sweden, Stockholms Län, South of Uppsala in forest                      |
| 9   | <i>Arianta arbustorum canigonensis</i>       | France, Aude, Massif du Canigou, Pla Guillem                            |
| 10  | <i>Arianta arbustorum repellini</i>          | France, Hautes Alpes, Pic des Chalances, Queyras                        |
| 11  | <i>Arianta arbustorum stenzii-arbustorum</i> | Austria, Ost-Tirol, Lienzer Dolomiten, 7 km S. of Lienz                 |
| 12  | <i>Arianta arbustorum styriaca</i>           | Austria, Oberösterreich, Dachstein, 4 km S of Obertraun, 'Krippenstein' |
| 13  | <i>Arianta arbustorum vareliensis</i>        | France, Alpes-Maritimes, near St. Etienne de Tinée (W of Bouseyas)      |
| 14  | <i>Arianta arbustorum vareliensis</i>        | France, Alpes-Maritimes, near St. Etienne de Tinée (W of Bouseyas)      |
| 15  | <i>Arianta arbustorum xatarti</i>            | Spain, Catalunya, Ripolles, Noucreous peak, 3 km NE of Nuria            |
| 16  | <i>Arianta chamaeleon</i>                    | Austria, Karawanken Bärenthal, Johannsenruhe                            |
| 17  | <i>Arianta chamaeleon</i>                    | Austria, Karawanken, S-side of Natschacher Sattel (N. of Bielschitz)    |
| 18  | <i>Arianta chamaeleon carnica</i>            | Italy, Udine, Karnic Alps, N of Rifugio Calvi, E. of Mount Perlba       |
| 19  | <i>Arianta chamaeleon carnica</i>            | Austria, Karnic Alps, Obere Valentin Alm (=12,5 km SE of St. Lorenzen)  |
| 20  | <i>Arianta chamaeleon subglobosa</i>         | Austria, Gailtaler Alps, S-slope Hohenrain (Dobratsch-massif)           |
| 21  | <i>Arianta schmidtii</i>                     | Slovenia, Kamniske Alpe, south slope of the Grintavec                   |
| 22  | <i>Ariantopsis pelia</i>                     | Bulgaria, Mount Vito                                                    |
| 23  | <i>Campylaea illyrica</i>                    | Slovenia, Planinska Jama near Planina                                   |
| 24  | <i>Campylaea lefeburiana</i>                 | Croatia, 21 km from road Bribir-Ledenice to Lic, along forest           |
| 25  | <i>Campylaea lefeburiana</i>                 | Slovenia, Nova Gorica                                                   |
| 26  | <i>Campylaea planospira</i>                  | Italy, Lucca, Serohio Valley                                            |
| 27  | <i>Campylaeopsis moellendorffii</i>          | Bosnia Herzegovina, Sutjeska valley, 3 km S of Tjentiste                |
| 28  | <i>Caracollina lenticula</i>                 | Italy, Lampedusa island, Punta Sottile residence                        |
| 29  | <i>Cattania faueri</i>                       | Greece, Macedonia, Mount Pangaeon, 23 km from Akrovouni                 |
| 30  | <i>Cattania faueri</i>                       | Greece, Macedonia, Kavala, Mount Pangaion (= Pangeo)                    |
| 31  | <i>Cattania kattingeri</i>                   | Greece, Macedonia, 6 km N of Prossotsani                                |
| 32  | <i>Cattania kattingeri</i>                   | Greece, Thracia, 2 km from road Xanthi-Stavroupoli in direction Ehinós  |
| 33  | <i>Cattania</i> n. sp. 1                     | Greece, Macedonia, 400 m S of Kehrokambos                               |
| 34  | <i>Cattania pseudocingulata</i>              | Greece, Macedonia, start canyon 2 km N of Neohori                       |
| 35  | <i>Cattania subaia</i>                       | Greece, Macedonia, summit NNE of Zarkadia                               |



| Elevation | ST/RMNH/CRC        | Collector | Date         | H3       | COI      | CytB     | 16S      |
|-----------|--------------------|-----------|--------------|----------|----------|----------|----------|
| 650 m     | 442/-/-            | EG        | 4.viii.1995  | JX156642 | AF296908 | -        | -        |
|           | 607/-/-            | EG        | 4.viii.1995  | -        | AF296914 | -        | -        |
| 510 m     | 440/-/-            | EG        | 7.vii.1998   | JX156643 | AF296925 | JX156879 | JX156945 |
|           | 600/-/-            | EG        | 25.viii.1999 | -        | AF296938 | -        | -        |
|           | 144/139193A/-      | EG        | 3.viii.1992  | -        | AF296924 | -        | JX156946 |
|           | 603/139193B/-      | EG        | 3.viii.1992  | -        | AF296940 | -        | JF717809 |
|           | 726/-/Winter_4645  | AJW       | 15.vi.1985   | -        | AF296945 | -        | JF717810 |
|           | 1239/-/-           | AJG       | 5.ix.2001    | -        | AF538294 | -        | JX156947 |
| 2250 m    | 1054/-/-           | AJW       | 15.vii.2000  | -        | AF538288 | JX156880 | JX156948 |
| 2430 m    | 361/-/-            | FPW       | 15.vii.1999  | JX156644 | AF296947 | JX156881 | -        |
| 2200 m    | 16/-/-             | EG        | 9.vii.1998   | JX156645 | AF296964 | JX156882 | JX156949 |
| 2100 m    | 454/-/-            | EG        | 7.vii.1998   | -        | AF296980 | -        | -        |
| 1470 m    | 1631/96616A/-      | G, R & F  | 30.viii.2002 | JX156646 | JX156780 | JX156883 | JX156950 |
| 1470 m    | 1632/96616B/-      | G, R & F  | 30.viii.2002 | -        | JX156781 | -        | -        |
| 2740 m    | 145/11241/JG.9802  | JG        | 1.viii.1998  | JX156647 | AF296993 | JX156884 | JX156951 |
| 1400 m    | 839/-/-            | PM        | 23.v.2000    | JX156648 | AF296994 | JX156885 | JX156952 |
| 1715 m    | 3345/-/8709        | EG        | 3.iv.1969    | JX156649 | JX156782 | -        | -        |
| 2300 m    | 3346/-/2426        | EG        | 1.ix.1972    | JX156650 | JX156783 | -        | -        |
| 1600 m    | 3347/93585/-       | EG        | 31.vii.2002  | JX156651 | JX156784 | JX156886 | -        |
| 1900 m    | 3344/-/8698        | EG        | 1.ix.1969    | JX156652 | JX156785 | -        | -        |
| 2450 m    | 725/-/-            | EG        | 22.vii.1981  | JX156653 | AF538286 | JX156887 | JX156953 |
|           | 3133/112256/X.4412 | LD & PS   | 8.vii.2004   | JX156654 | JX156786 | -        | JX156954 |
|           | -/99469/X.2755     | WJM       | 27.v.1984    | JX156655 | JX156787 | -        | -        |
| 900 m     | 2868/99379/X.458   | PS        | 3.viii.2002  | JX156656 | -        | -        | -        |
|           | -/99463/X.2753     | PS        | 5.ix.1986    | JX156657 | JX156788 | -        | -        |
| 75 m      | 3135/112260/X.4416 | PS        | 17.vii.2006  | JX156658 | JX156789 | -        | JX156955 |
| 500 m     | 3143/-/Winter_1485 | AJW       | 19.ix.1980   | JX156659 | -        | -        | -        |
|           | 2877/100400/-      | AWJ       | 2005         | JX156660 | JF717793 | JX156888 | JF717811 |
| 1600 m    | -/99438/X.1012     | WF & PS   | 23.iii.1989  | JX156661 | JX156790 | -        | -        |
| 1650 m    | 722/59633/-        | EG & DUW  | 18.v.1999    | JX156662 | AF296996 | -        | -        |
| 300 m     | -/99449/X.1021     | PS        | 8.v.1995     | JX156663 | JX156791 | -        | -        |
| 2000 m    | -/99440/X.1167     | PS        | 7.v.1995     | JX156664 | JX156792 | -        | -        |
| 350 m     | -/99447/X.1020     | PS        | 7.v.1995     | JX156665 | JX156793 | -        | -        |
| 300 m     | 2862/99443/X.1050  | PS & MS   | 20.v.1997    | JX156666 | JX156794 | -        | JX156956 |
| 500 m     | -/99442/X.1043     | PS        | 7.v.1995     | JX156667 | -        | -        | -        |



| Nr. | Species                                  | Locality data                                                          |
|-----|------------------------------------------|------------------------------------------------------------------------|
| 36  | <i>Cattania subaii</i>                   | Greece, Macedonia, W of Kozani, 2 km SSW of Xirolimni                  |
| 37  | <i>Cattania trizona</i>                  | Greece, Thessalia, Larissa, Olimbos (Mount Olympus)                    |
| 38  | <i>Cattania trizona</i>                  | Servia, Sicevacka Klissura near Ostrovica                              |
| 39  | <i>Causa holosericea</i>                 | Austria, Tirol, Stubaital                                              |
| 40  | <i>Causa holosericea</i>                 | Austria, Steiermark, E of Brandtriedl, N of Schladming                 |
| 41  | <i>Causa holosericea</i>                 | Austria, Steiermark, E of Brandtriedl, N of Schladming                 |
| 42  | <i>Causa holosericea</i>                 | Austria, Karawanken, NW of Bertahütte (= 25 km W of Ferlach)           |
| 43  | <i>Cepaea hortensis</i>                  | Netherlands, Limburg, Noorbeek, near camping 'Grensheuvel'             |
| 44  | <i>Cepaea nemoralis</i>                  | Netherlands, Zuid-Holland, Zwijndrecht                                 |
| 45  | <i>Chilostoma achates</i> (N Alps)       | Germany, Wimbachschloss (Wimbachtal), 8 km SW of Berchtesgaden         |
| 46  | <i>Chilostoma achates</i> (S Alps)       | Austria, Kärnten, Greifenburg                                          |
| 47  | <i>Chilostoma zonata adelozona</i>       | Italy, Bergamo, Mount Pressalano near Colere                           |
| 48  | <i>Cingulifera cingulatum baldensis</i>  | Italy, near lake Garda                                                 |
| 49  | <i>Cingulifera cingulatum baldensis</i>  | Italy, near lake Garda                                                 |
| 50  | <i>Cingulifera cingulatum colubrinum</i> | Italy, Brescia, Toscolano valley, 1.5 km N of main road in Toscolano   |
| 51  | <i>Cingulifera cingulatum frigidum</i>   | Italy, prov. Como, Rifugio Arnoldo Bogani, timberline above rifugio    |
| 52  | <i>Cingulifera cingulatum frigidum</i>   | Italy, prov. Como, Rifugio Arnoldo Bogani, timberline above rifugio    |
| 53  | <i>Cingulifera cingulatum gobanzi</i>    | Italy, Brescia, Valle di Vestino, E of Capovalle                       |
| 54  | <i>Cingulifera cingulatum gobanzi</i>    | Italy, Brescia, Valle di Vestino, E of Capovalle                       |
| 55  | <i>Cingulifera cingulatum peregrini</i>  | Austria, Tirol, Halltal, 9 km NE of Innsbruck                          |
| 56  | <i>Cingulifera cingulatum peregrini</i>  | Austria, Tirol, Halltal, 9 km NE of Innsbruck                          |
| 57  | <i>Cingulifera cingulatum preslii</i>    | Austria, Lienzer Dolomiten, Kerschbaumer Schutzhaus, 7 km S of Lienz   |
| 58  | <i>Cingulifera cingulatum preslii</i>    | Austria, Ost-Tirol, Gulitzenklamm                                      |
| 59  | <i>Cingulifera cingulatum preslii</i>    | Austria, Ost-Tirol, Gulitzenklamm                                      |
| 60  | <i>Cingulifera cingulatum preslii</i>    | Austria, Ost-Tirol, Gulitzenklamm                                      |
| 61  | <i>Cingulifera cingulatum preslii</i>    | Italy, Belluno, Ospitale di Cadore, Val Tovanella                      |
| 62  | <i>Cingulifera cingulatum preslii</i>    | Italy, Belluno, Ospitale di Cadore, Val Tovanella                      |
| 63  | <i>Cingulifera cingulatum preslii</i>    | Italy, Belluno, Valle d'Ampezzo, near Rio Felizon                      |
| 64  | <i>Cingulifera cingulatum preslii</i>    | Italy, Belluno, Valle d'Ampezzo, near Rio Felizon                      |
| 65  | <i>Cingulifera cingulatum preslii</i>    | Italy, Belluno, Valle d'Ampezzo, near Rio Felizon                      |
| 66  | <i>Cingulifera cingulatum preslii</i>    | Germany, Berchtesgaden, Wimbachtal Schloßwand                          |
| 67  | <i>Cingulifera cingulatum preslii</i>    | Germany, Berchtesgaden, Wimbachtal Schloßwand                          |
| 68  | <i>Cingulifera cingulatum preslii</i>    | Germany, Berchtesgaden, Wimbachtal Schloßwand                          |
| 69  | <i>Cingulifera cingulatum preslii</i>    | Germany, Berchtesgaden, Wimbachtal Schloßwand                          |
| 70  | <i>Cingulifera cingulatum tigrinum</i>   | Italy, Bergamo, Val Taleggio, gorge 4 km W of San Giovanni Bianco      |
| 71  | <i>Corneola desmoulinsii</i>             | Spain, Lerida, Rio Flamicell, 5 km NNW of La Pobla de Segur            |
| 72  | <i>Corneola desmoulinsii</i>             | Andorra, Canillo, N-slope                                              |
| 73  | <i>Corneola desmoulinsii</i>             | France, Bouches-du-Rhône, Bonnieux (ca. 60 km N of Marseille)          |
| 74  | <i>Corneola squamatinum</i>              | France, Pyrénées Orientales, Prats-de-Mollo-la-Preste, near river Tech |
| 75  | <i>Corneola squamatinum</i>              | France, Pyrénées Orientales, humid shrubs along the Tech la Preste     |



| Elevation | ST/RMNH/CRC         | Collector | Date         | H3       | COI      | CytB     | 16S      |
|-----------|---------------------|-----------|--------------|----------|----------|----------|----------|
| 700 m     | 2861/62482/-        | EG & DUW  | 23.v.1999    | JX156668 | JX156795 | -        | JX156957 |
| 70 m      | 724/-/-             | EG        | ca. 1992     | JX156669 | AF296998 | -        | -        |
|           | -/99441/X.2719      | WJM       | 14.v.1984    | JX156670 | -        | -        | -        |
| 1000 m    | 3358/-/6738         | VRA       | 28.viii.1957 | JX156671 | -        | -        | -        |
| 1500 m    | 4225/-/-            | AG & EG   | 19.v.1972    | JX156672 | -        | -        | -        |
| 1500 m    | 4226/-/-            | AG & EG   | 19.v.1972    | JX156673 | -        | -        | -        |
| 1150 m    | 4227/99568/-        | EG        | 25.vii.2002  | JX156674 | JX156796 | -        | -        |
|           | 4476/-/-            | DG        | 27.vi.2010   | -        | JX156797 | -        | JX156958 |
|           | 4372/-/-            | DG        | 17.v.2010    | JX156675 | JX156798 | JX156889 | JX156959 |
|           | 3285/114171/-       | DG        | 2.v.2009     | JX156676 | JX156799 | JX156890 | JX156960 |
|           | 2859/73686/-        | HM        | 2000         | JX156677 | JX156800 | JX156891 | JX156961 |
| 1700 m    | 2860/99468/X.1620   | WJM       | 19.vii.1986  | JX156678 | JX156801 | JX156892 | -        |
|           | 3821/114256A/-      | EJG       | 2009         | JX156679 | JX156802 | JX156893 | JX156962 |
|           | 3822/114256B/-      | EJG       | 2009         | JX156680 | JX156803 | JX156894 | -        |
| 125 m     | -/102871/EG.0605    | EG        | 29.iv.2006   | JX156681 | JX156804 | JX156895 | JX156963 |
|           | 3243/62194A/-       | WHN       | 8.vi.1973    | JX156682 | JX156805 | -        | -        |
|           | 3244/62194B/-       | WHN       | 8.vi.1973    | JX156683 | JX156806 | JX156896 | -        |
|           | 3947/-/1978-1       | EG        | 1978         | JX156684 | JX156807 | JX156897 | -        |
|           | 3948/-/1978-2       | EG        | 1978         | JX156685 | JX156808 | JX156898 | -        |
|           | 3283/114170A/-      | DG & MK   | 1.v.2009     | JX156686 | JX156809 | JX156899 | JX156964 |
|           | 3284/114170B/-      | DG & MK   | 1.v.2009     | JX156687 | JX156810 | JX156900 | JX156965 |
| 1800 m    | 2738/93542/EG.0201  | EG        | 19.vii.2002  | JX156688 | JX156811 | JX156901 | JX156966 |
| 930 m     | 3240/62180A/-       | WHN       | 17.vii.1968  | JX156689 | JX156812 | JX156902 | JX156967 |
| 930 m     | 3241/62180B/-       | WHN       | 17.vii.1968  | JX156690 | JX156813 | JX156903 | -        |
| 930 m     | 3242/62180C/-       | WHN       | 17.vii.1968  | JX156691 | JX156814 | -        | -        |
| 457 m     | 3247/62178A/-       | WHN       | 21.vii.1968  | JX156692 | JX156815 | -        | -        |
| 457 m     | 3248/62178B/-       | WHN       | 21.vii.1968  | JX156693 | JX156816 | -        | -        |
| 1300 m    | 3249/62177A/-       | WHN       | 18.vii.1968  | JX156694 | JX156817 | -        | -        |
| 1300 m    | 3250/62177B/-       | WHN       | 18.vii.1969  | JX156695 | JF717795 | -        | JX156968 |
|           | 3251/62177C/-       | WHN       | 18.vii.1970  | JX156696 | JX156818 | JX156904 | -        |
|           | 3790/-/ZI0072090A   | F         | 1986         | JX156697 | JX156819 | JX156905 | JX156969 |
|           | 3791/-/ZI0072090B   | F         | 1986         | JX156698 | JX156820 | JX156906 | JX156970 |
|           | 3792/-/ZI0072091A   | F         | 1987         | JX156699 | JX156821 | JX156907 | JX156971 |
|           | 3793/-/ZI0072091B   | F         | 1987         | JX156700 | JX156822 | JX156908 | JX156972 |
| 450 m     | -/102855/EG.0601    | EG        | 28.iv.2006   | JX156701 | JX156823 | JX156909 | JX156973 |
| 750 m     | 3356/-/Winter.7346  | AJW       | 14.vi.1989   | JX156702 | JX156824 | JX156910 | -        |
| 1560 m    | 4099/114300/-       | D         | 1.vii.1990   | JX156703 | JX156825 | -        | -        |
|           | 4234/-/Lenter.3787  | JCL       | 1.viii.1977  | JX156704 | -        | -        | -        |
| 1100 m    | 4232/-/Winter.5575B | AJW       | 21.vi.1987   | JX156705 | JX156826 | -        | JX156974 |
| 1160 m    | 4233/-/Winter.679   | AJW       | 7.ix.1980    | JX156706 | JX156827 | -        | JX156975 |



| Nr. | Species                               | Locality data                                                           |
|-----|---------------------------------------|-------------------------------------------------------------------------|
| 76  | <i>Corneola squamatinum</i>           | France, Pyrénées Orientales, Prats-de-Mollo-la-Preste, near river Tech  |
| 77  | <i>Cylindrus obtusus</i>              | Austria, Niederösterreich, Schneeberg massive, Waxriegel                |
| 78  | <i>Cylindrus obtusus</i>              | Austria, Ennstaler Alpen, Grosser Buchstein (NW of Gstatterboden)       |
| 79  | <i>Cylindrus obtusus</i>              | Austria, Rax, 200 m N of Karl Ludwighaus                                |
| 80  | <i>Delphinatia fontenillii alpina</i> | France, Hautes Alpes, Galibier                                          |
| 81  | <i>Delphinatia glaciale</i>           | France, dept. Hautes Alpes, Col Fromage, 13 km ENE of Guillestre        |
| 82  | <i>Dinarica pouzolzii</i>             | Montenegro, Skutari Lake area, Kurioci village                          |
| 83  | <i>Drobacia banatica</i>              | Rumania, Brasov, Bran                                                   |
| 84  | <i>Drobacia maeotica</i>              | Rumania, Komitat Bihor, Stana de Vale                                   |
| 85  | <i>Drobacia maeotica</i>              | Hungary, Gyulavar, Sitkai Forest                                        |
| 86  | <i>Faustina faustina associata</i>    | Rumania, Komitat Harghita, Mount Hasmas, Bicaz canyon                   |
| 87  | <i>Faustina faustina faustina</i>     | Rumania, Transsylvania, Torda, Tordai-Hasadek                           |
| 88  | <i>Faustina faustina faustina</i>     | Czech, NE Böhmen, Opcno, Schlosspark                                    |
| 89  | <i>Faustina faustina orba</i>         | Rumania, Komtitat Arges, Cheile Dambuicidaaei, Mount Piatra Craiului    |
| 90  | <i>Faustina kiralikoeica</i>          | Rumania, Brasov, Piatra Cralului                                        |
| 91  | <i>Helicigona lapicida andorrica</i>  | Spain, Girona, along river El Ter near Setcases                         |
| 92  | <i>Helicigona lapicida andorrica</i>  | Spain, Catalunya, Castellar de N'Hug, Fonis de Liobregat                |
| 93  | <i>Helicigona lapicida andorrica</i>  | Spain, Catalunya, Castellar de N'Hug, Fonis de Liobregat                |
| 94  | <i>Helicigona lapicida andorrica</i>  | Spain, Catalunya, Castellar de N'Hug, Fonis de Liobregat                |
| 95  | <i>Helicigona lapicida andorrica</i>  | Spain, prov. Lerida, S-bank of Rio Segre (15 km WNW of Seo de Urgel)    |
| 96  | <i>Helicigona lapicida lapicida</i>   | Belgium, Luxembourg, La Roche en Ardenne, chateau                       |
| 97  | <i>Helicigona lapicida lapicida</i>   | Germany, Nordrhein-Westfalen, walls of Blankenheim castle               |
| 98  | <i>Helicodonta obvolvata</i>          | Netherlands, Limburg, Magraten, Savelsbos                               |
| 99  | <i>Isoognomostoma isognomostomos</i>  | Suisse, Bern, Merligen, along the Grönbach                              |
| 100 | <i>Isoognomostoma isognomostomos</i>  | Germany, Rheinland-pfalz, 2 km ESE of Bollendorf (= near Ferschweiler)  |
| 101 | <i>Isoognomostoma isognomostomos</i>  | Trento-Alto Adige, between Predazzo and Bellamonte                      |
| 102 | <i>Isoognomostoma isognomostomos</i>  | Trento-Alto Adige, between Predazzo and Bellamonte                      |
| 103 | <i>Josephinella argentellei</i>       | Greece, Lakonia, Mount Taygetos, Agh. Elias                             |
| 104 | <i>Josephinella conemenosi</i>        | Greece, Achaia, near Sandomeri, Mount Skolus, W-slope                   |
| 105 | <i>Josephinella conviva</i>           | Greece, Achaia, Chelmos Mountains, Mount Kastro Grias, N of Kalavrita   |
| 106 | <i>Josephinella krueperi</i>          | Greece, Messenia, Platsa                                                |
| 107 | <i>Josephinella</i> n. sp. 1          | Greece, Arkadia, 1 km from Rises to Ag. Petros at bifurcation to Astros |
| 108 | <i>Josephinella</i> n. sp. 2          | Greece, Etoloarkarnani, 2 km from Antikofito to Nafraktos               |
| 109 | <i>Josephinella</i> n. sp. 3          | Greece, Epirus, Moni Pateron, W of Zitsa                                |
| 110 | <i>Josephinella</i> n. sp. 4          | Greece, Epirus, 13 km from Vroussina to Igoumenitsa                     |
| 111 | <i>Josephinella</i> n. sp. 5          | Greece, Arkadia, 4 km N of Dimitsana                                    |
| 112 | <i>Josephinella</i> n. sp. 6          | Greece, Epirus, Monodendri, Monastery Agh. Paraskevi                    |
| 113 | <i>Josephinella peritricha</i>        | Greece, Amaia, Moni Omblon, SW of Patras                                |
| 114 | <i>Josephinella phoea</i>             | Greece, Fokis, Parnass Ski Centre                                       |
| 115 | <i>Josephinella pindica</i>           | Greece, Thessalia, Pili, 18 km W Trikala, on Turkish bridge             |



| Elevation | ST/RMNH/CRC         | Collector | Date         | H3       | COI      | CytB     | 16S      |
|-----------|---------------------|-----------|--------------|----------|----------|----------|----------|
| 1100 m    | 4231/-/Winter.5575A | AJW       | 21.vi.1987   | JX156707 | JX156828 | -        | JX156976 |
| 1820 m    | 1052/-/-            | KE        | 6.ix.2000    | JX156708 | JX156829 | JX156911 | JX156977 |
| 2200 m    | 1238/-/-            | JGL       | 29.viii.2001 | JX156709 | JX156830 | JX156912 | JX156978 |
|           | 1313/92186/-        | KE        | 1.vii.2001   | JX156710 | JX156831 | JX156913 | -        |
| 2340 m    | 3134/112262/X.4418  | PS        | 23.vii.2006  | JX156711 | JX156832 | JX156914 | -        |
| 2325 m    | 2864/96591/X.686    | EG        | vii.1984     | JX156712 | JX156833 | -        | JX156979 |
|           | 3137/112261/X.4417  | PS        | 23.iii.2006  | JX156713 | JX156834 | JX156915 | -        |
| 740 m     | 3131/112258/X.4414  | PS        | 28.iv.2008   | JX156714 | JX156835 | -        | -        |
|           | 2746/99547/X.2648   | PS & MS   | 6.vii.2004   | JX156715 | JX156836 | JX156916 | JX156980 |
|           | -/99377/X.1371      | DM & PS   | 16.iv.1994   | JX156716 | -        | -        | -        |
| 900 m     | 2743/99410/X.2637   | MS & PS   | 3.iv.2004    | JX156717 | JX156837 | JX156917 | JX156981 |
|           | -/100305/X.2643     | ZF        | v.2005       | JX156718 | JX156838 | JX156918 | -        |
| 300 m     | 2871/99378/X.3253   | KL & PS   | 20.viii.2004 | JX156719 | JX156839 | -        | JX156982 |
|           | 2872/99406/X.2638   | MS & PS   | 2.vii.2004   | JX156720 | JX156840 | JX156919 | -        |
| 930 m     | 3132/112259/X.4415  | PS        | 29.iv.2008   | JX156721 | JX156841 | -        | -        |
| 1279 m    | -/99448/X.2904      | PS        | 8.ix.1983    | JX156722 | JX156842 | JX156920 | -        |
|           | 3652/116481A/-      | BK & NS   | 11.vi.2009   | JX156723 | JF717796 | JX156921 | JF717814 |
|           | 3653/116481B/-      | BK & NS   | 11.vi.2009   | JX156724 | JF717797 | JX156922 | JF717815 |
|           | 3654/116481C/-      | BK & NS   | 11.vi.2009   | JX156725 | JF717798 | JX156923 | JF717816 |
| 825 m     | 3357/-/Winter.3661  | JD & AJW  | 10-vii-1984  | JX156726 | -        | -        | -        |
|           | -/96401/X.319       | DG        | ix.2003      | JX156727 | JF717799 | JX156924 | JF717817 |
|           | -/99439/X.721       | PS        | 24.vii.1983  | JX156728 | JF717800 | JX156925 | JF717818 |
|           | 3789/139195/-       | DG & PC   | 30.vii.2009  | JX156729 | JF717801 | -        | JF717819 |
| 640 m     | 3362/-/Winter.372   | AJW       | 16.v.1980    | JX156730 | -        | -        | -        |
|           | 3363/-/-            |           | 2.vi.1984    | JX156731 | -        | -        | -        |
|           | 3823/139196/-       | DG & IMV  | 9.ix.2009    | JX156732 | JF17802  | JX156926 | JF17820  |
|           | 3824/139197/-       | DG & IMV  | 9.ix.2009    | JX156733 | JF17803  | JX156927 | JF17821  |
| 1900 m    | -/99376/X.1009      | PS        | 15.vii.1979  | JX156734 | -        | -        | -        |
|           | -/99385/X.739       | PS        | 16.iv.1988   | JX156735 | JX156843 | -        | -        |
|           | 2867/99407/X.737    | PS        | 18.iv.2003   | JX156736 | JX156844 | -        | -        |
|           | -/99375/X.1054      | LB & PS   | 16.ii.1981   | JX156737 | -        | -        | -        |
|           | 2742/99408/X.743    | LB & PS   | 6.x.1986     | JX156738 | JX156845 | JX156928 | JX156983 |
|           | -/99404/X.898       | WF & PS   | 2.x.1989     | JX156739 | JX156846 | -        | -        |
|           | -/99412/X.894       | PS        | 22.v.1991    | JX156740 | -        | -        | -        |
| 700 m     | -/99383/X.891       | AR & PS   | 13.iv.1988   | JX156741 | JX156847 | -        | -        |
|           | 2876/99397/X.744    | PS        | 15.iv.2003   | JX156742 | JX156848 | JX156929 | JX156984 |
|           | -/99403/X.892       | WF & PS   | 27.ix.1989   | JX156743 | JX156849 | -        | -        |
| 780 m     | -/99399/X.741       | AR & PS   | 16.iv.1988   | JX156744 | -        | -        | -        |
| 1950 m    | -/99453/X.901       | PS        | 12.iv.1986   | JX156745 | JX156850 | JX156930 | -        |
|           | 2873/99402/X.1163   | PS        | 23.iv.2003   | JX156746 | JX156851 | -        | -        |



| Nr. | Species                            | Locality data                                                              |
|-----|------------------------------------|----------------------------------------------------------------------------|
| 116 | <i>Josephinella reischuetzi</i>    | Greece, Epirus, Ados Canyon, bridge near Konitsa                           |
| 117 | <i>Josephinella vikosensis</i>     | Greece, Ioannina, N of monastery Paraskevi (= 0.2 km N of Monodendri)      |
| 118 | <i>Kollarix kollari</i>            | Serbia, Monastery Manasija                                                 |
| 119 | <i>Kosicia ambrosi</i>             | Italy, Belluno, Giaron - Arsie, on the right shore of Cismon               |
| 120 | <i>Kosicia ambrosi</i>             | Italy, Vicenza, Cismon del Grappa, crevice SW of village                   |
| 121 | <i>Kosicia intermedia</i>          | Austria, Karawanken, Hainschbauer - Hornik (= 9 km SSE of Ferlach)         |
| 122 | <i>Kosicia intermedia</i>          | Austria, Karnic Alps, Untere Valentin Alm (= 14 km SE of St. Lorenzen)     |
| 123 | <i>Kosicia zieglerei</i>           | Austria, Karawanken, SE of Trögernklamm (= 8 km SE of Eisenkappel)         |
| 124 | <i>Kosicia zieglerei</i>           | Austria, Karawanken, ENE of Klagenfurter Hütte (= 12 km SW of Ferlach)     |
| 125 | <i>Liburnica albanograeca</i>      | Greece, Epiros, Loutro (=5 km N of Amarantos)                              |
| 126 | <i>Liburnica dunjana</i>           | Albania, Periferi Nat, Lumi i Matit (Burrel-Milot), 11 km W of Ulez        |
| 127 | <i>Liburnica insolita insolita</i> | Croatia, Island of Korcula, Korcula-Town                                   |
| 128 | <i>Liburnica kleciaki</i>          | Croatia, Peninsula Peljesac, Mali Ston                                     |
| 129 | <i>Liburnica setigera setigera</i> | Albania, S of Lezhe at S-slope mountain                                    |
| 130 | <i>Liburnica setosa</i>            | Croatia, Lovrec near Imotski                                               |
| 131 | <i>Ljubotenia ljubetenensis</i>    | Kosovo, Metohija, Ljuboten, Globocica                                      |
| 132 | <i>Sabljaria stenomphala</i>       | Croatia, Krasno Polje                                                      |
| 133 | <i>Soosia diodonta</i>             | Rumania                                                                    |
| 134 | <i>Superba grisea</i>              | Albania, Periferi Skrapar, NE of Radesh, gorge along the road to Zaloshnje |
| 135 | <i>Superba kulmankana</i>          | Albania, Periferi Skrapar, Mount Tomorrit, 1 km NW of Maja e Kulmakut      |
| 136 | <i>Superba skipetarica</i>         | Albania, Periferi Berat, Mount Tomorrit, peak region, near bektashi teque  |
| 137 | <i>Superba skipetarica</i>         | Albania, Periferi Berat, Mount Tomorrit, 6 km N of Maja e Tomorrit         |
| 138 | <i>Thiessea arcadica</i>           | Greece, Insel Siros, mountain W of Ano Siros                               |
| 139 | <i>Thiessea euboeae</i>            | Greece, Evvoia (= Euboea), Mount Dirfos, 23 km NE of Khalkis (=Halkida)    |
| 140 | <i>Thiessea heldreichi</i>         | Greece, Atika, Athen, mountain Likavittos                                  |
| 141 | <i>Thiessea sphaeriosstoma</i>     | Greece, Thessalia, Magnissia, side-road to Kotes (= 30.5 km SE of Volos)   |
| 142 | <i>Thiessea sphaeriosstoma</i>     | Greece, Thessalia, Magnissia, 4 km WSW of Marathia (= 32 km SE of Volos)   |
| 143 | <i>Thiessea sphaeriosstoma</i>     | Greece, Thessalia, Magnissia, Agh. Kiriaki (= 32 km SSE of Volos)          |
| 144 | <i>Vidovicia caeruleans</i>        | Yugoslavia, Lika Hrvatska on sotony walls in upper part of Kartobag        |
| 145 | <i>Vidovicia caeruleans</i>        | Croatie, Karlobag, Stara Vrata-pas, 6.5 km above Karlobag                  |
| 146 | <i>Vidovicia caeruleans</i>        | Croatie, Mali Alan-pas                                                     |
| 147 | <i>Wladislavia sztolcmani</i>      | Bulgaria, Blagoevgrad, Pirin Planina, mount Vihren, 30 km N of Melnik      |
| 148 | <i>Wladislavia sztolcmani</i>      | Bulgaria, Blagoevgrad, Pirin Planina, mount Vihren, 30 km N of Melnik      |
| 149 | <i>Wladislavia sztolcmani</i>      | Bulgaria, Blagoevgrad, Pirin Planina, mount Vihren, 30 km N of Melnik      |

\* ST refers to SeqTrace database numbers used in the molecular laboratory

RMNH are registration numbers used by NCB Naturalis

CRC is the registration code provided by the collector (or owner of the specimen)

\*\* AG = Adri Gittenberger, AJG = Arjan Gittenberger, AJW = A.J. de Winter, AR = A. Riedel, AWJ = A.W. Janssen, BK = B. Kokshoorn, D = Deeleman, DG = D.S.J. Groenenberg, DM = Domokos, DUW = D.R. Uit de Weerd, EJG = E.J. Gittenberger, EG = E. Gittenberger, F = G. Falkner, FPW



| Elevation | ST/RMNH/CRC*           | Collector** | Date         | H3       | COI      | CytB     | 16S      |
|-----------|------------------------|-------------|--------------|----------|----------|----------|----------|
| 1100 m    | 2741 / 99387 / X.893   | AR & PS     | 12.iv.1988   | JX156747 | JX156852 | JX156931 | JX156985 |
|           | 2863 / 93624 / EG.9127 | EG & EJG    | 30.v.1991    | JX156748 | JX156853 | -        | -        |
|           | - / 99467 / X.2718     | WJM         | 12.v.1984    | JX156749 | JX156854 | JX156932 | -        |
| 315 m     | 3353 / 46399 / -       | WHN         | 21.vii.1968  | JX156750 | JX156855 | -        | -        |
| 275 m     | 3354 / 46403 / -       | WHN         | 22.vii.1968  | JX156751 | JX156856 | JX156933 | -        |
| 1050 m    | 2740 / 93576 / EG.0212 | EG          | 26.vii.2002  | JX156752 | JX156857 | JX156934 | JX156986 |
| 1325 m    | - / 93589 / EG.0216    | EG          | 1.viii.2002  | JX156753 | JX156858 | JX156935 | -        |
| 700 m     | - / 93546 / EG.0203    | EG          | 23.vii.2002  | -        | JX156859 | -        | -        |
| 1700 m    | 2739 / 93562 / EG.0206 | EG          | 24.vii.2002  | JX156754 | JX156860 | JX156936 | JX156987 |
| 1400 m    | 2745 / 99477 / X.890   | AR & PS     | 12.iv.1988   | JX156755 | JX156861 | JX156937 | JX156988 |
| 100 m     | 2870 / 100306 / X.53   | ZPE & ZF    | 27.vi.2003   | JX156756 | JX156862 | -        | JX156989 |
|           | - / 99476 / X.469      | WJM         | x.1978       | JX156757 | -        | -        | -        |
|           | - / 99474 / X.488      | WJM         | vi.1978      | JX156758 | -        | -        | -        |
|           | 2744 / 99470 / X.60    | FWS & PS    | 21.ix.1995   | JX156759 | JX156863 | -        | JX156990 |
|           | - / 99475 / X.487      | WJM         | 30.iv.1989   | JX156760 | JX156864 | -        | -        |
|           | - / 99436 / X.1777     | WJM         | vii.1979     | JX156761 | -        | -        | -        |
| 1000 m    | 3136 / 112257 / X.4413 | PS          | 29.vii.2002  | JX156762 | JX156865 | JX156938 | -        |
|           | 3130 / 112264 / -      |             | 18.iv.2007   | JX156763 | JX156866 | JX156939 | JX156991 |
| 830 m     | - / 100301 / X.58      | ZF          | 8.viii.2004  | JX156764 | JX156867 | -        | -        |
| 2070 m    | - / 100302 / X.57      | ZF          | 9.viii.2004  | JX156765 | JX156868 | -        | -        |
| 2270 m    | - / 100303 / X.39      | ZF          | 10.viii.2004 | JX156766 | JX156869 | -        | -        |
| 2090 m    | - / 100304 / X.37      | ZF          | 11.viii.2004 | JX156767 | JX156870 | -        | -        |
| 980 m     | 2866 / 99466 / X.906   | PS          | 4.v.1991     | JX156768 | -        | -        | -        |
|           | 4230 / 49141 / SK.9632 | ME & SK     | 24.v.1996    | JX156769 | -        | -        | -        |
|           | - / 99472 / X.747      | PS          | 22.iv.1988   | JX156770 | -        | -        | -        |
| 70 m      | 723 / - / -            | EG & DUW    | 28.v.1999    | JX156771 | AF296997 | JX156940 | -        |
| 15 m      | 4228 / - / -           | EG & DUW    | 28.v.1999    | JX156772 | JX156871 | JX156941 | -        |
| 75 m      | 4229 / - / -           | EG & DUW    | 28.v.1999    | JX156773 | JX156872 | JX156942 | -        |
| 50 m      | 3144 / - / Winter.5129 | AJW         | 3.vii.1983   | JX156774 | JX156873 | -        | -        |
|           | 3348 / 62202 / -       | WHN         | 12.vii.1974  | JX156775 | JX156874 | JX156943 | JX156992 |
| 1045 m    | 3349 / 62204 / -       | WHN         | 12.vii.1974  | JX156776 | JX156875 | JX156944 | JX156993 |
| 2550 m    | 4221 / - / -           | AR          | 24.vi.1977   | JX156777 | JX156876 | -        | -        |
| 2550 m    | 4222 / - / -           | AR          | 24.vi.1977   | JX156778 | JX156877 | -        | JX156994 |
| 2550 m    | 4223 / - / -           | AR          | 24.vi.1977   | JX156779 | JX156878 | -        | -        |

= F.P. Wesselingh, FWS = F. Welter-Schultes, G = Gargominy, HM = H. Menkhorst, IMV = I.M. Voermans, JCL = J.C. van Lenteren, JD = J.M.L. Donders, JG = J. Goud, JGL = J. Gould, KE = K. Edlinger, KL = Kouloch, LB = L. Butot, LD = L. Dedov, ME = M. van Eendenburg, MK = M. Kolet, MS = M. Szekeres, NS = N. Snijders, PC = P.J.L. Cuijpers, PM = P. Mildner, PS = P. Subai, R = T.E.J. Ripken, SK = S. de Kort, VRA = C.O. van Regteren Altena, WF = W. Fauer, WHN = W.H. Neuteboom, WJM = W.J. Maassen, ZF = Z. Fehér, ZPE = Z.P. Erös.



### Genetic distances

Even though the GTR model was selected for all datasets except *H3*, we decided to calculate uncorrected pairwise (*p*) distances (instead of GTR-corrected distances) with Paup 4.0b10 for Unix (Swofford, 2002). We calculated these distances, because we preferably show underestimated (the minimum amount of sequence divergence) instead of overestimated distances (GTR-corrected distances are generally larger). Sequence divergence percentages (as referred to in this study) were calculated as the uncorrected *p*-distance times 100.

## RESULTS

**Table 3.** Primer information.

| Primer  | Sequence                         | Marker        | Length* | Source         | AT**  |
|---------|----------------------------------|---------------|---------|----------------|-------|
| H3-F    | 5'-ATGGCTCGTACCAAGCAGACVGC-3'    | <i>H3</i>     | 328     | Colgan, 2000   | 57    |
| H3-R    | 5'-ATATCCTTRGGCATRATRGTGAC-3'    |               |         |                |       |
| L1490 F | 5'-GGTCAACAAATCATAAAGATATTGG-3'  | <i>COI</i>    | 655     | Folmer , 1994  | 45-50 |
| H2198 R | 5'-TAAACTTCAGGGTGACCAAAAAATCA-3' |               |         |                |       |
| MB-F    | 5'-TCCACTAATCACAARGATATTGGTAC-3' | <i>COI-mb</i> | 124     | Meusnier, 2008 | 50    |
| MB-R*** | 5'-GAAAATTATKACAAARGCATGAGC-3'   |               |         |                |       |
| 151-F   | 5'-TGTGGRGCNACYGTWATYACTAA-3'    | <i>Cyt B</i>  | 361     | Merrit, 1998   | 50    |
| 270-R   | 5'-AANAGGAARTAYCAYTCNGGYTG-3'    |               |         |                |       |
| Pal-F   | 5'-CGGCCGCCTGTTTATCAAAAACAT-3'   | <i>EF 1a</i>  | 404-    | Palumbi, 1991  | 50    |
| Pal-R   | 5'-GGAGCTCCGGTTTGAAGTCAGATC-3'   |               | 412     |                |       |

\* Amplicon length; excluding primer sequence

\*\* Annealing Temperature (°C)

\*\*\* Minibarcode Reverse primer was modified to be more specific for Ariantinae

In total 392 sequences were obtained: 138, 127, 66 and 61 for the markers *H3*, *COI*, *CytB* and *16S*, respectively. All sequences were deposited in GenBank; accession numbers are given in Table 2. MrBayes analyses of the separate datasets reached convergence (diagnostic stop value 0.009) for all the mitochondrial markers before the set maximum number of generations (10,000,000). Only the run for *H3* failed to converge. Consensus trees that show all compatible groups were calculated, discarding 20% of the obtained trees as 'burn-in'. The Bayesian phylogeny reconstructions based on the *H3*, *COI*, *CytB* and *16S* datasets are shown in Figs. 4-7, respectively. The ILD-test for the concatenated dataset did not indicate incongruence between the four partitions



( $P = 0.65$ ). Pairwise ILD-tests showed that the *H3* partition was most congruent with the other partitions (but then *H3* was also the least informative partition):  $P=0.99$  (*H3-COI*),  $P=0.75$  (*H3-CytB*) and  $P=0.99$  (*H3-16S*). *COI* and the other mitochondrial partitions were less congruent:  $P=0.49$  (*COI-CytB*) and  $P=0.54$  (*COI-16S*). The *CytB* and *16S* partitions were the least congruent ( $P=0.14$ ), but none of the partitions were significantly incongruent ( $P<0.05$ ) with one another.

Figures 8-A and -B depict the phylogenies obtained with MrBayes (p.158) and BEAST (p.159), respectively. Since both are Bayesian phylogenetic analyses, the

**Table 4.** Information content per dataset

| Dataset             | Taxa | nr Char | Constant | Inf. Char.* | Perc. inf.** |
|---------------------|------|---------|----------|-------------|--------------|
| <i>H3</i>           | 138  | 328     | 268      | 45          | 13.7         |
| <i>COI</i>          | 127  | 655     | 341      | 297         | 45.3         |
| <i>CytB</i>         | 66   | 361     | 130      | 218         | 60.4         |
| <i>16S</i>          | 61   | 403     | 177      | 196         | 48.6         |
| <i>Concatenated</i> | 80   | 1747    | 936      | 749         | 42.9         |

\* The number of parsimony informative characters

\*\* The percentage of informative characters; calculated as  $100 \times (\text{Inf. Char.} / \text{nr. Char})$

obtained phylogenies are very similar, though not completely identical (which is why they were depicted separately). The two observed differences were the sister-group relations within the clade *Delphinatia*, *Drobacia* and *Vidovicia* and the position of *Dinarica* relative to *Liburnica*. The phylogeny obtained with MrBayes showed a sister-group relation between *Vidovicia* and *Drobacia*, whereas BEAST showed a sister-group relation between *Vidovicia* and *Delphinatia*. The phylogeny obtained with BEAST also depicted *Dinarica* and *Liburnica* as sister-groups, while these genera turned out paraphyletic in the MrBayes analysis. Figure 8-C (p. 161) depicts the same phylogeny as Fig.8-B, but with divergences time estimates. Within this study posterior probabilities for Fig.8 will be cited as “8-A/8-B” and remarks on divergence time estimates automatically concern Fig. 8-C

A summary of the character statistics for each dataset (as calculated with Paup 4.0b10 for Unix; Swofford, 2002) is given in Table 4. An overview of the uncorrected *p*-distances between taxa that will be addressed in the discussion is given in Table 5.

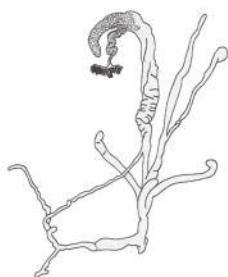
## DISCUSSION

For as much as the phylogenies of the separate markers (Fig. 4-7) allow a comparison (especially for the *CytB* and *16S* datasets, species are missing), there seems to be





1. *Arianta  
arbustorum*



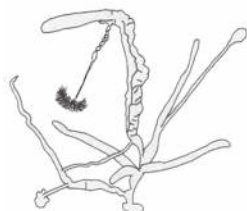
2. *Cyllindrus  
obtus*



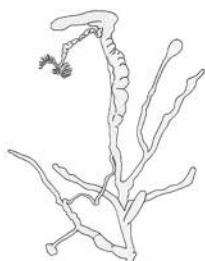
3. *Chilostoma  
adelozona*



4. *Cingulifera  
cingulatum*



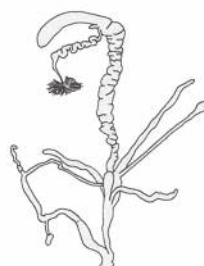
5. *Corneola  
squamatinum*



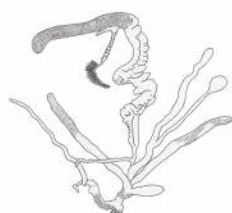
6. *Helicigona  
lapicida*



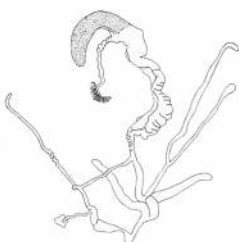
7. *Causa  
holosericea*



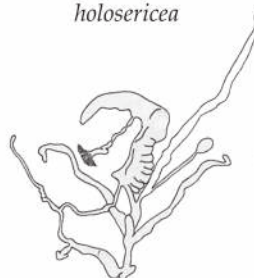
8. *Isognomostoma  
isognomostomos*



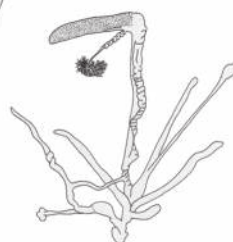
9. *Drobacia  
banatica*



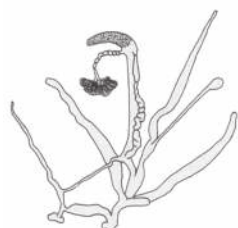
10. *Campylaeopsis  
moellendorffii*



11. *Vidovicia  
caeruleans*



12. *Delphinatia  
alpina*



13. *Kosicia  
intermedia*



14. *Campylaea  
planospira*

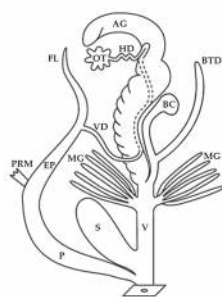


15. *Kollarix  
kollari*



16. *Thiessea  
sphaeristoma*



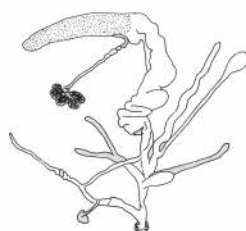


Genital morphology

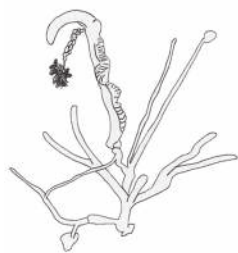
AG = albumen gland  
 BC = bursa copulatrix  
 BTD = diverticulum  
 EP = epiphallus  
 FL = flagellum  
 HD = hermaphroditic duct  
 MG = mucus glands  
 OT = ovotestis  
 P = penis  
 PRM = penis retractor muscle  
 S = dart sac (bursa telae)  
 V = vagina  
 VD = vas deferens



17. *Superba skipetarica*



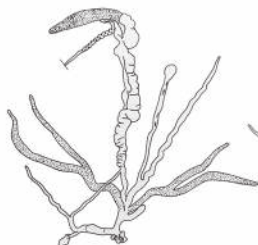
18. *Josephinella reischuetzi*



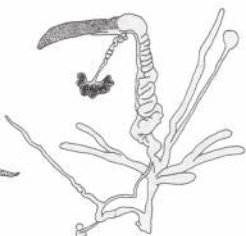
19. *Faustina faustina*



20. *Liburnica setosa*



21. *Superba kulmankana*



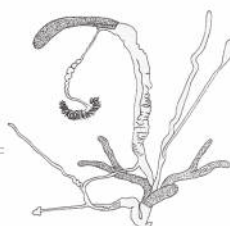
22. *Josephinella hemonica*



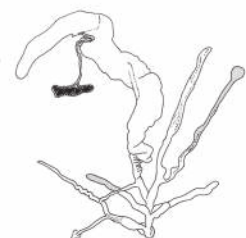
23. *Dinarica pouzolzii*



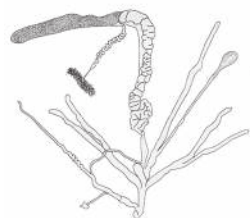
24. *Sabljaria stenomphala*



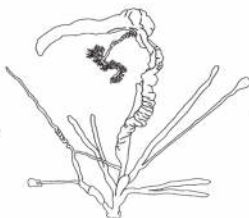
25. *Ljubotenja ljubetenensis*



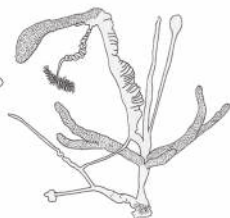
26. *Josephinella vikosensis*



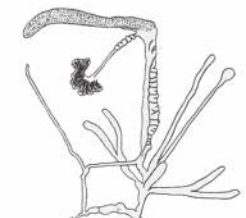
27. *Cattania trizona*



28. *Cattania inflata*



29. *Ariantopsis pelia*



30. *Wladislawia polinskii*



**Figure 3.** Genital anatomy for most of the currently recognized genera of Ariantinae. Left page, species with undivided or simple mucous gland; right page, (mostly) species with divided or forked mucous glands.

sufficient support for the majority of the genera to be recognized (such as *Arianta*, *Cattania*, *Corneola*, *Chilostoma*, *Cingulifera*, *Dinarica*, *Faustina*, *Helicigona*, *Josephinella*, *Kosicia* and *Liburnica*). However, apart from some exceptions (e.g. *Josephinella* & *Thiessea*, *Ariantopsis* & *Wladislawia*) all of the more basal relationships show very little support.

The *H3* dataset and corresponding phylogeny (Fig. 4) deviates clearly from the other datasets by its relatively low percentage of parsimony informative characters (Table 4). The *H3* dataset is very conservative and the phylogeny based on this dataset shows little resolution. Consequently, figure 4 (in most instances), does not discriminate between species within a genus, but *H3* can be useful for assigning ambiguous species (or subgenera) to genera (e.g. *Ariantopsis pelia*, *Campylaeopsis moellendorffii*, *Superba* spec., *Wladislawia sztolcmani*). Even though the Histone gene cluster is multicopy (slight intercopy variation might exist and *H3* pseudogenes have been reported; Rooney *et al.*, 2002), it has been used in higher-level phylogenetics (Armbruster, 2005; Colgan *et al.*, 2006 and references therein). For some genera in this study double peaks were observed at certain positions within the *H3* sequence. In the *H3* phylogeny this turned out to be a problem for the *Chilostoma*-*Cingulifera* clade. This group which is clearly shown monophyletic in the *COI* and *CytB* phylogenies, shows as a paraphyly in the phylogeny based on *H3*. It might be argued that the more extensive sampling within *C. (Cingulifera)* increased the chance of observing this apparent intercopy variation, but it was not observed in other genera for which multiple species were sequenced (e.g. *Arianta*, *Cattania*, *Josephinella* and *Liburnica*), either. In agreement with Colgan *et al.* (2000) we therefore conclude that intercopy variation in *H3* will not significantly interfere with the phylogeny reconstructions.

For only the *COI* dataset did we include mini-barcodes. Hajibabaei *et al.* (2006) showed (in silico) that these mini-barcodes (109 bp) were 3% less effective in

**Table 5.** Uncorrected *p*-distances for cited taxa.

| Between species:                           | And species:                             | <i>H3</i> | <i>COI</i> | <i>CB</i> | <i>16S</i> |
|--------------------------------------------|------------------------------------------|-----------|------------|-----------|------------|
| 19 <i>Arianta chamaeleon carnica</i>       | 78 <i>Cylindrus obtusus</i>              | 0.000     | 0.177      | 0.249     |            |
| 23 <i>Campylaea illyrica</i>               | 25 <i>Campylaea lefeburiana</i>          |           | 0.144      |           |            |
| 42 <i>Causa holosericea</i>                | 101 <i>Isognomostoma isognomostomos</i>  |           | 0.194      |           |            |
| 46 <i>Chilostoma achates</i>               | 45 <i>Chilostoma achates</i>             | 0.000     | 0.002      | 0.003     | 0.003      |
| 48 <i>Cingulifera cingulatum baldensis</i> | 59 <i>Cingulifera cingulatum preslii</i> |           | 0.073      | 0.069     |            |



| Between species:                            | And species:                                | H3    | COI   | CB    | 16S   |
|---------------------------------------------|---------------------------------------------|-------|-------|-------|-------|
| 50 <i>Cingulifera cingulatum colubrinum</i> | 58 <i>Cingulifera cingulatum preslii</i>    |       | 0.032 | 0.066 | 0.013 |
| 51 <i>Cingulifera cingulatum frigidum</i>   | 59 <i>Cingulifera cingulatum preslii</i>    |       | 0.147 |       |       |
| 52 <i>Cingulifera cingulatum frigidum</i>   | 59 <i>Cingulifera cingulatum preslii</i>    |       |       | 0.177 |       |
| 70 <i>Cingulifera cingulatum tigrinum</i>   | 50 <i>Cingulifera cingulatum colubrinum</i> |       | 0.145 | 0.216 |       |
| 70 <i>Cingulifera cingulatum tigrinum</i>   | 52 <i>Cingulifera cingulatum frigidum</i>   |       | 0.124 | 0.163 |       |
| 70 <i>Cingulifera cingulatum tigrinum</i>   | 59 <i>Cingulifera cingulatum preslii</i>    |       | 0.145 | 0.191 |       |
| 72 <i>Corneola desmoulinsi</i>              | 76 <i>Corneola squamatinum</i>              |       | 0.169 |       |       |
| 78 <i>Cylindrus obtusus</i>                 | 19 <i>Arianta chamaeleon carnica</i>        |       | 0.177 |       |       |
| 81 <i>Delphinatia glaciale</i>              | 80 <i>Delphinatia fontenillii alpina</i>    |       | 0.158 |       |       |
| 82 <i>Dinarica pouzolzii</i>                | 125 <i>Liburnica albanograeca</i>           |       | 0.195 | 0.260 |       |
| 83 <i>Drobacia banatica</i>                 | 125 <i>Liburnica albanograeca</i>           |       | 0.192 |       |       |
| 84 <i>Drobacia maeotica</i>                 | 83 <i>Drobacia banatica</i>                 |       | 0.099 |       |       |
| 84 <i>Drobacia maeotica</i>                 | 125 <i>Liburnica albanograeca</i>           |       | 0.205 |       |       |
| 87 <i>Faustina faustina faustina</i>        | 86 <i>Faustina faustina associata</i>       |       | 0.107 | 0.130 |       |
| 87 <i>Faustina faustina faustina</i>        | 124 <i>Kosicia zieleri</i>                  |       | 0.156 | 0.227 |       |
| 90 <i>Faustina kiralikoeica</i>             | 89 <i>Faustina faustina orba</i>            |       | 0.174 |       |       |
| 91 <i>Helicigona lapicida andorrica</i>     | 92 <i>Helicigona lapicida andorrica</i>     |       | 0.008 | 0.025 |       |
| 92 <i>Helicigona lapicida andorrica</i>     | 97 <i>Helicigona lapicida lapicida</i>      |       | 0.107 | 0.147 |       |
| 96 <i>Helicigona lapicida lapicida</i>      | 97 <i>Helicigona lapicida lapicida</i>      |       | 0.024 | 0.014 |       |
| 97 <i>Helicigona lapicida lapicida</i>      | 72 <i>Corneola desmoulinsii</i>             |       | 0.168 | 0.280 |       |
| 97 <i>Helicigona lapicida lapicida</i>      | 76 <i>Corneola squamatinum</i>              |       | 0.188 |       |       |
| 121 <i>Kosicia intermedia</i>               | 120 <i>Kosicia ambrosi</i>                  |       | 0.137 | 0.194 |       |
| 124 <i>Kosicia zieglerei</i>                | 120 <i>Kosicia ambrosi</i>                  |       | 0.148 | 0.197 |       |
| 124 <i>Kosicia zieglerei</i>                | 121 <i>Kosicia intermedia</i>               |       | 0.076 | 0.072 |       |
| 125 <i>Liburnica albanograeca</i>           | 129 <i>Liburnica setigera setigera</i>      | 0.000 | 0.058 |       |       |
| 125 <i>Liburnica albanograeca</i>           | 137 <i>Superba skipetarica</i>              |       | 0.018 |       |       |
| 126 <i>Liburnica dunjana</i>                | 129 <i>Liburnica setigera setigera</i>      | 0.000 | 0.063 |       |       |
| 131 <i>Ljubotenja ljubetenensis</i>         | 24 <i>Campylaea lefeburiana</i>             | 0.000 |       |       |       |
| 131 <i>Ljubotenja ljubetenensis</i>         | 25 <i>Campylaea lefeburiana</i>             | 0.000 |       |       |       |
| 132 <i>Sabljarja stenomphala</i>            | 82 <i>Dinarica pouzolzii</i>                | 0.015 | 0.163 | 0.235 |       |
| 132 <i>Sabljarja stenomphala</i>            | 125 <i>Liburnica albanograeca</i>           |       | 0.171 | 0.211 |       |
| 135 <i>Superba kulmankana</i>               | 129 <i>Liburnica setigera setigera</i>      |       | 0.069 |       |       |
| 135 <i>Superba kulmankana</i>               | 137 <i>Superba skipetarica</i>              | 0.000 | 0.021 |       |       |
| 142 <i>Thiessea sphaeriosoma</i>            | 111 <i>Josephinella n. sp. 5</i>            |       | 0.192 | 0.305 |       |
| 145 <i>Vidovicica caeruleans</i>            | 80 <i>Delphinatia fontenillii alpina</i>    |       | 0.173 | 0.247 |       |



the correct identification of closely related species (tested for a group of fish and butterflies) compared to the full-length barcode (654 bp). These authors also pointed out that mini-barcodes might be less useful to classify a specimen in a larger species assemblage. Although Meusnier *et al.* (2008) showed that the mini-barcodes work within a wide taxonomic range, we had to modify their reverse primer in order to make it work for Ariantinae (Table 3). Based on those taxa for which both a complete and a mini-barcode sequence were obtained (*C. (Cattania) faueri*, *Corneola desmoulinsii*, *Helicigona lapicida andorrca* and *Vidovicia caeruleans*), we conclude that the mini-barcodes (despite their short length) are placed correctly in the COI phylogeny (Fig. 5).

There are some peculiarities in, or even inconsistencies between, the phylogenies.

- Even though all of the obtained phylogenies show a sister-group relation between *Arianta* and *Cylindrus*, the phylogeny reconstruction based on COI does not. The latter phylogeny however does depict the genera *Arianta*, *Chilostoma*-*Cingulifera*, *Cylindrus* and *Helicigona* as a monophyletic group (albeit without support).
- The phylogeny based on *CytB* (Fig. 6) shows *Helicigona* in a basal position, whereas the other mitochondrial phylogenies indicate that *Helicigona*, *Chilostoma* and *Arianta* form a monophyletic group, which clearly does not represent a basal lineage within

**Table 6.** Classification of the Ariantinae according to this study

Family Helicidae  
Subfamily Ariantinae

| Genera               | Subgenera                                                   | Genera                                                  | Subgenera                           | Genera                              | Subgenera                          |
|----------------------|-------------------------------------------------------------|---------------------------------------------------------|-------------------------------------|-------------------------------------|------------------------------------|
| <i>Arianta</i>       |                                                             | <i>Corneola</i>                                         |                                     | <i>Kollarix</i>                     |                                    |
| <i>Campylaea</i>     | <i>Campylaea</i><br><i>Ljubotenina</i> *                    | <i>Cylindrus</i><br><i>Delphinatia</i>                  |                                     | <i>Liburnica</i>                    | <i>Liburnica</i><br><i>Superba</i> |
| <i>Cattania</i>      | <i>Ariantopsis</i><br><i>Cattania</i><br><i>Wladislawia</i> | <i>Dinarica</i>                                         | <i>Dinarica</i><br><i>Sabljarja</i> | <i>Thiessea</i><br><i>Vidovicia</i> |                                    |
| <i>Campylaeopsis</i> |                                                             | <i>Drobacia</i><br><i>Faustina</i><br><i>Helicigona</i> |                                     |                                     |                                    |
| <i>Causa</i>         |                                                             | <i>Isognomostoma</i><br><i>Josephinella</i>             |                                     |                                     |                                    |
| <i>Chilostoma</i> ** | <i>Chilostoma</i><br><i>Cingulifera</i>                     |                                                         |                                     |                                     |                                    |

\* with only a sequence for *H3*, our data are insufficient to either confirm or reject the existence of this subgenus.

\*\* our data indicate the presence of a third subgenus within *Chilostoma*.



the Ariantinae.

- In the 16S phylogeny (Fig. 7) the position of *Cepaea* (which served well as an outgroup for the other phylogenies) within the Ariantinae is definitely wrong, but has been reported before. Based on 28S sequences, Beese *et al.* (Fig. 2; 2009) also found *Cepaea* (as well as *Cornu aspersa*, *Helix pomatia* and *Theba pisana*) nested within the Ariantinae, but did not mention it in their discussion.

- Unlike the phylogenies based on COI and *CytB*, *Chilostoma* and *Cingulifera* are no longer sister-groups in the phylogeny reconstruction based on 16S due to the placement of the clade consisting of *Corneola*, *Helicigona* and *Vidovicia*.

Even though only a fraction of the known species of Ariantinae were included in this study, sequences were obtained for all the genera and subgenera enumerated by Zilch (1960; Table 1). This at least allows for a modest comparison between the existing classification, shell morphology, anatomy and the genetically characterized groups of Ariantinae. The here proposed classification of the Ariantinae is given in Table 6.

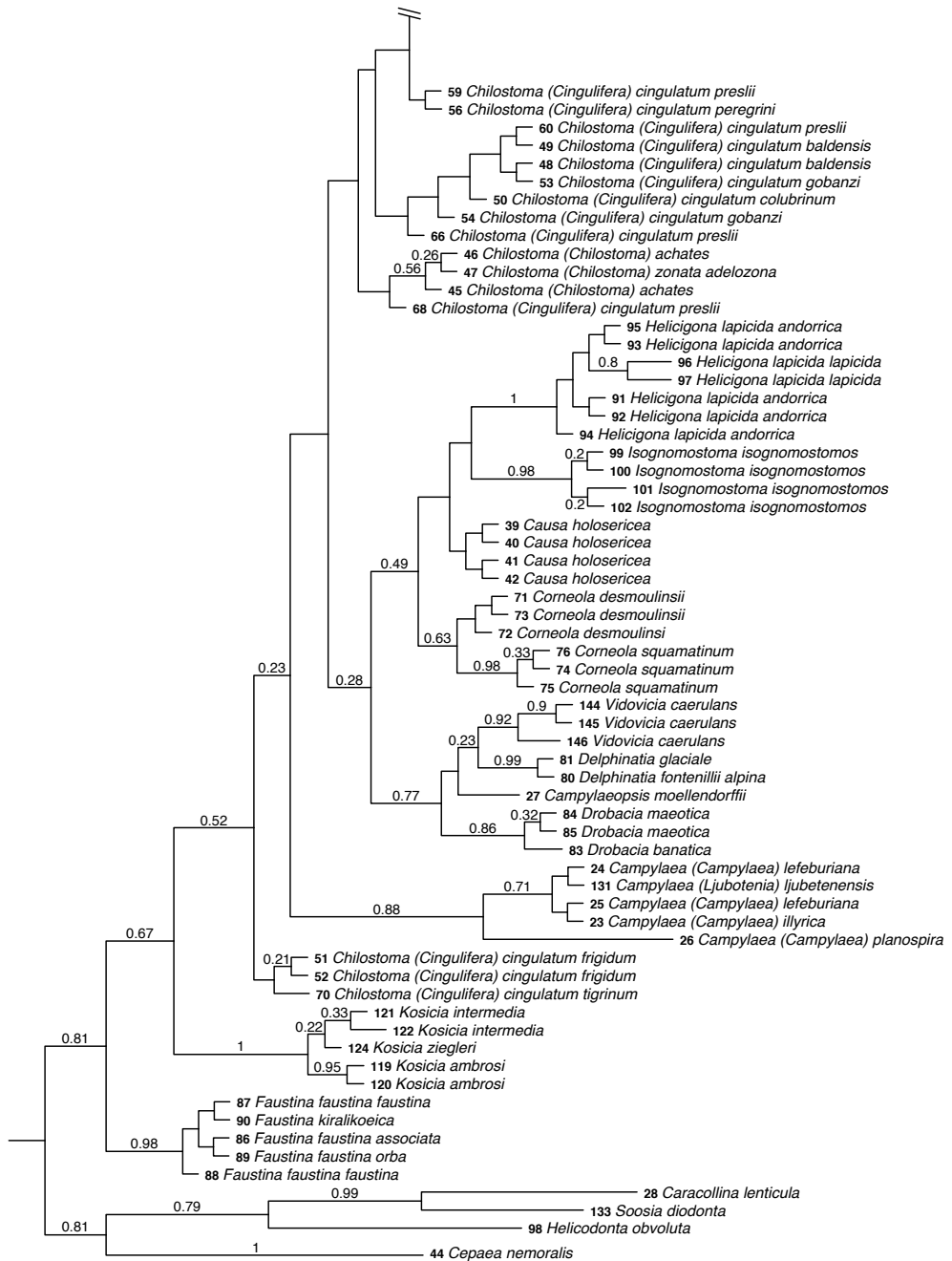
#### Genus *Arianta* Turton, 1831

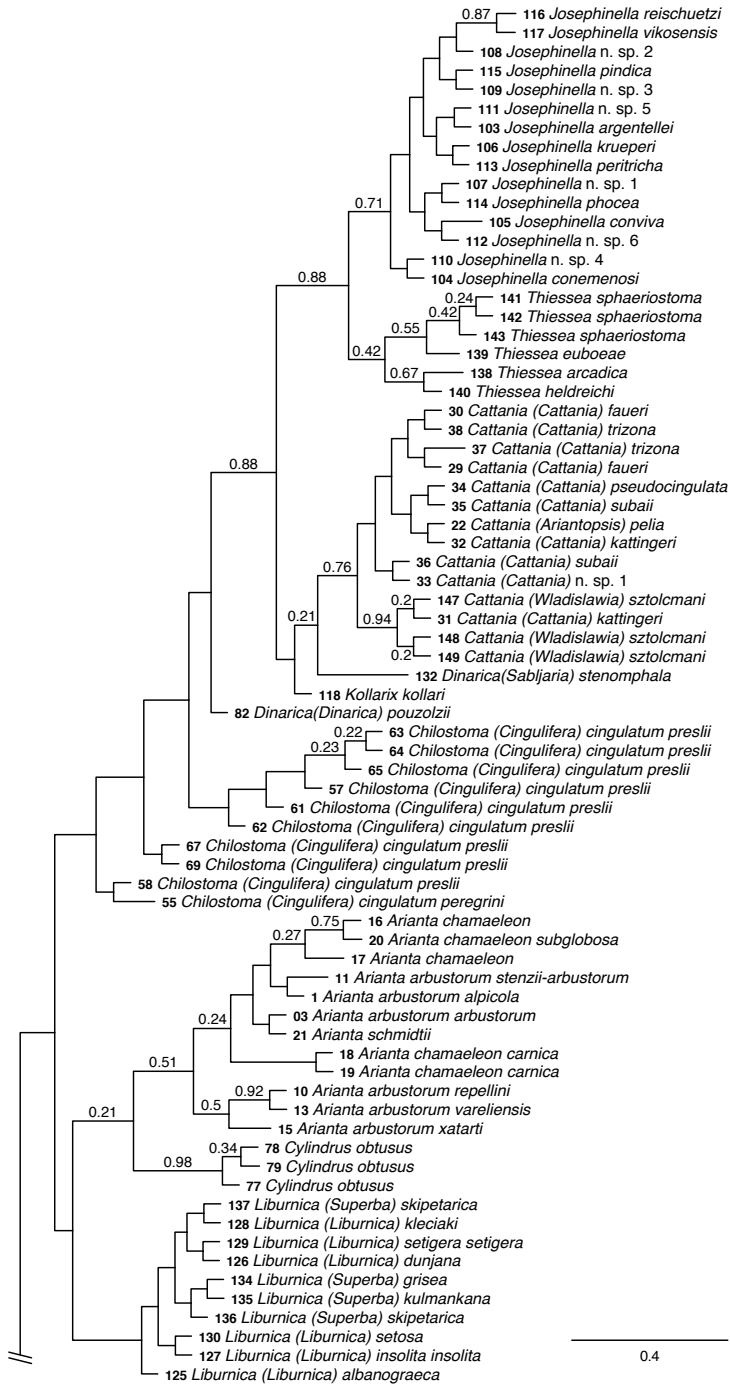
Type species: *Helix arbustorum* Linné, 1857. *Arianta arbustorum* (Fig. 1-1), has the largest distribution of all species within the subfamily Ariantinae. It is aberrant both in terms of shell morphology and in its ecological requirements, occurring independently of limestone from the lowland to high in the mountains. While nearly all Ariantinae have a depressed shell and an open umbilicus, *A. arbustorum* has a globular shell with a closed umbilicus (Gittenberger *et al.*, 2004). Other species of *Arianta*, such as *A. chamaeleon* (Pfeiffer, 1842; Fig. 1-2) and *A. schmidtii* (Rossmässler, 1836) retained the plesiomorphic, depressed shell phenotype. The mucous glands in the genital system of *Arianta* are always undivided (Fig. 3-1). Our data strongly support the monophyly of the genus *Arianta* (posterior probability > 0.98; Figs. 4, 6-8). Unexpectedly, the phylogenies for the concatenated dataset (Fig. 8-A,B) shows high support (post. prob. 1.0/1.0) for a sister-group relationship between *Arianta* and *Cylindrus* (Fig. 1-3); the split between both groups is estimated at 17.5 MYA. The position of the *Arianta*-*Cylindrus* clade relative to the other Ariantinae still is somewhat obscure. Figure 8 indicates (albeit with low support 0.76/0.64) that *Arianta*, *Cylindrus*, *Chilostoma* (including *Cingulifera*), *Corneola* and *Helicigona* form a derived clade in the phylogeny of Ariantinae. The age of the latter group is estimated at 21.4 MYA. Just like *Arianta*, the four lastmentioned genera all possess simple undivided mucous glands.

#### Distribution:

North and central Europe. From Iceland, Norway, Sweden, Nord-Ireland, Great Britain, and central France eastwards to the Baltic countries, Poland, Ukraine and Rumania (Carpathian mountain arc). Occurring in both lowland and montane habitats. The southern border of its distribution ranges from the North-Italian alps through Slovenia, Croatia, Serbia into Bulgaria (up to Stara Planina); it does not occur in the Iberian peninsula.







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**Figure 4.** Bayesian phylogeny reconstruction based on the *H3* dataset.

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Subgenus *Ariantopsis* Wagner, 1928 (monotypic), genus *Cattania*

Type species: *Helicigona (Arianta) pelia* Hesse, 1912. *Cattania (Ariantopsis)* is endemic to SW and W Bulgaria. *Cattania (Ariantopsis) pelia* (Fig. 1-30) superficially resembles *Arianta aethiops* (Bielz, 1851), which is not included in this study but shown to belong to *Arianta* by Gittenberger *et al.*, 2004. The latter species is only known from a few high montane areas in the southern Carpatian mountains in Rumania. The uncertainty regarding the taxonomic position of *C. (A.) pelia* is reflected by the different genera to which it had been ascribed, as e.g. *Arianta pelia* (Kroupa, 1994; Dedov, 1998), *Helicigona pelia* Hesse, 1912, *Chilostoma pelia* (Pall-Gergely, 2011) and *Faustina pelia* (Damjanov & Likharev, 1975). The phylogenies based on COI and 16S (Figs. 5 and 7) both show high support (post. prob. 0.92 and 1, respectively) for a sister-group relation between *C. (Ariantopsis)* and *C. (Wladislavia)*. The *H3* phylogeny is less explicit, probably because *C. (Wladislavia)* shows an ambiguity at position 194, where *C. (Ariantopsis)* does not. Besides that, the *H3* sequences for *C. (Wladislavia)* and *C. (Ariantopsis)* are identical. The age of the split between *C. (Ariantopsis)* and *C. (Wladislavia)* is estimated at ca. 3.1 MYA (1.5-5.1; Fig. 8). Moreover, the obtained phylogenies (no data for *CytB*) show that *C. (Ariantopsis)*, *C. (Wladislavia)* and *C. (Cattania)* represent a well supported clade (post. prob. 0.76, 1.0, 1.0 and 1.0/1.0; Figs. 4, 5, 7 and 8, respectively). As in *C. (Wladislavia)* (Fig. 3-30) and *C. (Cattania)* (Fig. 3-27 and 3-28), both divided and undivided mucous glands have been observed in *Ariantopsis* (Fig. 3-29). Since both *C. (Ariantopsis)* and *C. (Wladislavia)* are nested within *Cattania*, each of these groups (including *Cattania* s. str.) can be considered a subgenus of *Cattania*.

Distribution:

Endemic to SW and W Bulgaria. The eastern boundary of its distribution is situated near Plovdiv, the northern boundary is near Lakatnik in the Iskar-valley and its western distribution is delineated by Mount Vitosha and the Rila mountains.

Subgenus *Campylaea* Beck, 1837, genus *Campylaea*

Type species: *Helix hispana* Linnaeus, 1758. The data obtained for *Campylaea* (*Campylaea*) are limited; only for *C. (C.) planospira* Lamarck, 1801 (Fig. 1-13) sufficient sequences were obtained to include it in the combined analysis. For three other species *H3* sequences were obtained, whereas only for *C. (C.) illyrica* (Fig. 1-14) and *C. (C.) lefeburi* a COI sequence was obtained as well. The genus is shown to be monophyletic with reasonable support (post. prob. 0.88 and 0.85; Figs. 4 and 5, respectively), although *C. (C.) planospira* and the three other species together clearly represent separate lineages. The genital anatomy of *C. (Campylaea)* is rather indeterminate; within this genus the mucous glands can be undivided (Fig. 3-14), divided up to half of their length (Fig. 3-25) and even both can occur within a single



specimen. The phylogenies based on the concatenated dataset (Fig. 8A-B) suggests a sister-group relation between *Isognomostoma* and *C. (Campylaea)*, but without proper support (post. prob. 0.53/0.52). Unpublished morphological and anatomical data suggest that *C. (Ljubotenina) ljubetenensis* might represent an undescribed subgenus of *Campylaea* (Subai, in prep.), but our data are insufficient to give insight in the position of *Campylaea* relative to the other members of the subfamily Ariantinae.

**Distribution:**

Mainland Italy, including the island of Sicily, southern Germany (introduced), southern Austria (Kärnten, Steiermark, E side of Ost-Tirol), Slovenia up to W and N Croatia. *Campylaea illyrica* is the only species within this genus that also occurs along the SW border of Hungary, in northern Serbia and along the Danube in SW Rumania. One species, *C. schlaerotricha* (Bourguignat, 1870), lives in E Algeria.

**Genus *Campylaeopsis* Sturany & Wagner, 1914 (monotypic)**

Type species: *Helicigona moellendorffii* Kobelt, 1871. *Campylaeopsis moellendorffii* (Fig. 1-18) has a conspicuous shell with regularly arranged, widely spaced hairs. Here as well, the uncertainty of its taxonomic position is reflected by the different genera to which it had been ascribed, e.g. as *Helicigona moellendorffii* by Knipper (1939) and *Chilostoma (Josephinella) moellendorffii* by Bank (2012). Unfortunately the specimen of *C. moellendorffii* that we had at our disposal was relatively old (collected in 1980, Table 2) and we were merely able to amplify *Histone H3*. The position of that sequence in the *H3* phylogeny (Fig. 4) suggests that *C. moellendorffii* forms a clade together with the genera *Delphinatia*, *Drobacia* and *Vidovicia*; it occurs sympatrically with only the latter genus. *Campylaeopsis moellendorffii* always has undivided mucous glands (Fig. 3-10), which is the common character state in *Drobacia* (Fig. 3-9), *Vidovicia* (Fig. 3-11) and *Delphinatia* (Fig. 3-12) as well.

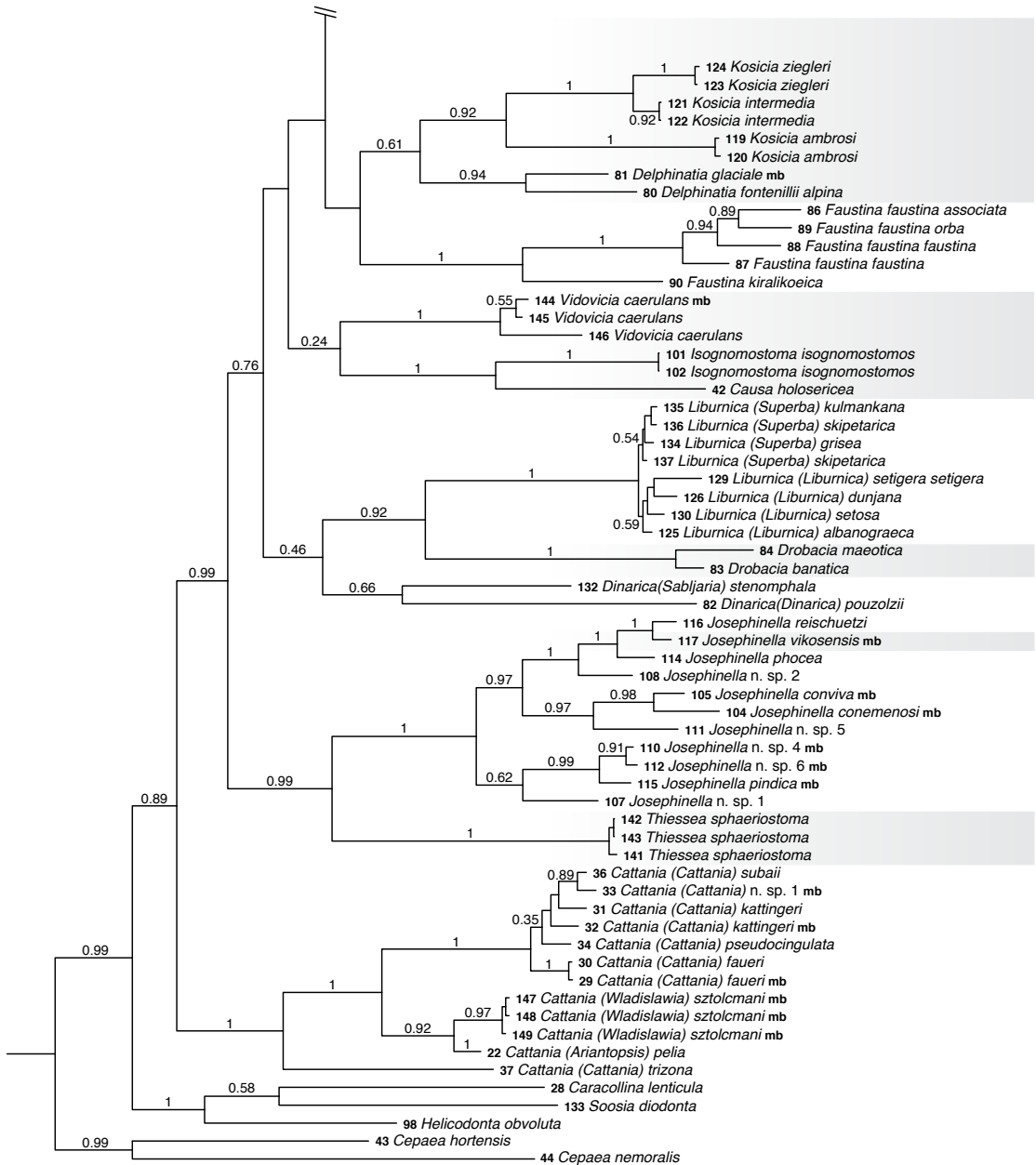
**Distribution:**

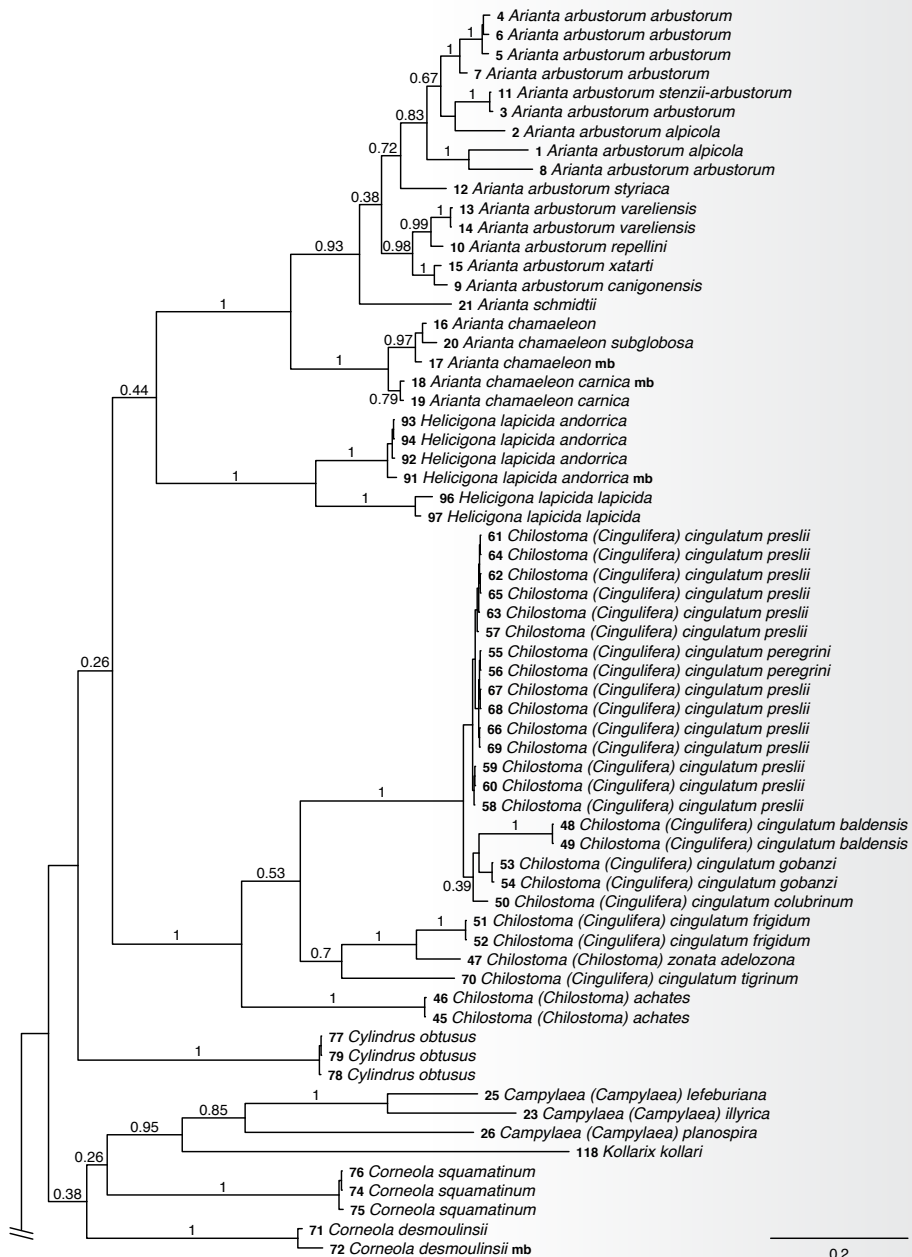
The mountains of Bosnia-Herzegovina and Montenegro.

**Subgenus *Cattania* Brusina, 1904, genus *Cattania***

Type species: *Helix trizona* Rossmässler, 1835. Our confined datasets (a restricted number of *COI* minibarcodes and no sequences for *CytB*) indicate that *Cattania* represents a well supported, monophyletic group (post. prob. of 0.76, 1.0, 1.0, 1.0/1.0; Figs. 4, 5, 7 and 8, respectively). The phylogeny based on *COI* (Fig. 5) shows that the sequence for *C. (C.) trizona* clearly deviates from those of the other *Cattania* species, representing a basal branch within this genus, but this is not confirmed by the more conservative phylogeny reconstruction based on *H3* (Fig. 4). The difference in *COI* might not as much reflect a special position of *C. (C.) trizona* within *Cattania*, as it underlines and confirms the positions of *C. (Ariantopsis)* and *C. (Wladislawia)*; Figs. 4, 5, 7 and 8. The phylogenies for *COI*, *16S* and the concatenated dataset (Figs. 5, 7 and 8) clearly depict *Cattania* as a basal lineage within the subfamily Ariantinae. The fact that this is not the case with the *H3* phylogeny may be attributable to the limited number of informative characters (Table 4). The age of *Cattania* (including the subgenera *Ariantopsis* and *Wladislawia*) is estimated at 9.1 MYA (Fig. 8), but might







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**Figure 5.** Bayesian phylogeny reconstruction based on the *COI* dataset.

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be higher. Insufficient sequence data were obtained to include the possibly even more basal lineage represented by *C. (C.) trizona*. The genital anatomy is diverse; the mucous glands are divided from about a third up to over half of their length (Fig. 3-27 and 3-28), but can also be undivided or trifurcate. Frequently, simple and divided mucous glands occur within a single specimen, which is with some regularity observed in other genera as well: e.g. *Liburnica skipetarica* (Fig. 3-17) and *Josephinella reischuetzi* (Fig. 3-18). Within *C. (Ariantopsis)* and *C. (Wladislawia)* the mucous glands are either simple or divided.

**Distribution:**

NE on the Balkan peninsula, dominantly represented in Rumania on the southern slopes of the Carpathians and in Dobrudscha, in E and S Serbia, everywhere in Bulgaria and Macedonia, in the E half of Montenegro, in N and E Albania and along the northern boundary of Greece.

**Genus *Causa* Schileyko, 1971 (monotypic)**

Type species: *Helix holosericea* Gmelin, 1791. *Causa holosericea* (Fig. 1-5) and *Isognomostoma isognomostomos* (Fig. 1-4) have always been considered closely related. Actually the former species was known as *Isognomostoma holosericea* until Schileyko (1971) introduced a new genus for it. The distinction between both genera, based on anatomical differences in the first place (Schileyko, 1971), is also supported by differences in shell morphology (compare Fig. 1-4 and 1-5) and is now underlined by genetic distances (Table 5) as well. Unfortunately we were not able to obtain *CytB* and *16S* sequence data for *C. holosericea*, so the species could not be depicted in the phylogeny of the concatenated dataset (Fig. 8). Despite the large genetic distance (19.4% based on *COI*; Table 5) between *C. holosericea* and *I. isognomostomos* the *COI* phylogeny shows high support (post prob. 1.0; Fig. 5) for a sister-group relation between both species. This relation could not be shown with *H3*; *Isognomostoma* and *Causa* represent a clade with *Helicigona* in the phylogeny based on *H3*. Although the latter phylogeny does not indicate a sister-group relation between *Isognomostoma* and *Causa*, the genetic distance between these two species is smaller than that between *Isognomostoma* and *Helicigona* (Fig. 4). The shape of the penis and the length of the flagellum of the epiphallus of *Causa* and *Isognomostoma* differ clearly; both species have undivided mucous glands (Figs. 3-7 and 3-8).

**Distribution:**

Alps, Sudetes and W Carpathians (Tatra mountains), isolated in S Germany (Franconian Jura).

**Subgenus *Chilostoma* Fitzinger, 1833, genus *Chilostoma***

Type species: *Glischrus (Helix) foetens* Studer, 1820. *Chilostoma* is a difficult genus to comprehend because a lot of (sub)genera have been placed under it. According to



Zilch (1960), *Campylaeopsis*, *Chilostoma*, *Cingulifera*, *Drobacia*, *Josephinella*, *Kosicia* and *Thiessea* are to be considered subgenera of *Chilostoma* (Table 1). There has been some debate (von Martens, 1870) about the identity of *Helix foetens*; nowadays it is either synonymised with *C. (Chilostoma) zonata* (Turner *et al.*, 1998) or it is considered a subspecies thereof (Bank, 2012). On the one hand, based on mitochondrial data (*COI* and *CytB*; Figs. 5 and 6), *C. (Chilostoma) zonata* (Fig. 1-7) somewhat unexpectedly groups with *C. (Cingulifera) frigidum* and *C. (Cingulifera) tigrinum* (Fig. 2-8). The phylogeny for *H3* on the other hand (Fig. 4) does show *C. (Chilostoma) zonata* close to *C. (Chilostoma) achates* (Fig. 1-6), as might be expected on the basis of the anatomy, but depicts the genus *Chilostoma* (if *C. (Cingulifera)* is to be included) as paraphyletic. Support for the monophyly of *Chilostoma*, including *C. (Cingulifera)*, is shown in the phylogenies for *COI*, *CytB* and the concatenated dataset (post. prob. of 1.0, 0.99 and 1.0/1.0; Fig. 5, 6 and 8, respectively). In the phylogeny based on 16S (Fig. 7) the monophyly is distorted by the placement of the group *Corneola*, *Vidovicia* & *Helicigona*. Despite the lack of clarity in Figs. 4 and 7, all phylogenies indicate the presence of three groups: a) *C. (Chilostoma) achates*, b) *C. (Chilostoma) frigidum/tigrinum* and c) *C. (Cingulifera)*. Because we did not obtain a 16S sequence for *C. (Chilostoma) zonata*, whereas the nuclear and the mitochondrial phylogenies contradict each other, our data are insufficient to confidently assign *C. (Chilostoma) zonata* to either group a or group b. In confirmation with the current classification, our data do show that *C. (Chilostoma) zonata* is not a member of group c *C. (Cingulifera)*.

Two representatives of *C. (Chilostoma) achates* are included, viz. (a) a specimen collected in the northern limestone Alps (Berchtesgaden, Bayern, Germany), and (b) a specimen from the southern limestone Alps (Greifenburg, Kärnten, Austria). The sequence divergence for *COI* and *CytB* between these snails is less than 0.3% each. Compared to a sequence divergence of 3.2-7.3% and 6.6-6.9% among subspecies of *C. (Cingulifera)* (group c) and even 12.4 and 16.3% between *C. (Cingulifera) cingulatum frigidum* and *C. (Cingulifera) c. tigrinum* (group b; see also sub *Cingulifera*), for the same markers (Table 5), there is no molecular support for a classification of the populations of *C. (Chilostoma) achates* from Berchtesgaden and Greifenburg with different subspecies (as was suggested by Falkner, 1998). *Chilostoma* is a sister-group (moderately supported: 0.67/0.8, Fig. 8) to the clade consisting of *Helicigona* and *Corneola*. The age of the genus is estimated at 11.6 MYA (Fig. 8). Within *Chilostoma* the mucous glands are always undivided (Fig. 3-3 and 3-4).

Distribution:

The Alps (SE France, SE Switzerland, N Italy), SE Germany and Austria S of the Donau-river.

### Subgenus *Cingulifera* Held, 1837, genus *Chilostoma*

Type species: *Helix cingulata* Studer, 1820. *Chilostoma (Cingulifera)* is a generally accepted subgenus of *Chilostoma* (Zilch, 1960; Bank, 2012). Taxonomically it contains just one species (*C. (Cingulifera) cingulatum* Studer, 1820; Fig. 1-9) but the subgenus is marked by a large number of alleged subspecies (> 30). Some of these have very noticeable shell characters, such as a striking staining pattern in *C. (Cingulifera)*



*cingulatum colubrinum* (De Cristofori & Jan, 1832), radial stripes in *C. (Cingulifera) c. baldensis* (Rossmässler, 1839) or prominent ribs in *C. (Cingulifera) c. gobanzi* (Frauenfeld, 1867) (Fig. 1-10). Our dataset contains seven of these supposed subspecies, viz. *C. (Cingulifera) c. baldensis*, *C. (Cingulifera) c. colubrinum*, *C. (Cingulifera) c. frigidum* (De Cristofori & Jan, 1832), *C. (Cingulifera) c. gobanzi*, *C. (Cingulifera) c. peregrini* Falkner, 1998, *C. (Cingulifera) c. preslii* (Rossmässler, 1836) and *C. (Cingulifera) c. tigrinum* (De Cristofori & Jan, 1832; Fig. 1-8). Even though the majority of the *Cingulifera* subspecies (including the nominate subspecies) were not included in our analysis, our preliminary data show that *C. (Cingulifera) c. frigidum* and *C. (Cingulifera) c. tigrinum* (group b, previous paragraph) do not belong to the lineage that contains the other here included subspecies of *C. (Cingulifera) cingulatum* (group c, previous paragraph). The minimum sequence divergence between groups b and c (based on the *p*-distance between *C. (Cingulifera) c. tigrinum* and *C. (Cingulifera) c. preslii* -59) is 14.5% and 19.1%, based on COI and CytB, respectively (Table 5). The relatively large genetic differentiation between groups b and c corresponds with the allopatric distribution of the species. *C. (Cingulifera) c. frigidum* and *C. (Cingulifera) c. tigrinum* (as well as *C. (Chilostoma) zonata*) occur east of lake Iseo (also known as lake Sebino), whereas the subspecies from group c are distributed to the east and northeast of that area. It is not unlikely that the glaciers that formed lake Iseo and lake Garda during the last glacial maximum formed a barrier between eastern and western refuges, resulting in allopatric differentiation. The age of the split between group b and c is estimated at 9.6 MYA (Fig. 8). The maximum observed sequence divergence within group c (between *C. (Cingulifera) c. baldensis* -48 and *C. (Cingulifera) c. preslii* -59) is 7.3% and 6.9% for COI and CytB, respectively (Table 5). Even though group b represents only two of the seven so-called subspecies of *C. (Cingulifera)*, the sequence divergence within that group (based on the *p*-distance between *C. (Cingulifera) c. tigrinum* and *C. (Cingulifera) c. frigidum*) is evidently larger: 12.4% and 16.3% for COI and CytB, respectively (Table 5). This could indicate a stronger isolation (or more rapid radiation) of subspecies in group b compared to group c.

The last *C. (Cingulifera)* subspecies to address here is *C. (Cingulifera) c. peregrini* Falkner, 1998. Falkner introduced this replacement name for *C. (Cingulifera) c. cingulina* (Strobel, 1844), not *Helix cingulina* Deshayes, 1839. Contrary to the prevailing view, he suggested at the same time that the northern alpine populations of *C. (Cingulifera) cingulata* might belong to two instead of only a single subspecies, viz. *C. (Cingulifera) c. peregrini* from near Innsbruck (Austria) and *C. (Cingulifera) c. preslii* from Berchtesgaden (Germany) and surroundings. However, the COI and CytB sequences for specimens from both northern alpine localities (55,56 vs. 66-69, Table 2) are identical and differ by less than 0.5 % (Table 5) from sequences of southern alpine, undisputed *C. (Cingulatum) c. preslii* (61-65, Table 2). The molecular data suggest that *C. (Cingulatum) c. peregrini* is a junior synonym of *C. (Cingulatum) c. preslii*, what implies that *C. (Cingulatum) c. preslii* has a disjunct range, occurring in both the northern and the southern limestone Alps. As mentioned in the previous paragraph, all representatives of *Chilostoma*, thus including *C. (Cingulifera)* (Fig. 3-4) have undivided mucous glands.





Figure 6. Bayesian phylogeny reconstruction based on the *CytB* dataset.



#### Distribution:

The Alps of SE France, S Switzerland, S Germany (partly introduced), S and W Austria and Italy, north of the line Napoli-Termoli.

#### Genus *Corneola* Held, 1838

Type species: *Helix cornea* Draparnaud, 1801. In recent classifications (Bank, 2001) *Corneola* is regarded as a subgenus of *Chilostoma*. *Corneola* contains four species: *Corneola acrotricha* Fisher, 1877, *C. crombezi* Bourguignat, 1880, *C. desmoulinsii* Farines, 1834 (Fig. 1-24) and *C. squamatinum* Rossmässler, 1835. Only *C. squamatinum* and *C. desmoulinsii* are included in this study. The material that we had at our disposal was collected in the period 1977-1999 and had an average age of 25 years. This caused severe problems for the amplification of some of the markers. Albeit with very little support (post. prob. 0.63), the *H3* phylogeny (Fig. 4) shows the combined *C. desmoulinsii* and *C. squamatinum* as a monophyletic group. In the *COI* phylogeny (Fig. 5) *Corneola* is paraphyletic, but again *C. desmoulinsii* and *C. squamatinum* belong to the same clade (albeit not exclusively). The fact that both species are not sister-groups in the *COI* analysis, might be due to the large genetic distance (16.9%; Table 5) between them. The phylogeny of the concatenated dataset nevertheless supports the monophyly of *Corneola*, as well as a sister-group relation with *Helicigona*, both with a post. prob. of 0.82 (Fig. 8). Together with *Chilostoma*, the latter genera form a clade which is a sister-group of the clade that holds *Arianta* and *Cylindrus* (see below). the age of the genus is estimated at 14.4 MYA (Fig. 8). Just like *Helicigona* and *Chilostoma*, *Corneola* has undivided mucous glands (Fig. 3-5).

#### Distribution:

*Corneola acrotricha*, *C. desmoulinsii* and *C. squamatinum* occur in the Pyrenees. *Corneola squamatinum* has a larger distribution area, extending further into southern and central France, along the Atlantic coast up to Brittany. *Corneola acrotricha* is known from the central Pyrenees, and *C. desmoulinsii* from the French and Spanish eastern Pyrenees. *Corneola crombezi* is restricted to the Alps in the department of Alpes-Maritimes (Falkner et al., 2002).

#### Genus *Cylindrus* Fitzinger, 1833 (monotypic)

Type species: *Pupa obtusa* Draparnaud, 1805. Among the (sub)genera of Ariantinae that can be distinguished by shell-morphology, *Cylindrus* without any doubt has by far the most remarkable and distinctive shell (Fig. 1-3). It is endemic to the high Alps of Austria and is generally only found at altitudes above 1600 m. The phylogeny of the concatenated dataset (Fig. 8) strongly supports a sister-group relation between *Cylindrus* and *Arianta* (post. prob. 1.0). Except for *COI* (Fig. 5), which puts *Cylindrus* basal to the genera *Arianta*, *Helicigona* and *Chilostoma*, all datasets indicate that *Cylindrus* and *Arianta* shared a unique common ancestor. The minimum sequence divergence based on the uncorrected *p*-distance between *Cylindrus obtusus* (78) and *Arianta chamaeleon* (19) is 17.7% and 24.9% for *COI* and *16S*, respectively (Table 5). The split between *Arianta* and *Cylindrus* is estimated at 17.5 MYA, which is much older than thus far indicated by the late Pleistocene fossils (see Material and Methods).



Just like *Arianta* (Fig. 3-1), *Cylindrus* has undivided mucous glands (Fig. 3-2).

Distribution:

Endemic to the Austrian Alps (between 1600 and 2500 m), known from Oberösterreich, Niederösterreich, Salzburg, Steiermark and Kärnten.

#### Genus *Delphinatia* Hesse, 1931

Type species: *Helix alpina* Michaud, 1831. *Delphinatia* is regarded as a subgenus of *Chilostoma* by Bank *et al.* (2001), but has been considered a subgenus of *Campylaea* as well (Zilch, 1960; Table 1). It contains two species (Falkner *et al.*, 2002), viz. *Delphinatia fontenillii* Michaud, 1829, with the subspecies *D. f. fontenillii* and *D. f. alpina* Férussac, 1822 (Fig. 1-11) and *Delphinatia glaciale* Férussac, 1832, which are both included in this study. The phylogeny of the concatenated dataset (Fig. 8) shows strong support for the monophyly of *Delphinatia* (post. prob. 1.0) and indicates a basal position of *Delphinatia* relative to those Ariantinae that possess undivided mucous glands. The concatenated dataset moderately supports a sister-group relation with *Vidoviccia* and *Drobacia* (post. prob. of 0.78), but except for *H3* the other datasets only show indirect support for such a relation. Due to the age of the available *Delphinatia* specimens (the youngest included specimen was 25 years old; Table 2), the obtained data were rather fragmentary (only a COI mini-barcode and no *CytB* sequence for *D. glaciale*; no 16S sequence for *D. f. alpina*). Based on the COI minibarcodes, the data indicate a sequence divergence between *D. f. alpina* and *D. glaciale* of 15.8%. The sequence divergence between *D. f. alpina* and *Vidoviccia* is 17.3% and 24.7% for COI and *CytB*, respectively (Table 5). There is some disagreement between the MrBayes and BEAST phylogenies regarding the sister-group of *Delphinatia* (Fig. 8). According to the MrBayes phylogeny it is *Drobacia*, whereas according to the BEAST phylogeny the sister-group of *Delphinatia* is *Vidoviccia*. Based on the latter analysis, the lineage consisting of *Delphinatia*, *Drobacia* and *Vidoviccia* arose about 22 MYA and *D. f. alpina* and *D. glaciale* have been separated for approximately 6.9 MY (Fig. 8).

In general the mucous glands of *Delphinatia* (Fig. 3-12) are undivided (which is always the case for *Drobacia*, Fig. 3-9 and *Vidoviccia*, Fig. 3-11), but sporadically one of them can be divided over up to 25-50% of its length.

Distribution:

Known from SE France, in the Alps (Hautes-Alpes, Haute-Savoie, Isère and Savoie), into adjacent Italian Alps (Alpi Cozie and Graie) (Gavetti *et al.*, 2008).

#### Subgenus *Dinarica* Kobelt, 1902, genus *Dinarica*

Type species: *Helix pouzolzii* Deshayes, 1830. *Dinarica* occurs along the eastern Adriatic coast and consists of at least the species *Dinarica (Dinarica) pouzolzii*, *D. (D.) serbica* Kobelt, 1872 and *D. (Sabljaria) stenomphala* Menke, 1830, with several subspecies. In this study only *D. (D.) pouzolzii* (Fig. 1-22) and *D. (S.) stenomphala* (Fig. 1-21) were included. Especially the taxonomic position of the latter species is unclear; it has been referred to as *Helicigona (Dinarica) stenomphala* by Knipper (1939) and as *Chilostoma (Josephinella) stenomphala* by Bank (2011). *Dinarica (S.) stenomphala* differs clearly from *D. (D.) pouzolzii* both in shell shape and genital anatomy



(compare Figs. 3-23 and 3-24), so that Brusina assigned it to another genus (*Sabljaria* Brusina, 1904). The phylogenies based on *COI*, *CytB* and the concatenated dataset support the monophyly of *Dinarica* (post. prob. 0.66, 1.0 and 1.0; Figs. 5, 6 and 8, respectively). The *H3* phylogeny (Fig. 4) also shows *D. (S.) stenomphala* and *D. (D.) pouzolzii* in relative close proximity, but indicates that *D. (S.) stenomphala*, unlike *D. (D.) pouzolzii*, shares a common ancestor with *Cattania*, *Josephinella* and *Thiessea* (post. prob. 0.88). For neither species of *Dinarica* did we obtain 16S sequences. Although the phylogenies for the concatenated dataset obtained with MrBayes and BEAST are very similar, they differ in the position of *Dinarica*. In the former, *Dinarica* is connected to a deeper node (closer to the root) than *Liburnica*, whereas in the latter *Dinarica* is the sister-group of *Liburnica*. For the most part the distribution areas of the species from the latter genera overlap. The *COI* and *CytB* phylogenies (Figs. 5 and 6) also depict a close relation between *Dinarica* and *Liburnica*, though with very little support (post. prob. of 0.46 and 0.48). The average sequence divergence between *Dinarica* and *Liburnica albanograeca* is 18.3% and 23.5% for *COI* and *CytB*, respectively. The sequence divergence within *Dinarica* (based on the two included species) is 16.3% and 23.5% for the same markers, respectively (Table 5). The oldest known fossils of *Dinarica* are from the "Sarmat", i.e. 12.7-11.6 MYA (see Material and Methods), whereas the split between *Dinarica* and *Liburnica* is estimated at 19.8 MYA (15.6-25.1). Both *Dinarica* and *Liburnica* predominantly have divided mucous glands (opposed to the large clade of Ariantinae that is dominated by undivided mucous glands; from *Drobacia* to *Arianta* in Fig. 8).

#### Distribution:

In countries along the N coast of the Adriatic sea, in SE Croatia, Bosnia-Herzegovina, S Serbia, Montenegro, Kosovo, Albania, the western border of Macedonia and NW Greece.

#### Genus *Drobacia* Brusina, 1904

Type species: *Helix banatica* Rossmässler, 1838. *Drobacia* is represented only in SW Rumania. Two species are known, viz. *D. banatica* (Fig. 1-26) and *D. maeotica* Wenz, 1926. Both of these are included in this study, but only for *D. maeotica* did we obtain enough data to include it in the concatenated dataset as well. The phylogeny of the datasets that represented both species (Figs. 4 and 5) shows reasonable support for the monophyly of the genus (post. prob. of 0.86 and 1.0, respectively). The relative position of *Drobacia* within the subfamily Ariantinae is somewhat ambivalent. Both the phylogeny based on *H3* and the concatenated dataset moderately support the monophyly of *Drobacia* with *Delphinatia* and *Vidovicia* (post. prob. 0.77 and 0.78/0.98; Figs. 4 and 8, respectively). Of the other datasets only *COI* (Fig. 5) provides appreciable support (post. prob. 0.92) for a sister-group relation between *Drobacia* and another group (*Liburnica*), but the sequence divergence between these genera is substantial (on average 19.8% for *COI*, between *Drobacia* and *L. (Liburnica) albanograeca*, Table 5). The *COI* sequence divergence within *Drobacia* is 9.9% (Table 5). The *CytB* and 16S phylogenies (Figs. 6 and 7) are less clear about the position of *Drobacia*. Figure 6 shows *Drobacia* and *Vidovicia* in close proximity of each other, whereas Fig. 7 does the



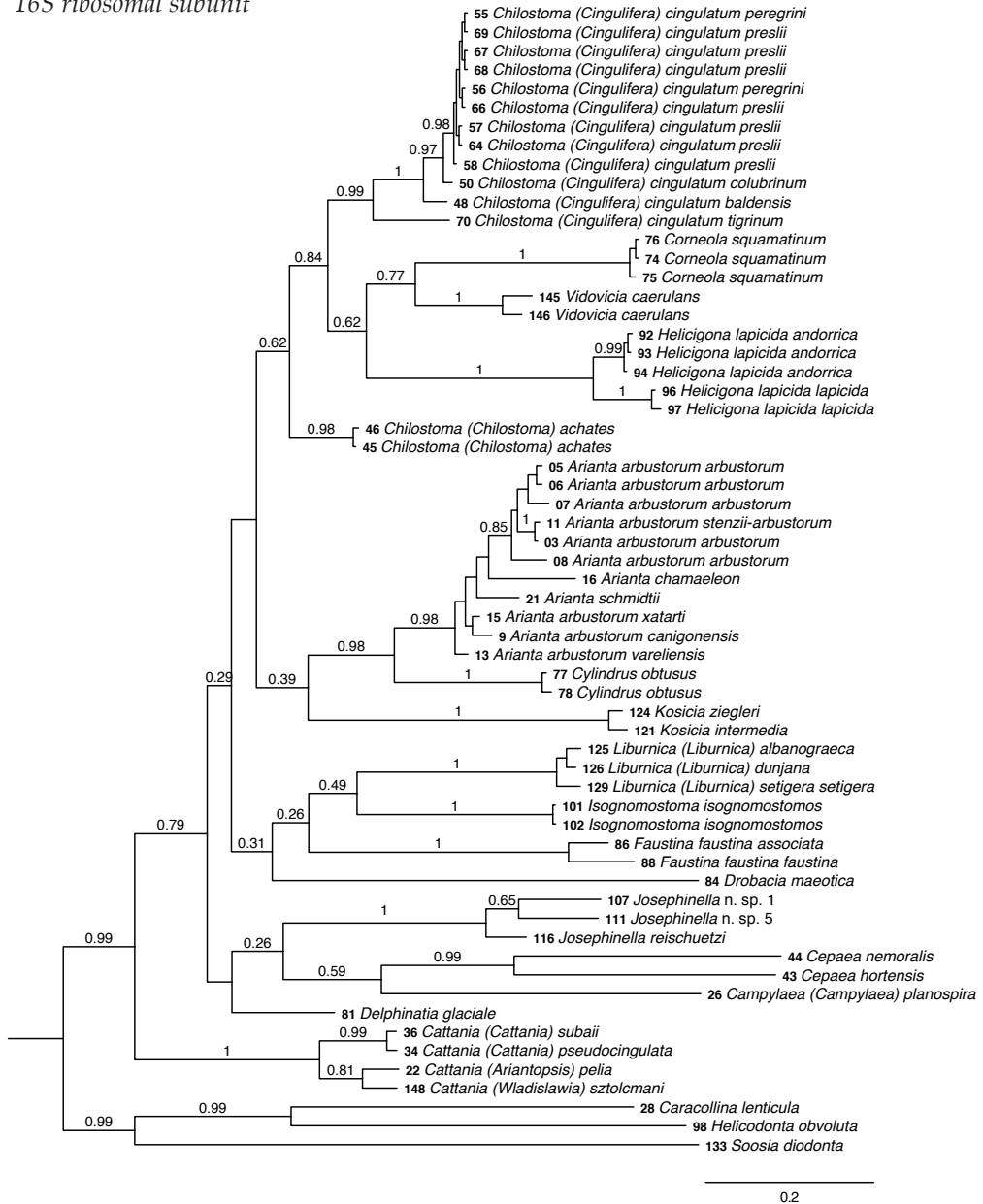


Figure 7. Bayesian phylogeny reconstruction based on the 16S dataset.



same for *Drobacia* and *Delphinatia*. This ambiguity is also reflected by the different analyses of the concatenated dataset; in the MrBayes phylogeny *Drobacia* is the sister-group of *Vidovicia*, whereas in the BEAST phylogeny it is the sister-group of the clade that contains *Delphinatia* and *Vidovicia* (Fig. 8). The most recent common ancestor of the *Drobacia*, *Delphinatia* and *Vidovicia* is estimated to have diverged about 22.0 MYA. Just like *Campylaeopsis* (Fig. 3-10), *Vidovicia* (Fig. 3-11) and (in most cases) *Delphinatia* (Fig. 3-12), *Drobacia* has undivided mucous glands (Fig. 3-9).

#### Distribution:

W and SW Rumania and locally in E Hungary. In the Pleistocene *Drobacia* reached as far as the Harz mountains in Thüringen, Germany (Jaeckel, 1962).

#### Genus *Faustina* Kobelt, 1904

Type species: *Helix faustina* Rossmässler, 1835. *Faustina* is mainly distributed in the Carpathians, but can also be found in NE Hungary. Of the five known species, three inhabit the North Carpathian mountains [*Faustina cingulella* Rossmässler, 1837, *Faustina rossmaessleri* (Pfeifer, 1842) and *Faustina faustina* (Rossmässler, 1835; Fig. 1-12)], while the other species [*Faustina barcensis* (von Kimakowicz, 1890) and *Faustina kiralikoeica* (von Kimakowicz, 1890)] only occur in the South Carpathians. There are about a dozen of alleged subspecies.

This study includes the nominate subspecies of *F. faustina*, two additional subspecies [*F. faustina associata* (Rossmässler, 1835) and *F. faustina orba* (von Kimakowicz, 1890)], and *F. kiralikoeica*. The monophyly of the genus could only be assessed based on the phylogenies for *H3* and *COI*, because these were the only markers for which sequences of both *F. faustina* and *F. kiralikoeica* were obtained; Figs. 4 and 5 show strong support for the monophyly of *Faustina* (post. prob. of 0.98 and 1.0, respectively). The phylogeny for the concatenated dataset (Fig. 8) shows moderate support for a sister-group relationship between *Faustina* and *Kosicia* (post. prob. 0.71/0.78). The maximum sequence divergence for *COI* observed within *Faustina* (between *F. faustina* and *F. kiralikoeica*) is 17.4% (Table 5). The sequence divergence even among subspecies of *F. faustina* can be quite high (10.7% and 13.0% between *F. f. faustina* and *F. f. associata*) for *COI* and *CytB*, respectively (Table 5). The divergence between *F. faustina orba* and *F. kiralikoeica* is larger than that between *F. faustina orba* and *Kosicia intermedia* (Table 5). For *Faustina* the fossil record indicates late Pliocene, 2.6-1.8 MYA, as the minimum age (see Material and Methods). The split between *Faustina* and *Kosicia* is estimated at 19.0 MYA (15.0-23.5). In *Faustina* both types of mucous glands are observed. *Faustina cingulata* and *F. rossmaessleri* have undivided glands, whereas they are divided up to half of their length in *F. faustina* (Fig. 3-19), *F. barcensis* and *F. kiralikoeica*. Within the large clade of Ariantinae that is dominated by the possession of undivided mucous glands (from *Drobacia* to *Arianta* in Fig. 8-B), *Faustina* is one of the few genera that also has divided mucous glands. Counterintuitively, the possession of divided mucous glands might actually represent a plesiomorphic condition within the subfamily Ariantinae. Future research will have to show whether *F. cingulata* and *F. rossmaessleri* (both have undivided mucous glands) represent derived lineages within *Faustina* indeed.



#### Distribution:

The Carpathian mountains in Czech, Slovakia, Poland, Ukraine and Rumania; also on some mountains in NE Hungary.

#### Genus *Helicigona* Férussac, 1821

Type species: *Helix lapicida* (Linnaeus, 1758). *Helicigona* is considered now a monotypic genus with only two subspecies, viz. *Helicigona l. lapicida* (Fig. 1-25) and *H. l. andorrica* Bourguignat, 1876. It has served as an umbrella (partly with *Chilostoma*) to enclose a number of (sub)genera (i.e. Hesse, 1931; Knipper, 1939; Subai, 1984). In a paper on the Ariantinae from Albania and Greek, *Dinarica* and *Cattania* were considered subgenera of *Helicigona* (Subai, 1995). Later on, *Thiessea* and *Liburnica* were classified as subgenera of *Helicigona* as well (Subai, 1996 and 2002). None of the obtained phylogenies (Fig. 4-8), however, provides evidence that any of the just mentioned genera is closely related to *Helicigona*, or should be considered a subgenus of it. Except for the phylogeny based on *CytB* (Fig. 6), in which the position of *Helicigona* is clearly incorrect, these genera all branch off basally of *Helicigona*. In fact, the phylogeny of the concatenated dataset (Fig. 8) only shows a sister-group relation between *Helicigona* and *Corneola* (post. prob. 0.82/0.68), but there is no justification to consider the latter a subgenus of *Helicigona*. The nominate subspecies, *H. lapicida lapicida*, has a wide distribution in N and W Europe, whereas *H. lapicida andorrica* is restricted to the Pyrenees of N Spain. The monophyly of *Helicigona* is beyond dispute (post. prob. 1.0 for all of the obtained phylogenies, Figs. 4-8). The observed sequence divergences (*COI* and *CytB*) within each subspecies of *Helicigona* are less than 2.5%, but between both subspecies (measured between *H. lapicida lapicida* -97 and *H. lapicida andorrica* -92) it reaches up to 10.7% and 14.7% for *COI* and *CytB*, respectively. The sequence divergence between *Helicigona* and *Corneola* is clearly larger (16.8% and 28.0% between *H. lapicida lapicida* -97 and *C. desmoulinii* -71 for the same markers, respectively; Table 5). The fossil record indicates a minimum age for *Helicigona* as "Pont", 5.3-3.6 MYA (see Material and Methods), whereas the split between *Helicigona* and *Corneola* is estimated at 17.0 MYA (13.3-20.7). Just like *Corneola*, but also *Arianta*, *Cylindrus*, *C. (Chilostoma)* and *C. (Cingulifera)* does the genital system of *Helicigona* contain undivided mucous glands (Fig. 8).

#### Distribution:

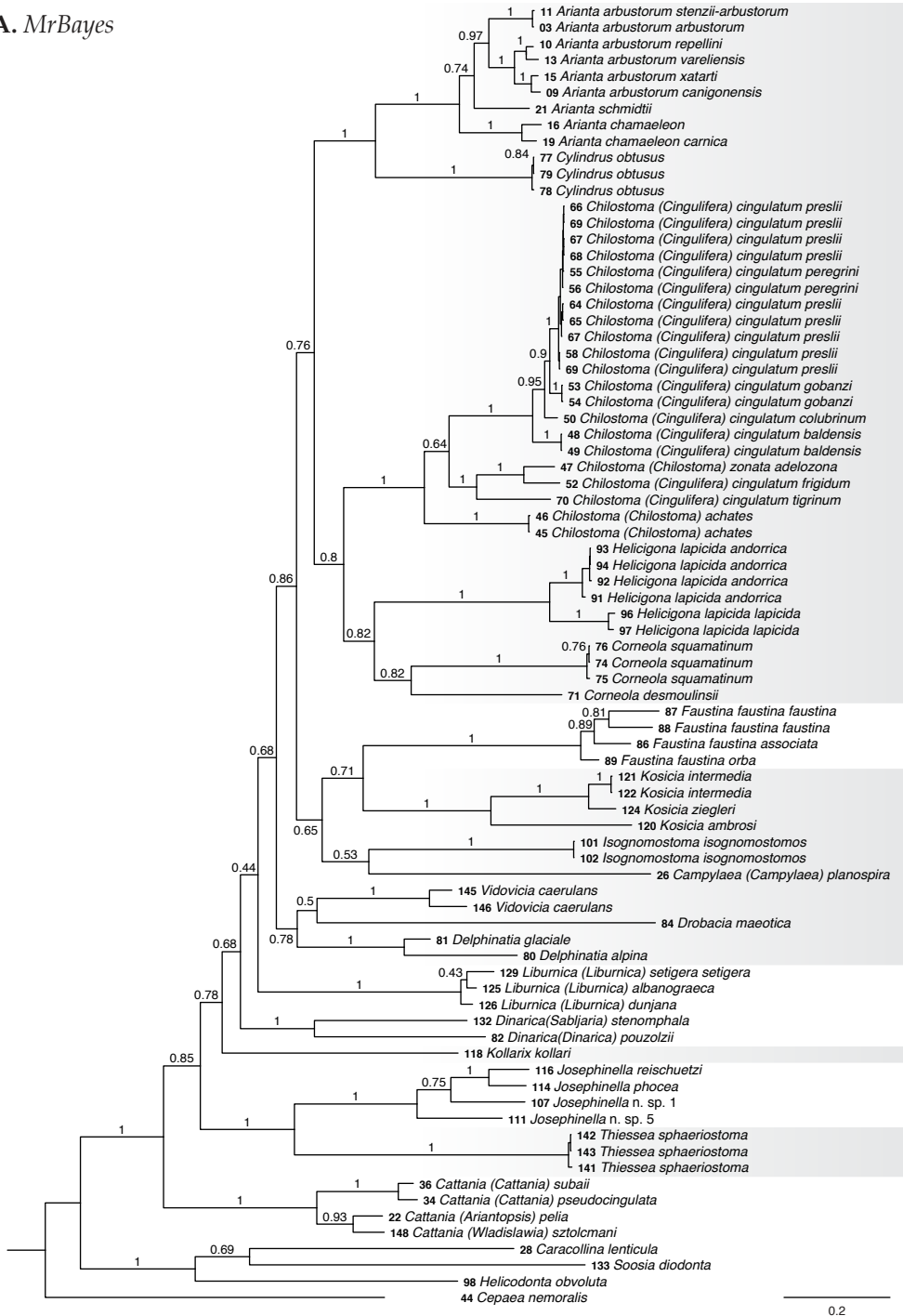
W and N Europe, from S Scandinavia and about the middle of England to the south up to S France. *Helicigona* occurs in the Pyrenees in N Spain (*H. (lapicida) andorrica*), Switzerland, Austria, Czech and W Poland.

#### Genus *Isognomostoma* Fitzinger, 1833 (monotypic)

Type species: *Helix personata* Lamarck, 1792 [= *Isognomostoma isognomostomos* (Schroter, 1784)] (Fig. 1-4). *Isognomostoma isognomostomos* and *Causa holosericea* (Fig. 1-5) have long been regarded as congeneric, but merit generic status according to shell morphology, genital anatomy and genetics (see the paragraph on *Causa*). Despite the lack of *CytB* and 16S sequences for *Causa*, the data for *H3* and *COI* show that *Isognomostoma* and *Causa* belong to the same monophyletic group (Figs.



# A. MrBayes

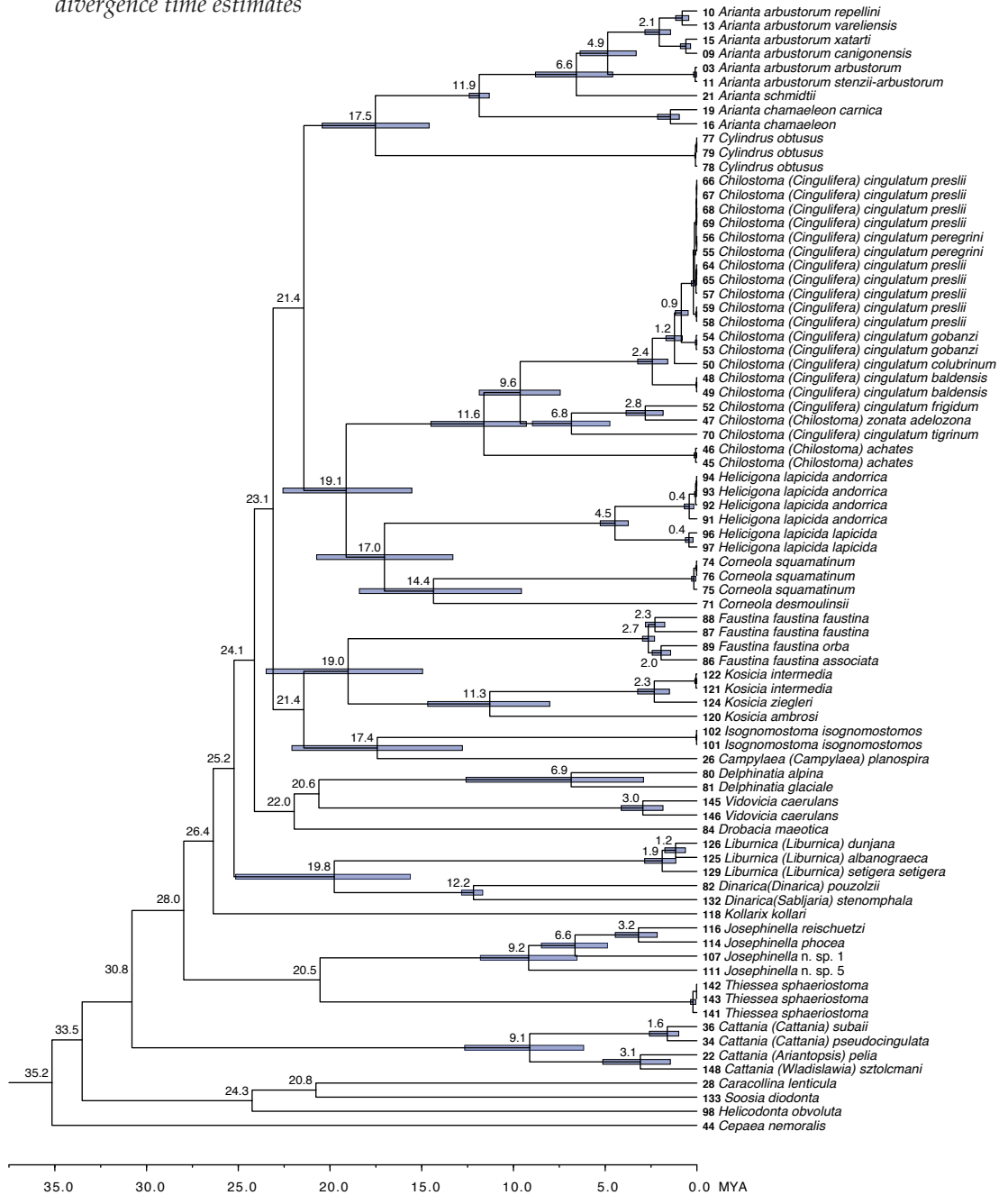


**B. BEAST**

10 *Arianta arbustorum repellii*  
13 *Arianta arbustorum vareliensis*  
15 *Arianta arbustorum xatarti*  
09 *Arianta arbustorum canigoniensis*  
03 *Arianta arbustorum arbustorum*  
11 *Arianta arbustorum stenzii-arbustorum*  
21 *Arianta schmidtii*  
19 *Arianta chamaeleon camica*  
16 *Arianta chamaeleon*  
77 *Cylindrus obtusus*  
79 *Cylindrus obtusus*  
78 *Cylindrus obtusus*  
66 *Chilostoma (Cingulifera) cingulatum preslii*  
67 *Chilostoma (Cingulifera) cingulatum preslii*  
68 *Chilostoma (Cingulifera) cingulatum preslii*  
69 *Chilostoma (Cingulifera) cingulatum preslii*  
56 *Chilostoma (Cingulifera) cingulatum peregrini*  
55 *Chilostoma (Cingulifera) cingulatum peregrini*  
64 *Chilostoma (Cingulifera) cingulatum preslii*  
65 *Chilostoma (Cingulifera) cingulatum preslii*  
57 *Chilostoma (Cingulifera) cingulatum preslii*  
59 *Chilostoma (Cingulifera) cingulatum preslii*  
58 *Chilostoma (Cingulifera) cingulatum preslii*  
54 *Chilostoma (Cingulifera) cingulatum gobanzi*  
53 *Chilostoma (Cingulifera) cingulatum gobanzi*  
50 *Chilostoma (Cingulifera) cingulatum colubrinum*  
48 *Chilostoma (Cingulifera) cingulatum baldensis*  
49 *Chilostoma (Cingulifera) cingulatum baldensis*  
52 *Chilostoma (Cingulifera) cingulatum frigidum*  
47 *Chilostoma (Chilostoma) zonata adelozona*  
70 *Chilostoma (Cingulifera) cingulatum tigrinum*  
46 *Chilostoma (Chilostoma) achates*  
45 *Chilostoma (Chilostoma) achates*  
94 *Helicigona lapidica andorrica*  
93 *Helicigona lapidica andorrica*  
92 *Helicigona lapidica andorrica*  
91 *Helicigona lapidica andorrica*  
96 *Helicigona lapidica lapidica*  
97 *Helicigona lapidica lapidica*  
74 *Corneola squamatinum*  
76 *Corneola squamatinum*  
75 *Corneola squamatinum*  
71 *Corneola desmoulinii*  
88 *Faustina faustina faustina*  
87 *Faustina faustina faustina*  
89 *Faustina faustina orba*  
86 *Faustina faustina associata*  
122 *Kosicia intermedia*  
121 *Kosicia intermedia*  
124 *Kosicia zieglerei*  
120 *Kosicia ambrosi*  
102 *Isognomostoma isognomostomos*  
101 *Isognomostoma isognomostomos*  
26 *Campylaea (Campylaea) planospira*  
80 *Delphinatia alpina*  
81 *Delphinatia glaciale*  
145 *Vidovicia caeruleans*  
146 *Vidovicia caeruleans*  
84 *Drobacia maetotica*  
126 *Liburnica (Liburnica) dunjana*  
125 *Liburnica (Liburnica) albanogracca*  
129 *Liburnica (Liburnica) setigera setigera*  
82 *Dinarica (Dinarica) pouzolzi*  
132 *Dinarica (Sabljaria) stenomphala*  
118 *Kollarix kollari*  
116 *Josephinella reischuetzi*  
114 *Josephinella phoece*  
107 *Josephinella n. sp. 1*  
111 *Josephinella n. sp. 5*  
142 *Thiessea sphaerostoma*  
143 *Thiessea sphaerostoma*  
141 *Thiessea sphaerostoma*  
36 *Cattania (Cattania) subai*  
34 *Cattania (Cattania) pseudocingulata*  
22 *Cattania (Ariantopsis) pelia*  
148 *Cattania (Wladislawia) sztolcmani*  
28 *Caracollina lenticula*  
133 *Soosia diodonta*  
98 *Helicodonta obvoluta*  
44 *Cepaea nemoralis*

4.0

C. BEAST  
divergence time estimates



4 and 5). On the one hand the phylogeny based on *H3* groups the latter genera with *Corneola* and *Helicigona*, albeit with low support (post. prob. 0.49; Fig. 4). The phylogeny based on *COI* on the other hand strongly supports a sister-group relation between *Isognomostoma* and *Causa* (post. prob. 1.0; Fig. 5). The taxonomic position of *Isognomostoma* (and *Causa*, by deduction) nevertheless remains unclear. Apart from *Causa*, most closely related to *Isognomostoma* based on *H3*, *COI*, *CytB* and *16S* are *Helicigona*, *Vidoviccia*, *Cylindrus* and *Cylindrus*, respectively (Figs. 4-7). The sister-group relationship between *Isognomostoma* and *Campylaea* in the phylogeny for the concatenated dataset (post. prob. 0.52/0.53; Fig. 8) is thus remarkable. The split between *Isognomostoma* and *Campylaea* is estimated at 17.4 MYA (12.8-22.1). In *Isognomostoma*, *Causa*, *Corneola* and *Helicigona* the mucous glands are undivided (Figs. 3-8, 3-7, 3-5 and 3-6, respectively). It is plausible therefore, to assume a closer relationship between these taxa (Fig. 4) than between *Isognomostoma* and *Campylaea* (Fig. 8), which is more diverse in genital anatomy (see the paragraph on *Campylaea*). Distribution:

In the mountains of central Europe, S of the line Eifel, Sauerland and the Harz mountains. From E France eastwards: Zwitterland, Austria, northern Italy, Slovenia, Croatia, Czech, Slovenia, S Poland (Carpathians), NE Hungary and Rumia.

#### Genus *Josephinella* Haas, 1936

Type species: *Helix hemonica* Thiesse, 1884 (Fig. 1-34). *Josephinella* contains at least 19 species, whereas >20 new species are to be described (Subai, in prep.). This genus has its main distribution area in Greece and the southern half of Albania. The material at our disposal was collected in the period 1979-2003 and, with an average age of 22 years, difficult to amplify. For *COI* this resulted in a comparatively high number of mini-barcode sequences (denoted by the MB suffix, Fig. 5) and for *CytB* and *16S* the number of obtained sequences was limited. Despite these impediments, all the reconstructed phylogenies show high support for the monophyly of the genus (post. prob. 0.71, 1.0, 1.0, 1.0 and 1.0/1.0; Figs. 4-8, respectively). In all phylogenies for which sequences of *Thiessea* were obtained (that is all markers except *16S*) that genus is shown to be the sister-group of *Josephinella*. Although this relation is beyond dispute (post. prob. 0.88, 0.99, 0.98 and 1.0/1.0; Figs. 4-6 and 8, respectively), the outcome is skewed by the fact that mitochondrial sequences were obtained only for *Thiessea sphaeriotoma*. Sequences for additional species of *Thiessea* (only *H3*) nevertheless show that that genus is monophyletic (Fig. 4). Therefore we expect the results for *T. sphaeriotoma* (Figs. 4-6 and 8) to be representative for that genus. Subai (1990) described *Chilostoma reischuetzi* and *Chilostoma vikosensis*, which were

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**Figure 8.** Phylogeny reconstructions based on the concatenated *H3-COI-CytB-16S* dataset. A and B depict posterior probabilities for a MrBayes (p. 158) and BEAST (p.159) analysis, respectively. C shows the same cladogram as B, but depicts divergence time estimates; bars indicate 95% confidence intervals (only shown for nodes estimated younger than 20 MYA).

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later on assigned to the genus *Superba* (Subai and Féher, 2006). Our results clearly show that these species belong to *Josephinella*, not *Superba*, and might represent a distinct clade within *Josephinella* (post. prob. 0.87 and 1.0; Figs. 4 and 5, respectively). *Josephinella* and *Thiessea* represent a basal lineage within the subfamily Ariantinae and are estimated to have split from each other 20.5 MYA (15.6-26.0). The mucous glands in *Josephinella* are generally divided half up to 2/3 of their length (Fig. 3-22); specimens with simple, or partly undivided (*J. vikosensis* and *J. reischuetzi*; Fig. 3-26 and Fig. 3-18) or trifurcate mucous glands are rare. Within the sister-group *Thiessea* the mucous glands are always undivided (Fig. 3-16).

**Distribution:**

Southern half of Albania, the SW border area of Slavian-Macedonia, the Ionian islands, mainland Greece and the Peloponnes. The E boundary of its distribution is W of the Axios river and the line Lárisa – Lamia – Livadiá - Nafplio

**Genus novum *Kollarix***

Type species: *Helix kollari* Pfeiffer, 1856. *Kollarix kollari* occurs in Serbia. It has a clearly depressed shell with hairs (Fig. 1-19). The species has been referred to as *Campylaea kollari* (Tomić, 1959), *Helicigona kollari* (Knipper, 1939; Maassen, 1985) and most recently as *Liburnica kollari* (Subai, 2002; Bank, 2012). In his treatise on *Liburnica*, Subai hypothesized that *K. kollari* might be the oldest representative of *Liburnica* (Subai, 2002). None of the obtained phylogenies (no information available for 16S) show *K. kollari* as a member of the otherwise monophyletic group *Liburnica*. In fact, *K. kollari* is shown to represent a distinct lineage within the Ariantinae, which could be older than *Liburnica* (the lineage is estimated to have diverged ca. 26.4 MYA; Fig. 8) and deserves generic status. The genital anatomy of *K. kollari* clearly differs from *Liburnica*; while divided mucous glands are characteristic for *Liburnica* (Figs. 3-20 and 3-21), the mucous glands of *K. kollari* are always undivided (Fig. 3-15).

**Distribution:**

Endemic to Serbia, S of the Donau river up to Aleksinac; between Šabac and Bor districts.

**Genus *Kosicia* Brusina, 1904**

Type species: *Helix intermedia* Pfeiffer, 1828. According to Zilch (1960) *Kosicia* can be regarded as a subgenus of *Chilostoma* (Table 1). *Kosicia* consists of three species, viz. *K. ambrosi* Strobel, 1851 (Fig. 1-15), *K. intermedia* (Fig. 1-16) and *K. ziegleri* Rossmässler, 1836 (Fig. 1-17), which are all included in this study. *Kosicia ambrosi* clearly differs from *K. intermedia* and *K. ziegleri*, both in shell morphology and distribution; *K. ambrosi* is smaller, has a pronounced continuous band on its shell and occurs in NE Italy, whereas the other species have a diffuse or discontinuous band and occur mainly in SE Austria (province of Kärnten, Karawanken Mts). This distinction is reflected in the obtained phylogenies (except of 16S, for which no sequence could be obtained). The sequence divergence between *K. intermedia* and *K. ziegleri* is 7.6% and 7.2% for COI and CytB, whereas the minimum distances between *K. ambrosi* and *K. intermedia* are 13.7% and 19.4% for the same markers, respectively (Table 5). Despite these



large distances within *Kosicia*, all the obtained phylogenies clearly show that these species form a clade (post. prob. 1.0, 0.92, 1.0, 1.0, 1.0/1.0; Figs. 4-8, respectively). The separation between *K. ambrosi* and the lineage *K. intermedia* – *K. ziegleri* is estimated at 11.3 MYA (8.0-14.7). *Kosicia ziegleri* can readily be distinguished from *K. intermedia* by the prominent ribs on its shell, by which it closely resembles *Josephinella vikosensis* (Fig. 1-33). Despite this fact, the origin of *K. intermedia* and *K. ziegleri* (Fig. 8) seems to date back to only c. 2.3 MYA (1.5-3.3). As pointed out under *Faustina*, there is moderate support (post. prob. 0.71/0.78; Fig. 8) for a sister-group relation between *Kosicia* and *Faustina*. The mucous glands in *Kosicia* are always undivided (Fig. 3-13).  
Distribution:

From the southeastern Alps in Austria (Kärnten), to NE Italy, N Slovenia and NW Croatia.

#### Subgenus *Liburnica* Kobelt, 1904, genus *Liburnica*

Type species: *Helix setosa* Férussac, 1832 (Fig. 1-20). With over 15 described species, *Liburnica* is one of the larger genera of Ariantinae. Although Zilch (1960) regards *Liburnica* as a subgenus of *Campylaea*, none of the obtained phylogenies indicate a close relationship between both taxa. All phylogenies (except for the one based on *CytB*, due to a lack of data) show high support for the monophyly of this genus (post. prob. 1.0; Figs. 4, 5, 7 and 8). The position of *Liburnica* relative to the other Ariantinae is less clear. Only the phylogeny for *COI* (Fig. 5) explicitly supports (post. prob. 0.93) a sister-group relationship between *Liburnica* and *Drobacia*. The next most closely related group in the latter phylogeny is *Dinarica* (post. prob. 0.26; Fig. 5), which is also shown in the phylogenies based on *CytB* and the concatenated datasets, albeit without proper support (post. prob. 0.48 and -/0.79; Figs. 6 and 8). The assumed split between *Liburnica* and *Dinarica* is estimated at 19.8 MYA (15.6-25.1). Within *Liburnica* shell morphology can be extremely variable. There exist clear differences in size, depression and shell sculpture; the shells may be smooth or covered with hairs (i.e. *Liburnica (Liburnica) setosa setosa* (Férussac 1832); Fig. 1-20), whereas riblets also occur (i.e. *L. (Superba) s. skipetarica* (Subai 1995); Fig. 1-23). Without *Josephinella reischuetzi* and *J. vikosensis*, (see sub *Josephinella*) *L. (Superba)* forms a clade within *Liburnica* (post. prob. 0.54; Fig. 5). This clade could neither be shown in the phylogeny for *H3*, due to the conservative nature of that marker (sequences for *L. (Liburnica)* and *L. (Superba)* were identical), nor in the phylogenies for *CytB* and *16S*, because no sequences of *L. (Superba)* were obtained for those markers. Consequently, *L. (Superba)* was not included in Fig. 8 and the age of this (sub)genus could not be estimated. *Liburnica* seems to have radiated only recently, the age of this genus (Fig. 8) is estimated at 1.9 MYA (1.2-2.9), and *L. (Superba)* is expected to be even younger. The upper 1/3 to 2/3 of the mucous glands in *L. (Liburnica)* are nearly always divided (Fig. 3-20 and 3-21); sporadically one of the glands can be undivided (Fig. 3-17).

Distribution:

The mountains along the NE coast of the Adriatic sea. Croatia, Bosnia-Herzegovina, Montenegro, Kosovo, Albania up to Epirus in NW Greece.



Subgenus novum *Ljubotenina*, genus *Campylaea*

*Helicigona* (*Arianta*) *ljubetenensis* Wagner, 1912, has long been regarded as a subspecies of *Helicigona* (*Cattania*) *trizona* (Knipper, 1939; Bank, 2012). After an anatomical study, Subai (in prep.) holds the view that this species differs sufficiently from *Cattania* to assign it to a separate genus. The specimen at our disposal was collected over thirty years ago (Table 2) and unfortunately did not allow the amplification of other markers than *H3*. Even though the data are preliminary, this sequence confirms the presumption that *H. (A.) ljubetenensis* does not belong to *Cattania*. In fact, the *H3* sequence is identical to the sequences of *C. lefeburiana* and *C. illyrica*, which is also surprising, since *H. (A.) ljubetenensis* occurs south of the known distribution area of *Campylaea*. We provisionally assign *H. (A.) ljubetenensis* to *Campylaea*, representing a new subgenus: *Ljubotenina*.

The mucous glands of *Campylaea* (*Ljubotenina*) *ljubetenensis* are always divided (Fig. 3-25).

Distribution:

Mount Ljuboten in the Šar mountain area, on the border between Kosovo and NW Macedonia.

Subgenus *Sabljaria* Brusina, 1904, genus *Dinarica*

Type species: *Dinarica* (*Sabljaria*) *stenomphala* (Menke, 1830). During our study we initially regarded *D. (S.) stenomphala* as a species of *Dinarica*, since it turned out to form a clade with *D. (D.) pouzolzii* (post. prob. 0.66, 1.0, 1.0/1.0; Figs. 5, 6 and 8, respectively). As pointed out under *Dinarica*, *D. (S.) stenomphala* differs considerably from *D. (D.) pouzolzii* in shell morphology (Fig. 1-21 vs. 1-22), genital anatomy (Fig. 3-23 vs. 3-24) and genetically. The sequence divergence between both species is 16.3% and 23.5%, for *COI* and *CytB*, respectively. This is comparable to the distances of 15.6% and 22.7% as shown between species of *Faustina* and *Kosicia* for the same markers, respectively (Table 5). Based on an anatomical study, Subai (in prep.) supports the view of Brusina (1904) that *D. (S.) stenomphala* represents a different group, to be called *Sabljaria*. For the moment being, we tentatively consider *Sabljaria* to be a subgenus of *Dinarica*.

Distribution:

Endemic to the Velebit mountains (coast of Croatia).

Subgenus *Superba* Subai & Fehér, 2006, genus *Liburnica*

Type species: *Helicigona skipetarica* (Subai, 1995). *Liburnica* (*Superba*) contains only three species, viz. *L. (S.) skipetarica* (Subai, 1995; Fig. 1-23), *L. (S.) grisea* (Subai & Fehér, 2006) and *L. (S.) kulmankana* (Subai & Fehér, 2006). *Josephinella reischuetzi* and *J. vikosensis* (Fig. 1-33), have also once been assigned to *L. (Superba)* by Subai & Fehér (2006), while referring to differences between these species from Greece and the other *L. (Superba)* species, which occur in Albania. This study confirms that observation (see sub *Josephinella*) and shows additionally that *L. (Superba)* (Subai & Fehér, 2006) is not monophyletic. The three Albanian species mentioned before belong to *Liburnica* (Fig. 4; see sub *Liburnica*), in which they represent a weakly supported clade (post.



prob. 0.54; Fig. 5). Consequently we disagree with the conclusion that *L. (Superba)* “constitutes a group that has to be separated from other groups of Ariantinae on the generic level” (Subai and Fehér, 2006). *Liburnica (Superba)* should not be given more than subgeneric status. Since too little sequences were obtained to include *L. (Superba)* in the concatenated dataset, the age of this subgenus could not be estimated, but it cannot be older than 1.9 MY (see sub *Liburnica*). The mucous glands of *L. (Superba)* can both be forked (Fig. 3-21) which is the most common state in *Liburnica* (Fig. 3-20), or one of them can be undivided (Fig. 3-17).

Distribution:

Albania, Tomor and Kulmakës mountains.

#### Genus *Thiessea* Kobelt, 1904

Type species: *Helix cyclobaris* Deshayes, 1839. *Thiessea* is represented in SE Greece and is generally considered a subgenus of *Chilostoma* (Zilch, 1960; Bank, 2012; Table 1). This view cannot be accepted. With at least 16 species, *Thiessea* represents one of the larger genera of Ariantinae, which is not even closely related to *Chilostoma* (Figs. 4, 5, 6 and 8). Unfortunately, the condition of our material (average age 16 years; Table 2) put severe restrictions on the data that could be obtained for *Thiessea*. For four of the five species included in this study, only *H3* could be amplified. The *H3* phylogeny does show *Thiessea* as a monophyletic group (Fig. 4), albeit without support (post. prob. 0.42). The phylogenies of the other datasets provide no information regarding this monophyly, because they only include *Thiessea sphaeriosoma* Bourguignat, 1857 (Fig. 1-35). In agreement with the current classification (Bank, 2012), the *H3* data show that *Thiessea arcadica* (Frauenfeld, 1867), *T. euboeae* (Frauenfeld, 1867), *T. heldreichi* (Pfeiffer, 1846) and *T. sphaeriosoma* indeed belong to this genus. Except for Fig. 7 (because no *Thiessea* sequences were obtained for 16S), the obtained phylogenies all show a well supported sister-group relation between *Thiessea* and *Josephinella* (post. prob. 0.88, 0.99, 0.98, 1.0/1.0; Figs. 4-6 and 8, respectively). The latter genera are estimated to have diverged ca. 20.5 MYA (15.6-26.0). They seem to represent one of the basal lineages within the phylogeny of the Ariantinae. The mucous glands in *Thiessea* are always undivided (Fig. 3-16; see Subai, 1996 for an overview).

Distribution:

Mainland SE Greece, NE Peloponnese, Aegean Islands and SW Turkey.

#### Genus *Vidoviccia* Brusina, 1904 (monotypic)

Type species: *Helix lactina* Rossmässler, 1837 [= *Vidoviccia caeruleans* (Pfeiffer, 1828)]. *Vidoviccia* inhabits the Dalmatian mountains of Croatia (below 1400 m) and is restricted to limestone. It has a typically depressed shell with a pointed apex and a fine to nearly rib-like radial sculpture (Fig. 1-29). The phylogeny of the concatenated dataset (Fig. 8) shows some support (post. prob. 0.78/0.98) for the monophyly of *Vidoviccia*, *Drobacia* and *Delphinatia*. The sister-group relationships of the genera within this clade nevertheless remain elusive: the phylogeny of the concatenated dataset shows a sister-group relation (post. prob. 0.5/-; Fig. 8) between *Vidoviccia* and *Drobacia* with the MrBayes analysis (Ronquist & Huelsenbeck, 2003), whereas



a sister-group relation (post. prob.  $\sim 0.4$ ; Fig. 8) between *Vidovicia* and *Delphinatia* is shown for the analysis with BEAST (Drummond *et al.*, 2012). The uncertainty of the position of *Vidovicia* is emphasized by the lack of support for a sister-group relation between *Vidovicia* and any of the other genera in the phylogenies for *H3*, *COI* and *CytB* (post. prob. 0.23, 0.24 and 0.16; Figs. 4-6, respectively). Only the phylogeny based on *16S* shows some support for a sister-group relation between *Vidovicia* and *Corneola* (post. prob. 0.77; Fig. 7), but that phylogeny is shown to hold other peculiarities as well (consider the position of *Cepaea*, as pointed out earlier). The mucous glands of *Vidovicia* are undivided (Fig. 3-11), as in *Drobacia* (Fig. 3-9) and *Delphinatia* (Fig. 3-12).

Distribution:

The Dalmatian mountains on the coast of Croatia, from the Velebit mountains in the NW to Peljesac peninsula in the SE.

Subgenus *Wladislawia* Wagner, 1928, genus *Cattania*

Type species: *Campylaea polinskii* Wagner, 1927. According to Zilch (1960), *Wladislawia* is a small subgenus of *Campylaea* (Table 1). The genus is represented by two species, viz. *Cattania (Wladislawia) polinskii* (Fig. 2-31) and *C. (W.) sztolcmani* Wagner, 1927 (Fig. 1-32). This study included only a few specimens of *C. (W.) sztolcmani*. The phylogeny for the concatenated dataset shows a well supported sister-group relationship between *C. (Wladislawia)* and *C. (Ariantopsis)* (post. prob. 0.93/1.0; Fig. 8). The split between the latter subgenera is estimated at ca. 3.1 MYA (1.5-5.1; Fig. 8, also see sub *Ariantopsis*). Moreover, all phylogenies (except *CytB* for which no data were obtained) show a close relation between *C. (Wladislawia)*, *C. (Ariantopsis)* and *C. (Cattania)*. Instead of being a subgenus of *Campylaea* (Table 2), *Faustina* (Damjanov & Likharev, 1975) or *Chilostoma* (Bank, 2012), our results indicate *C. (Wladislawia)*, as well as *C. (Ariantopsis)* and *C. (Cattania)*, should be considered a subgenus of *Cattania*. *Cattania (W.) polinskii* possesses forked mucous glands (Fig. 3-30), whereas in *C. (W.) sztolcmani* these glands are undivided.

Distribution:

Endemic to the Pirin mountains of SW Bulgaria (only found at high altitudes).

## CONCLUSION

In confirmation with Knipper (1939), this study shows that the genital morphology of the species and higher taxa of Ariantinae (Fig. 3) is very homogeneous, what is uncommon in pulmonates. With some exceptions, the shells are also rather monomorphic. When the mucous gland type is plotted on the phylogeny reconstructions of the *COI* and concatenated dataset (Figs. 5 and 8), the polarity of the character transformation can be determined. Obviously, forked or divided mucous glands is a common character within the basal lineages of Ariantinae, whereas in the more derived lineages (from *Causa* to *Arianta* in Fig. 5 and from *Drobacia* to *Arianta* in Fig. 8) nearly all genera possess simple or undivided mucous glands. This is also



seen within individual lineages that are primarily characterized by the possession of forked glands, such as *Cattania*, *Josephinella* and *Dinarica-Liburnica*, where undivided glands are found in the most derived species or groups within such clades (*C. (Wladislawia) sztolemanni*, *Josephinella vikosensis* and *L. (Liburnica) skipetarica*, respectively). Thus, the forked mucous glands represent the plesiomorphic condition versus the apomorphic undivided glands.

The phylogeny of the Ariantinae (Figs. 5 and 8), also sheds light on the biogeographic history of this subfamily. All the basal lineages within Ariantinae (*Cattania*, *Ariantopsis*, *Wladislawia*, *Thiessea*, *Josephinella*, *Dinarica* and *Liburnica*) are currently distributed in the Balkan peninsula, whereas most of the derived lineages (from *Drobacia* to *Arianta*; Fig. 8) are distributed more to the NW (Alps, Italy, Pyrenees, NW Europe, and the Carpathians). This points at a scenario in which the common ancestor originated in the Balkan peninsula, ca. 30 MYA (Fig. 8), and subsequently dispersed and radiated over Europe. The majority of the recognized genera of *Ariantinae* most likely originated between ca. 25-15 MYA. Despite low support for a number of branches and clear inconsistencies within sister-group relations, the molecular phylogenies presented here at least provide a less arbitrary measure for relatedness of the different taxa of Ariantinae than the hitherto used classifications. It is very unlikely that a morphological assessment of *Arianta* and *Cylindrus* would have led to the conclusion that these genera are each other's closest living relatives. What both genera do share however is the potential to develop shell morphological novelties that distinguish them from most of the other Ariantinae. Together, *Arianta* and *Cylindrus* apparently represent a lineage that was less restricted (from an evolutionary point of view) to changes in shell bauplan.

The taxonomy and systematics of the subfamily Ariantinae (Pulmonata, Helicidae) has been under debate for more than a century. Our data confirm that the subfamily originated in southern Europe (most likely the Balkan peninsula) and subsequently dispersed and radiated over Europe. Despite the fact that the shells within Ariantinae are very uniform, the (sub)genera that have been identified based on minute and subtle morphological differences, indeed represent distinct genetic groups.

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## REFERENCES

- BANK, R.A., BOUCHET, P., FALKNER, G., GITTENBERGER, E., HAUSDORF, B., VON PROSWITZ, T. & RIPKEN, T.E.J. 2001. Supraspecific classification of European non-marine Mollusca (CLECOM Sections I+II). *Heldia* 4, 77–128.



- BANK, R.A. 2012. Fauna Europaea version 2.4: Mollusca. <http://www.faunaeur.org>
- BEESE, K., ARMBRUSTER, G.F.J., BEIER K. & BAUR B. 2009. Evolution of female sperm-storage organs in the carrefour of stylommatophoran gastropods. *Journal of Zoological Systematics and Evolutionary Research* 47(1), 49–60. doi: 10.1111/j.1439-0469.2008.00491.x
- COLGAN, D.J., PONDER, W.F. & EGGLE, P.E. 2000. Gastropod evolutionary rates and phylogenetic relationships assessed using partial 28S rDNA and histone H3 sequences. *Zoologica Scripta* 29, 1, January 2000, p. 29–63
- DAMJANOV, S. & LIKHAREV, I. 1975. Fauna Bulgarica, Gastropoda terrestria, vol. IV. *Bulgarian Academy of Sciences, Sofia*, 425 pp.
- DEDOV, I.K. 1998. Annotated check-list of the Bulgarian terrestrial snails (Mollusca, Gastropoda). *Linzer biologische Beiträge* 30(2): 745–765.
- DARLU, P. & LECOINTRE, G. 2002. When Does the Incongruence Length Difference Test Fail? *Molecular Biology and Evolution* 19(4):432–437.
- DRUMMOND, A.J., SUCHARD, M.A., XIE, D. & RAMBAUT, A. 2012. Bayesian phylogenetics with BEAUti and the BEAST 1.7 *Molecular Biology and Evolution* 29 (8): 1969–1973
- FALKNER, G. 1998. Malakologische Neufunde und Forschungsprobleme in den Bayerischen Alpen und ihrem Vorland. In: W.H. Jung, *Naturerlebnis Alpen*: 89–124. Verlag Dr. Friedrich Pfeil, München.
- FALKNER, G., RIPKEN, T.E.J., & FALKNER, M. 2002. Mollusques continentaux de France. Liste de référence annotée et bibliographie. *Patrimoines naturels* 52, 350 pp.
- FOLMER, O., BLACK, M., HOEH, W., LUTZ, R. & VRIJENHOEK, R. 1994. DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. *Molecular Marine Biology and Biotechnology* 3: 294–299.
- FRANK C. 2006. Plio-pleistozäne und holozäne Mollusken Österreichs 2. In: Frank C, Grillitsch H, editors. Mitteilungen der prähistorischen Kommission. *Verlag der Österreichischen Akademie der Wissenschaften, Wien*. 62:397–861.
- GAVETTI, E., BIRINDELLI S., BODON, M. & MANGANELLI, G. 2008. Molluschi terrestri e d'acqua dolce della Valle di Susa. Monografia 44. Museo Regionale di Scienze Naturali, Torino, 274 pp.
- GERNHARD, T. 2008. The conditioned reconstructed process. *J Theor Biol* . 253(4):769–778.
- GITTENBERGER, E., PIEL, W.H. & GROENENBERG, D.S.J. 2004. The Pleistocene glaciations and the evolutionary history of the polytypic snail species *Arianta arbustorum* (Gastropoda, Pulmonata, Helicidae). *Molecular Phylogenetics and Evolution* 30: 64–73.  
doi:10.1016/S1055-7903(03)00182-9
- HAJIBABAEI, M., SMITH, M.A., JANZEN, D.H., RODRIGUEZ, J.J., WHITFIELD J.B. & HEBERT, P.D.N. 2006. A minimalist barcode can identify a specimen whose DNA is degraded. *Molecular Ecology Notes* 6: 959–964.
- HIPP, A.L., HALL, J.C. & SYTSMA, K. 2004. Congruence versus phylogenetic accuracy: revisiting the incongruence length difference test. *Systematic Biology* 53(1):81–89.
- JAECKEL, S.G.H. 1962. Ergänzungen und Berichtigungen zum rezenten und quartären Vorkommen der mitteleuropäischen Mollusken. In: A. Zilch and S.G.A. Jaeckel, Mollusken. - Die Tierwelt Mitteleuropas 2 (1) Ergänzung: 25–294.
- KNIPPER, H. 1939. Systematische, anatomische, ökologische und tiergeographische Studien a südosteuropäischen Heliciden. (Moll. Pulm.). *Archiv für Naturgeschichte* (N.F.) 8, 327–517.



- KROUPA, O. 1994. Zur Molluskenfauna Bulgariens. 3. Über *Arianta pelia* (Hesse, 1912) (Stylommatophora, Helicidae). *Nachrichtenblatt der Ersten vorarlberger malakologischen Gesellschaft* 2: 22-28.
- MAYR, E. 1942. Sytematics and the origin of species, from the viewpoint of a zoologist. Colombia University Press.
- MAASSEN, W.J.M. 1985. *Helicigona kollari* L. Pfeiffer, 1856. De Kreukel, 21 (1/2): 5-6. Amsterdam.
- MERRITT, T.J.S., SHI, L., CHASE, M.C., REX, M.A., ETTER, R.J. & QUATTRO, J.M. 1998. Universal cytochrome b primers facilitate intraspecific studies in molluscan taxa. *Molecular Marine Biology and Biotechnology* 7: 7-11.
- MEUSNIER, I., SINGER, G.A.C., LANDRY, J.F., HICKEY, D.A., HEBERT P.D.N. & HAJIBABAEI, M. 2008. A universal DNA mini-barcode for biodiversity analysis. *BMC Genomics* 9:214.
- NYLANDER, J.A.A. 2004. MrModeltest v2. Evolutionary Biology Centre, Uppsala University. Program distributed by the author.
- PALL-GERGELY, B. 2011. IUCN Red List of Threatened Species. Version 2011.2. [www.iucnredlist.org](http://www.iucnredlist.org)
- PALUMBI, S., MARTIN, A., ROMANO, S., MCMILLAN, W.O., STINE, L. & GRABOWSKI, G. 1991. The simple fools guide to PCR. Special Publ. Dept. Zoology, University of Hawaii, Honolulu.
- PLANET P.J. 2006. Tree disagreement: measuring and testing incongruence in phylogenies. *Journal of Biomedical Informatics* 39: 86–102.
- RONQUIST, F. & HUELSENBECK, J.P. 2003. MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics* 19, no. 12: 1572-1574.
- SCHILEYKO, A.A. 1971. (In Russian) A comparative anatomical study of some Ariantinae (Pulmonata, Helicidae). *Zoologicheskij Zhurnal* 50 (7): 990-1001.
- SUBAI, P. 1984. Über Bestrebungen zur Aufteilung der Subfamilie Ariantinae (= Helicigoninae, Campylaeinae). *Mitteilungen der Deutschen Malakozoologischen Gesellschaft* 37: 229-231.
- SUBAI, P. 1990. Beiträge zur Molluskenfauna des Epirus, I. Zwei neue *Chilostoma*-Arten aus Nord-Epirus (Gastropoda: Ariantinae). *Annalen des Naturhistorischen Museums in Wien* 91B; 243-247.
- SUBAI, P. 1995. Beitrag zur Kenntnis albanischer und griechischer Ariantinae (Gastropoda: Helicidae). *Annalen des Naturhistorischen Museums in Wien* 97 B, p.75-94.
- SUBAI, P. 1996. Revision der Ariantinae. 1. Die *Helicigona*-Untergattung *Thiessea* (Gastropoda: Pulmonata: Helicidae). *Archiv für Molluskenkunde* 126(1/2), p.1-49.
- SUBAI, P. 2002. Revision der Ariantinae. 2. Die *Helicigona*-Untergattung *Liburnica* (Gastropoda: Pulmonata: Helicidae). *Archiv für Molluskenkunde* 131(1/2), p.1-65.
- SUBAI, P. & FEHÉR, Z. 2006. Revision of the Ariantinae, 3. *Superba* n. gen., with the description of three new species. *Archiv für Molluskenkunde* 135(2), p.205-223.
- SWOFFORD, D.L. 2002. PAUP\*. Phylogenetic Analysis Using Parsimony (\* and other methods). Version 4.0b10. Sinauer Associates, Sunderland, Massachusetts.
- YULE, G.U. 1925. A Mathematical Theory of Evolution, based on the Conclusions of Dr. J.C. Willis, F.R.S. *Philosophical Transactions of the Royal Society B* 213: 21–87.
- ZILCH, A. 1960. Handbuch der Paläozoologie – Band 6, Teil 2, Berlin, 834 p.





CHAPTER 7

**A mouse's tail: how to settle an  
insurance dispute**

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## ABSTRACT

Over the last years insurance companies have shown an increased interest in identifications of pest species, either as a form of risk assessment or to address liability issues. This paper describes a case report of such a forensic insurance investigation. The names of the corporations involved have been withheld in compliance with a pre-existing confidentiality agreement.

A sea container containing sterile goods was shipped from China to the Netherlands. Upon inspection at its final destination, the contents were declared lost due to the presence of a dead mouse. Determination of which company should be held liable for this loss depended on where the mouse entered the container. The specimen was identified as belonging to the genus *Apodemus* (Muridae) based on morphology. Two species, *A. sylvaticus* and *A. flavicollis* occur in the Netherlands, two different species, *A. agrarius* and *A. draco*, occur in the relevant area in Eastern China. The distribution areas of these Dutch and Chinese *Apodemus* species do not overlap. Because the specimen was adolescent and partly mummified, key morphological characters for species-level identification were missing or not discernable. Published literature and sequence data available on GenBank showed that the four candidate species could be distinguished based on *Cytochrome B* barcode sequences. Given the decayed condition of the specimen, we expected possible DNA degradation. Therefore, both internal *CytB* primers (designed to amplify a short nucleotide sequence) and universal primers (which amplify a fourfold larger fragment) were employed. Remarkably, the primerset that was designed to amplify a short *CytB* sequence of *A. draco* amplified a well-studied pseudogene of *A. sylvaticus*. Both the *CytB* and the pseudogene sequence confirmed that the specimen in question is *A. sylvaticus*. Contamination of the sterile goods must therefore have taken place in the Netherlands.

## INTRODUCTION

A container with sterile goods from Eastern China, which was opened in the Netherlands after a four-month journey, contained a dead mouse (Fig. 1). The mouse, found among some packing material was partly mummified. Because of the mouse, the contents of the container were declared lost. Insurance company procedure required determination of which party was responsible for the loss. The insurance company asked a taxonomic identification service, the *Expertcentrum* of the Netherlands Centre for Biodiversity (NCB) Naturalis, if the specimen could be identified and whether, based on the geographic distribution of the species, it was possible to establish where the mouse had entered the container.

Based on morphology, the specimen was identified by taxonomists at NCB Naturalis as a wood or field mouse belonging to the genus *Apodemus*. Two species of this genus occur in The Netherlands: *A. sylvaticus* and *A. flavicollis*; China is home to another





**Figure 1.** Photograph of the adolescent, partly mummified *Apodemus* sp. specimen as it was found in the container.



eight *Apodemus* species: *A. agrarius*, *A. chevrieri*, *A. draco*, *A. latronum*, *A. pallipes*, *A. peninsulae*, *A. semotus* and *A. uralensis* (Smith and Xie, 2008). The container was loaded and shipped in an area of Eastern China, which falls within the distribution range of just two of the eight Chinese field mice (Fig. 2). This limited the number of candidate species to only four. The distribution areas of these four species do not overlap; the two Dutch species do not occur in China en vice versa.

The long-tailed wood mouse, *Apodemus sylvaticus*, is widespread in the West Palearctic (including North Africa) and occurs as far to the East as Nepal and the Altai Mountains in Mongolia, but it is not known from Eastern China (Corbet, 1978 and Smith and Xie, 2008). Of the four candidates it is the second most widely distributed species. Its sympatrically occurring and less opportunistic sister species, the yellow-necked wood mouse, *A. flavicollis*, seems to depend more heavily on forested habitats. In the Netherlands it is therefore only known from southern Limburg (van der Straeten, 1977; Broekhuizen *et al.* 1992). With a known distribution range from China to central Europe, the Eurasian striped field mouse, *A. agrarius*, is the most widely distributed species of *Apodemus* (Fig. 2). According to Corbet (1978), the western boundary of *A. agrarius* reaches from northern Italy to western Germany; it is not known to occur in the Netherlands (Fig. 2 shows a global outline of the distribution area of *A. agrarius*; for a more detailed map of the western distribution of this species, see Fig.1 of Aguilar *et al.*, 2008). The South China field mouse, *Apodemus draco*, represents several genetically distinct forms, which probably arose due to geographical isolation and adaptation to different climatical conditions during the Quaternary glaciations (Sakka *et al.*, 2010). Some of these forms (*A. d. draco*, *A. d. orestes* and *A. ilex*) have been regarded as different (sub)species (Liu *et al.*, 2004 and references therein) and do not occur in Eastern China (i.e. *A. ilex* from Yunnan), but *A. draco* does (reported from Anui and Fuijan by Liu *et al.*, 2004 and also in Smith and Xie, 2008 and Sakka *et al.*, 2010). The distribution areas of *A. flavicollis* (Corbet, 1978 and Michaux *et al.*, 2004) and *A. draco* (Corbet, 1978; Smith and Xie, 2008 and Sakka *et al.*, 2010) are restricted to Europe and China, respectively (Fig. 2).

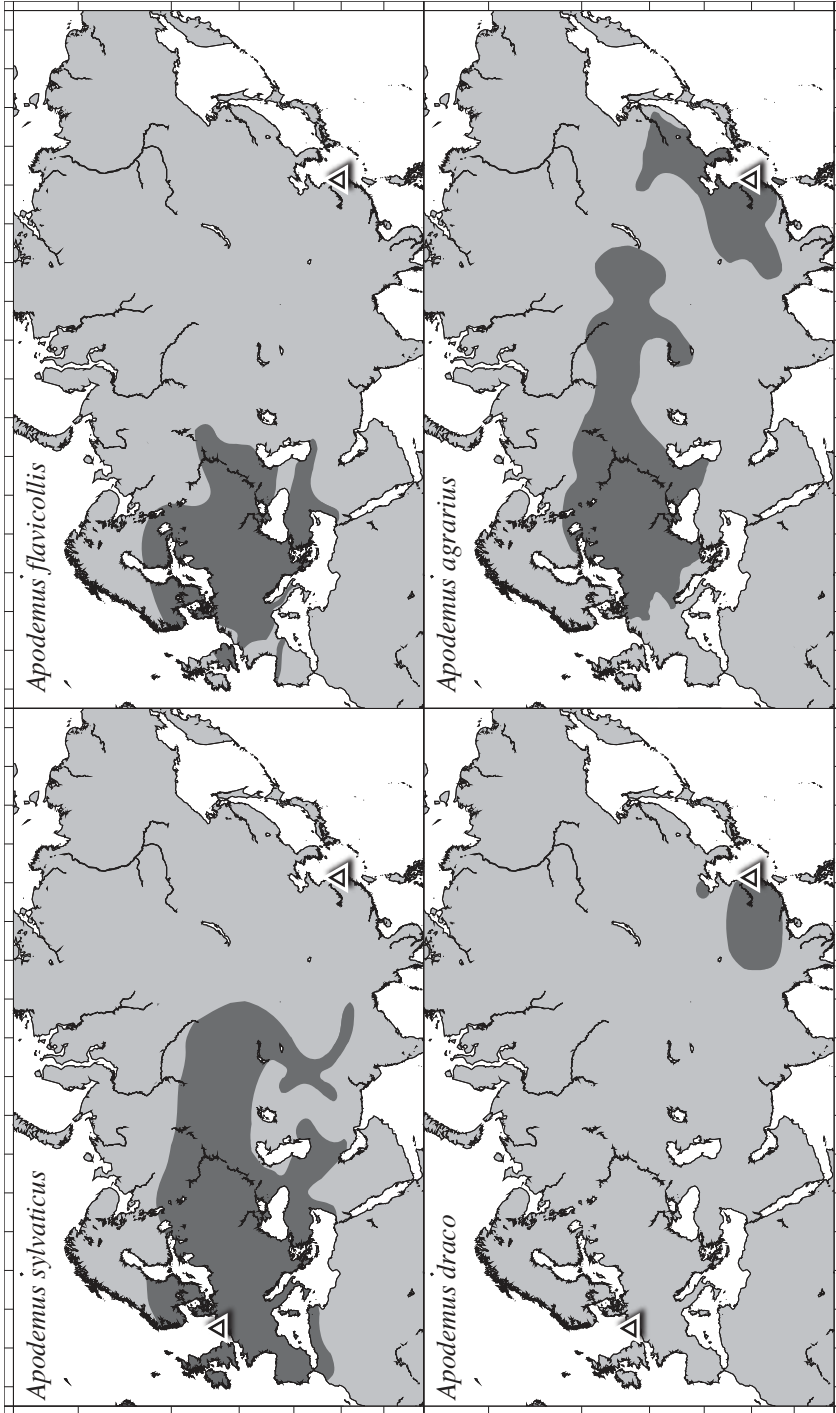
Based on the known distributions of the four species in question we knew that once our specimen was identified to the species level, its origin would be clear and the question of where it had entered the container would be answered. However, further identification based on morphological characters and comparison with specimens in the scientific collection of NCB Naturalis ended here because the specimen under

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**Figure 2.** Palearctic distribution maps (after Corbet, 1978; Michaux *et al.*, 2004 and 2005; Aguilar *et al.*, 2008; Smith and Xie, 2008 and Sakka *et al.*, 2010) of *Apodemus* species that are of relevance to this case report. The triangles indicate the approximate positions of the Chinese and Dutch harbours. For *A. flavicollis* and *A. agrarius* the triangles for the Dutch harbor were left out, to show the location of the western distributional boundaries relative to the Netherlands.

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investigation (Fig. 1) was not fully adult and partly mummified: distinguishing characters where either not yet present or not discernable. Consequently, we turned to molecular techniques to try to identify this specimen.

### MATERIAL AND METHODS

After being detected, the dead mouse was put in a freezer to prevent further decay and degradation of DNA. A small piece (approx. 4 mm) was cut off the tip of the tail for DNA extraction. DNA was extracted with a DNeasy blood and tissue kit (Qiagen) following the manufacturer’s protocol. A quick survey of available literature (Serizawa *et al.*, 2000; Suzuki *et al.*, 2003 and 2008; Liu *et al.*, 2004; Michaux *et al.* 2005; Dubey *et al.*, 2009 and Sakka *et al.*, 2010) and data stored in GenBank, showed that barcoding marker *Cytochrome B* would allow discrimination between the four *Apodemus* candidates. For amplification of *CytB*, universal primers L14724 and H15915 (Irwin *et al.*, 1991; Table 1) were ordered, but given the condition of the specimen, there was serious doubt that the DNA would not be too degraded to directly amplify a fragment of almost 1200 basepairs. Therefore, additional primer pairs were designed to target shorter sequences (ca. 300 bp) using Primer3 ([www.primer3.com](http://www.primer3.com)). These primer sets (Drac, Flav and Syl; Table 1), were based on sequences of *A. draco*, *A. flavicollis* and *A. sylvaticus*, respectively, and were expected to be more specific to each particular species. For the initial trial, we did not design primers based on a sequence of *A. agrarius*. PCRs (in 25 µl volumes) were done with each primer set (Table 1), using 0.4 µM of each primer, 1.5 mM MgCl<sub>2</sub>, 0.2 mM dNTP’s and 0.25 µl (1.25 units) Taq DNA polymerase (Qiagen). The reactions were cycled 40

**Table 1:** Primer information.  
‘Primer’ refers to the primer name or alias, ‘Length’ to the expected size (in bp) of the target amplicon and ‘ACCN’ refers to the GenBank accession number.

| Primer | Sequence (direction: 5’→ 3’) | Length | ACCN     | Reference:                    |
|--------|------------------------------|--------|----------|-------------------------------|
| L14724 | CGAAGCTTGATATGAAAAACCATCGTTG | 1192   | HQ158102 | Irwin <i>et al.</i> ,<br>1991 |
| H15915 | AACTGCAGTCATCTCCGGTTTACAAGAC |        |          |                               |
| DracF  | ACCTTCCTGCCCCATCTAAC         | 289    | HQ158103 | This study                    |
| DracR  | CCAATGTTTCATGTTTCTATGAATG    |        |          |                               |
| FlavF  | TCACTCTTTCATCGACTTACCA       | 294    | HQ158104 |                               |
| FlavR  | CATGTTTCTATGAAGGTGTATGATCC   |        |          |                               |
| SylF   | TCGACCTACCAGCTCCATCT         | 300    | HQ158105 |                               |
| SylR   | GAACTACACCAATGTTTCATGTTTC    |        |          |                               |

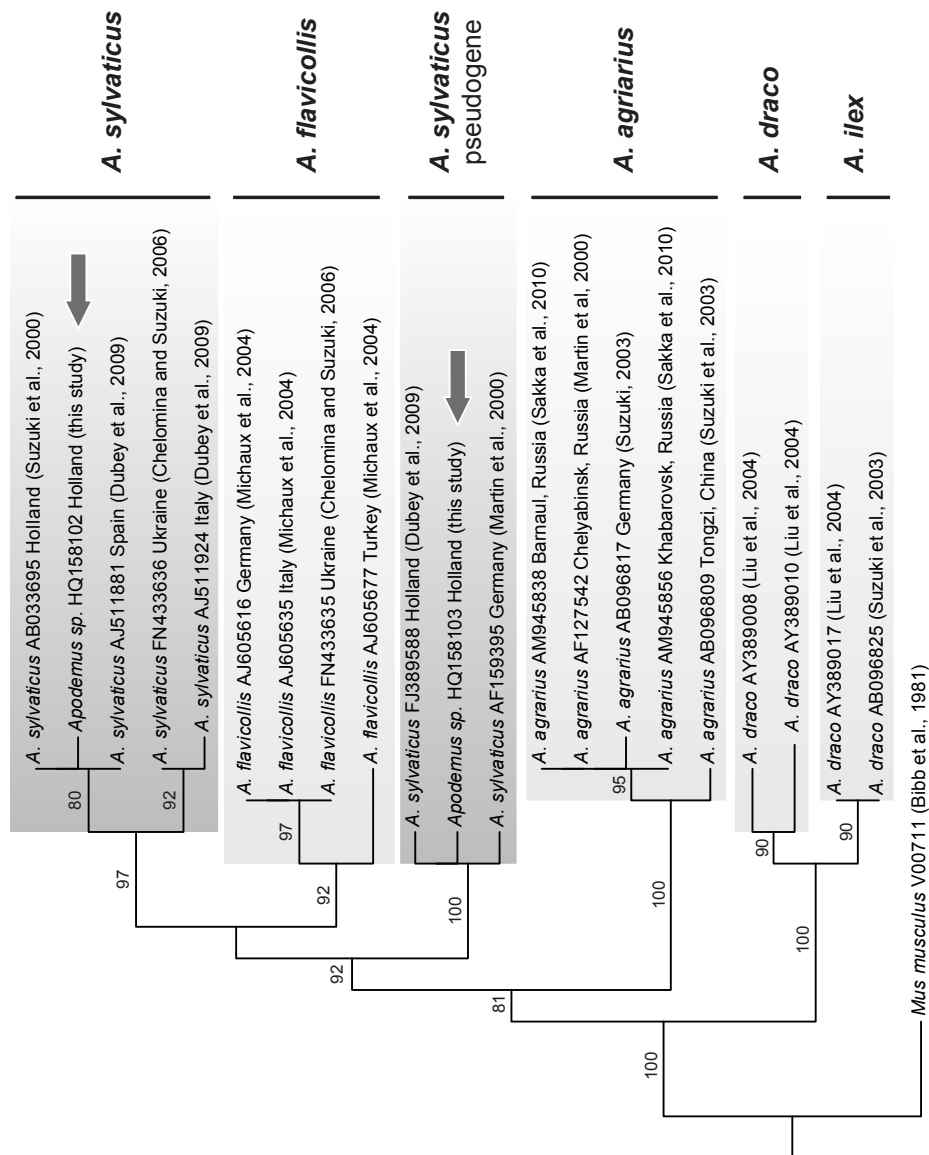


times with an annealing temperature of 50° C and the products were checked on a 1% agarose gel (0.5x TBE, stained with Ethidium Bromide). Subsequently the PCR products were sent to MacroGen Europe (Amsterdam), where they were purified with a Montage purification kit (Millipore) and sequenced in both directions (using the same primers that were used for PCR) on an ABI3730XL. The resulting forward and reverse sequences were assembled with Sequencher 4.10.1 (Gene Codes Corporation). A Blast search (<http://blast.ncbi.nlm.nih.gov/Blast.cgi>) was done to find the most similar sequences in GenBank. In case the barcodes would not univocally indicate the identity of our specimen, we wanted to show which group of *Apodemus* sequences was most similar. Therefore, we manually aligned our sequences in MacClade 4.08 (Maddison and Maddison, 2005) against a selection of *Cyt B* and *Cyt B* pseudogene (when available) sequences from the four candidate *Apodemus* species downloaded from GenBank. The aim of this selection was to have the distribution areas of these four species reasonably covered in order to show that each species is indeed distinct and monophyletic. A Neighbor-Joining tree (branch lengths proportional to genetic distance, Fig. 3) was constructed with Paup 4.0b10 (Swofford, 2003) to show the position of sequences from this study (indicated with grey arrows) relative to the existing data. The house mouse, *Mus musculus* (Bibb *et al.*, 1981) was used as outgroup. The branch support values were obtained by standard maximum likelihood (ML) bootstrapping (100 replicates, no model specified). GenBank accession-numbers of selected sequences as well as the provenance of the corresponding specimens and a reference to their original publication (Bibb *et al.*, 1981; Martin *et al.*, 2000; Suzuki *et al.*, 2000 and 2003; Liu *et al.*, 2004; Michaux *et al.*, 2004; Chelomina and Suzuki, 2006; Dubey *et al.*, 2009 and Sakka *et al.*, 2010) are shown next to the taxon-labels (Fig. 3). The sequences obtained with this study were deposited in GenBank and are available under accession-numbers HQ158102 - HQ158105. Sequences HQ158102, HQ158104 and HQ158105, amplified with the universal, Flav and Syl primers (table 1), respectively, were identical. Therefore only sequence HQ158102 was included in the analysis. Because HQ158103 and some of the selected sequences (FN433635, FN433636 and AF127542) were much shorter, the analysis was done with the first 325 bases of HQ158102 (HQ158103 was included completely).

## RESULTS

PCRs were successful with all of the tested primer sets (Table 1). As expected sequence HQ158102 was larger than the other sequences (Table 1), but not all of the obtained sequences were identical; sequence HQ158103 (amplified with primers DracF and DracR; Table 1) clearly differed from the other three. Blast searches with HQ158102 (length 1046 bp) and HQ158103 (length 244 bp) showed that these sequences were 99.2% identical with sequences AB033695 (1038/1046) and AF159395 (242/244), both belonging to a specimen of *A. sylvaticus* from the Netherlands (Suzuki *et al.*, 2000) and Germany (Martin *et al.*, 2000), respectively. The latter sequence (AF159395)





**Figure 3.** Neighbor-Joining tree (distance criterion) showing *Cytochrome B* sequences of the four species of *Apodemus* relevant to this case report, sequences of available *Cyt B* pseudogenes and sequences from this study (indicated by grey arrows). Values on branches indicate ML bootstrap support (100 replicates, no model selected).



was shown by Dubey *et al.* (2009) to actually belong to a cryptic pseudogene of *Cytochrome B*. Consistent with available literature (Serizawa *et al.*, 2000; Michaux *et al.*, 2002 and 2005; Suzuki *et al.*, 2003 and 2008 and Liu *et al.*, 2004) Figure 3 showed that *Cyt B* sequence data indeed separate the candidate species (even from distant localities within their respective distribution areas) in distinct groups. Both the *Cyt B* sequence and the nuclear pseudogene indicate that the specimen found in the sea container is *A. sylvaticus*.

## DISCUSSION

With a clear west-Palearctic distribution (Fig. 2), the eastern limit of *A. sylvaticus* does not extend beyond the Altai mountains in Mongolia (Corbet, 1978 and Sakka *et al.*, 2010), we can rule out the possibility that the specimen found in the container (Fig. 1) could have originated from Eastern China. Contamination of the sterile goods must therefore have taken place in the Netherlands, not China. Based on the distribution areas of the relevant *Apodemus* species (Fig. 2), the only disputable outcome might have been if the specimen was identified as *A. agrarius*, which occurs in both East Asia and Europe. Especially since its distribution ranges as far west as Germany, approximately 200 km from the Dutch border (Aguilar *et al.*, 2008 and Schlegel *et al.*, 2009). Even though the internal primers Drac and Flav were designed on sequences of *A. draco* and *A. flavicollis*, this study does not demonstrate whether these internal primers will actually work for these species in a PCR. In the absence of *A. draco* and *A. flavicollis* template, the internal primers attached to any best-fitting template; the *Cyt B* pseudogene of *A. sylvaticus* in the case of the Drac-primerset.

## ACKNOWLEDGEMENTS

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## REFERENCES

- AGUILAR, J.P., PÉLISSIE, T., SIGÉ, B. & MICHAUX, J.R. 2008. Occurrence of the stripe field mouse lineage (*Apodemus agrarius* Pallas 1771; rodentia; mammalia) in the Late Pleistocene of southwestern France. *Comptes Rendus Palevol* 7: 217-225.



- BIBB, M.J., VAN ETEN, R., WRIGHT, C.T., WALBERG, M.W. & CLAYTON, D.A. 1981. Sequence and gene organization of mouse mitochondrial DNA. *Cell* 26: 167-80.
- BROEKHUIZEN, S., HOEKSTRA, B., VAN LAAR, V., SMEENK, C. & THISSEN, J.B.M. 1992. Atlas van de Nederlandse zoogdieren. Stichting Uitgeverij Koninklijke Nederlandse Natuurhistorische Vereniging, Utrecht, pp. 286-288.
- CORBET, G.B. 1978. The mammals of the Palearctic Region: a taxonomic review. British Museum (Natural History), no. 788.
- CHELOMINA, G.N. & SUZUKI, H. 2006. Molecular evolution and phylogeography of west-Palearctic wood mice of the genus *Sylvaemus* based on the data on variation of nuclear (IRBP) and mitochondrial (cyt *b*) DNA genes. *Zoologicheskii Zhurnal* 85: 219-234.
- DUBEY, S., MICHAUX, J.R., BRÜNNER, H., HUTTERER, R. & VOGEL, P. 2009. False phylogenies on wood mice due to cryptic cytochrome-b pseudogene. *Molecular Phylogenetics and Evolution* 50: 633-641.
- IRWIN, D.M., KOCHER, T.D. & WILSON, A.C. 1991. Evolution of the Cytochrome b Gene of Mammals. *Journal of Molecular Evolution* 32: 128-144.
- LIU, X., WEI, F., LI, M., JIANG, X., FENG, Z. & HU, J. 2004. Molecular phylogeny and taxonomy of wood mice (genus *Apodemus* Kaup, 1829) based on complete mtDNA cytochrome b sequences, with emphasis on Chinese species. *Molecular Phylogenetics and Evolution* 33: 1-15.
- MADDISON, D.R. & MADDISON, W.P. 2005. MacClade 4.08. Sinauer Associates, Sunderland, Massachusetts.
- MARTIN, Y., GERLACH, G., SCHLÖTTERER, C. & MEYER, A. 2000. Molecular phylogeny of European Muroid rodents based on complete Cytochrome b sequences. *Molecular Phylogenetics and Evolution* 16: 37-47.
- MICHAUX, J.R., CHEVRET, P., FILIPPUCCI, M.G. & MACHOLAND, M. 2002. Phylogeny of the genus *Apodemus* with a special emphasis on the subgenus *Sylvaemus* using the nuclear IRBP gene and two mitochondrial markers: cytochrome b and 12S rRNA. *Molecular Phylogenetics and Evolution* 23: 123-136.
- MICHAUX, J.R., LIBOIS, R., PARADIS, E. & FILIPPUCCI, M.G. 2004. Phylogeographic history of the yellow-necked fieldmouse (*Apodemus flavicollis*) in Europe and in the Near and Middle East. *Molecular Phylogenetics and Evolution* 32: 788-798.
- MICHAUX, J.R., LIBOIS, R. & FILIPPUCCI, M.G. 2005. So close and so different: comparative phylogeography of two small mammal species, the Yellow-necked fieldmouse (*Apodemus flavicollis*) and the Woodmouse (*Apodemus sylvaticus*) in the Western Palearctic region. *Heredity* 94: 52-63.
- SAKKA, H., QUÉRÉ, J.P., KARTAVTSEVA, I., PAVLENKO, M., CHELOMINA, G., ATOPKIN, D., BOGDANOV, A. & MICHAUX, J.R. 2010. Comparative phylogeography of four *Apodemus* species (Mammalia: Rodentia) in the Asian Far East: evidence of Quaternary climatic changes in their genetic structure. *Biological Journal of the Linnean Society* 100: 797-821.
- SCHLEGEL, M., KLEMPA, B., AUSTE, B., BEMMANN, M., SCHMIDT-CHANASIT, J., BÜCHNER, T., GROSCHUP, M.H., MEIER, M., BALKEMA-BUSCHMANN, A., ZOLLER, H., KRÜGER, D.H. & ULRICH, R.G. 2009. Dobrava-Belgrade Virus Spillover Infections, Germany. *Emerging Infectious Diseases* 15: 2017-2020.



- SERIZAWA, K., SUZUKI, H. & TSUCHIYA, K. 2000. A phylogenetic view on species radiation in *Apodemus* inferred from variation of nuclear and mitochondrial genes. *Biochemical Genetics* 38: 27-40.
- SMITH, A.T. & XIE, Y. 2008. A guide to the mammals of China, Princeton University Press, pp. 252-256.
- SUZUKI, H., TSUCHIYA, K. & TAKEZAKI, N. 2000. A molecular phylogenetic framework for the Ryukyu endemic rodents *Tokudaia osimensis* and *Diplothrix legata*. *Molecular Phylogenetics and Evolution* 15: 15-24.
- SUZUKI, I.H., SATO, J.J., TSUCHIYA, K., LUO, J., ZHANG, Y., WANG, Y. & JIANG, X. 2003. Molecular phylogeny of wood mice (*Apodemus*, Muridae) in East Asia. *Biological Journal of the Linnean Society* 80: 469-481.
- SUZUKI, H., FILIPPUCCI, M.G., CHELOMINA, G.N., SATO, J.J., SERIZAWA, K. & NEVO, E. 2008. A biogeographic view of *Apodemus* in Asia and Europe inferred From nuclear and mitochondrial gene sequences. *Biochemical Genetics* 46: 329-346.
- SWOFFORD, D.L. 2003. Phylogenetic Analysis Using Parsimony (PAUP), version 4.0 beta 10, Massachusetts.
- VAN DER STRAETEN, E. 1977. *Apodemus flavicollis* (Melchior, 1834) in Nederland. *Lutra* 19.





CHAPTER 8

**On the identity of the first rhinoceros  
owned by the Dublin Zoo (†1865);  
genetic characterisation of a poorly  
preserved museum specimen**

GROENENBERG D.S.J.

DE COURSY C.

LINNIE M.

OOSTERWEGHEL L.

## ABSTRACT

Understanding past and present genetic diversity within endangered species is crucial for the identification of evolutionary significant units (ESUs) and subsequent conservational decisions. Direct access to genetic diversity of extinct populations can only be gained from (sub)fossils or specimens housed in natural history collections. With probably less than fifty extant specimens, the Javan rhinoceros, *Rhinoceros sondaicus*, is a critically endangered species. It is rare even in museum collections, thus each newly discovered specimen is of conservation importance. Although the Indian (*Rhinoceros unicornis*) and Javan rhinoceros differ in size, skinfolds and skin texture, the two have been confused on several occasions in the recorded history of both species. Examples can be found in textbooks, zoological gardens and museums. As for the latter, identification of mounted specimens can be compromised among others by poor preservation of the skin. An example of such an ambiguous specimen is the Dublin Zoo rhinoceros (†1865) housed in the Zoological Museum at Trinity College Dublin. Ever since it was mounted it bears a name plate that claims it represents a specimen of *R. unicornis*, but this is not necessarily supported by a number of morphological characters. With this study we determine the identity of this one-and-a-half century old specimen by DNA sequencing a fragment of mitochondrial *Cytochrome B*.

## INTRODUCTION

Throughout the nineteenth century, exotic animals were scarce and expensive. In the early years of the Dublin Zoo (opened 1831), animals such as elephants and rhinoceroses were hired from travelling menageries, acquired by gift or exchange from other zoos and, on rare occasions, purchased. In 1864, Sir Charles Trevelyan (1807-1886), finance minister in India and a friend of Dublin Zoo, donated a young, male rhinoceros. Together with a consignment of animals for the London Zoo, the rhinoceros was shipped from Barrackpore Park, Calcutta, to London. From there on it was shipped to the Dublin Zoo, where it arrived on the 3<sup>rd</sup> of August 1864 (Rookmaaker, 1994). Unfortunately the young rhinoceros never settled in Dublin; zoological records show that it was frequently 'crabby' and did not eat very well (de Courcy, 2009 and references therein); it died in horrific circumstances in April 1865. Generally deceased animals were sold by the Zoo to generate some additional income. In this case the carcass was bought by the Zoological Department of Trinity College Dublin for a post-mortem and anatomical study, after which the skin and skeleton were donated to the Zoological Museum of Trinity College, where they are still housed today.

The rhinoceros had been in captivity since at least October 1863 and Samuel Haughton identified it as a three year old in the autopsy report (Haughton, 1867). And this is where the questions arise. The mounted specimen in the Zoological



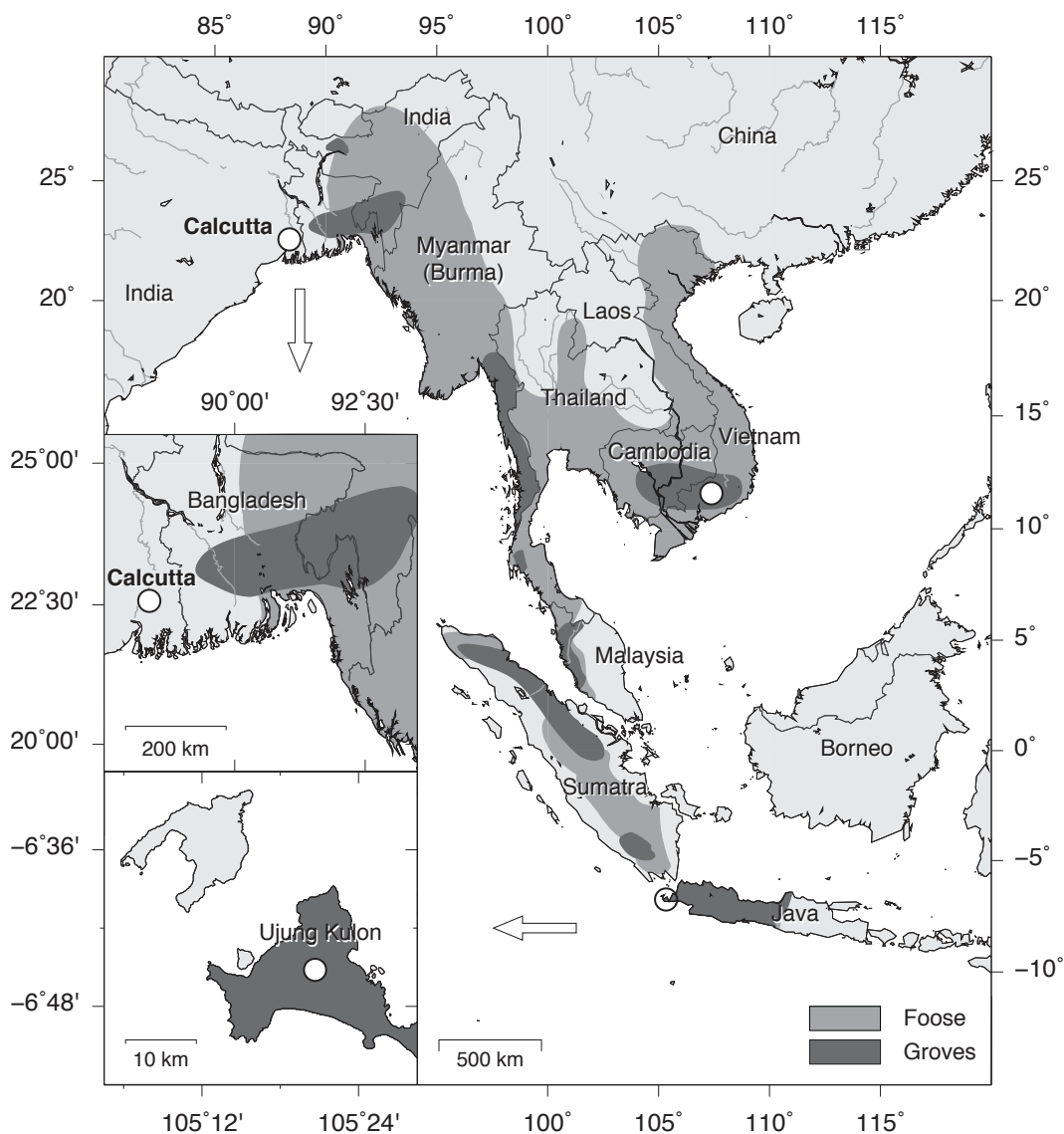
Museum in Trinity College, which has to be the Dublin Zoo rhinoceros seems remarkably small for an Indian rhinoceros, *Rhinoceros unicornis* [Lin. 1758] of this age. Given the rarity of the Javan rhinoceros in museum collections, it was felt worthwhile to investigate this specimen further.

Nowadays only a small group (probably less than 50 specimens) of Javan rhinoceros, *Rhinoceros sondaicus sondaicus* [Desmarest 1822] survive in Ujung Kulon (Java). The few ‘rediscovered’ Javan rhinoceroses, *Rhinoceros sondaicus annamiticus* [Heude, 1892] in Cat Tien (Vietnam; Polet *et al.*, 1999) were officially considered extinct in a recent WNF report (Brook *et al.*, 1011) after the last one was found dead in April 2010 as a result of poaching. When the Dublin rhinoceros was shipped in 1864 however, the Javan rhinoceros (*R. sondaicus inermis*) still inhabited the Sunderbunds of Bangladesh and *R. sondaicus* was occasionally kept in Barrackpore Park, Calcutta. Fig. 1 shows the historic distribution of *R. sondaicus* (after Foose and van Strien, 1997 and Groves and Leslie, 2011). Note that Groves and Leslie (2011) consider the historical distribution of *R. sondaicus* in Foose and van Strien (1997) to be overstated.

Unfortunately *R. unicornis* and *R. sondaicus* share a history of confusion; in 1959 Sody already listed four books that showed illustrations of *R. unicornis* accompanying descriptions of *R. sondaicus* and in 1966, Guggisberg wrote “A Javan rhinoceros shown in the Berlin Zoo sometimes in the last century was in fact an Indian one, while an Indian rhino which died in the Zoo of Adelaide, Australia, was found to belong to the Javan species”. More recently Rookmaaker concluded that the so-called Javan Rhinoceros that once lived in the Zoological garden of Liverpool likely to have been an Indian rhinoceros (Rookmaaker, 1993).

There are a number of clear morphological differences between adult *R. unicornis* and *R. sondaicus*, however. The Javan rhinoceros has a skin fold protruding around the shoulder that results in an additional skin shield (also referred to as nape shield or saddle), which lacks in *R. unicornis*. The skin folds on *R. sondaicus* are also shallower than that on *R. unicornis* and *R. sondaicus* does not develop a “bib” (pronounced in the adult Indian rhinoceros); nor does the *R. sondaicus* develop the deep cheek and neck folds that are seen in *R. unicornis* (Groves and Leslie, 2011). Moreover there is a difference in the form of the head (which is more slender in the Javan rhinoceros) and the overall size of *R. sondaicus* is less than that of the Great Indian. Another obvious distinction is the presence of a prominent horn in both sexes of *R. unicornis*, which can reach up to 2 feet (61 cm), whereas only the male *R. sondaicus* possesses a horn, which regularly does not exceed 10 inches (25 cm). According to Lydekker (1907) “the tail of *R. sondaicus* stands out quite distinct from the hind-quarters so that its whole extent is exposed in a side view”, whereas in *R. unicornis* “the tail is enclosed in a deep groove, in such a manner that only its terminal portion is visible in a side view” (Fig. 2). The skin of the Javan rhinoceros does not show the tubercles characteristic for *R. unicornis*, but has a cracked scaly (reticulated) appearance. Guggisberg (1966) describes it as a “mosaic-like pattern”, which indeed can be seen





**Figure 1.** Historic distribution of *Rhinoceros sondaicus* (after Foose and van Strien, 1997 and Groves and Leslie, 2011).

The three circles in the main picture represent Calcutta, the origin of the Dublin rhinoceros (†1865), Cat Loc (within Cat Tien National Park, Vietnam), where *R. sondaicus annamiticus* recently died out and Ujong Kulon National Park, where *R. sondaicus sondaicus* still survives.



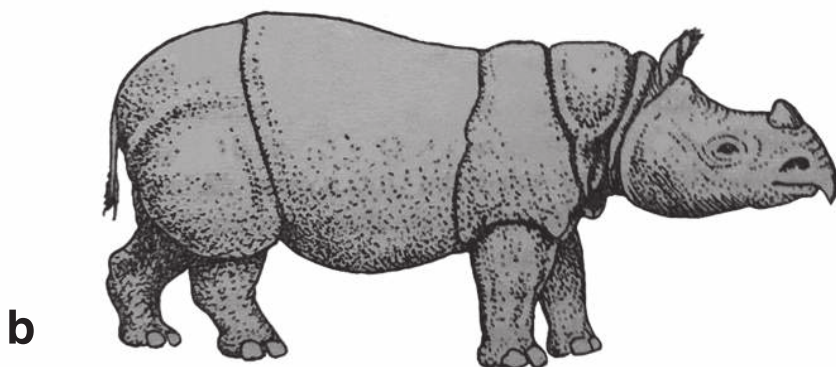
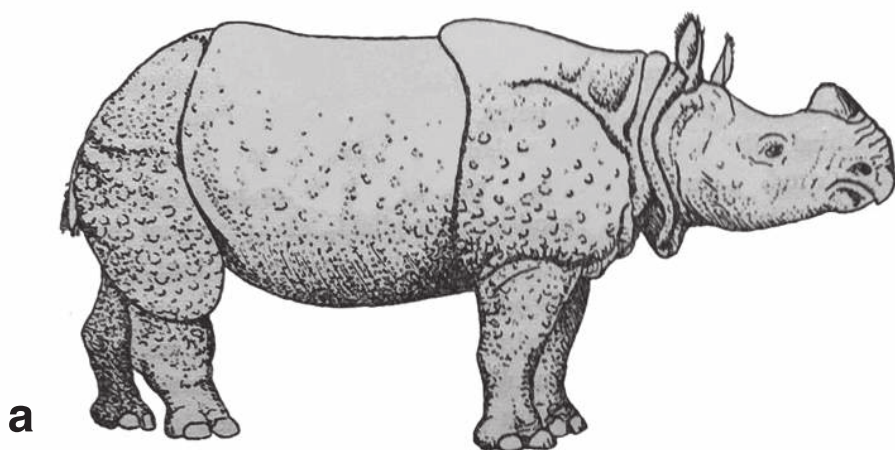
in the photographs of several specimens killed in Sumatra that illustrate an article published by JC Hazewinkel in 1933. Finally, the upper lip of *R. sondaicus* might be longer and more prehensile than that of *R. unicornis* (Sclater, 1874 and Groves and Leslie, 2011).

Despite these differences, confidently identifying the 1865 Dublin rhinoceros (Fig. 2) on sheer morphology turned out to be difficult for two main reasons. Firstly, the specimen is small for a three year old *R. unicornis*. Secondly, some of the before mentioned characters are absent or are not discernable. Thirdly, because of the poor way in which the skin was preserved, a number of details were either missing or ambiguous. Consequently only genetic analysis could result in a reliable identification of this specimen. Next to extracting DNA from the skin, we tried to extract DNA from the skeleton as well, because a priori it was unclear which body part (and hence preservation method) provided the best conditions for DNA conservation. Moreover if the analysis proved successful for both objects, it would allow us to see if the skin and skeleton indeed belonged to the same species. At first sight it seems hardly imaginable how the skin (Fig. 2) and skeleton (Fig. 3) could not belong to the same individual, given the age, size and origin of these body parts, but once separated, different parts can end up in different institutions (Rookmaaker, 1993) and objects displayed together today, do not necessarily share the same history.

## MATERIAL AND METHODS

Prior to sending samples to Leiden (the Netherlands) for molecular analysis permission was granted by the Irish CITES Management Authority (Fig. 4). In total eight samples were taken, four from the skin (under the head region, tail region, left rear toe and right front toe) and four from the skeleton (right shoulder blade, pelvic region, underside of the jaw and left rear heel). For the skin samples small pieces of tissue were cut from the mounted specimen using gloves and sterile blades; for the bone samples tiny holes were drilled in the skeleton of which drilling chips and dust were collected. DNA extractions were performed in a dedicated ancient DNA facility (LAF, Leiden), where no work on Rhinocerotidae was previously performed and which is physically isolated from the main laboratory (where post-PCR work was carried out). All samples were pulverized with an MM200 mixer mill (Retsch) using steel balls and grinding jars. DNA was extracted using the GuanidineThioCyanate (GuSCN) protocol described by Rohland and Hofreiter (2007). Final extraction volume was 40  $\mu$ l and DNA extraction and PCR blanks were included to monitor for contamination.

Based on available sequence data (GenBank) for both the Indian (Xu *et al.*, 1996) and Javan rhinoceroses (Tougard *et al.*, 2001) internal primer sets were designed using Primer 3 (Rozen and Skaletsky, 2000) to amplify small fragments (suitable for ancient DNA studies) of *Cytochrome B* (*Cyt B*). Primer-names and sequences can be found in Table 1; numbers relate to the position of these primers relative to GenBank



reference sequence X97336 (Xu *et al.*, 1996). Although the primers were designed to work on both *R. unicornis* and *R. sondaicus*, we selected the nucleotide found in *R. sondaicus* (FJ905815; Tougard *et al.*, 2001) for the minority of positions that were not identical. Initially we attempted to amplify the largest fragment (396 bp: 14613-F and 15008-R) followed by successively smaller fragments (186 bp: 14823-F and 15008-R; 74 bp: 14719-F and 14792-R), if PCRs would not succeed. PCR reactions were carried out on a PTC-200 thermocycler (MJ research) in 25  $\mu$ l volumes, using 2.5  $\mu$ l genomic DNA extract, 0.4  $\mu$ M of each primer, 1.5 mM MgCl<sub>2</sub> (in buffer), 0.2 mM dNTPs and 0.5  $\mu$ l Phire™ Hot Start DNA Polymerase (Finnzymes). The PCR thermal cycling profile started with 5 min. initial denaturation at 98°C, followed by 40 cycles of 5 sec. denaturation at 98°C, 20 sec. annealing at 60°C, 30 sec. extension at 72°C and a final extension of 1 min. at 72°C.

To establish the authenticity of ‘ancient’ DNA sequences, PCR products are generally cloned to distinguish between post-mortem changes (“damage induced errors”; Cooper A. and Poinair H.N., 2000) and genuine sequence data. Because the main objective of this study was identification of the 1865 specimen and *Cyt B* sequences of the Javan and Indian rhinoceros are sufficiently divergent (genetic distances are 8.6%, 10.9% and 15.4% for the 396, 186 and 74 bp fragments respectively; also see Tougard *et al.*, 2001 and Hsieh *et al.*, 2003), authenticity was not a significant issue here (as long as the recovered sequences would represent a species of *Rhinoceros*). To facilitate DNA sequencing only the smallest PCR product (74 bp) was cloned using a TOPO TA Clonig kit (Invitrogen) following the manufacturers instructions, albeit using only half of the prescribed reaction volumes. Colony PCRs were performed with the same primers (14719-F and 14792-R) and with M13 primers (M13/pUC-F and M13/pUC-R, see Table 1). All PCR products were sent to MacroGen Europe (Amsterdam) where they were purified with a Montage purification kit (Millipore) and sequenced in both directions (with the same PCR primers) on an ABI3730XL. The obtained forward and reverse sequences were assembled using Sequencher v.4.10.1 (GeneCodes Cooperation), checked for indels and stopcodons, and subsequently submitted to GenBank.

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**Figure 2.** Schematic overview of a) *Rhinoceros unicornis*, b) *Rhinoceros sondaicus* (both from Sody, 1959; used with permission) and a photo of c) the Dublin rhinoceros (†1865) as currently housed in the Trinity museum.

To display the juvenile Dublin rhinoceros as detailed as possible it is depicted roughly the same size as the Javan rhino. The relative size of *R. unicornis* compared to *R. sondaicus* is after Sody (1959).

---

**Table 1** Internal *Cytochrome B* primers designed for the genus *Rhinoceros* and the M13 primers used for colony PCR

| Primer name | Sequence (5'→ 3')        |
|-------------|--------------------------|
| 14613-F     | ATTACAAATCTCCTCTCAGCCATC |
| 14719-F     | TCCACTTCATCCTTCCCTTTATTA |
| 14792-R     | GGATCCTGTTTCGTGTAGGAATAG |
| 14823-F     | GACAAAATTCCATTTACCCCTTAC |
| 15008-R     | AGCAAATAGGAAATACCATTCTGG |
| M13/pUC-F   | GCCAGGGTTTTCCCAGTCACGA   |
| M13/pUC-R   | GAGCGGATAACAATTTACACAGG  |

RESULTS

The initial PCRs for the 396 bp fragment only worked for the right front toe sample. The amplification of the 187 bp fragment worked for the same sample as well as for the left rear toe sample (thus only skin samples). Of the skeleton samples only the sample from the pelvic region yielded an amplicon for the smallest (74 bp) fragment. The sequences have been deposited in GenBank under accession numbers JN935370-JN935374. Except for sequence JN935373 (colony PCR from the pelvic bone sample using the M13 primers; Table 1) which had one A to G substitution (corresponding to position 14753 in the reference sequence) the five obtained sequences were identical to each other as well as to the same *R. unicornis* reference sequence (X97336; Xu *et al.*, 1996).

DISCUSSION

Despite the fact that the DNA sequences unambiguously show that the 1865 Dublin Zoo rhinoceros represents *R. unicornis*, it is remarkable to see how very shallow (if visible at all) its skin folds are (especially in the neck area); this contrasts sharply with the prominent folds that are normally seen in this species. Also the skin is rather smooth hardly showing the large tubercles characteristic for the Great Indian. On the other hand the skin does not have the reticulate appearance typical for *R. sondaicus* either. From side view the tail does not stand out distinctly and indeed only the terminal portion was visible. The shape of the head is more difficult to appraise, especially since this specimen appears to be a juvenile. Compared to the rest of the body the head seems relatively small, also rather untypical for *R. unicornis*. The





**Figure 3.** Mounted skeleton most likely belonging to the Dublin rhinoceros (†1865).

upper lip is long and shaped into a point, seemingly more resembling *R. sondaicus* than *R. unicornis*. But if it is a juvenile male *R. unicornis*, the 1865 specimen would not have yet developed characteristics typical for adult males of this species, such as a pronounced “bib” and large horn. The overall size of the Dublin rhinoceros is unarguably small for a three year old Indian rhinoceros, even one that was malnourished (de Courcy, 2009 and references therein). Although the results of both skin and bone samples identify the specimen as *R. unicornis*, the *Cyt B* sequence alone does not provide enough information to genotype individuals. Together with the rather well documented history of this specimen, the results leave little doubt that the mounted skin and skeleton originate from the same specimen. Even though there is no evidence for lesions in ancient DNA that result in A to G substitutions (Stiller *et al.*, 2006; Lulueza-Fox *et al.*, 2009), sequence JN935373 provides yet another example that these substitutions are being observed. Nevertheless, we cannot rule out that these substitutions can be attributed to the type of polymerase (Stiller *et al.*, 2006) or reaction conditions used.

EUROPEAN COMMUNITY

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| ORIGINAL | 1. Holder                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>CERTIFICATE</b>                                                                                                                                                                                                                   |                                                                                                                                                               | No        |
|          | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Address                                                                                                                                                                                                                              | Not for use outside the European Community                                                                                                                    |           |
|          | Trinity College Dublin                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Zoology Department                                                                                                                                                                                                                   | ECR 022/10                                                                                                                                                    |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Trinity College Dublin<br>Dublin 2<br>Ireland                                                                                                                                                                                        | Council Regulation (EC) No 338/97 and Commission Regulation (EC) No 865/2006 on the protection of species of wild fauna and flora by regulating trade therein |           |
|          | 2. Authorised location for live specimens of Annex A species<br>N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3. Issuing Management Authority<br>Irish CITES Management Authority<br>National Parks & Wildlife Service<br>Department of Environment, Heritage & Local Government<br>7 Ely Place<br>Dublin 2<br>Ireland<br>e-mail: CITES@environ.ie |                                                                                                                                                               |           |
|          | 4. Description of specimens (incl. marks, sex/date of birth for live animals)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 5. Net mass (kg)                                                                                                                                                                                                                     | 6. Quantity                                                                                                                                                   |           |
|          | SKP<br>Small sample piece of Rhinoceros skin taken from circa 1860's taxidermy specimen, being sent to the Netherlands for DNA analysis to confirm species type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | N/A                                                                                                                                                                                                                                  | 1                                                                                                                                                             |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 7. CITES Appendix                                                                                                                                                                                                                    | 8. EC Annex                                                                                                                                                   | 9. Source |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | I                                                                                                                                                                                                                                    | A                                                                                                                                                             | W         |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10. Country of Origin<br>Unknown                                                                                                                                                                                                     |                                                                                                                                                               |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 11. Permit No                                                                                                                                                                                                                        | 12. Date of issue                                                                                                                                             |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | N/A                                                                                                                                                                                                                                  | N/A                                                                                                                                                           |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 13. Member State of import<br>N/A                                                                                                                                                                                                    |                                                                                                                                                               |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 14. Document No                                                                                                                                                                                                                      | 15. Date of issue                                                                                                                                             |           |
|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | N/A                                                                                                                                                                                                                                  | N/A                                                                                                                                                           |           |
|          | 16. Scientific name of species<br><i>Rhinoceros unicornis</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 17. Common name of species<br>Great Indian Rhinoceros                                                                                                                                                                                |                                                                                                                                                               |           |
|          | 18. It is hereby certified that the specimens described above:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|          | <input type="checkbox"/> were taken from the wild in accordance with the legislation in force in the issuing Member State<br><input type="checkbox"/> are abandoned or escaped specimens that were recovered in accordance with the legislation in force in the issuing Member State<br><input type="checkbox"/> are captive born-and-bred or artificially propagated specimens<br><input type="checkbox"/> were acquired in or introduced into the Community in compliance with the provisions of Council Regulation (EC) No 338/97<br><input type="checkbox"/> were acquired in or introduced into the Community before 1 June 1997 in accordance with Council Regulation (EEC) No 3626/82<br><input type="checkbox"/> were acquired in or introduced into the Community before 1 January 1984 in compliance with the provisions of CITES<br><input checked="" type="checkbox"/> were acquired in or introduced into the issuing Member State before the provisions of Regulations (EC) No 338/97 or (EEC) No 3626/82 or of CITES became applicable in this territory<br><input type="checkbox"/> are to be used for the advancement of science/breeding or propagation/research or education or other non-detrimental purposes |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|          | 19. This document is issued for the purpose of:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|          | <input type="checkbox"/> confirming that a specimen to be (re-) exported has been acquired in accordance with the legislation in force on the protection of the species in question<br><input checked="" type="checkbox"/> exempting Annex A specimens from the prohibitions relating to commercial activities listed in Article 8.1 of Regulation (EC) No 338/97<br><input type="checkbox"/> authorising the movement within the Community of a live Annex A specimen from the location indicated in the import permit or in any certificate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|          | 20. Special Conditions <u>Transaction Specific Certificate</u><br>This Certificate is valid solely for the purposes of one transaction. After this sample has been analysed in the Netherlands this Certificate will no longer be valid. (See section 4 above).<br><br><input type="checkbox"/> certificate valid only for holder named in box 1 (Issued under Article 48(1) (d) of Regulation (EC) No 865/2006)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |
|          | <div style="display: flex; justify-content: space-between; align-items: flex-end;"> <div> <p>Kevin Cahillane</p> <p>Name of issuing official</p> </div> <div> <p>Dublin, Ireland.</p> <p>10 March 2010</p> <p>Place and date</p> </div> <div>  <p>Signature and stamp</p> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                      |                                                                                                                                                               |           |

Annex I (d) - 30770-002



## CONCLUSION

Despite its small size and ambiguous morphology, the first rhinoceros owned by the Dublin Zoo was a specimen of *R. unicornis*, not *R. sondaicus*. Although this study unfortunately did not result in the discovery of another specimen of the increasingly threatened Javan rhinoceros (Brook, 2011), it shows the possibility of extracting genetic information from museum specimens and emphasizes the potential of these specimens as a tool in conservation of endangered species (Tracy and Jamieson, 2011). In cases where morphology based identifications are inconclusive, which (as shown here) could be due to poor preservation of collections specimens, molecular identifications can be of decisive importance.

## ACKNOWLEDGEMENTS

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## REFERENCES

- BROOK, S., VAN COEVERDEN DE GROOT, P., MAHOOD, S. & LONG, B. 2011. Extinction of the Javan Rhinoceros (*Rhinoceros sondaicus*) from Vietnam. WWF report.
- COOPER, A. & POINAR, H.N. 2000. Ancient DNA: Do it right or not at all. *Science* 289 (5482) p.1139.
- DE COURCY, C. 2009. The Dublin Zoo: an illustrated history. Harper Collins Publishers Limited, Cork. Pp.356. ISBN 978-1848890084.
- FOOSE, T.J. & VAN STRIEN, N.J. (1997) Asian Rhinos. Status Survey and Conservation Action Plan. IUCN/SSC Asian Rhino Specialist Group, IUCN, Gland, Switzerland.
- GROVES, C.P. & LESLIE jr., D.M. 2011. *Rhinoceros sondaicus* (Perissodactyla: Rhinocerotidae). *Mammalian species* 43 (887): 190-208.
- GUGGISBERG, C.A.W. 1966. S.O.S. rhino. Andre Deutsch, London.
- HAZEWINKEL, J.C. 1933. The one-horned Javanese rhinoceros found also in Sumatra. The illustrated London news. 1018-1019.
- HAUGHTON, S. 1867. On the muscular anatomy of the rhinoceros. *Proceedings of the Royal Irish Academy* 9: 515-526.

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**Figure 4.** CITES certificate which allowed the exportation of samples from the Dublin Zoo rhinoceros (†1865) from Ireland to the Netherlands for genetic analysis.

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- HSIEH, H.M., HUANG, L.H., TSAI, L.C., KUO, Y.C., MENG, H.H., LINACRE, A. & LEE, J.C. 2003. Species identification of rhinoceros horns using the *cytochrome b* gene. *Forensic Science International* 136, 1–11.
- LALUEZA-FOX, C., RÖMLER, H., CARAMELLI, D., STÄUBERT, C., CATALANO, G., HUGHES, D., ROHLAND, N., PILLI, E., LONGO, L., CONDEMI, S., DE LA RASILLA, M., FORTEA, J., ROSAS, A., STONEKING, M., SCHÖNEBERG, T., BERTRANPETIT, J. & HOFREITER, M. 2007. A melanocortin 1 receptor allele suggests varying pigmentation among Neanderthals. *Science* 318(5855): 1453–5.
- LYDEKKER, R. 1907. The game animals of India, Burma, Malaya, and Tibet, being a new and revised edition of 'The great and small game of India, Burma, and Tibet.' Rowland Ward, Limited, London, United Kingdom.
- POLET, G., TRAN, V.M., NGUYEN, X.D., BUI, H.M. & BALTZER, M. 1999. The Javan rhinoceros, *Rhinoceros sondaicus annamiticus*, of Cat Tien National Park, Vietnam: current status and management implications. *Pachyderm* 27, 34–48.
- ROHLAND, N. & HOFREITER, M. 2007. Ancient DNA extraction from bones and teeth. *Nature protocols* 2(7), p.1756–1762. doi:10.1038/nprot.2007.247
- ROOKMAAKER, L.C. 1993. The Mysterious "Liverpool Rhinoceros". *Der Zoologische Garten (N.F.)* 63, 4, p. 246–258.
- ROOKMAAKER, L.C., JONES, M.L., KLOES, H.G. & REYNOLDS, R.J. 1994. The rhinoceros in captivity: a list of 2439 rhinoceroses kept from Roman times to 1994. The Hague, SPB Academic Publishing.
- ROZEN, S. & SKALETSKY, H.J. 2000. Primer3 on the WWW for general users and for biologist programmers. In: Krawetz S, Misener S, editors *Bioinformatics Methods and Protocols: Methods in Molecular Biology*. Totowa, NJ: Humana Press. p. 365–386.
- SCLATER, P.L. 1874. Announcement of the arrival in the Society's menagerie of a Javan rhinoceros. *Proceedings of the Zoological Society of London* 1874:182–183.
- SODY, H.J.V. 1959. Das Javanische Nashorn *Rhinoceros sondaicus*, historisch und biologisch. *Zeitschrift für Säugtierkunde* 24(3–4).
- STILLER, M., GREEN, R.E., RONAN, M., SIMONS, J.F., DU, L., HE, W., EGHOLM, M., ROTHBERG, J.M., KEATES, S.G., OVODOV, N.D., ANTIPINA, E.E., BARYSHNIKOV, G.F., KUZMIN, Y.V., VASILEVSKI, A.A., WUENSCHALL, G.E., TERMINI, J., HOFREITER, M., JAENICKE-DESPRÉS, V. & PÄÄBO, S. 2006. Patterns of nucleotide misincorporations during enzymatic amplification and direct large-scale sequencing of ancient DNA. *Proceedings of the National Academy of Sciences USA* 103(37):13578–84.
- TOUGARD C., DELEFOSS T., HÄNNI C. & MONTGELARD C. 2001. Phylogenetic relationships of the five extant *Rhinoceros* species (Rhinocerotidae, Perissodactyla) based on mitochondrial *Cytochrome b* and 12S rRNA genes. *Molecular Phylogenetics and Evolution* 19(1), pp. 34–44.
- TRACY L.N. & JAMIESON I.G. 2011. Historic DNA reveals contemporary population structure results from anthropogenic effects, not pre-fragmentation patterns. *Conservation Genetics* 12(2), 517–526. DOI: 10.1007/s10592-010-0158-9



XU X., JANKE A. & ARNASON U. 1996. The complete mitochondrial DNA sequence of the greater Indian rhinoceros, *Rhinoceros unicornis*, and the Phylogenetic relationship among Carnivora, Perissodactyla, and Artiodactyla (+ Cetacea). *Molecular Biology and Evolution* 13(9):1167-73.



CHAPTER 9

**Ancient DNA elucidates the controversy  
about the flightless island hens  
(*Gallinula* sp.) of Tristan da Cunha**

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DEKKER R.W.R.J

GITTENBERGER E.

## ABSTRACT

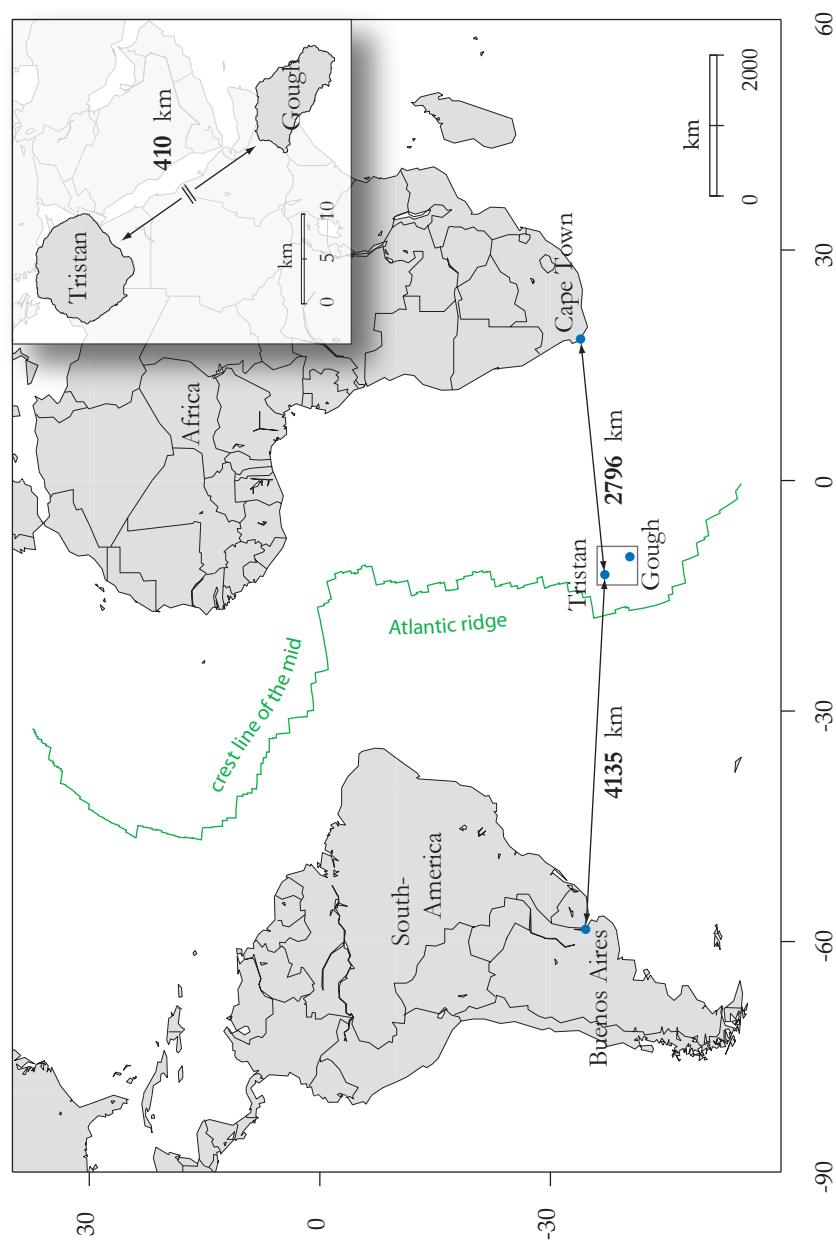
A persistent controversy surrounds the flightless island hen of Tristan da Cunha, *Gallinula nesiotis*. Some believe that it became extinct by the end of the 19th century. Others suppose that it still inhabits Tristan. There is no consensus about *Gallinula comeri*, the name introduced for the flightless moorhen from the nearby island of Gough. On the basis of DNA sequencing of both recently collected and historical material, we conclude that *G. nesiotis* and *G. comeri* are different taxa, that *G. nesiotis* indeed became extinct, and that *G. comeri* now inhabits both islands. This study confirms that among gallinules seemingly radical adaptations (such as the loss of flight) can readily evolve in parallel on different islands, while conspicuous changes in other morphological characters fail to occur.

## INTRODUCTION

Until recently it was assumed that the flightless moorhen of remote Tristan da Cunha in the southern Atlantic (Fig. 1), *Gallinula nesiotis* (Sclater, 1861), became extinct by the end of the 19th century (Beintema, 1972). A few decades after its description, a very similar moorhen that was also flightless namely *G. comeri* (Allen, 1892), was described from the island of Gough, ca. 400 km SE of Tristan. In the period between these descriptions *G. nesiotis* became rare (Milner and Brierly, 1869; Sperling, 1872) and by the turn of the century it had probably gone extinct (Nicoll, 1906; Nicoll, 1908). Authentic remnants are two skins and a skeleton in the Natural History Museum, Tring (Knox and Walters, 1994). Since unequivocal *G. nesiotis* had been collected only once from Tristan, and because of the presence of a healthy population of similar moorhens on the nearby island of Gough, some authors doubted whether an endemic moorhen had ever existed on Tristan (Broekhuysen and Macnae, 1949). Eber (Eber, 1961) compared Sclater's description of *G. nesiotis* from Tristan with her series of *G. comeri* from Gough and concluded that the differences fall within the range of variation of the latter. In her opinion it was very unlikely that moorhens from two islands in the same region would have independently lost the ability of flight, without differentiating in other characters. She suggested that Sclater's material might have been labelled inaccurately and that his specimens in fact also came from Gough. Consequently, Eber considered *G. comeri* a junior synonym of *G. nesiotis* (Eber, 1961) and controversy surrounded future illustrations of both taxa (Fig. 2).

Beintema (Beintema, 1972) mentioned that there are old records of moorhens for Tristan, demonstrating that such birds were truly indigenous there. In his view, skeletal measurements differ slightly between *G. nesiotis* and *G. comeri*. Furthermore, rails are known to rapidly lose the ability of flight as soon as they arrive on remote islands (Slikas, Olson and Fleischer, 2002). When in 1972 live moorhens were discovered on Tristan (Richardson, 1984), these birds were regarded as descendants





**Figure 1.** Map showing the location of Tristan da Cunha and Gough in the Mid Atlantic.



of a small number of individuals brought from Gough (Taylor and van Perlo, 1998). Alternatively, Beintema suggested that *G. nesiotis* might have been temporarily rare on Tristan, but not extinct, and that the moorhens found there today are descendents of the original island population. Here we address this question, making use of DNA analyses of authentic material of *G. nesiotis*, recent specimens of the moorhens from both Tristan and Gough and some geographically and taxonomically close other taxa of moorhens.

## MATERIAL AND METHODS

### *Taxa*

Tissues from fourteen gallinules and a coot were put at our disposal by various institutes. These include tissues of (I) 'recently' collected moorhens from Tristan da Cunha, (II) moorhens from Gough from the collection ZMA, (III) the 1864 specimen of *G. nesiotis* from the Natural History Museum, Tring, and (IV) a number of subspecies of *Gallinula chloropus* from South-America, Africa, Europe, Taiwan and Java from the National Museum of Natural History Naturalis, Leiden. The coot, *Fulica atra*, was used as outgroup. An overview of the used specimens, with taxon names, locality data, and year of acquisition, is given in Table 1.

### *DNA extraction*

DNA extractions on specimens from 1968 (Table 1) and older were carried out in a dedicated ancient DNA (aDNA) facility (LAF, Leiden, the Netherlands), which is physically isolated from the main laboratories. Before extractions took place, the extraction room was cleaned with a 0.05% bleach solution and the extraction-cabinet was decontaminated by turning on the UV lights at least 1 hour prior to the start of the extractions. No more than four extractions were done at once and negative controls were included with each set of extractions. Pipettes were cleaned with bleach and subsequently decontaminated (together with the disposables) by UV irradiation (UV linker). Tissues were cut into small pieces to enlarge the contact surface between tissue and buffer. Total genomic DNA was extracted with a DNeasy Tissue Kit (Qiagen) using a prolonged incubation (24 hours). Proteinase K was added twice, once at the start and after 6 hours of incubation. To concentrate the extract, elution volume was decreased to 40 µl. Extractions on recently collected specimens (1993-2005) were done in a common lab, also using a DNeasy Tissue Kit (Qiagen).

### *PCR and sequencing*

PCRs were never performed in the aDNA facility and amplicons were never stored in this building. PCRs on the extract of the 1864 specimen were duplicated in different laboratories that are physically separated from each other as well as from the aDNA facility. In none of these labs had ever been worked on any species of gallinule before. Fragments from three non-adjacent mitochondrial gene regions (679 basepairs in total; primersites excluded) were amplified by PCR: the D-loop, tRNA-Lysine/ATP8 and cytochrome *b*. The length of these fragments (primer sites included) was





**Figure 2.** Illustrative stamps, issued in 1987 and 2005.

(A) 2005: Text and illustrations belong together and are correct. (B) 1987: In Gough *G. comeri* occurs, not *G. nesiotis*; both names should not be synonymized. (C) 2005: The text correctly indicates *G. nesiotis* as from Tristan, but the bird itself most probably belongs to *G. comeri*, introduced from Gough, since *G. nesiotis* is now extinct on Tristan and not available to be pictured anymore.

234, 236 and 375 bp, respectively. Primer sequences and references are described in Table 2. For *G. nesiotis* and (most) specimens of 1960 and older, cytochrome *b* could not be amplified directly using primers L14841 and H15149 (Kocher, *et al.*, 1989). Presumably because the DNA within these specimens got too degraded over time. Therefore, internal primers were designed (Table 2) to amplify this fragment in two overlapping parts: L14841- Rev219 (219 bp) and Fwd141- H15149 (249 bp). The primers for the D-loop (CR-OU-D-F and CR-OU-D-R) are ‘gallinule-specific’. For the coot, *Fulica atra*, the same region had to be amplified with other primers: CR-175-F and 12S-29-R (Table 2). PCRs were done using a standard Taq DNA Polymerase kit (Qiagen). Reaction volume was 25  $\mu$ l and PCR conditions were 0.4  $\mu$ M of each primer, 0.2 mM dNTP’s and 5 units of Taq DNA Polymerase. For amplification of



**Table 1.** Taxa and collection information.

| Species                              | Locality          | Registration number | Year | Institute |
|--------------------------------------|-------------------|---------------------|------|-----------|
| (1) <i>G. comeri</i>                 | Tristan da Cunha  | AB1759              | 1993 | AJB *     |
| (2) <i>G. comeri</i>                 | Tristan da Cunha  | AB1760              | 1993 | AJB *     |
| (3) <i>G. comeri</i>                 | Gough Island      | ZMA 14695 (no. 7)   | 1960 | ZMA *     |
| (4) <i>G. comeri</i>                 | Gough Island      | ZMA 14696 (no. 14)  | 1960 | ZMA *     |
| (5) <i>G. nesiotis</i>               | Tristan da Cunha  | 1864.7.30.1         | 1864 | BM *      |
| (6) <i>G. chloropus galeata</i>      | Suriname          | RMNH 53835          | 1968 | NNM *     |
| (7) <i>G. chloropus galeata</i>      | Suriname          | RMNH 53836          | 1968 | NNM *     |
| (8) <i>G. chloropus brachyptera</i>  | S.W. Africa       | Cat. no. 21         | 1867 | NNM *     |
| (9) <i>G. chloropus brachyptera</i>  | S.W. Africa       | Cat. no. 22         | 1867 | NNM *     |
| (10) <i>G. chloropus brachyptera</i> | Tanzania          | RMNH 43858          | 1965 | NNM *     |
| (11) <i>G. chloropus orientalis</i>  | Cheribon, Java    | RMNH 26803 (no. 94) | 1925 | NNM *     |
| (12) <i>G. chloropus indica</i>      | Chang Hwa, Taiwan | RMNH 53054 (no. 12) | 1968 | NNM *     |
| (13) <i>G. chloropus chloropus</i>   | The Netherlands   | DG2073              | 2005 | NNM *     |
| (14) <i>G. chloropus chloropus</i>   | The Netherlands   | DG2077              | 2005 | NNM *     |
| (15) <i>Fulica atra</i>              | The Netherlands   | DG2071              | 2005 | NNM *     |

\* AJB = personal collection A.J. Beintema, ZMA = Zoologisch Museum Amsterdam, BM = Natural History Museum Tring, NNM = Nationaal Natuurhistorisch Museum, Naturalis.

the cytochrome *b* and ATP8 regions, the final concentration of  $\text{MgCl}_2$  was 2.5 mM. For amplification of the D-loop fragment no  $\text{MgCl}_2$  was added (1.5 mM was already in the Qiagen PCR-buffer). Thermocycling conditions were 3 min. at 94°C (initial denaturation), followed by 40 cycles (15 sec. at 94°C, 30 sec. at AT°C and 40 sec. at 72°C) and final extension 5 min. at 72°C. Where AT is the annealing temperature for each primer set; 50°C for both cytochrome *b* and ATP8, 55°C for the D-loop fragment and 57°C for reamplification of cloned products (see next paragraph).

All PCR products from *G. nesiotis* and a number of PCR products from the D-loop of selected taxa (Table 3) were cloned using either pGEM®-T Easy Vector Systems from Promega or Topo TA Cloning® from Invitrogen. At least three colonies were picked per plate and used to initiate reamplifications ("colony PCRs") with primers 21M13\_F and 21M13\_R (Table 4). Reamplified products were cleaned using a Nucleospin® kit (Macherey-Nagel). Subsequently these products were sequenced either in-house on a Megabace™ 1000 DNA Analysis System (Amersham), or on a 3730xl DNA analyzer (Applied Biosystems) at Macrogen Inc. (Korea) using only primer 21M13\_F. All other PCR products were cleaned (same procedure) and



**Table 2.** PCR and sequencing primers.

| Primer name | Primer sequence (5' to 3')          | Target                           | Reference              |
|-------------|-------------------------------------|----------------------------------|------------------------|
| L14841      | AAAAAGCTTCCATCCAACATCTCAGCATGATGAAA | cytochrome <i>b</i>              | Kocher, 1989           |
| H15149      | AAACTGCAGCCCCTCAGAATGATATTTGCCTCA   | cytochrome <i>b</i>              | Kocher, 1989           |
| Fwd141      | CCACACATGCCGCAACGTACAATA            | cytochrome <i>b</i>              | This study             |
| Rev219      | GCAGATGAAGAAGAATGAGGCTCC            | cytochrome <i>b</i>              | This study             |
| L9051       | CAGCACTAGCCTTTTAAG                  | tRNA-Lys / ATP8                  | Slikas, 2002           |
| H9241       | TTGGTCGAAGAAGCTTAGGTTCA             | tRNA-Lys / ATP8                  | Slikas, 2002           |
| CR-OUd-F    | CCAAGTGTTAATAGTATATGAGCTTACTCC      | D-loop                           | This study             |
| CR-OUd-R    | TGATACATTTTGATTGTTTGGTATGAA         | D-loop                           | This study             |
| CR-175-F    | GAGCATACTATTGGTTGACGTGAG            | D-loop                           | This study             |
| 12S-29-R    | TTTCACTGGAGTGCGGATACTTGCAT          | D-loop                           | This study             |
| 21M13_F     | TGTAACACGACGGCCAGT                  | pCR 2.1-TOPO<br>M13 priming site | TOPO TA<br>Cloning kit |
| 21M13_R     | CAGGAAACAGCTATGACC                  | pCR 2.1-TOPO<br>M13 priming site | TOPO TA<br>Cloning kit |

**Table 3.** Number of colonies sequenced per target region per taxon.

|      | Taxon                    | Year | D-loop   | tRNA-Lysine / Atp8 | cytochrome <i>b</i> |
|------|--------------------------|------|----------|--------------------|---------------------|
| (3)  | <i>G. comeri</i>         | 1960 | 4        | -                  | -                   |
| (5)  | <i>G. nesiotis</i>       | 1864 | 8 [5/3]* | 14 [7/7]*          | 23 [6/5-6/6]*       |
| (6)  | <i>G. c. galeata</i>     | 1968 | 3        | -                  | -                   |
| (7)  | <i>G. c. galeata</i>     | 1968 | 4        | -                  | -                   |
| (9)  | <i>G. c. brachyptera</i> | 1867 | 5        | -                  | -                   |
| (10) | <i>G. c. brachyptera</i> | 1965 | 3        | -                  | -                   |
| (11) | <i>G. c. orientalis</i>  | 1925 | 5        | 4                  | 7 [7-0]*            |
| (12) | <i>G. c. indica</i>      | 1968 | 4        | 5                  | 7 [5-2]*            |

\* Within square brackets are the number of colonies sequenced for each PCR product. Of taxa numbers (1), (2), (4), (8), (13), (14) and (15) the PCR products were not cloned, but sequenced directly.



sequenced directly (both directions) with their respective PCR primers (Table 2). A summary of the specimens and the number of colonies sequenced per target region is given in Table 3. Sequences were assembled using Sequencher version 4.2 (Gene Codes Corporation) and aligned manually using MacClade version 4.08 (Maddison and Maddison, 2005). The sequences were deposited in GenBank (accession numbers EF681971-EF682015).

### *Phylogenetic analysis*

For phylogenetic analyses, all sequences (all regions; consensus sequences when products were cloned) were put in a single datamatrix (Dataset S4; an ILD-test showed no incongruence between the regions,  $p = 0.971$ ) and *Fulica atra* was designated as outgroup. To get branch support values, we performed phylogenetic analyses with three methods: Neighbour-Joining and Maximum Likelihood using PAUP version 4.0b2a (Swofford, 1998) and Bayesian analysis with Mr.Bayes version 3.1.2 (Ronquist and Huelsenbeck, 2003). For the NJ analysis, we performed a bootstrap analysis (1000 replicates, optimality criterion set to distance) and calculated a 50% majority rule consensus cladogram (Fig. 3). For the ML analysis, the HKY+G model was selected by Modeltest ver. 3.7 (Posada and Crandall, 1998) with the following parameters: Tratio = 10.660, gamma shape parameter = 0.0941, base frequencies A = 0.3473, C = 0.3004, G = 0.1341, T = 0.2182 and proportion of invariable sites (pinvar) = 0. A bootstrap analysis (1000 replicates, 5 random additions per bootstrap replicate and TBR branch swapping) was performed and a 50% majority rule consensus tree was calculated (Fig. 4). For the MrBayes analysis, the best-fit model for each partition (four partitions: tRNA-Lysine and ATP8 were considered as two partitions) was selected by hLRT in MrModeltest ver. 2.2 (Nylander, 2004): D-loop (HKY+Y), tRNA-lysine (HKY), ATP8 (GTR+I) and cytochrome *b* (GTR+I). A dirichlet (1,1,1,1) prior was specified on the state frequencies for all partitions, except for the tRNA-Lysine partition, where the frequencies were equal. All partitions had different rates for transition and transversions (nst = 2), except for tRNA-Lysine (nst = 1). Among-site rate variation was equal for tRNA-Lysine, gamma-distributed for both D-loop and ATP8 and for cytochrome *b* a proportion of the sites was invariant. Two runs (set up for 10 000 000 generations) were performed simultaneously (4 chains per run) in MrBayes ver. 3.1.2 (Posada and Crandall, 1998) and the convergence diagnostic was set to 0.009. A Markov chain Monte Carlo (MCMC) analysis was done with swapfreq = 2, temp = 0.002 and samplefreq = 100; convergence was reached after 165 000 generations. The trees of both runs (3302 in total) were combined (2702: burnin was set to 300) and a 50% majority rule consensus tree (contype = halfcompat) was calculated (Fig. 5).

### *Genetic distances.*

Genetic distances (absolute number of changes and uncorrected “p” distances) were calculated with Paup ver. 4.0b2a (Swofford, 1998) based on the combined dataset (Dataset S4; see Supporting informationN).



**Table 4.** Genetic distances.

|                          | comeri (1) | comeri (2)  | comeri (3)  | comeri (4)  | nesiotis (5) | galeata (6) | galeata (7) |
|--------------------------|------------|-------------|-------------|-------------|--------------|-------------|-------------|
| <i>G. comeri</i>         | (1)        |             |             |             |              |             |             |
| <i>G. comeri</i>         | (2)        | 0 / 0.000   |             |             |              |             |             |
| <i>G. comeri</i>         | (3)        | 0 / 0.000   | 0 / 0.000   |             |              |             |             |
| <i>G. comeri</i>         | (4)        | 0 / 0.000   | 0 / 0.000   | 0 / 0.000   |              |             |             |
| <i>G. nesiotis</i>       | (5)        | 21 / 0.031  | 21 / 0.031  | 21 / 0.031  |              |             |             |
| <i>G. c. galeata</i>     | (6)        | 40 / 0.059  | 40 / 0.059  | 40 / 0.059  | 41 / 0.061   |             |             |
| <i>G. c. galeata</i>     | (7)        | 41 / 0.060  | 41 / 0.060  | 41 / 0.060  | 40 / 0.059   | 3 / 0.004   |             |
| <i>G. c. brachyptera</i> | (8)        | 20 / 0.035  | 20 / 0.035  | 20 / 0.035  | 24 / 0.042   | 42 / 0.073  | 40 / 0.070  |
| <i>G. c. brachyptera</i> | (9)        | 22 / 0.038  | 22 / 0.038  | 22 / 0.038  | 24 / 0.042   | 44 / 0.076  | 42 / 0.073  |
| <i>G. c. brachyptera</i> | (10)       | 22 / 0.032  | 22 / 0.032  | 22 / 0.032  | 25 / 0.037   | 43 / 0.063  | 42 / 0.062  |
| <i>G. c. orientalis</i>  | (11)       | 19 / 0.033  | 19 / 0.033  | 19 / 0.033  | 21 / 0.037   | 40 / 0.070  | 40 / 0.070  |
| <i>G. c. indica</i>      | (12)       | 21 / 0.031  | 21 / 0.031  | 21 / 0.031  | 23 / 0.034   | 46 / 0.068  | 45 / 0.066  |
| <i>G. c. chloropus</i>   | (13)       | 21 / 0.031  | 21 / 0.031  | 21 / 0.031  | 24 / 0.035   | 44 / 0.065  | 43 / 0.063  |
| <i>G. c. chloropus</i>   | (14)       | 22 / 0.032  | 22 / 0.032  | 22 / 0.032  | 25 / 0.037   | 44 / 0.065  | 43 / 0.063  |
| <i>F. atra</i>           | (15)       | 104 / 0.154 | 104 / 0.154 | 104 / 0.154 | 100 / 0.148  | 93 / 0.138  | 94 / 0.139  |

|                          | brachypt. (8) | brachypt. (9) | brachypt. (10) | orientalis (11) | indica (12) | chlorop. (13) | chlorop. (14) |
|--------------------------|---------------|---------------|----------------|-----------------|-------------|---------------|---------------|
| <i>G. c. brachyptera</i> | (9)           | 6 / 0.010     |                |                 |             |               |               |
| <i>G. c. brachyptera</i> | (10)          | 0 / 0.000     | 6 / 0.010      |                 |             |               |               |
| <i>G. c. orientalis</i>  | (11)          | 17 / 0.030    | 15 / 0.026     | 17 / 0.030      |             |               |               |
| <i>G. c. indica</i>      | (12)          | 5 / 0.009     | 7 / 0.012      | 5 / 0.007       | 18 / 0.031  |               |               |
| <i>G. c. chloropus</i>   | (13)          | 1 / 0.002     | 5 / 0.009      | 1 / 0.001       | 16 / 0.028  | 4 / 0.006     |               |
| <i>G. c. chloropus</i>   | (14)          | 4 / 0.007     | 4 / 0.007      | 4 / 0.006       | 15 / 0.026  | 5 / 0.007     | 3 / 0.004     |
| <i>F. atra</i>           | (15)          | 93 / 0.163    | 91 / 0.160     | 103 / 0.152     | 94 / 0.166  | 105 / 0.155   | 103 / 0.152   |

Cell values show: absolute number of changes / uncorrected “p” distances.



#### *Pairwise relative rates test.*

With *Fulica atra* specified as outgroup, a Pairwise Relative Rate Test (Muse and Weir, 1992) as implemented in HyPhy (Pond, Frost and Muse, 2005) using the HKY model (as specified by Modeltest ver. 3.7 (Posada and Crandall, 1998) was performed on the combined dataset (S4; Supporting Information).

## RESULTS

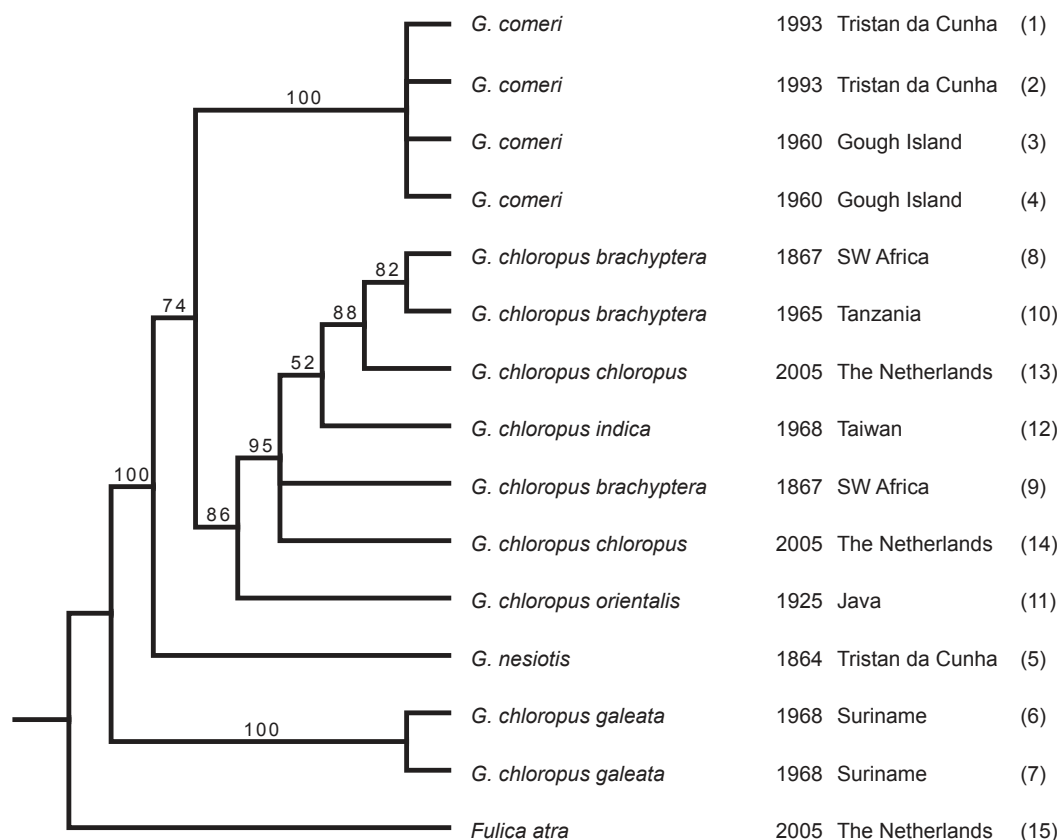
Alignments of all cloned sequences of *G. nesiotis* (two independent amplifications per target region) are shown in Dataset S1-3 (Supporting informationN). On the one hand, none of the sequences of genuine, historical *G. nesiotis* was identical to those of *G. comeri* and, on the other hand, all sequences of the moorhens collected in Tristan da Cunha in 1993, are identical to those found for specimens of *G. comeri* from Gough, dated 1960. The genetic distances between *G. nesiotis* and the other gallinules, are of the same magnitude as the distances between *G. comeri* and the other gallinules (Table 4). A pairwise relative rates test did not reveal significantly different substitution rates for any of the lineages ( $P > 0.18$ ). Of the selected markers, most variation was detected in the control region (D-loop) sequences, viz. 9.6% total sequence heterogeneity (TSH) for *G. nesiotis* compared with *G. comeri* (Binladen, *et al.*, 2006; Gilbert, *et al.*, 2003) versus 2.1% and 0.3% TSH for tRNA-Lys / ATP synthase subunit 8 (ATP8) and cytochrome *b*, respectively.

The results of phylogenetic analyses (Neighbour-Joining, Maximum Likelihood and Bayes) based on a combined dataset (all taxa, all regions, Dataset S4) are shown in Fig. 3-5. In these cladograms *Gallinula nesiotis* and *G. comeri* form a clade with the moorhens of Africa/Eurasia, whereas the other taxa that were investigated are less closely related.

## DISCUSSION

Our results show that genuine *G. nesiotis*, identified on the basis of historical material from the island of Tristan da Cunha, differs genetically from *G. comeri*, which has been described from the island of Gough. For each marker the sequence of *G. nesiotis* differs from that of *G. comeri*, as well as from all other *Gallinula* taxa that were analysed, but the amount of variation differs strongly between the regions studied. The position of *G. nesiotis* on the cladograms (Fig. 3-5) makes sense biologically. Apparently, *G. nesiotis* became extinct on Tristan and *G. comeri* from Gough was introduced there, resulting in the current situation with *G. comeri* occurring on both islands. This implies that modern illustrations of so-called *G. nesiotis* from Tristan (Fig. 2) probably show introduced *G. comeri* from Gough.

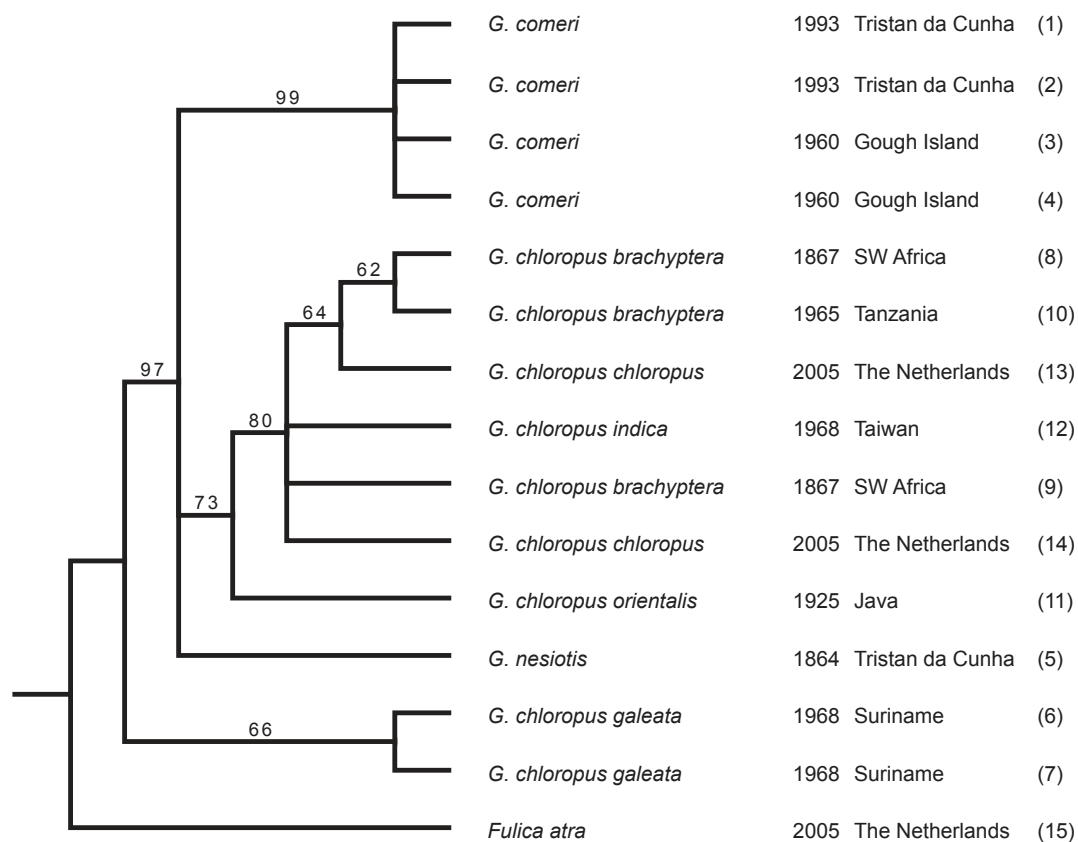




**Figure 3.** Bootstrap 50% majority rule consensus NJ tree. Values indicate bootstrap support.

The difference between *G. nesiotis* and *G. comeri* is most conspicuous in the D-loop sequence. This is a marker from a non-coding region, which makes it more difficult to exclude it as a potential pseudogene (Perna and Kocher, 1996). In some cases preferential amplification of numt (nuclear mitochondrial insertion) sequences has been observed (Collura and Stewart, 1995; Sorenson and Quinn, 1998), but in most ancient DNA studies only occasional co-amplification of numts has been reported (Kolokotronis, MacPhee and Greenwood, 2007; Orlando, *et al.*, 2003). This is not surprising since (particularly) in ancient DNA samples, mitochondrial DNA will be in excess over nuclear DNA. Consequently the incidence of numts should be reduced in such samples. To minimize the chance of amplifying numts, we did not use blood as a source for DNA-extractions in our recently collected material (Quinn, 1992; Sorenson and Quinn, 1998); in the older specimens it was not possible anyway.

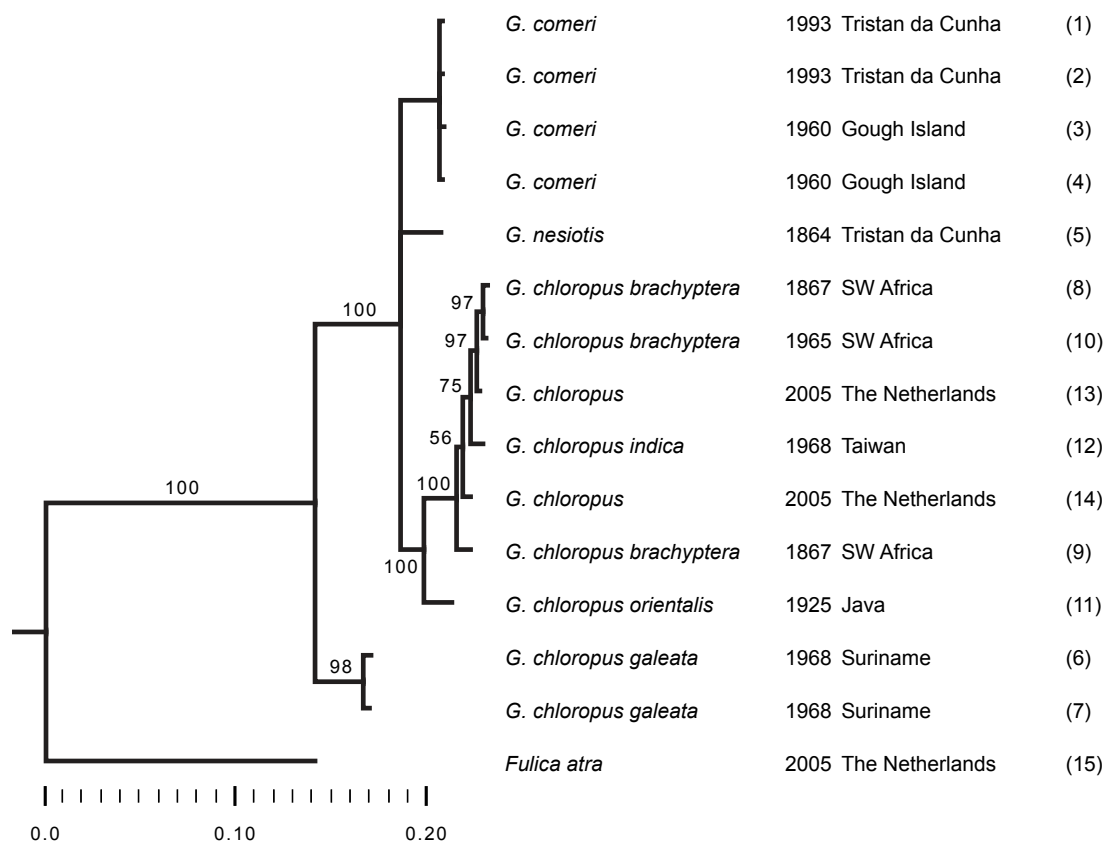




**Figure 4.** Bootstrap 50% majority rule consensus ML tree. Values indicate bootstrap support.

Because ‘universal’ primers may also be particularly prone to amplification of numts (Sorenson and Quinn, 1998), the primers for the D-loop were made ‘gallinule-specific’. They did not even work for the closely related coot, *Fulica atra*. Products that were sequenced directly (both strands) showed only one signal, whereas multiple signals can be expected if both the target product and a numt would have been amplified. Interclone variation was low (Dataset S1-3; Supporting informationN). Only one sequence (*G. nesiotis*, marker ATP8) out of 96 clones (Table 3) could clearly be identified as a numt (Dataset S2; Supporting informationN). No stop-codons or frame-shift mutations were observed for the coding-region datasets (ATP8 and cytochrome *b*). No obvious deviations in either substitution rate (pairwise relative rate test) or base composition, like a decrease in GC content (Bensasson, *et al.*, 2001; Casane, *et al.*, 1997), were observed.





**Figure 5.** Bayes 50% majority rule consensus ('halfcompat') phylogram (branch lengths are proportional to the expected number of substitutions). Values indicate branch support by Bayesian inference.

All sequences of recently collected moorhens from Tristan were identical to those of *G. comeri* from Gough and should be considered conspecific therefore. Cross contamination is very unlikely, since specimens from Tristan and Gough were amplified in different PCR-batches and contamination was not detected in other, partly much older specimens. Most probably the sequences are identical because *G. comeri* was introduced only recently on Tristan. Genetic variation within island populations is generally small compared to mainland populations (Eldridge, *et al.*, 1999; Frankham, 1997; Frankham, 1998). For example, the giant tortoises (*Aldabrachelys*) of Mahé (Seychelles) and Mauritius (Mascare islands) still have identical sequences compared to those of Aldabra, from where they were shipped since the 1820s or earlier (Austin, Arnold and Bour, 2003). *G. comeri* may have been



introduced on Tristan somewhere in the mid 1950s (Richardson, 1984). Assuming a generation time of two years, as known for *Gallinula chloropus*, *G. comeri* would only have had about only 20 generations (40 years) to differentiate on Tristan.

Both the genetic distances (Table 4) and the fact that *G. nesiotis* and *G. comeri* form a clade with the investigated moorhens of Africa/Eurasia (Fig. 3-5), suggest that the ancestor(s) of these island gallinules originated from Africa and not America, as suggested by Eber (Eber, 1961). Our data do not allow us to distinguish between a single dispersal event to the archipelago, followed by allopatric differentiation, or two separate introductions from the continent to both Tristan and Gough.

As is inevitably the case with isolated island populations, the question of whether *G. nesiotis* and *G. comeri* were reproductively isolated under natural circumstances cannot be answered. Our limited data from a small number of specimens and sequences of only the mitochondrial lineage are insufficient to demonstrate hybridisation. Even though island populations generally show lower genetic variation than related mainland populations (Frankham, 1997), the genetic distances between *G. nesiotis* and *G. comeri* are of at least the same magnitude as those found between taxa that figure as subspecies of *G. chloropus* in the literature (Fig. 5, Table 4). Therefore, we propose that the extinct moorhen of Tristan and the moorhens that live on Gough and Tristan today be regarded as subspecies, viz. *G. n. nesiotis* and *G. n. comeri*, respectively. This is in conformity with two recent, general checklists of the birds of the world (del Hoyo, Elliott and Sargatal, 1996; Dickinson, 2003) and a detailed monograph of the rails of the world (Taylor and van Perlo, 1998), but is different from a morphological study by Eber (Eber, 1961) in which both taxa are considered synonyms.

## SUPPORTING INFORMATION\*

**Dataset S1** Cloning results of *G. nesiotis* for ATP8.  
doi:10.1371/journal.pone.0001835.s001

**Dataset S2** Cloning results of *G. nesiotis* for D-loop.  
doi:10.1371/journal.pone.0001835.s002

**Dataset S3** Cloning results of *G. nesiotis* for cytochrome b.  
doi:10.1371/journal.pone.0001835.s003

**Dataset S4** Combined dataset showing all taxa and all markers (D-loop, ATP8 and cytochrome b) used in this study.  
doi:10.1371/journal.pone.0001835.s004

\* available online; datasets are served from the PLoS ONE website.



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## REFERENCES

- ALLEN, J.A. 1892 Description of a new Gallinule from Gough island. *Bulletin of the American Museum of Natural History* 4: 57-58.
- AUSTIN, J.J., ARNOLD, E.N. & BOUR, R. 2003 Was there a second adaptive radiation of giant tortoises in the Indian Ocean? Using mitochondrial DNA to investigate speciation and biogeography of Aldabrachelys (Reptilia, Testudinidae). *Molecular Ecology* 12: 1415-1424.
- BEINTEMA, A.J. 1972 The history of the Island Hen (*Gallinula nesiotis*), the extinct flightless gallinule of Tristan da Cunha. *Bulletin of the British Ornithologists' Club* 92: 106-113.
- BENSASSON, D., ZHANG, D.-X., HARTL, D.L. & HEWITT, G.M. 2001 Mitochondrial pseudogenes: evolution's misplaced witnesses. *Trends in Ecology and Evolution* 16: 314-321.
- BINLADEN, J., WIUF, C., GILBERT, M.T.P., BUNCE, M., BARNETT, R., LARSON, G., GREENWOOD, A.D., HAILE, J., HO, S.Y.W., HANSEN, A.J. & WILLERSLEV, E. 2006 Assessing the fidelity of ancient DNA sequences amplified from nuclear genes. *Genetics* 172: 733-741.
- BROEKHUYSEN, G.J. & MACNAE, B. 1949 Observations on the birds of Tristan da Cunha Islands and Gough Island in February and early March, 1948. *Ardea*: 97-113.
- CASANE, D., BOISSINOT, S., CHANG, B.H.J., SHIMMIN, L.C. & LI, W.H. 1997 Mutation Pattern Variation Among Regions of the Primate Genome. *Journal of Molecular Evolution*, 45: 216-226.
- COLLURA, R.V. & STEWART, C.B. 1995 Insertion and duplications of mtDNA in the nuclear genomes of Old World monkeys and hominoids. *Nature* 378: 485-489.
- DEL HOYO, J., ELLIOTT, A. & SARGATAL, J. 1996 *Handbook of the birds of the world* 3. Lynx, Barcelona.
- DICKINSON, E.C. 2003 *The Howard and Moore complete checklist of the birds of the world*. Christopher Helm, London.
- EBER, G. 1961 Vergleichende untersuchungen am flugfähigen Teichhuhn *Gallinula chl. chloropus* und an der flugunfähigen Inselralle *Gallinula nesiotis*. *Bonner Zoologische Beiträge* 12: 247-315.



- ELDRIDGE, M.D.B., KING, J.M., LOUPIS, A.K., SPENCER, P.B.S., TAYLOR, A.C., POPE, L.C. & HALL, G.P. 1999 Unprecedented Low Levels of Genetic Variation and Inbreeding Depression in an Island Population of the Black-Footed Rock-Wallaby. *Conservation Biology* 13: 531-541.
- FRANKHAM, R. 1997 Do island populations have less genetic variation than mainland populations? *Heredity* 78: 311-327.
- FRANKHAM, R. 1998 Inbreeding and Extinction: Island Populations. *Conservation Biology* 12: 665-675.
- GILBERT, M.T.P., WILLERSLEV, E., HANSEN, A.J., BARNES, I., RUDBECK, L., LYNNERUP, N. & COOPER, A. 2003 Distribution patterns of postmortem damage in human mitochondrial DNA. *American Journal of Human Genetics* 72: 32-47.
- KNOX, A.G. & WALTERS, M.P. 1994 Extinct and endangered birds in the collections of the Natural History Museum. *British Ornithologist's Club Occasional Publications* 1.
- KOCHER, T.D., W.K. THOMAS, A. MEYER, S.V. EDWARDS, S. PAABO, F.X. VILLABLANCA & WILSON, A.C. 1989 Dynamics of Mitochondrial-DNA Evolution in Animals - Amplification and Sequencing with Conserved Primers. *Proceedings of the National Academy of Sciences of the USA* 86: 6196-6200.
- KOLOKOTRONIS, S.O., MACPHEE, R.D.E. & GREENWOOD, A.D. 2007 Detection of mitochondrial insertions in the nucleus (NuMts) of Pleistocene and modern muskoxen. *BMC Evolutionary Biology* 7: 67. DOI:10.1186/1471-2148-7-67
- MADDISON, D.R. & MADDISON, W.P. 2005 MacClade 4: Analysis of phylogeny and character evolution. Version 4.08. In). Sunderland, Massachusetts.
- MILNER, J. & BRIERLY, O.W. 1869 *The Cruise of H.M.S. 'Glatea', Captain H.R.H. the Duke of Edinburgh, K.G., in 1867-1868*. W.H. Allen, London.
- MUSE, S.V. & WEIR, B.S. 1992 Testing for equality of evolutionary rates. *Genetics* 132: 269-276.
- NICOLL, M.J. 1906 On the Birds collected and observed during the Voyage of the 'Valhalla'. *Ibis* 8: 666-712.
- NICOLL, M.J. 1908 *Three Voyages of a Naturalist; being an Account to Many Little-Known Islands in the Three Oceans visited by the 'Valhalla'*. R.Y.S. Witherby & Co., London.
- NYLANDER, J.A.A. 2004 MrModeltest 2.2. Program distributed by the author. In). Uppsala University: Evolutionary Biology Centre.
- ORLANDO, L., LEONARD, J.A., THENOT, A., LAUDET, V., GUERIN, C. & HANNI, C. 2003 Ancient DNA analysis reveals woolly rhino evolutionary relationships. *Molecular Phylogenetics and Evolution* 28: 485-499.
- PERNA, N.T. & KOCHER, T.D. 1996 Mitochondrial DNA - Molecular fossils in the nucleus. *Current Biology* 6: 128-129.
- POND, S.L.K., FROST, S.D.W. & MUSE, S.V. 2005 HyPhy: hypothesis testing using phylogenies. *Bioinformatics* 21: 676-679.
- POSADA, D. & CRANDALL, K.A. 1998 MODELTEST: testing the model of DNA substitution. *Bioinformatics* 14: 817-818.
- QUINN, T.W. 1992 The genetic legacy of Mother Goose-phylogeographic patterns of lesser snow goose *Chen caerulescens caerulescens* maternal lineages. *Molecular Ecology* 1: 105-117.



- RICHARDSON, M.E. 1984 Aspects of the ornithology of the Tristan da Cunha group and Gough Island, 1972–1974. *Cormorant* 12: 123–201.
- RONQUIST, F. & HUELSENBECK, J.P. 2003 MrBayes 3: Bayesian phylogenetic inference under mixed models. *Bioinformatics* 19: 1572–1574.
- SCLATER, P.L. 1861 On the Island-Hen of Tristand d’Acunha. *Proceedings of the Zoological Society London*: 260–263.
- SLIKAS, B., OLSON, S.L. & FLEISCHER, R.C. 2002 Rapid, independent evolution of flightlessness in four species of Pacific Island rails (Rallidae): an analysis based on mitochondrial sequence data. *Journal of Avian Biology* 33: 5–14.
- SORENSEN, M.D. & QUINN, T.W. 1998 Numts: A challenge for avian systematics and population biology. *Auk* 115: 214–221.
- SPERLING, R.M. 1872 Letter on Tristan da Cunha. *Ibis* 3: 74–79.
- SWOFFORD, D.L. 1998 PAUP\* Phylogenetic Analysis Using Parsimony (\* and Other Methods). In). Sinauer Associates, Sunderland, MA.
- TAYLOR, B. & VAN PERLO, B. 1998 *Rails: A guide to the Rails, Coots, Crakes and Gallinules of the world*. Pica Press, Sussex.





CHAPTER 10

**Molecular identification of the type-  
specimen of the golden-striped salamander  
(*Chioglossa lusitanica*) Bocage, 1864  
leads to the description of a new  
subspecies**

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## ABSTRACT

Morphological and genetic data for the Iberian golden-striped salamander, *Chioglossa lusitanica*, demonstrate the existence of two groups with southern and northern ranges, connected by a zone of intergradation in central Portugal. Because reproductive isolation between them is incomplete we consider the groups to be subspecies. The type locality of *C. lusitanica* (Buçaco near Lousa) is situated inside the mixed zone. This necessitates identification of the nominotypical subspecies. We sequenced a fragment of mitochondrial DNA (*Cytochrome B*) from one of the species' syntypes and we determined what position over a latitudinal transect maximizes the morphological discrimination between the groups. Both approaches indicate that *C. lusitanica* from Buçaco represents the southern subspecies. A new subspecies of *C. lusitanica* is described from a northern locality (Valongo near Porto in north-western Portugal). A lectotype is designated for *Chioglossa lusitanica*.

## INTRODUCTION

The Golden-striped salamander, *Chioglossa lusitanica* Bocage, 1864 (Figure 1) occurs along mountain brooks of low and medium high altitudes in areas with high precipitation in the north-western corner of the Iberian Peninsula. *Chioglossa lusitanica* is the single extant representative of the genus *Chioglossa* that forms a monophyletic group with *Mertensiella*, another monotypical genus for *M. caucasica* (Waga, 1876) from the Caucasus. *Chioglossa lusitanica* and *M. caucasica* are streamside salamanders characterized by a slender built with short appendages and a long tail. They belong to the "true" salamanders within the family Salamandridae together with the sister group composed of the more stoutly built and less aquatic species in the genera *Salamandra* and *Lyciasalamandra* (Veith *et al.* 1998; Sequeira *et al.* 2006; Steinfartz *et al.* 2006; Weisrock *et al.* 2006).

Studies on the ecology and natural history of *C. lusitanica* (Arntzen 1981; Vences 1990) were followed by the mapping and modelling of its distribution (Teixeira *et al.* 2001; Arntzen 2006; Arntzen and Teixeira 2006) and by studies on intraspecific variation of morphological and genetic characters (Alexandrino *et al.* 2005, 2006; Sequeira *et al.* 2005). This revealed the presence of two different forms of *C. lusitanica* that are roughly distributed to the south and the north of the river Mondego in central Portugal, respectively. At the enzyme locus Pgm-1 different alleles predominate south and north of the river whereas the loci Adh, Pep-C and Pep-D show a full allelic transition at c. 10 km south of the Mondego (Alexandrino *et al.* 2000; Sequeira *et al.* 2005). Mitochondrial DNA haplotypes show a full transition at c. 10 km north of the Mondego (Sequeira *et al.* 2005). The colour pattern of the salamander shows the tendency of a clinal variation, with the dorsal golden-striped aspect (Figure 1) increasing from south to north. The variability in colour pattern is, however, higher in the Mondego area than at either side of the intergradation area, which has been associated with the merger of different genotypes, i.e., intraspecific





**Figure 1:** *Chioglossa lusitanica longipes* Arntzen 2007, Rio Tendi valley, Asturias, Spain. Photograph by Frank Deschandol.

hybridisation (Alexandrino *et al.* 2005, 2006). A morphometric transition takes the form of a stepped cline, with a complex pattern of variation around the western Mondego area (Alexandrino *et al.* 2005, 2006). Finally, an ecological analysis suggests slightly different habitat characteristics for the two forms, with a harder regime of lower annual temperature and lower humidity for the more mountainous southern form (Arntzen and Alexandrino 2004). Ecographical modelling indicates that the transition between the ecological regimes coincides with the river Mondego in western Portugal and is located c. 25 km north of the Mondego in central Portugal. The data summary in Table I supports the recognition of a northern and a southern group of *C. lusitanica*. The substantial width of the intergradation zone for multiple character states and the absence of syntopic populations suggest that the forms are not, or not strongly, genetically isolated from one another. We, therefore, consider these groups not to be separate species, but subspecies of *C. lusitanica*. Before Arntzen *et al.* (2007) could formally describe the newly recognized subspecies the following had to be ascertained : i) what is the type locality of *C. lusitanica* and ii) does the type material represent the northern or the southern subspecies ? For a start, uncertainty exists about what constitutes the official description of *C. lusitanica*, because Bocage published two descriptions, both in 1864. As pointed out by Arntzen (1999), the



description in the “Revue et Magazin de Zoologie Pure et Applique” (Bocage 1864a) has priority over that in the “Proceedings of the Zoological Society, London” (Bocage 1864b). Bocage’s material was sent to him by a Mr Rosa in two shipments from: a) the vicinity of Coimbra “ ... de Coïmbra en mai de l’année dernière [1863]. ... Ils ont été rencontrés aux environs de cette ville ...” and b) from Buçaco “... dans les derniers jours de janvier [1864], sur la montagne du Bussaco [Buçaco], à 5 lieues de Coïmbra”. So, whereas the origin of the second lot is known (namely Buçaco), the precise origin of the first shipment is uncertain and depends on what Bocage meant with “the vicinity of Coimbra”. The known type material is limited to two adult syntypes that are kept at the Museum National d’Histoire Naturelle, Paris (MNHNP 1182 and MNHNP 1182A). This material was in the first shipment and we follow Thireau (1986) in assuming that it originates from Buçaco that is, hence, the type locality.

By coincidence, Buçaco is located in the area where diagnostic character states provide contrasting signatures. In other words, whether the Buçaco population represents the southern or the northern subspecies depends on the data considered

**Table 1.** Discriminating character states in southern versus northern populations of *Chioglossa lusitanica* and the corresponding allocation of populations along a south to north axis.

| Characters          | Predominant character-state, or group |           | Population <sup>a</sup> |   |                |   |   |    |    |                    | References <sup>e</sup> |      |
|---------------------|---------------------------------------|-----------|-------------------------|---|----------------|---|---|----|----|--------------------|-------------------------|------|
|                     | South                                 | North     | 1-5 <sup>b</sup>        | 6 | 7 <sup>c</sup> | 8 | 9 | 10 | 11 | 12-20 <sup>d</sup> |                         |      |
| Nuclear enzyme loci |                                       |           |                         |   |                |   |   |    |    |                    |                         |      |
| <i>Adh</i>          | allele 2                              | allele 1  | S                       | N | N              | N | N | N  | N  | N                  | N                       | 1, 2 |
| <i>Pep-C</i>        | allele 2                              | allele 1  | S                       | N | N              | N | N | N  | N  | N                  | N                       | 1, 2 |
| <i>Pep-D</i>        | allele 2                              | allele 1  | S                       | N | N              | N | N | N  | N  | N                  | N                       | 1, 2 |
| <i>Pgm-1</i>        | allele 3F                             | allele 1F | S                       | N | N              | N | N | N  | N  | N                  | N                       | 1, 2 |
| Mitochondrial DNA   |                                       |           |                         |   |                |   |   |    |    |                    |                         |      |
| <i>Cyt B</i>        | Clade 3-1                             | Clade 3-2 | S                       | S | m              | N | N | N  | N  | N                  | N                       | 1, 3 |
| Morphology          |                                       |           |                         |   |                |   |   |    |    |                    |                         |      |
| classification      | Souther                               | Northern  | S                       | S | S              | a | N | N  | N  | N                  | N                       | 4    |
| Ecology             |                                       |           |                         |   |                |   |   |    |    |                    |                         |      |
| conditions          | Mild                                  | Harsh     | S                       | S | a              | N | N | N  | N  | N                  | N                       | 5    |

<sup>a</sup> S = Southern; N = Northern; a = ambiguous; m = mixed. <sup>b</sup> South of Mondego. <sup>c</sup> Type locality of *Chioglossa lusitanica*. <sup>d</sup> North of Douro. <sup>e</sup> Alexandrino *et al.* (2000); 2 = Sequeira *et al.* (2005); 3 = Alexandrino *et al.* (2002); 4 = present study; 5 = Arntzen & Alexandrino (2004).



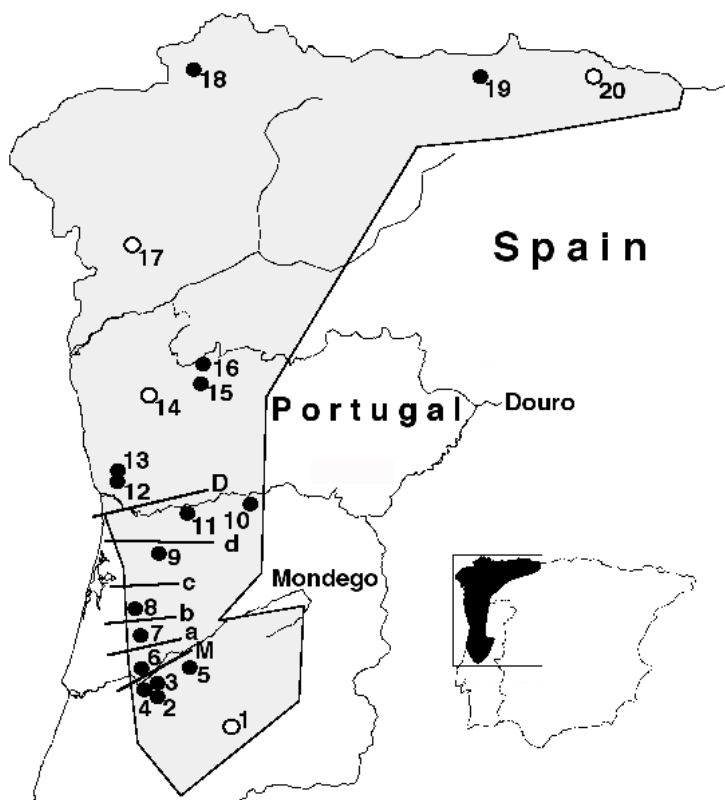
(Table 1). Prior to the subspecies description (Arntzen *et al.*, 2007), we set out to answer the following questions : i) where is the border between the subspecies located as seen from morphology, and ii) what are the morphological and molecular characteristics of the type material ?

## MATERIAL AND METHODS

Three hundred and fifty eight adult *C. lusitanica* (183 males and 175 females) from 16 localities covering the entire range of the species were measured alive and released in the locality where they were found (Figure 2; Alexandrino *et al.* 2005). We also studied the two syntypes from the Museum National d'Histoire Naturelle, Paris (MNHNP 1182 is a female and MNHNP 1182A is a male) and 54 ethanol preserved specimens from Valongo near Porto in north-western Portugal (localities 12 and 13; 30 males and 24 females). This material was available from the zoological museums in Amsterdam, Leiden, Lisbon and Madrid. The measurements taken were: snout-vent length (SVL), head length (HL), head width (HW), forelimb length (FLL), hind limb length (HLL), third finger length (TFL) and fourth toe length (FTL; see Alexandrino *et al.* 2005). The data were ln-transformed to increase statistical normality and to reduce distortion effects from allometric relationships. In a further treatment, the standardized residuals were calculated for the regression of ln<character> on ln<SVL> to reduce the impact of overall size in the analysis. Significant morphological differentiation between the sexes has already been demonstrated (Alexandrino *et al.* 2005) and because we are primarily interested in the spatial pattern of morphological variation (and not sexual dimorphism), data for males and females were analyzed separately. Salamanders from Valongo that at time of measuring were alive (group a, n=54) or ethanol preserved (group b, n=54), displayed significant differences between groups for ln-transformed data and not for size-adjusted data (results not shown). Preserved specimens were therefore only included in the analysis of size-adjusted morphometric data. Statistical analyses were carried out with SPSS 14 (SPSS 2005).

To investigate whether the northern or the southern subspecies of *C. lusitanica* would incorporate the type locality we used three approaches. First, we investigated the syntype MNHNP 1182A for subspecies-specific characters states at the mitochondrial *Cytochrome B* gene. A tiny tissue fragment was taken from the already broken tail of this syntype, with permission of the curator. The molecular work was carried out in a laboratory that is dedicated to the analysis of "ancient" DNA (Leiden aDNA Facility, The Netherlands). Two primer pairs were developed that each would amplify short sections of the *Cyt B* gene with respectively three and two diagnostic characters discerning between the southern and the northern haplotypes (Alexandrino *et al.* 2002). Primers CB159\_F 5'-AACCCCTCAAACATCTCATATT-3' and CB159\_R 5'-TAATTAACGTCTCGGCAGATATGG-3' amplify a 159 bp fragment that starts at position 9 of known sequences (GenBank accession numbers AF329285-AF329314; Alexandrino *et al.* 2002). Primers CB220\_F 5'-CCACCGTAATCACAAATCTCCT-3'





**Figure 2:** Approximate range of the Golden-striped salamander, *Chioglossa lusitanica* in the Iberian Peninsula (insert) with 16 populations that were subject to morphometric analysis (solid dots). Numbers 5-13 and letters a-d, D and M refer to putative subspecies borders that were designed to evaluate the congruence of range and subspecies morphology. Four populations with just molecular genetic data available are shown by open dots. Localities are as follows : 1, Muradal, Foz de Giraldo - N 40° 0' 23.6", W 7° 41' 49.8"; 2, Lousã, C. Pêra - N 40° 5' 28.7", W 8° 12' 4.1"; 3, Lousã, Fiscal - N 40° 6' 50.4", W 8° 13' 27.6"; 4, Lousã, Vilarinho - N 40° 7' 9.5", W 8° 12' 32.5"; 5, Açor, Margaraça - N 40° 13' 14.0", W 7° 55' 7.4"; 6, Várzeas - N 40° 14' 53.7", W 8° 22' 32.4"; 7, Buçaco - N 40° 21' 51.6", W 8° 21' 7.9"; 8, Saide - N 40° 26' 46.2", W 8° 19' 27.5"; 9, Covelo - N 40° 46' 37.1", W 8° 12' 47.3"; 10, Tarouca - N 41° 1' 38.3", W 7° 47' 12.1"; 11, Montemuro - N 41° 2' 33.1", W 8° 3' 57.6"; 12, Valongo, Silveirinha - N 41° 10' 45.9", W 8° 29' 57.3"; 13, Valongo, Águas Férreas - N 41° 11' 18.3", W 8° 29' 18.4"; 14, Bom Jesus - N 41° 33' 25.7", W 8° 22' 26.5"; 15, Cabreira - N 41° 39' 39.6", W 8° 1' 24.6"; 16, Gerês - N 41° 45' 26.8", W 8° 8' 45.4"; 17, Pontevedra - N 42° 30' 5.7", W 8° 28' 53.5"; 18, Caaveiro - N 43° 26' 3.9", W 8° 2' 54.9"; 19, Salas - N 43° 23' 40.7", W 6° 15' 22.8"; 20, Cuera - N 43° 17' 45.7", W 4° 35' 56.9"



and CB220\_R (5'-TCTGGGTTAGAGGGTAGTCCAG-3') amplify a 220 bp fragment, starting at position 371. The PCR-reactions were done twice, in laboratories that are physically separated from one another as well as from the ancient DNA facility. None of these laboratories had previously seen work on *C. lusitanica* DNA.

The tissue was cut as small as possible to enlarge the tissue-solution contact surface. DNA was extracted using the DNeasy® Tissue Kit (Qiagen). Tissue incubation time was 14 hours in buffer ATL and 20 µl of proteinase K was added twice (once at the start and once after 4 hours of incubation). DNA was eluted in 40 µl of buffer AE. PCR-reactions were done using a standard Taq DNA polymerase kit from Qiagen. Reaction volume was 25 µl and PCR conditions were 0.2 mM DNTs, 0.4 µM of each primer, 5 U of Taq DNA polymerase (Qiagen) and no additional MgCl<sub>2</sub> (1.5 mM MgCl<sub>2</sub> is present in PCR-buffer). Thermocycling conditions were 3 min. at 94°C for initial denaturation followed by 40 cycles (15 sec. at 94°C, 30 sec. at 50°C and 40 sec. at 72°C) and final extension 5 min. at 72°C. PCR products were cleaned using a Nucleospin® kit (Macherey-Nagel) and cleaned products were cloned using the Topo TA Cloning® system (Invitrogen). White colonies were picked and used to initiate reamplifications with primers 21M13\_F 5'-TGTAACGACGGCCAGT-3' and 21M13\_R 5'-CAGGAAACAGCTATGACC-3'. PCR, thermocycling and product cleaning were as described above, except that the annealing temperature was raised to 57°C. Sequencing was performed on a Megabace™ 1000 DNA Analysis System (Amersham).

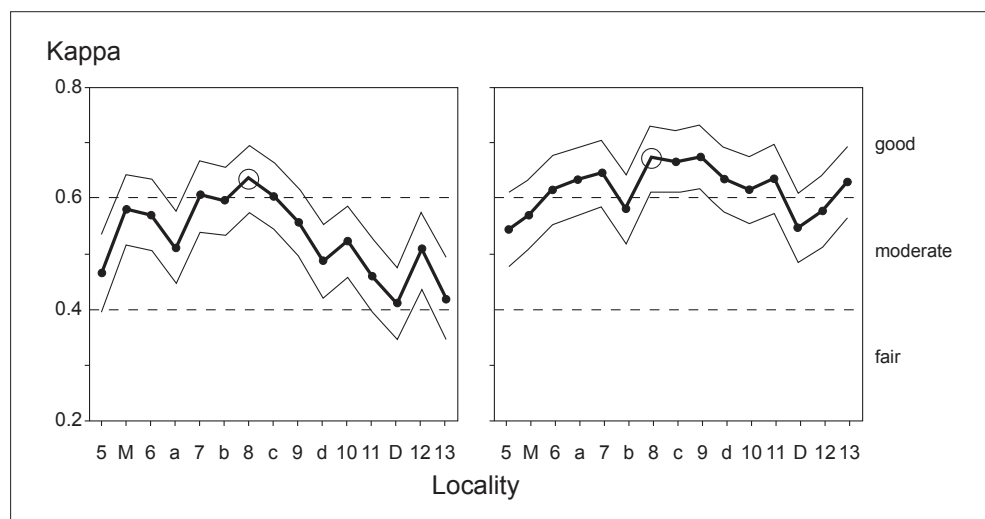
Secondly, we addressed the question of what position of the subspecies border, from a morphological perspective, would yield the most convincing classification between the northern and the southern subspecies. We varied the position of the putative range border across central and northern Portugal from positions 5, M (= Mondego river), 6, a, 7, b, 8, c, 9, d, 10, 11, D (= river Douro), 12 and 13 (see Figure 2) and subjected the data with these a priori defined groups to discriminant analysis. If population localities were taken as the subspecies border, local individuals were excluded and subjected to an a posteriori classification. Following this approach we determined what subspecies the two syntype specimens would represent. The congruence among the morphology and geography based classifications was expressed by kappa ( $\kappa$ ; Cohen 1960) and its asymptotic standard error. All analyses and calculations were performed in SPSS. Following Altman (1991) the strength of agreement is classified as "fair" for  $0.2 < \kappa < 0.4$ , "moderate" for  $0.4 < \kappa < 0.6$  and "good" for  $0.6 < \kappa < 0.8$ .

## RESULTS

The 159 bp fragment of the Cyt B gene was successfully amplified while amplification of the 220 bp fragment failed, possibly due to DNA degradation. Two subsamples of the 159 bp fragment were sequenced in both directions and three subsamples were sequenced using only primer 21M13\_F (GenBank EF122233-EF122237). Unexpected character states found are a T for a C at position 71 in one clone (GenBank EF122236)



and a C for a T at position 124 in another (GenBank EF122235), possibly due to postmortem damage (Lindahl 1993; Hofreiter et al. 2001; Gilbert et al. 2003; Stiller et al. 2006). Both substitutions would in vivo represent amino acid replacements. Character states at the subspecies diagnostic positions 54, 102 and 114 are T, A and T, respectively. The mtDNA of the type specimen MNHNP 1182A is the one otherwise found over the southern part of the *C. lusitanica* range.



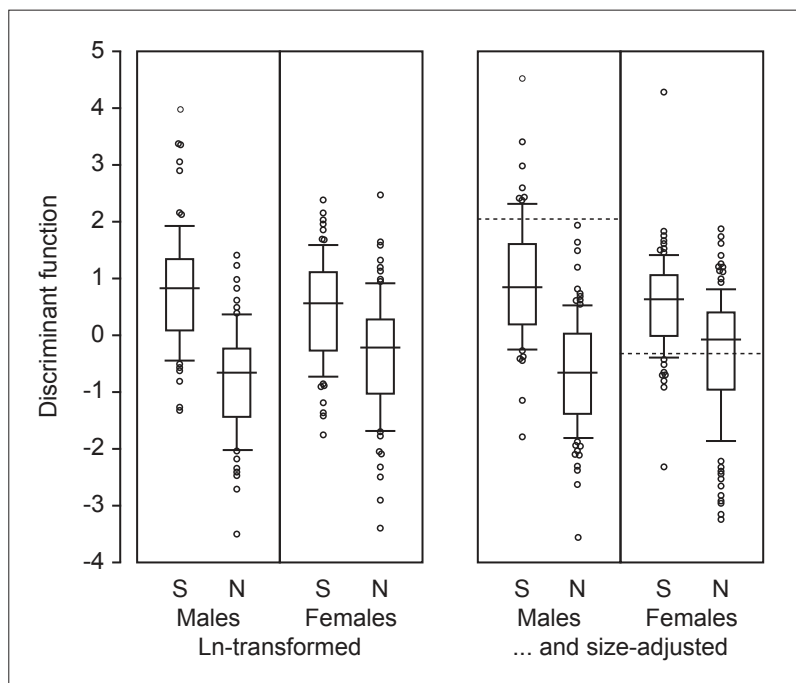
**Figure 3:** Congruence between morphometry- and geography-based classifications of the southern and northern subspecies of *Chioglossa lusitanica* across a latitudinal transect in central and northern Portugal, for males (left) and females (right). Numbers 5-13 and letters a-d, M and D refer to putative subspecies borders (see Figure 2). Congruence is expressed by kappa ( $\kappa$ )  $\pm$  asymptotic standard error. Following Altman (1991)  $\kappa$ -scores of  $< 0.4$  indicate a “fair” congruence between classifications; with  $0.4 < \kappa < 0.6$  congruence is “moderate” and with  $\kappa > 0.6$  congruence is “good”. The most convincing discrimination from morphology is achieved when the border between the southern and the northern population is set at population 8 (open round symbol).

The raw morphometric data are summarized in Table 2 and in Alexandrino *et al.* (2005). For males the congruence between the morphometry- and geography-based subspecies classifications is mostly moderate ( $0.4 < \kappa < 0.6$ ; Figure 3). A bell-shaped curve is obtained with lower values at the fringes than in the middle. In females the congruence is consistently higher than for males and the bell-shaped aspect of the graph is less prominent. To investigate the possibility that the bell shapes would represent a mid-domain effect operating through unbalanced sample sizes, we repeated the analysis with a weighing scheme in place, such that sample sizes



for the northern and southern groups were effectively equal. Results obtained were similar to those in Figure 3. Considering both sexes together  $\kappa$  is highest if locality 8 is taken as the subspecies border.

The male and female syntypes had a posteriori probability for representing the southern subspecies of 0.95 and 0.36, respectively (Figure 4).



**Figure 4:** Discriminant function scores for the southern (S) and northern (N) subspecies in males and females of *Chioglossa lusitanica* with the range border at locality 8. The boxes and bars represent 50% and 80% of the data, respectively and round symbols are outliers, for ln-transformed (left panel) and ln-transformed, size-adjusted data (right panel). Interrupted lines represent the discriminant functions for the two *C. lusitanica* syntypes.

## DISCUSSION

The level of genetic differentiation at enzyme and mtDNA loci, combined with the observed mtDNA phylogeographic pattern over the species range, support the largely independent evolution of two forms of *Chioglossa lusitanica*. Average genetic distance between the northern and the southern form is  $D_{\text{Nei}} = 0.15$ , measured over a set of 22 enzyme electrophoretic loci. The average *Cyt B* sequence divergence is



**Table 2.** Morphometric data for ethanol-preserved specimens in the museum collections of Amsterdam (ZMA), Leiden (RMNH R and A), Lisbon (MB) and Madrid (MNCN).

| Museum | Coll. nr. | Sex    | SVL  | HL   | HW  | FLL  | HLL  | TFL | FTL | Remarks                            |
|--------|-----------|--------|------|------|-----|------|------|-----|-----|------------------------------------|
| RMNH   | 40113     | male   | 44.9 | 10.4 | 7.4 | 11.1 | 13.1 | 2.2 | 3.4 |                                    |
| ZMA    | 7307a     | female | 42.9 | 10.0 | 6.3 | 10.1 | 11.6 | 2.5 | 3.5 |                                    |
|        | 7307b     | female | 41.9 | 10.1 | 6.0 | 9.3  | 11.9 | 2.2 | 3.1 |                                    |
|        | 7307c     | female | 43.3 | 9.3  | 6.2 | 9.6  | 12.3 | 2.6 | 3.2 |                                    |
|        | 7307d     | male   | 44.6 | 10.7 | 7.0 | 10.9 | 13.5 | 2.8 | 3.7 |                                    |
|        | 7307e     | female | 41.2 | 10.1 | 6.3 | 10.2 | 12.3 | 2.5 | 3.1 |                                    |
|        | 7340a     | male   | 43.0 | 9.9  | 7.1 | 10.9 | 13.0 | 2.6 | 3.3 |                                    |
|        | 7340b     | female | 43.1 | 10.3 | 7.0 | 10.0 | 12.1 | 2.8 | 3.2 |                                    |
|        | 7340c     | female | 44.1 | 10.1 | 6.9 | 8.6  | 12.2 | 2.2 | 3.0 |                                    |
|        | 7340d     | male   | 42.5 | 10.3 | 6.9 | 10.1 | 12.4 | 2.4 | 3.3 |                                    |
|        | 7340e     | female | 43.8 | 9.5  | 6.8 | 9.8  | 12.5 | 3.0 | 3.2 |                                    |
|        | 7340f     | male   | 42.0 | 9.5  | 6.7 | 9.7  | 12.2 | 2.9 | 3.4 |                                    |
|        | 7340g     | male   | 43.7 | 9.8  | 6.8 | 11.5 | 12.8 | 2.7 | 3.8 | toe 20 missing                     |
|        | 7340h     | male   | 40.4 | 9.2  | 6.6 | 9.4  | 12.6 | 2.6 | 3.5 |                                    |
|        | 7340i     | female | 44.1 | 10.4 | 6.9 | 10.0 | 13.6 | 2.7 | 3.4 |                                    |
|        | 7340j     | female | 45.6 | 10.9 | 7.4 | 9.8  | 13.2 | 2.3 | 3.5 |                                    |
|        | 7402      | female | 44.7 | 10.3 | 6.7 | 11.1 | 13.1 | 2.6 | 3.2 |                                    |
|        | 7482      | male   | 40.6 | 9.5  | 6.9 | 9.7  | 12.2 | 2.6 | 2.9 | toe 70 missing                     |
|        | 7588a     | male   | 44.4 | 9.9  | 7.0 | 11.2 | 13.4 | 3.0 | 3.5 | toe 20 missing                     |
|        | 7588b     | female | 45.6 | 9.9  | 6.9 | 10.1 | 13.2 | 2.4 | 3.1 |                                    |
|        | 7588c     | female | 43.7 | 9.8  | 6.9 | 9.6  | 13.0 | 2.6 | 3.3 |                                    |
|        | 7669      | male   | 41.6 | 9.2  | 6.1 | 11.0 | 12.6 | 2.4 | 3.6 | tail half broken,<br>toe 6 missing |
|        | 8086a     | female | 46.0 | 10.8 | 6.9 | 10.8 | 12.7 | 3.1 | 3.7 | autotomized tail                   |
|        | 8086b     | female | 42.8 | 10.1 | 7.0 | 10.6 | 12.1 | 2.8 | 3.7 |                                    |
|        | 8086c     | male   | 41.4 | 9.9  | 6.8 | 10.5 | 11.9 | 2.4 | 3.0 |                                    |
|        | 8086d     | male   | 43.4 | 10.6 | 7.0 | 9.9  | 11.6 | 2.5 | 3.3 |                                    |
|        | 8086e     | male   | 42.4 | 10.1 | 7.0 | 10.7 | 12.4 | 2.5 | 3.7 |                                    |
|        | 8086f     | male   | 41.4 | 9.6  | 6.6 | 10.4 | 12.3 | 2.6 | 3.4 |                                    |
|        | 8086g     | male   | 40.4 | 9.4  | 6.4 | 9.6  | 12.9 | 2.5 | 3.0 |                                    |



| Museum | Coll. nr. | Sex    | SVL  | HL   | HW  | FLL  | HLL  | TFL | FTL | Remarks                                      |
|--------|-----------|--------|------|------|-----|------|------|-----|-----|----------------------------------------------|
|        | 8086h     | male   | 42.1 | 9.0  | 6.2 | 11.2 | 12.7 | 2.5 | 3.2 |                                              |
|        | 8086i     | female | 41.9 | 9.9  | 6.7 | 10.3 | 12.5 | 2.8 | 3.6 |                                              |
|        | 8086j     | male   | 43.0 | 10.3 | 7.0 | 10.1 | 12.7 | 2.9 | 3.3 |                                              |
|        | 8086k     | male   | 41.2 | 9.4  | 6.6 | 9.6  | 12.3 | 2.9 | 3.1 |                                              |
| RMNH   | 18796     | male   | 44.9 | 10.0 | 7.3 | 10.0 | 13.7 | 2.4 | 3.3 | tail damaged at base                         |
|        | 18797     | male   | 44.7 | 10.5 | 7.6 | 10.9 | 13.0 | 2.8 | 3.3 |                                              |
|        | 18798     | male   | 44.6 | 10.2 | 7.5 | 11.0 | 12.5 | 2.9 | 3.6 |                                              |
|        | 18799     | male   | 42.9 | 9.4  | 7.2 | 10.9 | 11.9 | 2.9 | 3.3 | toe 8 missing, toe 2 short                   |
|        | 18800     | female | 48.7 | 11.4 | 7.7 | 10.7 | 12.7 | 2.9 | 3.4 | toe 10 missing, toes 2, 3 and 7 short        |
|        | 18801     | male   | 41.6 | 9.7  | 6.5 | 10.6 | 11.9 | 3.4 | 3.2 | toe 40 short                                 |
|        | 18802     | female | 44.0 | 10.1 | 7.0 | 9.6  | 12.4 | 3.3 | 3.7 | autotomized tail                             |
|        | 18803     | male   | 43.8 | 9.9  | 7.3 | 11.4 | 12.8 | 2.7 | 3.4 | toe 2 missing                                |
|        | 18804     | female | 43.9 | 9.8  | 7.1 | 9.7  | 12.1 | 2.5 | 3.4 |                                              |
|        | 18805     | male   | 44.0 | 10.5 | 7.2 | 10.5 | 12.3 | 2.8 | 3.3 |                                              |
|        | 40114     | male   | 46.0 | 10.7 | 7.0 | 11.1 | 13.1 | 2.7 | 3.7 |                                              |
|        | 40115     | female | 46.5 | 10.5 | 7.6 | 10.2 | 12.6 | 2.5 | 3.2 |                                              |
|        | 40116     | female | 45.4 | 9.9  | 7.2 | 10.0 | 11.9 | 2.1 | 2.8 |                                              |
| MB     | 06-450-01 | male   | 45.9 | 9.9  | 7.4 | 11.0 | 12.6 | 3.0 | 3.2 | toe 4 bifurcating at tip, toes 3 and 6 short |
|        | 06-450-02 | female | 47.3 | 10.7 | 7.3 | 11.0 | 12.7 | 2.3 | 3.5 |                                              |
|        | 06-450-03 | female | 45.8 | 10.7 | 6.8 | 11.1 | 13.8 | 2.8 | 3.5 |                                              |
|        | 06-450-04 | male   | 45.6 | 10.2 | 7.3 | 10.3 | 12.4 | 2.8 | 2.8 |                                              |
| MNCN   | 43641     | male   | 47.0 | 11.0 | 7.6 | 10.2 | 13.0 | 2.3 | 3.3 |                                              |
|        | 43642     | male   | 47.2 | 11.1 | 7.7 | 10.0 | 12.7 | 2.7 | 2.8 | several toes at forefeet incomplete          |
|        | 43643     | female | 47.7 | 10.8 | 7.3 | 11.0 | 12.6 | 2.4 | 3.1 |                                              |
|        | 43644     | female | 44.2 | 10.2 | 6.9 | 11.2 | 13.4 | 2.8 | 3.4 |                                              |

Presented are the averages (in mm) of three measurements by a single observer (JWA) of snout-vent length (SVL), head length (HL), head width (HW), forelimb length (FLL), hind limb length (HLL), third finger length (TFL) and length of the fourth toe (FTL). Under "remarks" fingers and toes are numbered as in the Hero (1989) scheme. Missing fingers or toes may be due to marking for ecological research. Some specimens are slit open at ventral side for the purpose of stomach content and female fecundity analyses (Arntzen 1979, 1981).



$d = 1.4\%$ . Following a “molecular clock” argument these data place the onset of the divergence at 1.5 - 3 million years (Alexandrino et al. 2000, 2002). Despite past geographical isolation - probably lasting throughout most of the Pleistocene - and a full transition over four enzyme genetic markers (Table I), *C. lusitanica* populations show admixture within the contact zone. This suggests i) the absence of genetic barriers to neutral diffusion of alleles across the zone, ii) the absence of strong selection against hybrid genotypes (Sequeira et al. 2005) and, iii) that the genetic cohesiveness between the two forms of *C. lusitanica* is maintained by population level processes associated with restricted gene flow across the Mondego river valley. The moderate level of genetic differentiation and the lack of genetic isolation justify the recognition of the northern and the southern form of *C. lusitanica* at the level of subspecies. The population of Buçaco is located within the hybrid zone, right in the middle of a steep mtDNA transition and somewhere near the centre of smooth morphological and ecological transitions. We therefore consider the Buçaco population of mixed origin, even though diagnostic enzyme loci would classify it unambiguously with the northern subspecies. The mitochondrial DNA sequence data show that the syntype MNHNP 1182A from Buçaco has the character states that are found in – and hence represents – the southern form. The morphological analysis also indicates that *C. lusitanica* from Buçaco represents the southern form. Consequently, the new taxon will be a subspecies for which the name-bearing type should be sought in the northern part of the species range and preferably well away from the zone of subspecific intergradation. The formal description of this taxon with type locality Valongo near Porto in northwestern Portugal is given in Appendice I of Arntzen et al. 2007. The nominotypical subspecies (with the trinomial “*lusitanica*”) will hence have its range within the boundaries of historical Lusitania, i.e., the province of the Roman empire with that name, whereas the new subspecies with a range over northern Portugal, Galicia and Asturias and largely outside Lusitania, will carry a different trinomial.

It is fortunate that the one syntype for which we obtained permission to use a tiny fragment of tissue for molecular genetic analysis – the one with a tail already broken – was a male, because morphometric discrimination of the subspecies is more apparent in males than in females (Fig 3). This is reflected in the more clear-cut discriminant scores at the population level and at the level of the individual syntypes also (Fig. 4). It must be noted that the northern and southern haplotypes occur in approximately the same frequency in the Buçaco population (Alexandrino et al. 2002). It is therefore not inconceivable that the syntype that we did not study for its mitochondrial DNA carries not the southern but the northern haplotype. In order to prevent future destabilisation of the nomenclature, we designate syntype MNHNP 1182A as lectotype and MNHNP 1182 as paralectotype of *C. lusitanica*.

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## REFERENCES

- ALEXANDRINO, J., FROUFE, E., ARNTZEN, J.W. & FERRAND, N. 2000. Genetic subdivision, glacial refugia and postglacial recolonization in the golden-striped salamander, *Chioglossa lusitanica* (Amphibia: Urodela). *Molecular Ecology* 9: 771-781.
- ALEXANDRINO, J., ARNTZEN, J.W. & FERRAND, N. 2002. Nested clade analysis and the genetic evidence for population expansion in the phylogeography of the golden-striped salamander, *Chioglossa lusitanica* (Amphibia: Urodela). *Heredity* 88: 66-74.
- ALEXANDRINO, J., FERRAND, N. & ARNTZEN, J.W. 2005 Morphological variation in two genetically distinct groups of the golden-striped salamander, *Chioglossa lusitanica* (Amphibia: Urodela). *Contributions to Zoology* 74: 213-222.
- ALEXANDRINO, J., TEIXEIRA, J., ARNTZEN, J.W. & FERRAND, N. 2006. Historical biogeography and conservation of the golden-striped salamander (*Chioglossa lusitanica*) in northwestern Iberia: integrating ecological, phenotypic and phylogeographic data. In: Weiss S, Ferrand N, editors. *Phylogeography of southern European refugia*. Springer: Dordrecht, The Netherlands. p 189-205.
- ALTMAN, D.G. 1991. *Practical statistics for medical research*. London: Chapman and Hall.
- ARNTZEN, J.W. 1979. *De levenswijze van de goudstreepsalamander Chioglossa lusitanica* Bocage, 1864 [MSc.-thesis]. Amsterdam: University of Amsterdam.
- ARNTZEN, J.W. 1981. Ecological Observations on *Chioglossa lusitanica* (Caudata, Salamandridae). *Amphibia-Reptilia* 1: 187-203.
- ARNTZEN, J.W. 1999. *Chioglossa lusitanica* Bocage 1864 - der Goldstreifen-salamander. In: Grossenbacher K, Thiesmeier B, editors. *Handbuch der Reptilien und Amphibien Europas*. Wiesbaden: Akademische Verlagsgesellschaft. p. 301-321.
- ARNTZEN, J.W. 2006. From descriptive to predictive distribution models: a working example with Iberian amphibians and reptiles. *Frontiers in Zoology* 3(8). doi: 10.1186/1742-9994-3-8.
- ARNTZEN, J.W. & ALEXANDRINO, J. 2004. Ecological modelling of genetically differentiated units of the Iberian endemic Golden-striped salamander, *Chioglossa lusitanica*. *Herpetological Journal* 14: 137-141.
- ARNTZEN, J.W. & TEIXEIRA, J. 2006. History and new developments in the mapping and modelling of the distribution of the golden-striped Salamander, *Chioglossa lusitanica*. *Zeitschrift Feldherpetologie*, supplement 10: 113-126.
- BOCAGE, DU B. [JV]. 1864a. Note sur un nouveau Batracien du Portugal, *Chioglossa lusitanica*, et sur une Grenouille nouvelle de l'Afrique occidentale. *Revue et Magazin de Zoologie Pure et Applique* 2e série 16: 248-254.
- BOCAGE, DU B. JV. 1864b. Notice sur un Batracien nouveau du Portugal (*Chioglossa lusitanica*,



- Nob.) Proceedings Zoological Society London 1864:264-265.
- COHEN, J. 1960. A coefficient of agreement for nominal scales. *Educational Psychology Measurement* 20: 37-46.
- GILBERT, M.T.P., HANSEN, A.J., WILLERSLEV, E., RUDBECK, L., BARNES, I., LYNNERUP, N. & COOPER, A. 2003. Characterization of genetic miscoding lesions caused by postmortem damage. *American Journal Human Genetics* 72: 48-61.
- HERO, J.M. 1989. A simple code for toe clipping anurans. *Herpetological Review* 20: 66-67.
- HOFREITER, M., JAENICKE, V., SERRE, D., VON HAESELER, A. & PAABO, S. 2001 DNA sequences from multiple amplifications reveal artifacts induced by cytosine deamination in ancient DNA. *Nucleic Acids Research* 29: 4793-4799.
- LINDAHL, T. 1993 Instability and decay of the primary structure of DNA. *Nature* 362: 709-715.
- SEQUEIRA, F., ALEXANDRINO, J., ROCHA, R., ARNTZEN, J.W. & FERRAND, N. 2005 Genetic exchange across a hybrid zone within the Iberian endemic golden-striped salamander, *Chioglossa lusitanica*. *Molecular Ecology* 14: 245-254.
- SEQUEIRA, F., FERRAND, N. & HARRIS, D.J. 2006 Assessing the phylogenetic signal of the nuclear  $\beta$ -Fibrinogen intron 7 in salamandrids (Amphibia: Salamandridae). *Amphibia-Reptilia* 27: 409-418.
- SPSS. 2005. SPSS for Windows. Release 14. Chicago: SPSS Inc.
- STEINFARTZ S., VICARIO S., ARNTZEN J.W. & CACCONE A. 2006. A Bayesian approach on molecules and behavior: reconsidering phylogenetic and evolutionary patterns of the Salamandridae with emphasis on Triturus newts. *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* 308B:139-162 doi 10.1002/jez.b.21119
- STILLER, M., GREEN, R.E., RONAN, M., SIMONS, J.F., DU, L., HE, W., EGHOLM, M., ROTHBERG, J.M., KEATS, S.G., OVODOV, N.D., ANTIPINA, E.E., BARYSHNIKOV, G.F., KUZMIN, Y.V., VASILEVSKI, A.A., WUENSCHALL, G.E., TERMINI, J., HOFREITER, M., JAENICKE-DESPRES, V. & PAABO, S. 2006. Patterns of nucleotide misincorporations during enzymatic amplification and direct large-scale sequencing of ancient DNA. *Proceedings National Academy Sciences USA* 103: 13578-13584, 14977.
- TEIXEIRA, J., FERRAND, N. & ARNTZEN, J.W. 2001. Biogeography of the golden-striped salamander *Chioglossa lusitanica*: a field survey and spatial modelling approach. *Ecography* 24: 618-624.
- THIREAU, M. 1986. Catalogue des types d'Urodèles du Muséum national d'Histoire Naturelle. Revue critique. Paris: Laboratoire de Zoologie (Reptiles et Amphibiens) du Muséum national d'Histoire Naturelle. p. 1-97.
- VEITH, M., STEINFARTZ, S., ZARDOYA, R., SEITZ, A. & MEYER, A. 1998. A molecular phylogeny of 'true' salamanders (family Salamandridae) and the evolution of terrestriality of reproductive modes. *Journal Zoological Systematics Evolutionary Research* 36: 7-16.
- VENCES, M. 1990. Untersuchungen zur Ökologie, Ethologie und geografischen Variation von *Chioglossa lusitanica* BOCAGE, 1864. *Salamandra* 26: 267-297.
- WEISROCK, D.W., PAPENFUSS, T.J., MACEY, J.R., LITVINCHUK, S.N., POLYMENI, R., UGURTAS, I.H., ZHAO, E., JOWKAR, H. & LARSON, A. 2006. A molecular assessment of phylogenetic relationships and lineage accumulation rates within the family Salamandridae (Amphibia, Caudata). *Molecular Phylogenetics and Evolution* 41: 368-383.







CHAPTER 11

**The complete mitogenome of**  
***Cylindrus obtusus***  
**(Helicidae, Ariantinae)**  
**using Illumina**  
**next generation sequencing**

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## ABSTRACT

This study describes how the complete mitogenome of a terrestrial snail, *Cylindrus obtusus* (Draparnaud, 1805) was sequenced without PCRs from a collection specimen that had been in 70% ethanol for 8 years. The mitogenome was obtained with Illumina GAIIx shot gun sequencing. Although the used specimen was collected relatively recently and kept in a DNA-friendly preservative (not formalin as frequently used with old museum specimens), we believe that the exclusion of PCRs as facilitated by NGS (Next Generation Sequencing) removes a great obstacle in DNA sequencing of collection specimens. A brief comparison is made between our Illumina GAIIx approach and a similar study that made use of the Roche 454-FLX platform. The mtDNA sequence of *C. obtusus* is 14,610 bases in length (about 0.5 kb larger than other stylommatophoran mitogenomes reported hitherto) and contains the 37 genes (13 protein coding genes, two rRNAs and 22 tRNAs) typical for metazoans. Except for a swap between the position of tRNA-Pro and tRNA-Ala, the gene arrangement of *C. obtusus* is identical to that reported for *Cepaea nemoralis*. The ‘aberrant’ rearrangement of tRNA-Thr and COIII compared to that of other Sigmurethra (and the majority of gastropods), is not unique for *C. nemoralis* (subfamily Helicinae), but is also shown to occur in *C. obtusus* (subfamily Ariantinae) and might be a synapomorphy for the family Helicidae. Natural history collections potentially harbor a wealth of information for the field of evolutionary genetics, but it can be difficult to amplify DNA from such specimens (due to DNA degradation for instance). Because NGS techniques do not rely on primer-directed amplification (PCR) and allow DNA to be fragmented (DNA gets sheared during library preparation), NGS could be a valuable tool for retrieving DNA sequence data from such specimens. A comparison between Illumina GAIIx and Roche 454 platform suggests that the former might be more suited for de novo sequencing of mitogenomes.

## INTRODUCTION

Although Next-Generation Sequencing (NGS) techniques advanced rapidly over the last years and sequencing of entire mitochondrial genomes (mitogenomes) has consequently become more common (Saitoh *et al.* 2006; Grande *et al.* 2008; Medina *et al.* 2011), knowledge about stylommatophoran (‘terrestrial snails’) mitogenomes seems to have advance at a slower pace. For over two decades, the complete mitogenomes of only two stylommatophorans, *Cepaea nemoralis* (Terret *et al.* 1996) and *Albinaria caerulea* (Hatzoglou *et al.* 1995), had been known. Of a third species, *Euhadra herklotsi*, most of the mitogenome has been covered, so comparisons of mitochondrial gene arrangements could be made (Yamazaki *et al.* 1997; Boore 1999; Boore *et al.* 2004), but a complete sequence for that mitogenome is still missing (Yamazaki *et al.* 1997; Grande *et al.* 2002; Simison and Boore 2008). Sequencing mitogenomes has been quite cumbersome because enrichment of the mitochondrial fraction (e.g. physical isolation of mitochondria, cloning of large mitochondrial fragments, long range,

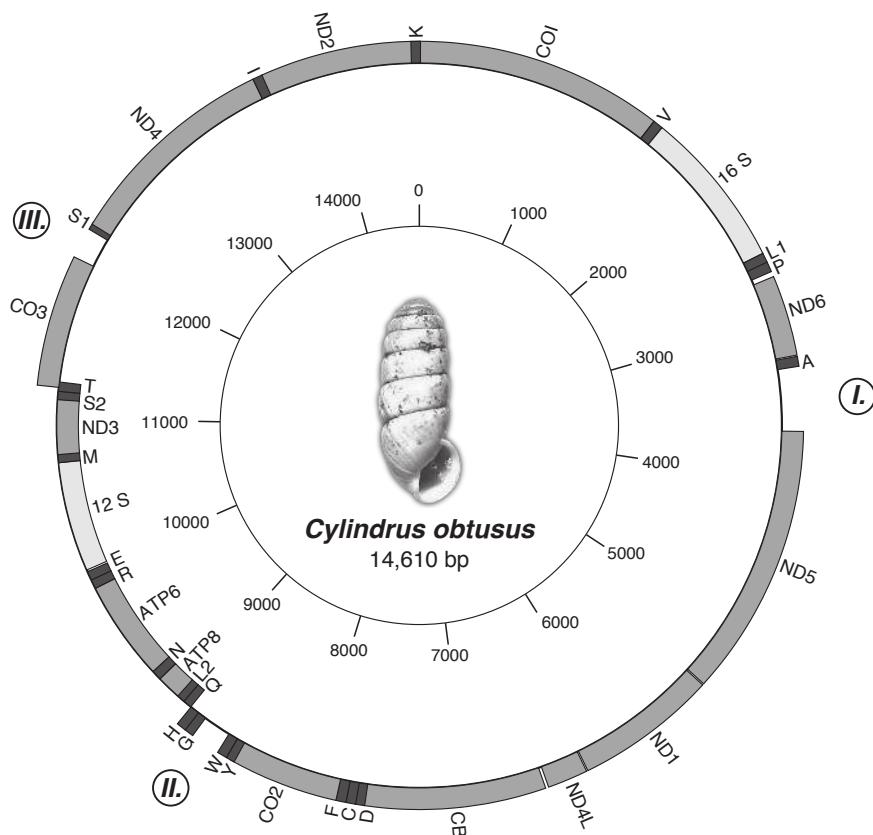


simplex or multiplex PCR) was quite laborious (Broughton *et al.* 2001; Miya *et al.* 2003; Zhang *et al.* 2003; Boore *et al.* 2004; Saitoh *et al.* 2006; Grande *et al.* 2008; Morin *et al.* 2010; Ki *et al.* 2010; Medina *et al.* 2011). Moreover, the throughput of traditional (Sanger) sequencing is limited and sequencing of larger (> 1kb) fragments is often delayed by the development of internal primers ('primer walking'). With NGS technologies, primer-directed amplification is no longer necessary and genome sequencing, mitogenomes in particular because of the small size and high-copy-number, has become fast and easy.

Nearly all traditional (Sanger) (Broughton *et al.* 2001; Miya *et al.* 2003; Zhang *et al.* 2003; Boore *et al.* 2004; Saitoh *et al.* 2006; Grande *et al.* 2008; Morin *et al.* 2010; Ki *et al.* 2010; Medina *et al.* 2011) and still some NGS (Jex *et al.* 2008) approaches to the sequencing of mitogenomes still rely on long PCR amplification. Due to the often degraded state of the DNA, this by and large excludes the use of specimens from natural history collections. An alternative is the hybridization capture approach (Knapp and Hofreiter 2010), but this requires *a priori* sequence knowledge in order to design probes. Although enrichment of the mitochondrial fraction by long PCR will increase the chance of obtaining a complete mitogenome (and facilitate the use of multiple specimens if sequence tags are exploited; Meyer *et al.* 2007), it is not essential to NGS (e.g. Miller *et al.* 2009; Dietz *et al.* 2010; Feldmeyer *et al.* 2010). In fact, the first step in NGS library preparation is fragmentation of the DNA. Consequently, DNA sequence data might be obtained from specimens of natural history collections with NGS, where PCR-based approaches fail. Whether NGS will allow, e.g., the recovery of complete mitogenomes from collection specimens, will depend on various parameters such as: the extent of DNA degradation, the ratio of nuclear to extrachromosomal DNA (which depends on the size of the genomes as well as on the type of tissue selected) and the number and length of the obtained sequences (dependent on the selected NGS platform). To the best of our knowledge, mitogenomes from NGS studies are thus far obtained either with the use of a long PCR enrichment procedure (Jex *et al.* 2008) prior to the NGS run, or with traditional Sanger sequencing after the run (to close the gaps remaining after assembly of the mitogenome, or to get an acceptable coverage: Dietz *et al.* 2010; Feldmeyer *et al.* 2010; Cui *et al.* 2009). Since each of these approaches relies on PCR, both can be impracticable for (fragmented) DNA retrieved from museum specimens. With this study we wanted to test whether it would be feasible (using the Illumina GAIIx platform) to obtain a complete mitogenome, without PCR, from a single museum specimen.

We selected *C. obtusus* because it is an interesting species from both a morphological and a biogeographic point of view. *C. obtusus* is endemic to the Austrian Alps where it can be found in calcareous areas (Duda *et al.* 2010) at altitudes nearly always above 1600 m. It has a disjunct distribution, probably mirroring the small insular areas ('nunataks'; Krajčok 1998) in which it survived the last glacial maximum (LGM). Except for some Late Pleistocene specimens (Frank 2006) there is no fossil





**Figure 1:** Map of the mitochondrial genome of *C. obtusus* (GenBank accession nr. JN107636).

Genes on the outer circle are transcribed clockwise; genes on the inner circle are transcribed counterclockwise. TRNAs are denoted by their one-letter abbreviations. Regions I, II and III are regions that could not be assigned to any mitochondrial gene; region III could be the most likely location for the mitochondrial control region (see Discussion)

record for *C. obtusus*. *Cylindrus* constitutes a monotypic genus. Within the helicid subfamily Ariantinae it is aberrant by being the only species with a cylindrical shell (Fig. 1); other members of this speciose subfamily have broadly depressed (e.g. *Campylaea*, *Helicigona* and *Chilostoma*) or globular (*Arianta arbustorum*) shells (Fig. 2 in Gittenberger *et al.* 2004).



This study shows that NGS can aid in the retrieval of sequence data (here a complete mitogenome) without using PCRs. Due to DNA degradation, PCRs are often a bottleneck for museum specimens. Based on our results for a specimen that has been in 70% ethanol for 8 years, we plea that NGS could be a promising technique for obtaining sequence data from museum specimens. We report the third complete mitogenome for a species of terrestrial snail ever published and compare our Illumina GAIIX strategy for sequencing mitogenomes with a similar study (Feldmeyer *et al.* 2010) in which the 454 platform of Roche was deployed.

## MATERIAL AND METHODS

### 1. Collection and preservation

Specimens of *C. obtusus* were collected by J. Gould in 2001 in Großer Buchstein (3.5 km NW of Gstatteboden), Ennstaler Alpen (Austria; 47°37' N 14°36' E) at an elevation of 2,200 m. After collection the specimens were drowned in water and subsequently placed in ethanol 70%. Finally they were stored in the molluscan wet collection of NCB Naturalis under collection number RMNH. MOL. 144846.

### 2. DNA extraction and quality assessment

DNA was extracted from a single specimen with a DNeasy Blood and Tissue kit (Qiagen). Apart from using a total of 40 µl of Proteinase K and overnight lysis, the manufacturer's instructions were followed. The DNA concentration of the extract was measured on a Nanodrop 1000 spectrophotometer (Thermo Scientific) and checked on an agarose gel.

### 3. Confirmation of NGS output

Because only a small number of *C. obtusus* (microsatellite) sequences were present in GenBank (Arthofer *et al.* 2010), we tried to sequence *COI* and *CytB* which would allow the identification of mitochondrial contig sequences (expected GAIIX output). To do so, the following primers were selected: L1490 & H2198 for *COI* (Folmer *et al.* 1994) and UCYTB151F & UCYTB270R for *CytB* (Merritt *et al.* 1998). PCRs were performed in 25 µl volumes using 1.5 mM MgCl<sub>2</sub>, 0.2 mM dNTPs, 0.4 mM of each primer and 0.25 µl (1.25 units) of Taq DNA polymerase (Qiagen). A thermoprofile of 3 min. at 94°C, followed by 40 cycles of 15 sec. at 94°C, 30 sec. at 50°C and 40 sec. at 72°C, and a final extension of 5 min. at 72°C was used for both markers. The PCR products were sent to Macrogen Europe (Amsterdam) where they were purified with a Montage purification kit (Millipore) and sequenced in both directions (using the same primers that were used for PCR) on an ABI3730XL. Contigs of forward and reverse sequences were assembled with Sequencher v. 4.10.1.

### 4. Assessing limitations of the DNA extract: Long PCR

Although the aim of this study is to assess the possibility of sequencing a complete mitogenome without PCRs from a collection specimen, the underlying assumption



is that enrichment of the target sequence(s) by primer-directed amplification will be difficult or impossible for these kinds of object. To test this assumption, we tried to enrich the mitochondrial fraction of the obtained DNA extract with long PCR. To increase the chance of successfully amplifying the complete mitochondrion, two *Cylindrus* specific primer-sets (A and B) were designed that each amplified (roughly) half of the mitochondrion. These two primer-sets were tested with the “Expand Long Template PCR System” of Roche (Cat. No. 11 681 834 001), following the manufacturer’s protocol. Primer-sets A and B were designed with Primer3 (Rozen and Skaletsky 2000) and face outward of the obtained *COI* and *CytB* sequences:

A-Cobt-COI-F 5’-TTACAAC TATTTT TAATATGCGTTCTCCT-3’ &  
A-Cobt-CB-R 5’-CGACGAGAAATAAAACATTTAACATAACTA-3’

B-Cobt-CB-F 5’-TACCTTTTGTGATTAGTGTTTGTGTAT-3’ &  
B-Cobt-COI-R 5’-TATTATTTATCCGGGGAAACCTTATATC-3’

We assumed that the orientation of *COI* and *CytB* would be identical to that of *C. nemoralis*.

#### 5. GAIx library preparation

For DNA extracts from fresh tissues, the first step in library preparation is fragmentation of (genomic or PCR amplified) DNA. For extracts from museum specimens, DNA can already be expected to be fragmented, which would make this step unnecessary or even detrimental (Knapp and Hoffreiter 2010 and references therein). The extent of DNA degradation will depend heavily on the preservation history. Based on the quality assessment of our DNA extract and the adverse effect that improperly fragmented DNA has on GAIx runs, we decided to follow a general library preparation procedure, for which we used the NEBNext™ DNA Sample Prep Kit (E6000-L, New England BioLabs). The DNA extract was randomly sheared with a nebulizer (K7025-05, Invitrogen) for 6 min. at 2.4 bar (35 psi) to obtain fragments in the range of 200–600 nucleotides. Fragments with a length of ca. 300 bp (insert-length without adaptors approx. 200 nucleotides) were extracted from an MS8-agarose gel (Talron Biotech. L.T.D.) with a Zymoclean Gel DNA Recovery Kit (Zymo Research, Orange, CA, USA). For all subsequent steps the NEBNext™ DNA Sample Prep Kit protocol was followed.

#### 6. GAIx run and data analysis

The prepared library was run at BaseClear B.V. (The Netherlands) on a single lane of a GAIx flow cell. CLC Genomics Workbench version 4.0 (CLC Bio, Cambridge, MA, USA) was used to filter the output and for *de novo* assembly.

#### 7. Annotation of the mitogenome

The contig sequence was annotated, based on similarity with available sequences in GenBank (Blast searches; <http://www.ncbi.nlm.nih.gov:80/BLAST/>), using pair-



wise alignments with, in particular, *C. nemoralis* and with the organellar genome annotation program DOGMA (Wyman *et al.* 2004). In order for DOGMA to detect all potential tRNAs (22 expected) we ultimately set the COVE-score to zero, but even then still some were missing. Therefore, the programs tRNAscan (Lowe and Eddy 1997) and ARWEN (Laslett and Canbäck 2008) were used as well.

## RESULTS

### *DNA extraction*

The DNA extract for the selected *C. obtusus* specimen had a yield 28  $\mu\text{g}$  (200  $\mu\text{l}$  buffer AE; concentration 139.8 ng/ $\mu\text{l}$ ) and degradation as judged on the agarose gel was limited.

### *PCR assessment*

The PCRs for the 396 bp and 705 bp (including primer sequences) fragments of *CytB* and *COI*, respectively, worked on the first attempt and were directly sequenced. We failed to make the long PCRs work; with our *C. obtusus* specific primers (see Methods: step 4), nor with any combination of the “universal” primers that successfully amplified the shorter *COI* and *CytB* sequences (L1490 & UCYTB270R; UCYTB151F & H2198).

### *Illumina GAIIX run*

The GAIIX run resulted in 34,174,164 reads with an average read length of 50 nucleotides. Of these, 685,537 reads were overlapping and used for a *de novo* assembly (CLC Bio version 4.0). This resulted in 740 contigs with a total length of 478,878 bp. The largest contig had a coverage of 26.65  $\times$ . The latter contig was identified as the mitogenome of *C. obtusus* based on the expected length (roughly 14 kb), the presence of the *COI* and *CytB* sequences (see Methods: step 3) and on similarity with mitochondrial sequences from other stylommatophorans as resulting from Blast searches.

### *Initial assignment of PCGs and rRNAs*

Twelve of the expected 13 protein-coding genes (PCGs), as well as both of the ribosomal RNAs (rRNAs) were recognized by DOGMA (Wyman *et al.* 2004); *ATP8* had to be located based on a pairwise alignment with *C. nemoralis* and *A. caerulea*. Although the gene arrangement (of the PCGs and rRNAs) as assigned by DOGMA seemed correct (compared to the gene arrangement for *C. nemoralis*), the program had difficulties determining the gene boundaries (most likely due to the absence of similar sequences on GenBank). Since we lack data from peptide sequencing for any of the PCGs, the putative gene boundaries (Table 1) were determined based on pairwise alignments with the amino acid sequences of *C. nemoralis*, *A. caerulea* and *E. herklotsi*. Nine of the PCGs start with a common initiation codon (ATA 5 $\times$ ; ATG 4 $\times$ ); the other four start with less common (but not unique for invertebrates) initiation



**Table 1.** Summary of the mitochondrial genome content of *C. obtusus*.

| Name             | One letter code | Position |      |            |        | Nr. of amino acids | Anti-codon | Inferred initiation codon | Inferred termination codon |
|------------------|-----------------|----------|------|------------|--------|--------------------|------------|---------------------------|----------------------------|
|                  |                 | from     | to   | Size       | Strand |                    |            |                           |                            |
| <i>COI</i>       |                 | 1        | 1527 | 1527       | H      | 508                |            | TTG                       | TAA                        |
| <i>tRNA-Val</i>  | V               | 1529     | 1589 | 61         | H      |                    | TAC        |                           |                            |
| <i>16 S</i>      |                 | 1590     | 2572 | 983        | H      |                    |            |                           |                            |
| <i>tRNA-Leu</i>  | L <sub>1</sub>  | 2573     | 2635 | 63         | H      |                    | TAG        |                           |                            |
| <i>tRNA-Pro</i>  | P               | 2632     | 2694 | 63         | H      |                    | TGG        |                           |                            |
| <i>ND6</i>       |                 | 2727     | 3224 | 498        | H      | 165                |            | ATA                       | TAG                        |
| <i>tRNA-Ala</i>  | A               | 3232     | 3293 | 62         | H      |                    | TGC        |                           |                            |
| <b>Region I</b>  |                 | 3294     | 3688 | <b>395</b> |        |                    |            |                           |                            |
| <i>ND5</i>       |                 | 3689     | 5368 | 1680       | H      | 559                |            | TTG                       | TAG                        |
| <i>ND1</i>       |                 | 5377     | 6249 | 873        | H      | 290                |            | ATA                       | TAG                        |
| <i>ND4L</i>      |                 | 6258     | 6507 | 250        | H      | 83                 |            | ATA                       | T                          |
| <i>CB</i>        |                 | 6526     | 7651 | 1126       | H      | 375                |            | ATG                       | T                          |
| <i>tRNA-Asp</i>  | D               | 7637     | 7696 | 60         | H      |                    | GTC        |                           |                            |
| <i>tRNA-Cys</i>  | C               | 7692     | 7752 | 61         | H      |                    | GCA        |                           |                            |
| <i>tRNA-Phe</i>  | F               | 7752     | 7814 | 63         | H      |                    | GAA        |                           |                            |
| <i>COII</i>      |                 | 7814     | 8500 | 687        | H      | 228                |            | ATG                       | TAG                        |
| <i>tRNA-Tyr</i>  | Y               | 8475     | 8528 | 54         | H      |                    | GTA        |                           |                            |
| <i>tRNA-Trp</i>  | W               | 8529     | 8590 | 62         | H      |                    | TCA        |                           |                            |
| <b>Region II</b> |                 | 8591     | 8771 | <b>181</b> |        |                    |            |                           |                            |
| <i>tRNA-Gly</i>  | G               | 8772     | 8832 | 61         | H      |                    | TCC        |                           |                            |
| <i>tRNA-His</i>  | H               | 8828     | 8889 | 62         | H      |                    | GTG        |                           |                            |
| <i>tRNA-Gln</i>  | Q               | 8891     | 8949 | 59         | L      |                    | TTG        |                           |                            |
| <i>tRNA-Leu</i>  | L <sub>2</sub>  | 8947     | 9004 | 58         | L      |                    | TAA        |                           |                            |
| <i>ATP8</i>      |                 | 8999     | 9157 | 159        | L      | 52                 |            | GTG                       | TAG                        |
| <i>tRNA-Asn</i>  | N               | 9158     | 9222 | 65         | L      |                    | GTT        |                           |                            |
| <i>ATP6</i>      |                 | 9222     | 9873 | 652        | L      | 216                |            | ATG                       | T                          |
| <i>tRNA-Arg</i>  | R               | 9874     | 9935 | 62         | L      |                    | TCG        |                           |                            |
| <i>tRNA-Glu</i>  | E               | 9933     | 9992 | 60         | L      |                    | TTC        |                           |                            |



Continued

|                   |                |       |       |      |   |     |     |     |     |
|-------------------|----------------|-------|-------|------|---|-----|-----|-----|-----|
| 12 S              |                | 10004 | 10717 | 714  | L |     |     |     |     |
| tRNA-Met          | M              | 10718 | 10779 | 62   | L |     | CAT |     |     |
| ND3               |                | 10770 | 11126 | 357  | L | 118 |     | ATA | TAA |
| tRNA-Ser          | S <sub>2</sub> | 11123 | 11177 | 55   | L |     | TGA |     |     |
| tRNA-Thr          | T              | 11177 | 11237 | 61   | L |     | TGT |     |     |
| COIII             |                | 11202 | 12014 | 813  | H | 270 |     | ATG | TAA |
| <b>Region III</b> |                | 12015 | 12203 | 189  |   |     |     |     |     |
| tRNA-Ser          | S <sub>1</sub> | 12204 | 12257 | 54   | H |     | GCT |     |     |
| ND4               |                | 12240 | 13568 | 1329 | H | 442 |     | ATC | TAA |
| tRNA-Ile          | I              | 13569 | 13631 | 63   | H |     | GAT |     |     |
| ND2               |                | 13628 | 14606 | 979  | H | 326 |     | ATA | T   |
| tRNA-Lys          | K              | 14559 | 7     | 59   | H |     | TTT |     |     |

codons (TTG 2×; ATC 1× and GTG 1×). For four PCGs (unrelated to the four just mentioned) we had to infer that they ended with a truncated termination codon (that is, the stop codon is most likely generated by posttranscriptional polyadenylation; Hatzoglou *et al.* 1995; DeJong *et al.* 2004).

The conserved regions at the beginning and end of both rRNAs (described in Fig. 4 in Hatzoglou *et al.* 1995) were found in the contig sequence of *C. obtusus* as well. However, for both *A. caerulea* and *C. nemoralis*, the annotation of *rrnS* (12S) and *rrnL* (16S) extends beyond these conserved regions. Even though the exact gene boundaries of these ribosomal genes need to be confirmed by transcript mapping, the sequence data for *A. caerulea* and *C. nemoralis* show little to no space between the rRNAs and the surrounding tRNAs. Consequently, we based the putative boundaries of 12S and 16S (Table 1) on alignments with sequences of the just mentioned species for those genes and the position of the surrounding tRNAs (described in the next paragraph).

#### Annotation of tRNAs

Because of the low COVE-score, half of the tRNAs assigned by DOGMA (17 out of 34) were false positives, but even with these relaxed settings tRNA-P(Pro), -G(Gly), -S<sub>2</sub>(Ser), -I(Ile) and -K(Lys) were missed. None of the missing tRNAs could be detected with tRNAscan. Using the least restrictive parameters, tRNAscan yielded no false positives, but merely detected four tRNAs (-L<sub>1</sub>(Leu), -N(Asn), -M(Met) and -T(Thr)) that were already assigned by DOGMA. Of the three programs tested, only Arwen found 20 of the 22 tRNAs at the cost of just one false positive. When the output of ARWEN and DOGMA was combined, all tRNAs except tRNA-G(Gly) were assigned. Apart from a swap between tRNA-P and tRNA-A(Ala), the gene order



for *C. obtusus* is identical to that of *C. nemoralis*. Alignments that included sequences of both *A. caerulea* and *C. nemoralis* (Yamazaki *et al.* 1997; Boore 1999) showed that tRNA-G is located between tRNA-W(Trp) and tRNA-H(His). In the annotated mitochondrion of *C. obtusus* (Fig. 1), there is indeed an unassigned region between tRNA-W and tRNA-H. An alignment with sequences of *C. obtusus*, *A. caerulea* and *C. nemoralis* showed a conserved sequence, TACCTTCCAAG (8797-8809) within this non-annotated region of *C. obtusus*, which represents the anticodon loop and part of the anticodon stem of tRNA-G. Consequently, this alignment was used to infer the approximate location of tRNA-G, even though the secondary structure for this tRNA could not be predicted. Additionally, the location of each tRNA was confirmed by the presence of the anti-codon. A map of the mitogenome of *C. obtusus* is depicted in figure 1. A summary of the mitochondrial genome content is given in Table 1. The corresponding annotated sequence was deposited in Genbank under accession number JN107636.

## DISCUSSION

At present, NGS techniques are being used increasingly in mitogenomic studies. One of the commonly used platforms is Roche 454 (Jex *et al.* 2008; Cui *et al.* 2009; Dietz *et al.* 2010; Feldmeyer *et al.* 2010). The ability to generate longer reads (which facilitates *de novo* assembly) and the reduction in sequence costs as compared to Illumina GAIIX, undoubtedly has added to the popularity of this platform. An overview of the costs and read lengths of these (and other) NGS platforms is given in Glen *et al.* (2011). When sequencing mitogenomes from museum specimens (Miller *et al.* 2009; Willerslev *et al.* 2009) the ability to sequence longer DNA fragments likely is of no advantage and such studies generally rely on platforms that are optimized for short fragments, such as Illumina GAIIX. Despite the momentum that NGS has provided for ancient DNA research (Knapp and Hofreiter 2010), the number of mitogenomic studies that actually use these new techniques to exploit natural history collections still is rather limited. This paper provides an example of how NGS technology can be used to retrieve genetic information from a museum specimen. We show that it is feasible to sequence a complete mitogenome, without PCR, from a snail that has been in 70% ethanol for eight years.

Thus far, to the best of our knowledge, only Feldmeyer *et al.* (2010) made use of an NGS platform to sequence the complete mitogenome of another gastropod. In that study, a similar approach (no prior enrichment of the mitochondrial fraction) was taken, albeit with freshly collected specimens and a different NGS platform (Roche 454-FLX). Despite the commonalities, there were some noteworthy differences between both studies as well. Firstly, Feldmeyer *et al.* (2010) used 13 specimens to obtain a complete mitogenome, whereas our results were obtained from a single specimen. The rationale behind using this many specimens is not given by the latter authors; perhaps they wished to account for intraspecific heterogeneity between the selected



Alanine  
(A)

```

      C
      G-C
      A-T
      A-T
      A-T
      T-A
      A-T
      A-T      T
        T      TTC
        T      !!! C
AA      A      AAG
A      TTTC      T T
T      +!!!      T
      A      GAAG      G
      A      A      A
      G-CC
      C-G
      T-A
      A-T
      G-C
      T      C
      T      A
      TGC

```

Arginine  
(R)

```

          C
          A
        A-T
        T-A
        T-A
        G-C
        T . T
        T . T
        G-C
          A
        CTCT
          C
          T
    C
    A GTGA
    C !+!
    CGCC
    C
    C
    T-AC
    C-G
    A-T
    A-T
    G-C
    T
    T
    T
    CG
  
```

Asparagine  
(N)

```

          T
        T-A
        A-T
        T-A
        A-T
        G-C
        A-T
        T+G
        A-T
          T
          T
        TGTG
    A      T      T
  A  TTGC      !!!!! T
C  !!!!!      T  C
A  AAGC      C
    A      G      G
          C CA
          C-G
          T-A
          A-T
          A-T
          C  A
          T  A
          GTT

```

Aspartate  
(D)

```

      T
      T-A
      T-A
      T.T
      T-A
      A-T
      T-A
      C-G      T
      T      TGG
      A      A      !!! + T
T TTTG      ACT
T !!!!      T A
      AAAC      A
      A      G      A
      T+GC
      C-G
      G-C
      G+T
      C-G
      T      A
      T      A
      GTC

```

Cysteine  
(C)

```

      T
      A-T
      T-A
      A-T
      A-T
      T-A
      A-T
      A-T
      T      T
      T      GCG
      A      +11 G
      T      TGC
      C TTTG      A T
      C !!!!!      A
      T AAAC      A
      A      A
      A      T-AC
      A-T
      A-T
      A-T
      G-C
      T      A
      T      A
      GCA

```

Glutamate  
(E)

```

      T
      A-T
      T-A
      T-A
      T-A
      C-G
      T . T
      G-C      C
      T      CT
A      A      ! ! C
A AATG      GA T
T ! ! ! ! T G
T TTAC      T
A      T G
      C TA
      T-A
      A-T
      C-G
      C-G
      T C
      T G
      TTC

```

Glutamine  
(Q)

G  
 T-A  
 T-A  
 C-G  
 T-A  
 T-A  
 T-A  
 T-A  
 G  
 TGT  
 G  
 TGTG  
 A  
 C +!!!!  
 A  
 GCAC  
 G  
 C  
 A  
 G  
 G-C-A  
 G-C  
 A-T  
 A-T  
 A-T  
 A  
 G  
 T  
 G  
 TTG

Histidine  
(H)

H)

```

      C
      T-A
      T-A
      G+T
      C-G
      A-T
      C-G
      G-CAG
      T
    TA   A   TA
  G   TAGG   T
    +!!!   C
  T   GTCC   C
    GTA   G   A
      T+GAAG
      T-A
      A-T
      G-C
      C-G
      T   C
      T   G
      GTG

```

Isoleucine  
(I)

```

ucine      G
           T-A
           G+T
           T-A
           G-C
           A-T
           G-C
           G-C
           C-G      A
           C      TCTG
           G      ! ! ! ! T
T GCC      AGAC
G ! ! !      G G
G CGG      A
           A      T
           T-AG
           A-T
           T+G
           T-A
           G-C
           T      T
           T      G
           GAT

```

Leucine  
(L.)

```

      A
    T-A
    G+T
    A-T
    T-A
    A-T
    A-T
    T-A
      T
        T
    AG   G   TTG
A   ACG   !!! A
T   !!!   AAC A
A   TGC   T   A
      A     A   T
          T-AT
          T-A
          T-A
          G-C
          A-T
            T   T
            T   G
            TAG

```

Leucine  
(L<sub>2</sub>)

```

C
T..T
T..T
A..T
T..A
A..T
G..C      T
      T      TTC
AG      G      TTT T
C      ACG      AAG
A      TTT      T      A
A      TGC      A
      A      T
      T..AT
      T..A
      A..T
      G..C
      C..G
      T      C
      T      G
      TAA

```

Lysine  
(K)

ne

```

      T
    T+G
  T+G
    T C
  T+G
    T.T
  T.T
    A-T      T
      T      CCT
    G      G    !!! T
  G CCG      GGA
G  !!!      C  T
T GGC      A
      A      C
        A-TG
        T-A
        A-T
        G-C
        A-T
      T  A
      T  A
      TTT
```

Methioinine  
(M)

```

          C
          A
        C-G
        T-A
        T-A
        T-A
        A-T
        A-T
        A-T
          T
          C
          C
    A      T      C
    A      T      A
    A TCGA      GG G
    C !!!!!      T T
    A AGCT      A
    A      G      A
          T-AA
          T-A
          G-C
          G-C
          G-C
          C      A
          T      A
          CAT

```

Phenylalanine  
(F)

```

nylalanine      A
                  T-A
                  G-C
                  A-T
                  T+G
                  T-A
                  T-A
                  T+G
                  A-T      C
                    T      TA
CA              A      A      ! ! C
A      TTCG      AT T
C      ! ! ! !      C A
      A AAG      A
      A      G      G
                  T-AA
                  T-A
                  G-C
                  C-G
                  T+G
                  T      G
                  T      G
                  GAA

```

Proline  
(P)

```

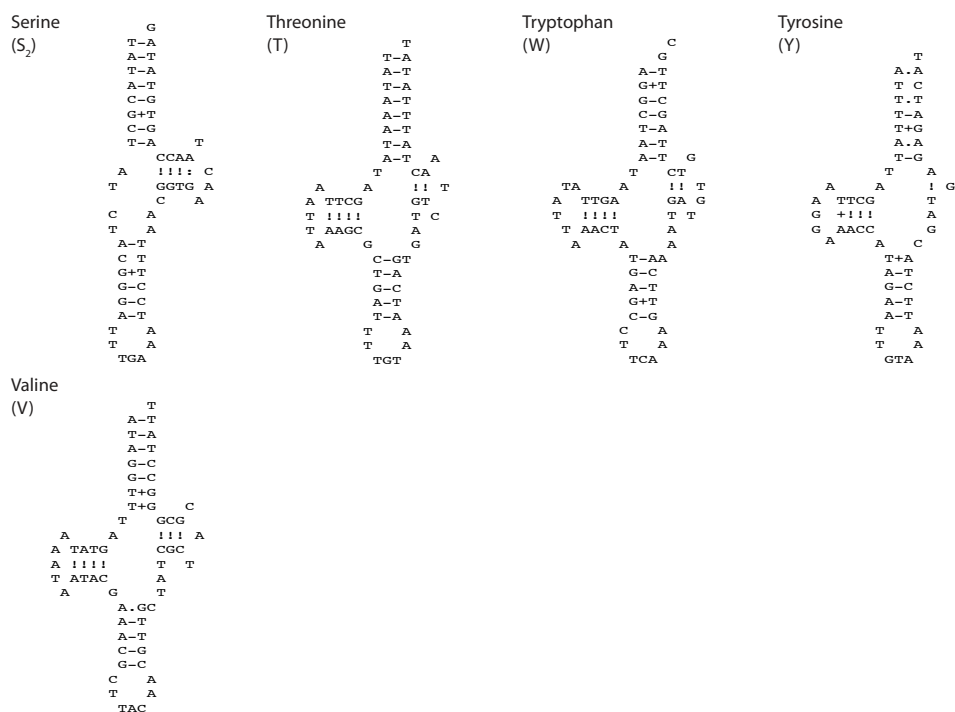
      A
    T-A
  T-A
A-T
A-T
A-T
G-C      A
      C   ATCT
     A    !!!!! A
G TATG   TAGA T
G !!!!  C      G
    ATAT   A
  A        G
          G.GT
        C-G
        A-T
        A-T
        C-G
      T      A
      T      TGG

```

Serine  
(S.)

A  
 A-T  
 G-C  
 A-T  
 G+T  
 A-T  
 A-T  
 T+G  
 T  
 CCTT  
 A 11+1 A  
 A GGGG C  
 T  
 A  
 A T  
 T  
 T-A  
 T-A  
 G+T  
 C-G  
 G+T  
 G-C  
 C A  
 T A  
 GCT





**Figure 2:** Potential secondary structures of 21 inferred tRNAs of *C. obtusus* mtDNA.

populations. It had nothing to do with the ‘sensitivity’ of the different platforms; the amount of DNA that Feldmeyer *et al.* (2010) used for their sample preparation (6  $\mu$ g), was roughly similar to what was used in this study (5  $\mu$ g). The ability to obtain a complete mitogenomic sequence from a single specimen obviously is an advantage. Secondly, in contrast to our GAIIX run, the 454 run did not cover the complete mitogenome, requiring the design of additional primers and Sanger sequencing to close the three gaps that were left over after assembly. This is also observed in other mitogenomic studies in which the 454 platform was used without prior enrichment of the mitochondrial fraction (Cui *et al.* 2009; Dietz *et al.* 2010). After filtering, the 454 run resulted in 114 reads with an average length of 318 nt (Feldmeyer *et al.* 2010) that could be assigned to the mitochondrial genome, compared to 7,808 reads (out of 34,174,164) with an average length 50 nt obtained with GAIIX. Thus the 454 run resulted in 36,252 mitogenomic nucleotides, whereas our GAIIX run yielded 390,400. The maximum coverage obtained with 454 and GAIIX is 2.6 $\times$  and 26.7 $\times$  for *R. balthica* and *C. obtusus* for mitogenome sizes of 13,993 bp and 14,610, respectively. Generally longer reads as obtained with the 454 platform facilitate *de novo* assembly and might be preferred when little *a priori* sequence information is available (Simon *et al.* 2009

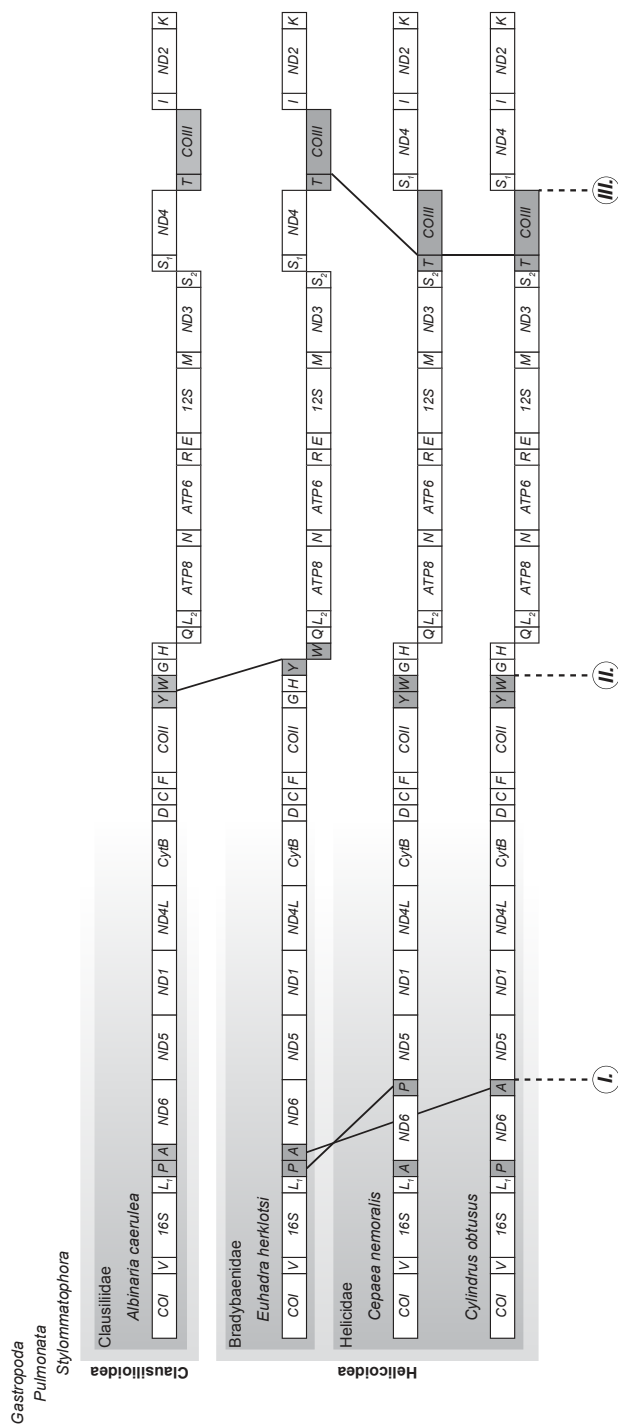


and references therein). When reconstructing the just mentioned mitogenomes the sheer number of short reads generated by the GAIIX platform outcompeted the smaller number of longer reads as obtained with 454 sequencing.

Identification of tRNAs in nematodes (Watanabe *et al.* 1994) and stylommatophorans (Hatzoglou *et al.* 1995; Terret *et al.* 1996; Yamazaki *et al.* 1997) can be difficult because the standard cloverleaf secondary structure may not be present (T or D arms can be lacking; see tRNA-H, -S<sub>1</sub> and -S<sub>2</sub> in Fig. 2) and pulmonate tRNAs can undergo post-transcriptional processing (Yokobori and Pääbo 1995; Boore 1999). In a number of instances different tRNAs were predicted in the same approximate nucleotide region, depending on the algorithm used. The hypothetical tRNAs differed a few nucleotides in length, or were shifted a few bases, or both, causing a slight shift in the anti-codon region. Despite the fact that only one of the tRNAs will be real, both algorithms correctly point to (roughly) the same nucleotide region for the placement of a tRNA. In other instances, the different algorithms predicted tRNAs on exactly the same position but on opposite strands, causing the tRNAs to be the reverse-complement of each other (Morrison 2010). Examples of the latter within *C. obtusus* are predictions by DOGMA of tRNA-W and tRNA-C on position 3632-3694 and 3232-3293, for which ARWEN predicted tRNA-P and tRNA-A, respectively. For both tRNAs, only those predicted by ARWEN were in agreement with existing annotations for related species in GenBank.

Although gene rearrangements are common in the mitogenomes of molluscs (Boore 1999; Boore *et al.* 2004) and gastropods in particular (Grande *et al.* 2008; Medina *et al.* 2009; Rawling *et al.* 2010), little is known about the mitochondrial gene organisation of terrestrial snails. The mitochondrial gene arrangements depicted by Yamazaki *et al.* (1997) and Boore *et al.* (2004) show less similarity between *C. nemoralis* and *E. herklotsi* (both species belonging to the Helicoidea), than between each of them and *A. caerulea* (a species belonging to the Clausilioidea). Based on a three-taxon statement Yamazaki *et al.* (1997) concluded that the rearrangement of the tRNAs between *COII* and *ATP8* represented a derived state in *E. herklotsi*. Similarly, they concluded that the positions of tRNA-P and the gene-region tRNA-T/*COIII* represented a derived state in *C. nemoralis*. By comparing these gene rearrangements with the gene order observed in *C. obtusus*, we can get some insight in the evolution of the mitochondrial gene order of the Helicoidea. Figure 3 gives an overview of the gene organisation of the four stylommatophoran mitogenomes currently known. Starting at *COI* the first observed rearrangement is the relocation of tRNA-P within *C. nemoralis*. Although the location of tRNA-P in *C. obtusus* is the same as that found within the other stylommatophorans, the gene region itself seems to be of high potential for rearrangements within the Helicidae. In *C. obtusus* we see that instead of tRNA-P, tRNA-A has been relocated between *ND6* and *ND5*. Thus within the Stylommatophora, the relocation of a tRNA from the L<sub>1</sub>A<sub>1</sub>P-region to between *ND6* and *ND5* seems to be a synapomorphy for the Helicidae. The second gene rearrangement (Fig. 3) is found in *E. herklotsi* within a series of tRNAs located between





**Figure 3.** Comparison of the gene order of the four known stylonmatophoran mitogenomes.

For each mitogenome genes above the horizontal line are transcribed from left to right; genes below the horizontal line are transcribed from right to left. TRNAs are denoted by their one-letter abbreviations. The diagonal lines indicate mitochondrial gene rearrangements between the specified taxa. In the gene-map of *C. obtusius*, *I-III* indicate the three unassigned regions (see Fig. 1 and Discussion). Gene sizes are not drawn to scale.

*COII* and *ATP8*. Despite the fact that these tRNAs (Y,W,G,H,Q,L<sub>2</sub>) have rearranged frequently during gastropod evolution (Grande *et al.* 2008; Medina *et al.* 2011), the gene order seems rather conserved within the Eupulmonata. The five eupulmonates currently listed in the genome database of NCBI and *C. obtusus* all show the order Y,W,G,H,Q,L<sub>2</sub>. Therefore, we endorse the conclusion of Yamazaki *et al.* (1997) that the arrangement of these tRNAs as observed in *E. herklotsi* can be considered a derived state (and possibly represents an apomorphy for the Bradybaenidae). Presence of the ancestral tRNA arrangement in both *C. obtusus* and *C. nemoralis*, suggests that that gene order has not changed in the Helicidae. The last gene rearrangement is the relocation of the gene-region tRNA-T/*COIII* in *C. nemoralis* from between *ND4* and *ND2* to between *ND3* and *ND4*. Exactly the same rearrangement is also observed in *C. obtusus*, likely indicating an apomorphy for the Helicidae.

Most metazoan mitogenomes possess a single major non-coding region presumed to contain the signals for replication and transcription. This region is usually referred to as the control region (Wolstenholme 1992). In some groups of invertebrates, such as gastropods (Hatzoglou *et al.* 1995; Grande *et al.* 2008) and spiders (Masta and Boore 2004) the mitogenomes can be very compact, hardly leaving any non-coding regions of significant length. Although the mitogenome of *C. obtusus* (14,610 bp) is still compact compared to that of other Metazoa (approx. 15-24 kb; Dellaporta *et al.* 2006), it is about half a kb larger than the mitogenomes of *C. nemoralis* (14,100 bp) and *A. caerulea* (14,130 bp). The schematic overview of this mitogenome (Fig. 1) shows three unassigned regions (indicated as *I*, *II* and *III*) with a length of 394, 181 and 189 nt, respectively. When the mitochondrial gene order of *C. obtusus* is compared with that of other stylommatophorans currently known (Fig. 3), it becomes clear that region *I* and *III* coincide with the transposition of tRNA-A and the region tRNA-T/*COIII*. As for region *II*, no gene rearrangement was observed in *C. obtusus* and neither *C. nemoralis* nor *A. caerulea* have any unassigned sequence between tRNA-W and tRNA-G. But then *E. herklotsi* does show a gene rearrangement in that region (Fig. 3). In stylommatophorans (or all gastropods for that matter) the location of the mitochondrial control region is still a subject of debate. Given the absence of region *II* in other stylommatophorans, we believe that of the three non-coding regions, region *II* is the least likely location for the control region. As for region *I*, Grande *et al.* (2002) suggested that the region between *ND6* and *ND5* might contain recognition signals for transcription in the nudibranch *Roboastra europaea*. Except for *Onchidella celtica*, which, like *C. obtusus*, has its longest non-coding sequence here (Grande *et al.* 2008), most of the heterobranch gastropods sequenced thus far show very little unassigned sequence between *ND6* and *ND5*. Therefore we assume that region *I* is not the most likely location for the control region either. Within the heterobranch gastropods the region between *COIII* and tRNA-I is most often cited as the potential location for the control region (Kurabayashi and Ueshima 2000; Grande *et al.* 2002; Grande *et al.* 2008; Medina *et al.* 2011). Because of the shown transposition of tRNA-T/*COIII* within *C. nemoralis* and *C. obtusus* (Fig. 3), this region was transposed as well. Thus for those species (and likely all Helicidae), the



potential control region (the region adjacent to *COIII*) is not located between *COIII* and tRNA-I, but between *COIII* and tRNA-S<sub>1</sub>. For *A. caerulea* and *Pupa strigosa*, the presence of 10 nt (or 20 nt if the 5 nt overlap with tRNA-I is included) and 25 nt palindromes, respectively (Hatzoglou *et al.* 1995; Kurabayashi and Ueshima 2000), has been implicated to function as a bidirectional promotor for this putative control region. Within *C. obtusus*, *C. nemoralis* and *E. herklotsi*, the palindromes found in this region were never larger than six nucleotides. We were unable to confidently align the putative control region for the four mentioned stylommatophorans. The length of this region was longer in the Helicidae (*C. obtusus* = 189 nt; *C. nemoralis* = 158 nt) than in the other two stylommatophorans (*A. caerulea* = 42 nt; *E. herklotsi* = 43 nt), which is most certainly related to the transposition of tRNA-T/*COIII*. Besides the apparent consistency of the presence of a non-coding region adjacent to *COIII*, there still is little to go on for recognition of the control region in heterobranch gastropods. Given the limited number of available stylommatophoran (and even eupulmonate) mitogenomes (GenBank), we consider more extensive genomic comparisons (such as Negrisolo *et al.* 2011) to be premature, based on these data.

Although this study illustrates the potential of NGS to obtain genetic information from museum specimens, there are some caveats that need to be addressed. As for the results of the long PCR, we did not have a recently collected specimen of *C. obtusus* at our disposal. Therefore, we have not shown that the failure of the long PCR was caused by the fact that we used an ‘aged’ collection specimen. Otherwise, no products would have been obtained with the long PCRs either, based, as they were, on the universal primers, which were shown to work with *C. obtusus*. The fact that an NGS approach worked well in this study, also does not imply that NGS approaches will always be fruitful when used on collection specimens. A vast number of parameters such as the effect of fixatives, time (the “age” of specimens) and preservation history were not assessed in this study. Also annotation of a mitogenome largely based on similarity with available sequences in GenBank (as opposed to transcript mapping or peptide sequencing) is hazardous. Closely related species might not be present and existing annotations are not guaranteed to be flawless (Harris 2003; Groenenberg *et al.* 2011). Another hypothetical problem is that due to the relative short length of the obtained sequences (50 nucleotides on average) repeats within the mitogenome could be missed. The mitogenomes of gastropods however are very compact, none of the stylommatophorans sequenced thus far show such repeats and the length of the complete mt sequence is similar to that of other stylommatophorans. We therefore assume that the mitogenome of *C. obtusus* presented here is complete. Based on the sheer number of sequences generated with GAIx and 454, we are convinced that without PCRs it will be more difficult to obtain a complete mitogenome with the latter platform (despite the longer reads). Even though our GAIx approach indeed is reasonably similar to the 454 approach described by Feldmeyer *et al.* (2010), the comparison will not be conclusive as long as the total genome sizes of *C. obtusus* and *R. balthica* are unknown. The average genome size (<http://www.genomesize.com>) of available stylommatophorans (2.86 c or 2.79



GB) compared to basommatophorans (1.34 c or 1.31 GB) nevertheless suggest that it will be more difficult to obtain the mitogenome of *C. obtusus* than of *R. balthica*. It is likely that the assembly of mitogenomes will benefit from the advances in NGS technologies (e.g. the Illumina HiSeq platform), as well as from the promising arrival of third generation sequencers (e.g. the PacBio RS platform).

## CONCLUSION

On a par with previous studies (Miller *et al.* 2009; Willerslev *et al.* 2009), this study shows that NGS can aid in the retrieval of mitogenomes from museum specimens. Although sequencing of mitogenomes by means of NGS without enrichment procedure is very inefficient (only 0.02% of the reads from our GAIIX run were used for assembly of the mitogenome), it eliminates the use of PCRs which is often a bottleneck for degraded DNA samples. Without prior enrichment of the mitochondrial fraction, the GAIIX platform (Illumina) might be better suited for *de novo* sequencing of mitogenomes than the 454 platform (Roche). Besides being much faster than conventional sequencing (which generally results in 2× coverage), sequencing of mitogenomes by means of NGS also yields higher confidence estimates (on average 26× times coverage, in this study). Except for a swap between tRNA-P and tRNA-A, the mitochondrial gene arrangement of *C. obtusus* is identical to that of *C. nemoralis*. Within the Helicidae the region tRNA-L<sub>1</sub>,P,A might be a hot spot for transposition of genes (in particular to the region between ND6 and ND5). The location of tRNA-T/COIII between ND3 and ND4 (instead of between ND4 and ND2) might be an apomorphy for the family Helicidae. We hope that the results of this study will aid to future studies on stylommatophoran evolution and the phylogeny of the subfamily Ariantinae in particular.

## ACKNOWLEDGEMENTS

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## REFERENCES

- ARTHOFFER, W., CADAHÍA, L. & KRUCKENHAUSER, L. 2009. Ten new microsatellite loci for analysis of genetic diversity in isolated populations of the Alpine land snail *Cylindrus obtusus*. *Conservation Genetics* 11: 1115-1118.
- BOORE, J.L. 1999. Animal mitochondrial genomes. *Nucleic Acids Research* 27: 1767-1780.
- BOORE, J.L., MEDINA, M. & ROSENBERG, L.A. 2004. Complete sequences of the highly



- rearranged molluscan mitochondrial genomes of the scaphopod *Graptacme eborea* and the bivalve *Mytilus edulis*. *Molecular Biology and Evolution* 21: 1492-1503.
- BROUGHTON, R.E., MILAM, J.E. & ROE, B.A. 2001. The complete sequence of the Zebrafish (*Danio rerio*) mitochondrial genome and evolutionary patterns in vertebrate mitochondrial DNA. *Genome research* 11: 1958-1967.
- CUI, Z., LIU, Y., LI, C.P., YOU, F. & CHU, K.H. 2009. The complete mitochondrial genome of the large yellow croaker, *Larimichthys crocea* (Perciformes, Sciaenidae): Unusual features of its control region and the phylogenetic position of the Sciaenidae. *Gene* 432: 33-43.
- DE JONG, R.J., EMERY, A.M. & ADEMA, C.M. 2004. The mitochondrial genome of *Biomphalaria glabrata* (Gastropoda: Basommatophora), intermediate host of *Schistosoma mansoni*. *Journal of Parasitology* 90: 991-997.
- DELLAPORTA, S.L., XU, A., SAGASSER, S., JAKOB, W., MORENO, M.A., et al. 2006. Mitochondrial genome of *Trichoplax adhaerens* supports Placozoa as the basal lower metazoan phylum. *Proceeding of the National Academy of Sciences USA* 103: 8751-8756.
- DIETZ, L., MAYER, C., ARANGO, C.P. & LEESE, F. 2010. The mitochondrial genome of *Colossendeis megalonyx* supports a basal position of Colossendeidae within the Pycnogonida. *Molecular Phylogenetics and Evolution* 58: 553-558.
- DUDA, M., KRUCKENHAUSER, L., HARING, E. & SATTMANN, H. 2010. Habitat requirements of the pulmonate land snails *Trochulus oreinos oreinos* and *Cylindrus obtusus* endemic to the Northern Calcareous Alps, Austria. *Eco.mont* 2: 2.
- FELDMAYER, B., HOFFMEIER, K. & PFENNINGER, M. 2010. The complete mitochondrial genome of *Radix balthica* (Pulmonata, Basommatophora), obtained by low coverage shot gun next generation sequencing. *Molecular Phylogenetics and Evolution* 57: 1329-1333.
- FOLMER O., BLACK, M., HOEH, W., LUTZ, R. & VRIJENHOEK, R. 1994. DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. *Molecular Marine Biology and Biotechnology* 3: 294-9.
- FRANK, C. 2006. Plio-pleistozäne und holozäne Mollusken Österreichs 2, pp. 397-861 in Mitteilungen der prähistorischen Kommission 62, edited by Frank C. & Grillitsch H. Verlag der Österreichischen Akademie der Wissenschaften, Wien.
- GITTENBERGER E., PIEL, W.H. & GROENENBERG, D.S.J. 2004. The Pleistocene glaciations and the evolutionary history of the polytypic snail species *Arianta arbustorum* (Gastropoda, Pulmonata, Helicidae). *Molecular Phylogenetics and Evolution* 30: 64-73.
- GLEN T.C. 2011. Field guide to next-generation DNA sequencers. *Molecular Ecology Resources* 11: 759-769.
- GRANDE C., TEMPLADO, J., CERVERA, J.L. & ZARDOYA, R. 2002. The complete mitochondrial genome of the nudibranch *Roboastra europaea* (Mollusca: Gastropoda) supports the monophyly of opisthobranchs. *Molecular Biology and Evolution* 19: 1672-1685.
- GRANDE, C., TEMPLADO, J. & ZARDOYA, R. 2008. Evolution of gastropod mitochondrial genome arrangements. *BMC Evolutionary Biology* 8: 61.
- GROENENBERG, D.S.J., NEUBERT, E. & GITTENBERGER, E. 2011. Reappraisal of the "Molecular phylogeny of Western Palaearctic Helicidae s.l. (Gastropoda: Stylommatophora)": when poor science meets GenBank. *Molecular Phylogenetics and Evolution* 61: 914-923.
- HARRIS, D.J. 2003. Can you bank on GenBank? *Trends in Ecology and Evolution* 18: 317-319.



- HATZOGLU, E., RODAKIS, G.C. & LECANIDOU, R. 1995. Complete sequence and gene organisation of the land snail *Albinaria coerulea*. *Genetics* 140: 1353-1366.
- JEX, A. R., HU, M., LITTLEWOOD, D.T.J., WAESCHENBACH, A. & GASSER, R.B. 2008 Using 454 technology for long-PCR based sequencing of the complete mitochondrial genome from single *Haemonchus contortus* (Nematoda). *BMC Genomics* 9: 11.
- KI, J.S., HWANG, D.S., PARK, T.J., HAN, S.H. & LEE, J.S. 2010. A comparative analysis of the complete mitochondrial genome of the Eurasian otter *Lutra lutra* (Carnivora; Mustelidae). *Molecular Biology Reports* 37: 1943-1955.
- KRAJICK, K. 1998. Nunataks, icebound islands of life. *National Geographic* 194: 60-71.
- KNAPP, M. & HOFREITER, M. 2010. Next Generation Sequencing of Ancient DNA: Requirements, Strategies and Perspectives. *Genes* 1: 227-243.
- KURABAYASHI, A. & UESHIMA, R. 2000. Complete sequence of the mitochondrial DNA of the primitive opisthobranch *Pupa strigosa*: Systematic implications of the genome organization. *Molecular Biology and Evolution* 17: 266-277.
- LASLETT, D. & CANBÄCK, B. 2008. ARWEN: a program to detect tRNA genes in metazoan mitochondrial nucleotide sequences. *Bioinformatics* 24: 172-175.
- LOWE, T. M. & EDDY, S.R. 1997. tRNAscan-SE: a program for improved detection of transfer RNA genes in genomic sequence. *Nucleic Acids Research* 25: 955-964.
- MASTA, S.E. & BOORE, J.L. 2004. The complete mitochondrial genome sequence of the spider *Habronattus oregonensis* reveals rearranged and extremely truncated tRNAs. *Molecular Biology and Evolution* 21: 893-902.
- MEDINA, M., LAL, S., VALLÈS, Y., TAKAOKA, T.L., BENOÎT, A. *et al.* 2011. Crawling through time: Transition of snails to slugs dating back to the Paleozoic, based on mitochondrial phylogenomics. *Marine Genomics* 4: 51-59.
- MERRITT, T.J.S., SHI, L., CHASE, M.C., REX, M.A., ETTER, R.J. *et al.* 1998. Universal cytochrome b primers facilitate intraspecific studies in molluscan taxa. *Molecular Marine Biology and Biotechnology* 7: 7-11.
- MEYER, M., STENZEL, U., MYLES, S., PRÜFER, K. & HOFREITER, M. 2007. Targeted high-throughput sequencing of tagged nucleic acid samples. *Nucleic Acids Research* 35: 15.
- MILLER, W., DRAUTZ, D.I., JANECKA, J.E., LESK, A.M., RATAN, A. *et al.* 2009. The mitochondrial genome sequence of the Tasmanian tiger (*Thylacinus cynocephalus*). *Genome Research* 19: 213-220.
- MIYA, M., TAKESHIMA, H., ENDO, H., ISHIGURO, N.B., INOUE, J.G. *et al.* 2003. Major patterns of higher teleostean phylogenies: a new perspective based on 100 complete mitochondrial DNA sequences. *Mol. Phylogenet. Evol.* 26: 121-138.
- MORIN, P.A., ARCHER, F.I., FOOTE, A.D., VILSTRUP, J., ALLEN, E.E. *et al.* 2010. Complete mitochondrial genome phylogeographic analysis of killer whales (*Orcinus orca*) indicates multiple species. *Genome Research* 20: 908-916.
- MORRISON, D.A. 2010. How and where to look for tRNAs in Metazoan mitochondrial genomes, and what you might find when you get there. Online Archive arXiv.org, ID:arXiv:1001:3813v1 [q-bio-GN]
- NEGRISOLO, E., BABBUCCI, M. & PATARNELLO, T. 2011. The mitochondrial genome of the ascalaphid owlfly *Libelloides macaronius* and comparative evolutionary mitochondriomics of neuropterid insects. *BMC Genomics* 12: 221.



- RAWLING, T.A., MACINNIS, M.J., BIELER, R., BOORE, J.L. & COLLINS, T.M. 2010. Sessile snails, dynamic genomes: gene rearrangements within the mitochondrial genome of a family of caenogastropod molluscs. *BMC Genomics* 11: 440.
- ROZEN, S. & SKALETSKY, H.J. 2000. Primer3 on the WWW for general users and for biologist programmers, pp. 365-386 in *Bioinformatics Methods and Protocols: Methods in Molecular Biology*, edited by Krawetz S. & Misener S. Humana Press, Totowa, NJ.
- SAITOH, K., SADO, T., MAYDEN, R.L., HANZAWA, N., NAKAMURA, K. *et al.* 2006. Mitogenomic evolution and interrelationships of the Cypriniformes (Actinopterygii: Ostariophysi): the first evidence toward resolution of higher-level relationships of the world's largest freshwater fish clade based on 59 whole mitogenome sequences. *Journal of Molecular Evolution* 63: 826-841.
- SIMISON, W.B. & BOORE, J.L. 2008. Molluscan Evolutionary Genomics, pp. 447-461 in: *Phylogeny and evolution of the Mollusca*, edited by Ponder W. F. & Lindberg D.R. University of California Press, Berkeley.
- SIMON, S.A., ZHAI, J., NANDETY, R.S., MCCORMICK, K.P., ZENG, J. *et al.* 2009. Short-read sequencing technologies for transcriptional analyses. *Annual Review of Plant Biology* 60: 305-33
- TERRETTI, J.A., MILES, S. & THOMAS, R.H. 1996. Complete DNA sequence of the mitochondrial genome of *Cepaea nemoralis* (Gastropoda: Pulmonata). *Journal of Molecular Evolution* 42: 160-168.
- YAMAZAKI, N., UESHIMA, R., TERRETTI, J., YOKOBORI, S., KAIFU, M. *et al.* 1997. Evolution of pulmonate gastropod mitochondrial genomes: comparisons of gene organizations of *Euhadra*, *Cepaea* and *Albinaria* and implications of unusual tRNA secondary structures. *Genetics* 145: 749-758.
- YOKOBORI, S. & PÄÄBO, S. 1995. Transfer RNA editing in land snail mitochondria. *Proceedings of the National Academy of Sciences USA* 92: 10432-10435.
- WATANABE, Y., TSURUI, H., UEDA, T., FURUSHIMA, R., TAKAMIYA, S. *et al.* 1994. Primary and higher order structures of nematode (*Ascaris suum*) mitochondrial tRNAs lacking either the T or D stem. *Journal of Biological Chemistry* 269: 22902-22906.
- WILLERSLEV, E., GILBERT, M.T.P., BINLADEN, J., HO, S.Y.W., CAMPOS, P.F. *et al.* 2009. Analysis of complete mitochondrial genomes from extinct and extant rhinoceroses reveals lack of phylogenetic resolution. *BMC Evolutionary Biology* 9: 95.
- WOLSTENHOLME, D.R. 1992 Animal mitochondrial DNA: structure and evolution. *International Review of Cytology* 141: 173-216.
- WYMAN, S.K., JANSEN, R.K. & BOORE, J.L. 2004. Automatic annotation of organellar genomes with DOGMA. *Bioinformatics* 20: 3252-3255.
- ZHANG P., CHEN, Y., ZHOU, H., WANG, X. & QU, L. 2003. The complete mitochondrial genome of a relic salamander, *Ranodon sibiricus* (Amphibia: Caudata) and implications for amphibian phylogeny. *Molecular Phylogenetics and Evolution* 28: 620-626.





## CURRICULUM VITAE

Dirk Schilman Jakob Groenenberg (Dick) werd op 29 juli 1977 geboren te Heerjansdam. Na de basisschool (Koningin Juliana School, eveneens te Heerjansdam) had Dick besloten dat studieboeken een noodzakelijk kwaad waren, waar hij zo min mogelijk mee van doen moest hebben. Een logische gevolg hiervan was zijn keuze om naar de 'ambachtsschool' (L.T.S. - lager technisch onderwijs) in, toen nog buurgemeente, Zwijndrecht te gaan. Ondanks dat het eerste jaar goed werd doorlopen, bood de opleiding hem onvoldoende uitdaging. Daarop besloot Dick naar het Walburgcollege (eveneens in Zwijndrecht) te gaan, waar zes jaar later het Atheneum diploma werd behaald. In datzelfde jaar, 1996, werd begonnen aan de studie Biologie aan de Rijksuniversiteit Leiden, waarop in 1997 het propedeuse-diploma werd behaald. Als differentiatierichting werd Moleculaire Biologie gekozen, waarbij de fascinatie voor evolutionaire vraagstukken toch groter bleek dan voor celbiologische processen. Tijdens de specialisatie resulteerde dit in een stage bij de sectie Theoretische Biologie en Fylogenetische Systematiek bij Prof. Dr. E. Gittenberger aan het toenmalige Instituut voor Evolutionaire en Ecologische Wetenschappen (E.E.W.). Met Dr. W.H. Piel als dagelijkse begeleider werd onderzoek gedaan naar de moleculaire fylogenie van het landslakkengenus *Arianta*, hetgeen resulteerde in een publicatie in het tijdschrift *Molecular Phylogenetics & Evolution* (Gittenberger *et al.*, 2004). Aan het eind van zijn studie heeft Dick een half jaar als analist voor de moleculaire labs van het zojuist genoemde E.E.W. gewerkt, waarna hij in 2001 afstudeerde (specialisatie: moleculaire diersystematiek). In 2002 kwam hij als projectmedewerker (moleculair lab) in dienst bij museum Naturalis, waar hij sindsdien werkzaam is. In 2007 werd Dick bij hetzelfde museum aangesteld als onderzoeker en sinds 2010 is hij formeel toegelaten als promovendus aan de faculteit der Wiskunde & Natuurwetenschappen van de Universiteit Leiden. Dit proefschrift is een reflectie van het onderzoek dat Dick, met name in de periode vanaf 2007, voor het museum (het huidige Naturalis Biodiversity Center) heeft uitgevoerd.



## PUBLICATIONS

### *In Prep.*

GROENENBERG, D.S.J., SUBAI P & GITTENBERGER, E. Molecular phylogeny of the Ariantinae (Gastropoda, Pulmonata, Helicidae), a new approach to an old problem.

### *Submitted.*

LAMMERS Y., KREMER, D., BRAKEFIELD, P., GROENENBERG, D.S.J., PIROVANO, W. & SCHILTHUIZEN, M. SNP genotyping for detecting the 'rare allele phenomenon' in hybrid zones

### *Accepted*

GROENENBERG, D.S.J., DE COURCY, C., LINNIE, M. & OOSTERWEGHEL, L. 2012 On the identity of the first rhinoceros owned by of the Dublin Zoo (†1865); genetic characterisation of a poorly preserved museum specimen. *Biology and Environment: Proceedings of the Royal Irish Academy*. doi: 103318/BIOE.2012.18

### **2012**

NIEUKERKEN VAN, E.J., DOORENWEERD, C., STOKVIS, F.R. & GROENENBERG, D.S.J. 2012. DNA barcoding of the leaf-mining moth subgenus *Ectoedemia* s. str. (Lepidoptera: Nepticulidae) with COI and EF1- $\alpha$ : two are better than one in recognising cryptic species. *Contributions to Zoology* 81 (1) 1-23 (2012)

GROENENBERG, D.S.J., PIROVANO W., GITTENBERGER E. & SCHILTHUIZEN M. 2012. The complete mitogenome of *Cylindrus obtusus* (Helicidae, Ariantinae) using Illumina Next Generation Sequencing. *BMC Genomics* 13:114  
doi:10.1186/1471-2164-13-114

### **2011**

GROENENBERG, D.S.J., NEUBERT E. & GITTENBERGER E. 2011. Reappraisal of the "Molecular phylogeny of Western Palaearctic Helicidae s.l. (Gastropoda: Stylommatophora)": when poor science meets GenBank. *Molecular Phylogenetics and Evolution* 61(3):914-23.  
doi:10.1016/j.ympev.2011.08.024

SCHILTHUIZEN M., SCHOLTE C., VAN WIJK R.E.J., DOMMERSHUIJZEN J., HORST, VAN DER D., MEIJER ZU SCHLOCHTERN M., LIEVERS R. & GROENENBERG D.S.J. 2011. Using DNA-barcoding to make the necrobiont beetle family Cholevidae accessible for forensic entomology. *Forensic Science International* 210: 91–95.



GROENENBERG, D.S.J., WESSELINGH, F.P., RAJAGOPAL S., JANSEN J.M., BOS M.M., VELDE VAN DER G., GITTENBERGER E., HOEKSEMA B.W., RAAD H. & HUMMEL H. 2011. On the identity of broad-shelled mussels (Mollusca, Bivalvia, Mytilus) from the Dutch delta region. *Contributions to Zoology* 80 (2) 95-106.

GROENENBERG, D.S.J. & DEKKER, R.W.R.J. 2011. A mouse's tail: how to settle an insurance dispute. *Forensic Science International* 207, e24–e27.  
doi:10.1016/j.forsciint.2010.11.022

#### 2010

BREURE, A.S.H., GROENENBERG, D.S.J. & SCHILTHUIZEN, M. 2010. New insights in the phylogenetic relations within the Orthalicoida (Gastropoda, Stylommatophora) based on 28S sequence data. *Basteria* 74 (1-3): 25-31.

#### 2009

GROENENBERG, D.S.J., GOUD, J., DE HEIJ, A. & GITTENBERGER, E. 2009. Molecular Phylogeny of North Sea Sepiolinae (Cephalopoda: Sepiolidae) reveals an overlooked Sepiola species. *Journal of Molluscan studies* 75 (4): 361-369.  
doi: 10.1093/mollus/eyp032

#### 2008

GROENENBERG, D.S.J., BEINTEMA, A.B., DEKKER, R.W.J.R. & GITTENBERGER E. 2008. Ancient DNA elucidates the controversy about the Flightless Island Hens (*Gallinula* sp.) of Tristan da Cunha. *PLoS ONE* 3(3): e1835.  
doi:10.1371/journal.pone.0001835

REEMER, M., GROENENBERG, D.S.J., VAN ACHTERBERG, C. & PEETERS, T.M.J. 2008. Taxonomic assessment of *Andrena rosae* and *A. stragulata* by DNA-sequencing (Hymenoptera: Apoidea: Andrenidae). *Entomologia Generalis* 31(1): 21-32.

#### 2007

OFWEGEN, VAN L.P. & GROENENBERG, D.S.J. 2007. A centuries old problem in nephtheid taxonomy approached using DNA data (Coelenterata: Alcyonacea). *Contributions to Zoology* 76, p. 153-178.

ARNTZEN, J.W., GROENENBERG, D.S.J., ALEXANDRINO J., SEQUEIRA F. & FERRAND N. 2007. Geographical variation in the Golden-striped salamander, *Chioglossa lusitanica* Bocage, 1864 and the description of a newly recognized subspecies. *Journal of Natural History* 41(13-16): 925-936.



## 2006

GITTENBERGER, E., GROENENBERG, D.S.J., KOKSHOORN, B. & PREECE, R. C. 2006. Molecular trails from hitch-hiking snails. *Nature* 439, 409-409.  
doi:10.1038/439409a

UIT DE WEERD, D.R., GROENENBERG, D.S.J., SCHILTHUIZEN, M. & GITTENBERGER, E. 2006. Reproductive character displacement by inversion of coiling in clausiliid snails (Gastropoda, Pulmonata). *Biological Journal of the Linnean Society* 88, 155-164.

## 2004

GITTENBERGER, E., GROENENBERG, D.S.J. & WESSELINGH, F. 2004. Mysterious mussels, *Mytilus* spec. (Bivalva, Mytilidae) in the Netherlands. *Vita Malacologica* 2: 50-51.

GITTENBERGER, E., PIEL, W.H. & GROENENBERG, D.S.J. 2004. The Pleistocene glaciations and the evolutionary history of the polytypic snail species *Arianta arbustorum* (Gastropoda, Pulmonata, Helicidae). *Molecular Phylogenetics and Evolution* 30, 64-73.



## PHOTO CREDITS

### Cover illustrations:

|                             |                                                                            |
|-----------------------------|----------------------------------------------------------------------------|
| <i>Cylindrus obtusus</i>    | D.S.J. Groenenberg & E. Gittenberger                                       |
| <i>Sepiola atlantica</i>    | J. Goud                                                                    |
| <i>Gallinula comeri</i>     | A.J. Beintema                                                              |
| <i>Apodemus sylvaticus</i>  | P. van Hoof ( <a href="http://www.paulvanhoof.nl">www.paulvanhoof.nl</a> ) |
| <i>Rhinoceros unicornis</i> | D.S.J. Groenenberg                                                         |
| <i>Helicopsis striata</i>   | D.S.J. Groenenberg & E. Gittenberger                                       |

|                                       |                                      |
|---------------------------------------|--------------------------------------|
| Tristan moorhen stamp                 | Tristan da Cunha Post Office         |
| <i>Arianta arbustorum</i>             | D.S.J. Groenenberg & E. Gittenberger |
| <i>Chioglossa lusitanica longipes</i> | F. Deschandol                        |
| <i>Mytilus edulis</i>                 | F.P. Wesselingh                      |
| <i>Andrena rosae</i>                  | J. Visser                            |

|                             |                    |
|-----------------------------|--------------------|
| Author 'on expedition'      | M. Kolet           |
| <i>Rhinoceros unicornis</i> | D.S.J. Groenenberg |

### Chapter icons:

|            |                                                                            |
|------------|----------------------------------------------------------------------------|
| Chapter 1  | D.S.J. Groenenberg                                                         |
| Chapter 2  | H. Hillewaert (wikipedia; CC-BY-SA-3.0)                                    |
| Chapter 3  | D.S.J. Groenenberg & E. Gittenberger                                       |
| Chapter 4  | F.P. Wesselingh                                                            |
| Chapter 5  | J. Visser                                                                  |
| Chapter 6  | D.S.J. Groenenberg & E. Gittenberger                                       |
| Chapter 7  | P. van Hoof ( <a href="http://www.paulvanhoof.nl">www.paulvanhoof.nl</a> ) |
| Chapter 8  | D.S.J. Groenenberg                                                         |
| Chapter 9  | A.J. Beintema                                                              |
| Chapter 10 | F. Deschandol                                                              |
| Chapter 11 | T. Meijer & D.S.J. Groenenberg.                                            |



*Illustrations within chapters:*

- p.31     References are given in Fig. 1; the arms of *Sepiola pfefferi* by J. Goud  
p.32     References are given in Fig. 2  
p.58     Fig. 1 compilation of shells by D.S.J. Groenenberg & E. Gittenberger  
p.80     Fig. 1 *Mytilus* specimens by F.P. Wesselingh  
p.112    Fig. 5 *Andrena* specimens by M. Reemer  
p.122    Fig. 1 compilation of shells by D.S.J. Groenenberg & E. Gittenberger  
p.173    Fig. 1 *Apodemus sylvaticus* by S. van der Mije & E. Kruidenier  
p.188    Fig. 2ab *Rhinoceros unicornis* / *sondaicus* after Sody (1959)  
          Fig. 2c *R. unicornis* -skin by D.S.J. Groenenberg  
p.199    Fig. 3 *R. unicornis* -skeleton by D.S.J. Groenenberg  
p.201    Fig. 2 stamps by Tristan da Cunha Post Office  
p.217    Fig. 1 *Chioglossa lusitanica longipes* by F. Deschandol  
p.234    Fig. 1 *Cylindrus obtusus* by T. Meijer (wikipedia; CC-BY-SA-3.0)





D.S.J. Groenenberg (Dick) studied biology at Leiden University. As an MSc student he worked on the molecular phylogeny of the European land snail genus *Arianta*. After he graduated in 2001, specializing in molecular systematics, he was appointed as a technician by the Dutch Natural History Museum, Naturalis, in Leiden. Although Dick worked on a wide diversity of animal groups, the terrestrial snails, particularly the subfamily Ariantinae, fascinated him the most (Chapters 3, 6 & 11). In 2007, Dick was appointed by the same museum as a scientific researcher, and the work described in this thesis mainly results from the subsequent period.

Stuffed birds, mounted skeletons, pickled molluscs, and pinned insects: some of the many objects in natural history collections. These began as eclectic collections of curious objects (rariteitenkabinetten), but later, Linnaeus' *Systema Naturae* ordered them into natural groups. The delineation of these groups has been a matter of debate ever since. Traditionally, groups were separated based on habitus and anatomical affinities. Nowadays, groups are designed to reflect their historical origins, and this history is illuminated by studying their DNA. This thesis explores the congruence and conflict between morphological and molecular taxonomy of natural history collections.

